



HYDERABAD GROWTH CORRIDOR LIMITED

HGCL Office Building, 2nd floor, Khajaguda Road,

Adjacent to ORR, Nanakramguda,

Hyderabad-500104

NAME OF THE WORK: Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of cumulative 13 MW (AC) Solar PV Power Plant on rooftop of Cycle track at 2 stretches along the ORR with 5 years Plant O&M.

Bid Document

Notice Inviting Tender

For

**Design, Engineering, Supply, Construction, Erection,
Testing & Commissioning of cumulative 13 MW (AC) Solar
PV Power Plant on rooftop of Cycle track at 2 stretches
along the ORR with 5 years Plant O&M**

At

**Hyderabad Growth Corridor Limited
(HGCL), Hyderabad.**

**BID NOTICE No.178/CGM(T)/HGCL/DGM(Elec)/Cycle
Track/2022-23/second call/Dated:24.04.2023**

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DISCLAIMER

1. Though adequate care has been taken while preparing the tender document, the bidder(s) shall satisfy themselves that the document is complete in all respect. Intimation regarding any discrepancy shall be given to the office of Employer immediately. If no intimation is received from any bidder within **7 (Seven) days from the date of issuance of Tender documents**, it shall be considered that the document is complete in all respect and has been received/ acknowledged by the bidder(s).
2. Hyderabad Growth Corridor Limited (HGCL) reserves the right to modify, amend or supplement this document.
3. While this tender document has been prepared in good faith, neither HGCL nor their employees or advisors make any representation or warranty, express or implied, or accept any responsibility or liability, whatsoever, in respect of any statements or omissions herein, or the accuracy, completeness or reliability of information, and shall incur no liability under any law, statute, rules or regulations as to the accuracy, reliability or completeness of this document, even if any loss or damage is caused by any act or omission on their part.
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Place: Hyderabad

Date:24/04/2023

SECTION - I

INVITATION FOR BIDS (IFB)

**(DOMESTIC COMPETITIVE BIDDING)
(DOMESTICALLY MANUFACTURED SOLAR PV MODULES)
(SINGLE STAGE TWO PART BIDDING)
UNDER E-TENDERING**

INVITATION FOR BIDS (IFB)

FOR

**DESIGN, ENGINEERING, SUPPLY, CONSTRUCTION, ERECTION, TESTING & COMMISSIONING
OF 13 MW (AC) Solar PV Power Plant on rooftop of Cycle track at 2 stretches along the ORR
with 5 years Plant O&M**

HYDERABAD GROWTH CORRIDOR LIMITED (HGCL), HYDERABAD

**TOTAL PROJECT CAPACITY OF 13 MW (AC) IS DIVIDED INTO 02 DIFFERENT STRETCHES
ALONG THE ORR AS PER BELOW TABLE:**

S NO	STRETCH DETAILS	SITE CAPACITY (MW)	LOCATIONS
1	STRETCH – 1	3MW (2MW Captive+ 1MW Net Metering)	Nanakramguda Junction to TSPA Junction
2	STRETCH – 2	10MW (9MW Captive+ 1MW Net Metering)	Jayabheri,Kokapet to Edunagulapally Junction

DOMESTIC CONTENT REQUIREMENT:

THE RFS MANDATES USE OF SOLAR PHOTOVOLTAIC (SPV) MODULES MANUFACTURED DOMESTICALLY AS PER SPECIFICATIONS AND TESTING REQUIREMENTS FIXED BY MNRE. FOR THE PROJECTS TO BE IMPLEMENTED UNDER THIS TENDER, SOLAR MODULES USED IN THE SOLAR POWER PROJECTS MUST BE MADE IN INDIA, PREFERABLY FROM THE MANUFACTURERS LISTED IN ALMM LIST, PUBLISHED & UPDATED BY MNRE FROM TIME TO TIME. IN CASE OF CRYSTALLINE SILICON TECHNOLOGY, ALL PROCESS STEPS AND QUALITY CONTROL MEASURES INVOLVED IN THE MANUFACTURE OF THE SOLAR CELLS AND MODULES STARTING FROM WAFERS UNTIL FINAL ASSEMBLY OF THE SOLAR CELLS INTO MODULES SHALL BE PERFORMED AT THE WORKS OF PV MANUFACTURERS IN INDIA.

ELIGIBILITY CONDITIONS SET FORTH IN THE TENDER DOCUMENT ARE AS PER 13 MW (AC) CUMULATIVE CAPACITY BASIS ONLY. BIDDERS ARE REQUIRED TO QUOTE ACCORDINGLY FOR THE COMPLETE CUMULATIVE CAPACITY OF 13 MW (AC). THE TOTAL CAPACITY OF 13 MW WILL BE AWARDED TO A SINGLE SUCCESSFUL BIDDER ONLY.

THE PROJECT BOQ (BILL OF QUANTITIES) O&M PRICE WILL BE CALCULATED BY MULTIPLYING PERCENTAGE QUOTED BY THE L1 BIDDER AND TOTAL WORK ORDER VALUE WILL BE FINALIZED. CONTRACT AGREEMENT WILL BE RELEASED BY THE HGCL TO THE SINGLE SUCCESSFUL BIDDER ONLY.

EMD CAN BE SUBMITTED BY THE BIDDERS JOINTLY FOR THE COMPLETE 13 MW PROJECT CAPACITY. HOWEVER, THE SUCCESSFUL BIDDER WILL BE REQUIRED TO SUBMIT PERFORMANCE SECURITIES & ADVANCE BANK GUARANTEES (IF AVAILED) FOR CONTRACT AGREEMENTS WITH THE SUCCESSFUL BIDDER. HGCL RESERVES THE OPTION TO REDUCE ANY OF THE ABOVE-MENTIONED CAPACITY FROM THE OVERALL CUMULATIVE CAPACITY BEFORE/AFTER AWARD OF CONTRACT AGREEMENT. ACCORDINGLY, THE TOTAL PROJECT PRICE WILL BE PROPORTIONATELY REDUCED & THE FINAL PROJECT PRICE ARRIVED AS SUCH AFTER REDUCTION WILL BE THE FINAL PROJECT PRICE OF THE CONTRACT AGREEMENT.

REGARDING PPP:

WRT MINISTRY OF NEW & RENEWABLE ENERGY (MNRE) ORDER NO 283/22/2019-GRID SOLAR DATED 23RD SEP 2020 FOR THE PUBLIC PROCUREMENT (PREFERENCE TO MAKE IN INDIA) INCLUDING ANY AMENDMENTS, TO PROVIDE FOR PURCHASE PREFERENCE (LINKED WITH LOCAL CONTENT) IN RESPECT OF RENEWABLE ENERGY (RE) SECTOR & DEPARTMENT FOR PROMOTION OF INDUSTRY AND INTERNAL TRADE (DPIIT) NOTIFICATION NO. P-45021/2/2017-PP (BE-LI) DATED 4TH JUNE, 2020 INCLUDING ANY AMENDMENTS,

THE PROCUREMENT OF SOLAR MODULES & SOLAR INVERTERS FOR THE SUBJECT TENDER WILL BE DONE FROM CLASS I LOCAL SUPPLIERS ONLY, WHEREIN, A CLASS-I LOCAL SUPPLIER' MEANS A SUPPLIER OR SERVICE PROVIDER, WHOSE GOODS, SERVICES OR WORKS OFFERED FOR PROCUREMENT, HAS LOCAL CONTENT EQUAL TO OR MORE THAN 50%, AS DEFINED UNDER THE ABOVE SAID ORDERS.

REGARDING ALMM:

THE PROVISIONS AS CONTAINED IN THE O.M. DATED 25.01.2023 ISSUED BY MNRE ON THE SUBJECT "APPROVED MODELS AND MANUFACTURERS OF SOLAR PHOTOVOLTAIC MODULES (REQUIREMENT OF COMPULSORY REGISTRATION) ORDER, 2019-IMPLEMENTATION-REG." AND ITS SUBSEQUENT AMENDMENTS AND CLARIFICATIONS, SHALL BE APPLICABLE FOR THIS TENDER. THE CELLS AND MODULES USED IN THE PROJECT UNDER THIS TENDER SHALL BE SOURCED ONLY FROM THE MODELS AND MANUFACTURERS INCLUDED IN THE LIST-I (SOLAR PV MODULES) OF ALMM ORDER, ANNEXURE-I OF THE OM, ISSUED BY MNRE ON 25TH JANUARY 2023 WHICH WILL BE UPDATED BY MNRE FROM TIME TO TIME. THE MODULES TO BE PROCURED FOR THIS PROJECT, SHALL BE FROM THE LIST-I OF THE ALMM ORDER APPLICABLE ON THE DATE OF INVOICING OF SUCH MODULES. BIDDERS SHALL ALSO PROVIDE A DISCLOSURE REGARDING THAT THEY ARE AWARE OF BINDING PROVISIONS OF THE ALMM ORDER AND THE LIST(S) THERE UNDER, WHILE QUOTING FOR THE TENDER.

(SINGLE STAGE TWO PART BIDDING)

Under e-Tendering

- 1.0 The Hyderabad Growth Corridor Limited (HGCL) is a company owned by the HMDA, Government of Telangana.
- 2.0 Hyderabad Growth Corridor Limited (HGCL), envisages the execution of Green Energy by way of implementing Solar PV Power Plant along the ORR Service Road at HGCL in Hyderabad. This would not only be a leap towards the use of Green Energy for HGCL but also be means of self-reliant and Economical Energy which would capitalize the abundance of Solar Energy.
- 3.0 The selection of the Contractor for the “Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant on rooftop of Cycle track at 2 stretches along the ORR Service Road with 5 years Plant O&M at Hyderabad Growth Corridor Limited, Telangana, India” has been decided to call tenders by Hyderabad Growth Corridor Limited.
- 4.0 Hyderabad Growth Corridor Limited (HGCL), invites bids from eligible bidders to participate in Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant on rooftop of Cycle track at 2 stretches along the ORR with 5 years Plant O&M at Hyderabad Growth Corridor Limited, Telangana, India”.
- 5.0 **VOID**
- 6.0 The complete Bidding Documents are available at <http://eprocurement.telangana.gov.in>. Of Telangana state portal and at tender.telangana.gov.in as well as on HGCL's website www.hmda.gov.in. However, for the purpose of participation, the official copy of the bidding documents shall only be downloaded from HMDA's e-tendering portal at www.hmda.gov.in as per the provisions available therein. Accordingly, the online bid also has to be uploaded by the respective bidders at <http://eprocurement.telangana.gov.in> only & no other mode of participation is permitted for this tender document other than Portal.
- 7.0 Interested bidders have to necessarily register themselves on the portal <http://eprocurement.telangana.gov.in> through M/s Vupadhi Techno Services Pvt. Ltd, Hyderabad to participate in the bidding under this invitation for bids. It shall be the sole responsibility of the interested bidders to get themselves registered at the aforesaid portal for which they are required to contact M/s Vupadhi Techno Services Pvt. Ltd. 1st Floor, Ramky Grandiose, Sy.No:136/2&4, Gachibowli, Hyderabad-500032.Telangana State. to complete the registration formalities. All required documents and formalities for registering on ETI are mentioned in the subsequent bidding documents.

They may obtain further information regarding this IFB from the registered office of HGCL at the address given on the Bid Information Sheet from 10:00 hours to 17:00 hours on all working days till the last date of the Bid Submission.

For proper uploading of the bids on the portal namely <http://eprocurement.telangana.gov.in> (hereinafter referred to as the 'portal'), it shall be the sole responsibility of the bidders to apprise themselves adequately regarding all the relevant procedures and provisions as detailed in the portal as well as by contacting M/s Vupadhi Techno Services Pvt. Ltd directly, as and when required, for which contact details are also mentioned on the Bid Information Sheet. The Employer in no case shall be responsible for any issues related to timely or properly uploading/ submission of the bid in accordance with the relevant provisions of Section II - ITB of the Bidding Documents.

- 8.0 While submitting/ uploading the bids, the system through portal asks to key in the pass-phrase for encryption of the documents. The pass-phrase is required by Employer for opening the bids (Separate for both First Parts as well as Second Parts). **The same may be submitted on the portal as per the provisions existing for submission of the pass- phrase and as per the details given in ITB.**

In the event, HGCL is unable to open the Bids with the given pass-phrase provided by the bidders, Employer on its discretion may give an option through the portal, to the bidder to open its bid as per provisions available on the portal. However, Employer shall not be responsible if bid could not be opened within reasonable time for whatsoever reason. In such a case, the bid shall be sent unopened to 'Archive' on the portal and shall not be considered at all any further.

- 9.0 A Single Stage Two Part Bidding Procedure will be adopted and will be processed as detailed in the Bidding Documents. Bidding will be conducted through the competitive bidding procedures as per the provisions of ITB/ BDS and the contract shall be executed as per the provisions of the Contract. It shall be noted that the respective rights of the Employer and the Bidder/ Contractor shall be governed by this Bidding Documents and Contract signed between the HGCL and the Contractor for the package.

- 10.0 Bidders should submit their bid proposal online complete in all aspect on or before last date and time of Bid Submission as mentioned on ETS Portal of at <http://eprocurement.telangana.gov.in> and as indicated in the Bid Information Sheet.

- 11.0 Bidder shall submit bid proposal along with non-refundable Tender Processing Fees, Earnest Money Deposit (EMD) if applicable/ Bid Securing Declaration complete in all respect as per the Bid Information Sheet. Techno-Commercial bids will be opened as per the Bid Information Sheet in **online** presence of authorized representatives of bidders who wish to be present **online**. Bid proposals received without the prescribed Tender Processing Fees and Earnest Money Deposit (EMD, if applicable) or Bid Securing Declaration will be rejected. **In the event of any date indicated is a declared Holiday, the next working day shall become operative for the respective purpose mentioned herein.**

- 12.0 Tender documents which include Eligibility Criteria, Technical Specifications, various Conditions of Contract, Formats etc. can be downloaded from ETS Portal of at <http://eprocurement.telangana.gov.in>. **It is mandatory to download official copy of Tender Document from Portal to participate in the Tender.** Any amendment(s)/ corrigendum(s)/ clarification(s) with respect to this Tender shall be uploaded on ETI site. The Bidder should regularly check for any Amendment(s)/ Corrigendum(s)/ Clarification(s) on the above mentioned ETI site. The same may also be uploaded on HMDA website www.hmda.gov.in also. **However, in case of any discrepancy, the information available on ETI site shall prevail.**

13.0 The Bidder selected shall be responsible for the performance of the following scope of work (more detailed in this tender document):

- (i) Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant on rooftop of cycle track along the ORR at Hyderabad Growth Corridor Limited, Telangana, India”.
- (ii) Comprehensive operation & maintenance of the Solar PV Power Plants for 5 (Five) years as mentioned in detailed scope of work, after successful commissioning and performance demonstration, as detailed in technical specification, including supply and storage of all mandatory spare parts, consumables, repairs/ replacement of any defective equipment, etc.

The above scope of work is indicative and the **detailed scope of work** is given in the Scope of Work and Technical Specification (Section - VII) of the Tender Documents.

14.0 **EMD/Bid Securing Declaration shall be enclosed in a sealed Part and shall be submitted in the office of Employer (offline) whose mailing address is mentioned in the Bid Information Sheet.**

15.0 Performance Securities as per this tender document shall be furnished by the successful bidder after issuance of Notification of Award (NOA)/ Letter of Intent (LOI)/ Letter of Award (LOA) by the HGCL.

16.0 The detailed Qualifying Requirements (QR) are given in the section, “Annexure to BDS”.

17.0 ***Employer/ HGCL reserves the right to cancel/ withdraw this invitation for bids without assigning any reason and shall bear no liability whatsoever consequent upon such a decision.***

INTERPRETATIONS

1. Words comprising the singular shall include the plural & vice versa.
2. An applicable law shall be construed as reference to such applicable law including its amendments or re-enactments from time to time.
3. A time of day shall save as otherwise provided in any agreement or document be construed as a reference to Indian Standard Time.
4. Different parts of this contract are to be taken as mutually explanatory and supplementary to each other and if there is any differences between or among the parts of this contract, they shall be interpreted in a harmonious manner so as to give effect to each part.

The table of contents and any headings or sub headings in the contract has been inserted for case of reference only & shall not affect the interpretation of this agreement.

BID INFORMATION SHEET

The brief details of the tender are as under:

A)	NAME OF WORK/ BRIEF SCOPE OF WORK/ JOB	Design & engineering, procurement & supply of equipment and materials, testing at manufacturers works, multi – level inspections, packing and forwarding, supply, receipt, unloading and storage at site, associated civil works, services, permits, licenses, installation and incidentals, insurance at all stages, erection, testing and commissioning of 13 MW (AC) Grid Interactive Solar PV Power Plant and performance demonstration with associated equipment and materials on turnkey basis at 2 stretches of Hyderabad Growth Corridor Limited, (HGCL), in Telangana State along with 5 (Five) years comprehensive operation and maintenance from the date of Operational Acceptance. 1.1 Design, Procurement & Supply, and erection of the following, in all respect: 1.2 Erection of Solar panels on existing already erected module mounting structures and fasteners. 1.3 All power conditioning systems including junction boxes, Inverters/ PCU, DC and AC circuit breaker(s). 1.4 All associated electrical works and equipment required for interfacing line/ cable (i.e. transformer(s) – power and auxiliary, breakers, isolators, lightning arrestor(s), LT/other panels, protection system, cables, metering etc., but not limited to) as per technical specifications. 1.5 Design, supply, erection, testing & commissioning defined in scope of work/ TS as per project requirement and associated switchgear equipment and metering equipment as per technical specification and state regulations. 1.6 All associated civil works, including design and Engineering, in the requisite project land as required for development of this Solar PV Power Plant 1.7 Construction of pipeline for Cleaning of Solar PV Project 1.8 Providing of Vehicles for cleaning, if required 1.9 Setting up of a comprehensive Fire Protection system/ Fire Extinguishers can be provided as per the Hazardous area classification for the site 1.10 Supply of mandatory spares & special tools and tackles 1.11 Demonstration of performance of the plant as per the requirement specified in the bidding documents.
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		1.12 Comprehensive operation & maintenance of the SPV plant for 5 (Five) years as mentioned in detailed scope of work from the date of Operational Acceptance, whichever is later, as detailed in technical specification, including supply and storage of all spare parts, consumables, repairs/ replacement of any defective equipment etc. 1.13 Obtaining all associated statutory and regulatory compliances and approvals for successful construction, commissioning and operation of plant 1.14 The detailed scope of work is given in Section VII: Scope of Work, specifications and Drawings of this bidding 1.15 documents.
(B)	TENDER NO. & DATE	BID NOTICE No.178/CGM(T)/HGCL/DGM(Elec)/Cycle Track/2022-23/second call/Dated:24.04.2023
(C)	SOURCE OF FUNDS	Employer/HGCL as defined in the Bid Data Sheets (BDS) intends to finance the package through domestic funding and own resources.
(D)	TYPE OF TENDER	"Single Stage Two part "online Bidding system.
(E)	BIDDERS ELIGIBLE FOR BIDDING	This bid is open to bidders from within the HGCL's/Employer's country only.
(F)	COMPLETION/ CONTRACT PERIOD	As mentioned in Special Conditions of Contract (SCC).
(G)	COST OF BIDDING DOCUMENTS	Free of cost
(H)	TENDER PROCESSING FEE to TSTS	0.04% of ECV payable to TSTS
(I)	EARNEST MONEY DEPOSIT (EMD)	1% OF THE ECV
(J)	CONTRACT PERFORMANCE SECURITY	Yes, 10% of ECV and returned after successful completion of 5 years O&M
(K)	DATE, TIME OF SITE VISIT	Site Visit: Prospective Bidders are advised to visit the proposed Land site in all working days between 11:00 AM to 4 :00 PM, to study the actual conditions and go through the plans/ drawings connected to the present scope of work etc including power evacuation system and Project capacities and get acquainted with the same. For seeking visit of site or any clarifications bidders may contact the DGM(Ele), HGCL or AGM(Ele) Contact No: 9502737918 or 8688147565

(L)	Start Date for Down loading of Tender Documents	<u>25.04.2023</u>
(M)	Last DATE FOR ONLINE BID Submission	<u>03.05.2023, 15:00 HRS onwards</u>
(N)	Technical Bid Opening	<u>03.05.2023, 15:30 HRS onwards</u>
(O)	FINANCIAL BID OPENING	To be intimated subsequent to the shortlisting of Techno Commercial Bids on e-market place
	TECHNICAL QUERIES CONTACT DETAILS	Sri. P. Sampath Kumar Deputy General Manager (Electrical), Hyderabad Growth Corridor Limited HGCL Office Building, 2nd Floor, Khajaguda Road, Adjacent to ORR, Nanakramguda, Hyderabad -500104. Contact No. 9502737918 E mail:- dgmelehgcl@gmail.com
(P)	CONTACT DETAILS OF ETI	M/s Vupadhi Techno Services Pvt. Ltd. 1 st Floor, Ramky Grandiose, Sy.No:136/2&4, Gachibowli, Hyderabad- 500032. Contact No.: +91 40-33762222, 23 Email: tseprocsupport@vupadhi.co Website: www.vupadhi.com

- 1.0 Bids must be submitted strictly in accordance with Clause no. 11 of ITB depending upon Type of Tender as mentioned at Clause no. (D) of Bid Information Sheet. The IFB is an integral and inseparable part of the Tender document.
- 2.0 Bidder(s) are advised to bid strictly as per terms and conditions of the tender documents and notto stipulate any deviations/ exceptions.
- 3.0 Any bidder, who meets the Qualifying Requirement and wishes to quote against this tender, may download the complete bidding document along with its amendment(s) if any from Portal of (<http://eprocurement.telangana.gov.in>) and submit their Bid complete in all respect as per terms & conditions of Tender Document on or before the due date of bid submission.
- 4.0 No Manual / Hard Copy of the Bid shall be acceptable.
- 5.0 Clarification(s)/ Corrigendum(s) if any shall also be available on above referred websites.

- 6.0 HGCL will release NOA/ LOI/ LOA to the successful bidder. The NOA/ LOI/ LOA shall be awarded for the work Design & engineering, procurement & supply of equipment and materials, testing at manufacturers works, multi – level inspections, packing and forwarding, supply, receipt, unloading and storage at site, associated civil works, services, permits, licenses, installation and incidentals, insurance at all stages, erection, testing and commissioning of 13 MW (AC) Grid Interactive Solar PV Power Plant and performance demonstration with associated equipment and materials on BOQ basis at 2 stretches of Hyderabad Growth Corridor Limited, (HGCL), in Telangana State along with 5 (Five) years comprehensive operation and maintenance from the date of Operational Acceptance.
- 7.0 However, the above-mentioned contract award methodology may be modified/ changed based on specific project requirements and upon sole discretion of HGCL.

SECTION - II

INSTRUCTION TO BIDDERS (ITB)

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Preamble

This part (Section - II) of the Tender Documents provides the information necessary for bidders to prepare responsive bids, in accordance with the requirements of the Employer/ HGCL. It also provides information on bid submission and uploading the bid on portal <http://eprocurement.telangana.gov.in>, bid opening, evaluation and on contract award. This Section (Section - II) contains provisions that are to be used unchanged unless Section - III (Bid Data Sheets) and Section - V (Special Conditions of Contract), which consists of provisions that supplement, amend, or specify in detail, information or requirements included in ITB and that are specific to each procurement, states otherwise.

However, provisions governing the performance of the Contractor, payments under the contract or matters affecting the risks, rights and obligations of the parties under the contract are not included in this section but instead under Section - IV (General Conditions of Contract) and/ or Section - V (Special Conditions of Contract).

Bidders may note that the respective rights of the Employer/ HGCL and Bidders/ Contractors shall be governed by the Tender Documents and Contracts signed between the Employer/ HGCL and the Contractor. The provisions of Tender Documents shall always prevail over any other documents in case of contradiction.

Further in all matters arising out of the provisions of this Tender document, the laws of the Union of India and the courts of Hyderabad shall have the exclusive jurisdiction in all matters arising there under.

[A] - GENERAL

1 SCOPE OF BID

- 1.1 The Employer/ HGCL, wishes to receive Bids as described in the Bidding documents/ Tenderdocuments issued by Employer/ HGCL.
- 1.2 SCOPE OF BID: The scope of work/ Services shall be as defined in the Tender documents.
- 1.3 The successful bidder will be expected to complete the entire scope of work within the period stated in Section - V, Special Conditions of Contract.
- 1.4 Day' means 'Calendar Day' and 'Singular' also means 'Plural'.

2 ELIGIBLE BIDDERS

- 2.1 The Bidder shall not be under a declaration of ineligibility by Employer for Corrupt/ Fraudulent/Collusive/ Coercive practices, as defined in "Instructions to Bidders [ITB], Clause no. 39" (Action in case Corrupt/ Fraudulent/ Collusive/ Coercive Practices).
- 2.2 The Bidder (either individually or as a JV consortium or any of the participating members of the JV Consortium) shall not have been debarred by Employer/ HGCL/ Ministry of New & Renewable Energy (MNRE) or any other ministries and / or any other Government Department, Agencies or CPSUs from future bidding due to "poor performance" or "corrupt and fraudulent practices" or any other reason in the past.

If the tender documents were issued inadvertently/ downloaded, offers submitted by such bidders shall not be considered for opening/ evaluation/ Award and will be returned to such bidders.

It is the sole responsibility of the Bidder to have informed HGCL about any change in status of the declaration (if any) prior to award of contract, the same has to be informed promptly to Employer/ HGCL by the bidder.

It shall be the sole responsibility of the bidder to inform Employer/ HGCL in case the bidder is debarred from bidding by Employer/ HGCL or Public Sector Project Management Consultant. Concealment of the facts shall tantamount to misrepresentation of facts and shall lead to action against such Bidders as per clause no. 39 of ITB.

- 2.3 The Bidder should not be under any liquidation court receivership or similar proceedings on the due date of submission of bid.

In case there is any change in status of the declaration prior to award of contract, the same has to be promptly informed to Employer/ HGCL by the bidder.

It shall be the sole responsibility of the bidder to inform Employer/ HGCL in case the bidder is under any liquidation court receivership or similar proceedings on due date of submission of bid and during the course of finalization of the tender. Concealment of the facts shall tantamount to misrepresentation of facts and shall lead to action against such Bidders as per clause no. 39 of ITB.

- 2.4 Bidder shall not be affiliated with an entity:
- (i) that has provided consulting services related to the work to the Employer/ HGCL during the preparatory stages of the work or of the project of which the works/ services forms a part of or
 - (ii) that has been hired (proposed to be hired) by the Employer/ HGCL as an Engineer/ Consultant for the contract.
- 2.5 Neither the entity [appointed as the Project Management Consultant (PMC)/Consultancies for a contract] nor its affiliates/ JV's/ Subsidiaries shall be allowed to participate in the tendering process.
- 2.6 Pursuant to qualification criteria set forth in the bidding document, the Bidder shall furnish all necessary supporting documentary evidence to establish Bidder's claim of meeting qualification criteria as mentioned in the Annexure to BDS.

3 BIDS FROM JOINT VENTURE CONSORTIUM

- 3.1 Unless otherwise specified in the Bid Data sheet (BDS), Bids from Joint Venture Consortium are allowed. However, the eligibility criteria along with other terms and conditions (as mentioned in clause no. 1.4 of Annexure to BDS) shall be complied upon.

4 NUMBER OF BIDS PER BIDDER

- 4.1 Unless otherwise specified in the Bid Data sheet (BDS), a Bidder shall submit only 'one [01] Bid' in the same Bidding Process. A Bidder who submits or participates in more than 'one [01] Bid' will cause all the proposals in which the Bidder has participated to be disqualified.

5 COST OF BIDDING & TENDER PROCESSING FEE

5.1 COST OF BIDDING

The Bidder shall bear all costs associated with the preparation and submission of the Bid including but not limited to Bank charges, all courier charges including taxes & duties etc. incurred thereof. Further, Employer/ HGCL will in no case, be responsible or liable for these costs, regardless of the outcome of the bidding process.

5.2 TENDER PROCESSING FEE (NON-REFUNDABLE):- 0.04% of ECV payable to TSTS

5.3 The EMD, if applicable is exempted for MSME Vendors registered under NSIC/ Udyog Aadhaar/DIC Category only. In order to avail the exemption in EMD in case of consortium/ JV, all the members should be registered as MSME Vendors under NSIC/ Udyog Aadhaar Category/DIC.

5.4 In case of any discrepancy/ non-submission of online bid documents by the bidder, the tender processing fee will be deemed as bidder's consent for participation in the bidding process. Henceforth, the tender processing fee shall be retained by Employer and shall not be returned under any circumstances. No plea in this regard shall be entertained by the Employer/ HGCL. However, EMD, if applicable will be returned in this case.

5.5 In the event of a particular tender being cancelled at any stage, the tender processing fee will be refunded to the concerned bidders without any interest charges within 30 days from the date of notification of cancellation of tender. No plea in this regard shall be entertained by the Employer/ HGCL. EMD, if applicable will also be returned in this case

6 SITE VISIT

6.1 The Bidder is advised to visit and examine the site of works and its surroundings and obtain for itself on its own responsibility all information that may be necessary for preparing the Bid and entering into a Contract for the required job. The costs of visiting the site shall be borne by the Bidder.

6.2 The Bidder or any of its personnel or agents shall be granted permission by the Employer/ HGCL to enter upon its premises and land/Area for the purpose of such visits, but only upon the express conditions that the Bidder, its personnel and agents will release and indemnify the Employer/ HGCL and its personnel, agents from and against all liabilities in respect thereof, and will be responsible for death or injury, loss or damage to property, and any other loss, damage, costs, and expenses incurred as a result of inspection.

6.3 The Bidder shall not be entitled to hold any claim against Employer/ HGCL for non-compliance due to lack of any kind of pre-requisite information as it is the sole responsibility of the Bidder to obtain all the necessary information with regard to site, surrounding, working conditions, weather etc. on its own before submission of the bid.

[B] - BIDDING DOCUMENTS

7 CONTENTS OF TENDER DOCUMENTS

- 7.1 The contents of Tender Documents are those stated below, and should be read in conjunction without payment of requisite Tender Processing Fee will be treated as non-responsive and with any 'Addendum/ Corrigendum' issued in accordance with "ITB: Clause-9":

➤	Section-I	:	Invitation for Bids [IFB]
➤	Section-II	:	Instructions to Bidders [ITB]
➤	Section-III	:	Bid Data Sheets [BDS]
			Annexure to Bid Data Sheets [Qualifying Requirements]
➤	Section-IV	:	General Conditions of Contract [GCC]
➤	Section-V	:	Special Conditions of Contract [SCC]
➤	Section-VI	:	Sample Forms and Formats ie., EMD, Additional Security etc.
➤	Section-VII	:	Scope of Work & Technical Specifications
➤	Section-VIII	:	List of approved makes
➤	Section-IX	:	Special conditions for O&M
➤	Section-X	:	Special conditions for payment of O&M
➤	Section-XI	:	Service levels for O&M
➤	Section-XII	:	Drawings
➤	Section-XIII	:	Bill of Quantities (Price-Bid)
➤	Section-XIV	:	Any other document listed as forming part of the contract.

- 7.2 The Bidder is expected to examine all instructions, forms, terms & conditions in the Tender Documents. The entire Tender Documents together with all its amendments, clarifications and attachments thereto, shall be considered to be read, understood and accepted by the Bidders. Failure to furnish all information required by the Tender Documents or submission of a Bid not substantially responsive to the Tender Documents in every respect will be at Bidder's risk and may result in the rejection of his Bid. However, Employer/HGCL at its sole discretion may seek clarifications from the Bidders to adjudge the exact content and facts of the Tender Documents.

8 CLARIFICATION OF TENDER DOCUMENTS

- 8.1 A prospective Bidder requiring any clarification(s) of the Tender Documents may notify Employer in writing by E-mail or at Employer's mailing address indicated in the Bid Information Sheet no later than 02 (Two) working days after the pre-bid meeting (in cases where pre-bid meeting is scheduled) or 3 (Three) days prior to the bid closing date (in cases where pre-bid meeting is not held). Employer reserves the right to ignore the bidders request for clarification if received beyond the aforesaid period. Employer may respond in writing to the request for clarification. Employer's response including an explanation of the query, but without identifying the source of the query will be uploaded on Web Portal of <http://eprocurement.telangana.gov.in> and/ or Employer's website www.hmda.gov.in.

- 8.2 Any clarification or information required by the Bidder but same not received by the Employer at clause 8.1 above is liable to be considered as "no clarification/ information required".
- 8.3 Clarifications sought by the bidders are to be mandatorily submitted in the HGCL provided format only provided along with the Tender document. Pre-Bid queries submitted in any other format will not be considered.

9 AMENDMENT OF TENDER DOCUMENTS

- 9.1 At any time prior to the 'Bid Due Date', Employer/ HGCL may, for any reason, whether at its own initiative or in response to a clarification requested by a prospective Bidder, modify the Tender Documents by addendum/ corrigendum.
- 9.2 Any addendum/ corrigendum thus issued shall be part of the Tender Documents and shall be hosted on ETS Portal of <http://eprocurement.telangana.gov.in> and/ or Employer's website www.hmda.gov.in. Bidders have to take into account all such addendum/ corrigendum before submitting their bid.
- 9.3 The Employer, if consider necessary, may extend the date of submissions of Bid in order to allow the Bidders a reasonable time to furnish their most competitive bid taking into account the amendment issued thereof.

[C] - PREPARATION OF BIDS

10 LANGUAGE OF BID:

The bid prepared by the bidder and all correspondence/ drawings and documents relating to the bid exchanged by bidder and Employer/ HGCL shall be written in English language alone. Any printed literature furnished by the bidder may be written in another language as long as accompanied by an ENGLISH translation duly authenticated by the Chamber of Commerce/ Certified Translator of bidder's country, in which case, for the purpose of interpretation of the bid, the ENGLISH translation shall govern.

In the event of submission of any document/ certificate by the Bidder in a language other than English, the English translation of the same duly authenticated by Chamber of Commerce/ Certified Translator of Bidder's country shall be submitted by the Bidder.

11. DOCUMENTS COMPRISING THE BID

The bid shall be submitted by the Bidder under "Single Stage - Two Part" procedure of bidding. Under this procedure, the bid submitted by the Bidder in two Parts - First Part (also referred to as Techno - Commercial Part) and Second Part (also referred to as Price Part) shall comprise of the following documents:

I. Offline Documents: Deleted

Offline documents of the bid shall comprise of following documents to be submitted in sealed Part, as part of First Part. **The Part shall bear {the name of Tender, the Tender No. and the words 'DO NOT OPEN BEFORE' (due date & time)}.**

Contact Persons Name: P. Sampath Kumar (DGM/Elc)

- (a) Original Non-Refundable Tender Processing Fee as per clause no. 05 of ITB
- (b) Original Non-Refundable Cost of Tender Document, if applicable
- (c) 'Covering Letter' on Bidder's 'Letterhead' (in Original) clearly specifying the enclosed contents, as per 'Form F-0'
- (d) EMD, in original as per Clause 16 of ITB as per 'Form F-4A' or as prescribed.
- (f) Power of Attorney for authorized signatory in non-judicial stamp paper (as per 'Form F-24')
- (g) Copy of Board Resolution
- (h) The Pass-Phrase to decrypt the relevant Bid-Parts (for both Techno-Commercial and Financial) in separate sealed Parts before the start date and time of the Tender Opening Event (TOE)

Bidder shall also upload the scanned copies of all the above-mentioned original documents during online Bid Submission as a part of First Part.

"Bidder should explicitly note that no offline documents are to be submitted as part of Second Part".

II. Online Documents/Soft Copy

Online documents/Soft copy of the bid shall comprise of following documents to be uploaded on the portal <http://eprocurement.telangana.gov.in> as per provisions therein.

II (a) As Part-I

- (a) The Electronic Form of the bid for First Part (Techno-Commercial), as available on the portal, shall be duly filled.
- (b) Scanned copies of all the above-mentioned original documents during online Bid Submission as a part of First Part and Bid Form for first Part.
- (c) Certificate of Incorporation
- (d) 'Bidder's General Information', as per 'Form F-1'.
- (e) 'No Deviation Confirmation', as per 'Form F-6'.
- (f) 'Bidder's Declaration regarding Banning, Liquidation etc.', as per 'Form F-7'
- (g) 'Declaration on restriction on procurement from certain countries.', as per 'Form F-8'
- (h) 'Declaration regarding the procurement of Solar Inverters & Solar Modules from class I local suppliers, as per 'Form F-12'
- (i) 'Bidders Experience as per 'Form F-13'

- (j) Disclosure for ALMM compliance as per 'Form F-14'
- (k) Format of Chartered Accountant certificate for financial capability of the bidder as per 'Form F-16'
- (l) 'E-Banking Format as per 'Form F-19'
- (m) 'Shareholding Certificate' as per 'Form F-23'
- (n) 'Format for Cyber Security Agreement' as per 'Form F-25'
- (o) 'Integrity Pact Format' as per 'Form F-26'
- (p) Documents in accordance with the "Qualifying Requirements (QR)" establishing the qualification
- (q) Document showing annual turnover for the financial years as required in Qualifying Requirements (QR) such as annual reports, profit and loss account, net worth etc. along with information as sought in enclosed Format F-16
- (r) Tender Document. (Only First and Last Pages of Original Tender Document duly sealed and signed/ digitally signed and all pages of amendments to Tender Documents duly sealed and signed/ digitally signed by the Authorized Signatory).

II (b) As Second Part

- (a) The Electronic Form of the bid for Second Part (Price - Part), as available on the portal, shall be duly filled. "Termed as **ELECTRONIC FORM**"
- (b) Main Price Bid comprising of Schedule No 1 to 4 of the Price Schedule (available in Section - VIII, SOR), duly completed, sealed and signed/ digitally signed shall be uploaded. "Termed as MAIN BID".

12 PRICE BID:

- 12.1 Unless stated otherwise in the Tender Documents, the Contract shall be for the whole works as described in Tender Documents, based on the rates and prices submitted by the Bidder and accepted by the Employer/ HGCL. The prices quoted by the Bidders should indicate clearly the Goods & Service Tax (GST) components as also mentioned under the SOR.
- 12.2 Prices must be filled in format for "Schedule of Rates [SOR]/ Price Schedule [PS]/ Bid Prices" enclosed as part of Tender documents under Section - VIII. If quoted in separate typed sheets, such bids may be rejected.
- 12.3 Bidder shall quote for all the items of "SOR/ PS" after careful analysis of cost involved for the performance of the completed item considering all parts of the Tender Document. In case any activity though specifically not covered in description of item under "SOR/ PS" but is required to complete the works as per Specifications, Scope of Work/ Service, Standards, "GCC", "SCC" or any other part of Bidding Document, the prices quoted shall be deemed to be inclusive of cost incurred for such activity.
- 12.4 All Goods & Service Tax (GST) components [applicable for both Centre and state] payable by the Contractor under the Contract, or for any other cause, shall be mentioned as per the SOR formats Schedule Nos 1-4.

12.5 Prices quoted by the Bidder, shall remain FIRM and Fixed and valid until completion of the Contract and will not be subject to variation on any account.

12.6 In case of any variation (positive/ negative) in existing rates of taxes/ duties/ levies or a new tax/ duty/ levy is introduced or any existing tax/ duty/ levy is abolished or application of any Tax in the course of the performance of this Contract, which will/ may impact the overall pricing in connection with performance of the Contract, an equitable adjustment of the Contract Price shall be made to factor any such change by addition to the Contract Price or deduction therefrom, as the case may be. All these adjustments would be carried out by considering the base price of taxes equivalent to the amount mentioned under taxes and duties column of the SOR/ PS.

The term Change in Law shall refer to the occurrence of any of the following events pertaining to this project only after the last date of the bid submission, including (i) the enactment of any new law; or (ii) an amendment, modification or repeal of an existing law; or (iii) any change in the rates of any Taxes including any duties and cess or introduction of any new tax made applicable for setting up the project.

However, Change in Law shall not include (i) any change in taxes on corporate income or (ii) any change in any withholding tax on income or dividends distributed to the shareholders of the Contractor, or (iii) any change on account of regulatory measures by the Appropriate Commission.

12.7 The Bidder shall quote the prices in 'figures' & words. There should not be any discrepancy between the prices indicated in figures and the price indicated in words. In case of any discrepancy, the same shall be dealt as per clause no. 30 of ITB.

12.8 Bidder need to submit the detailed break-up of Goods & Service Tax (GST) (applicable for both Central and State) in the SOR formats viz Schedule Nos 1-4. This data is required to ascertain the

- a) Computation of taxes assumed at the time of bidding.
- b) The total impact due to revision in applicable tax rate or introduction of new tax, if any.

Bidders are required to ascertain the correctness of amount related to Goods & Service Tax (GST) as mentioned in the SOR/ PS as on the date of techno-commercial bid submission as it will impact the Price assessment part at the time of evaluation of price bid.

13 Goods & Service Tax (GST)

Contractor shall mandatorily obtain the registration under GST Law at Central level and/or in respective State as may be required. Further, Contractor shall mandatorily file returns under GST before their due date & comply with the requirements of the Law within timelines. Before releasing the

payment to the Contractor. HGCL shall not be responsible for any delay in payment release to the contractor in case the GST compliance is not fulfilled from the contractor side in any manner.

Contractor shall be responsible to comply with all the requirements of applicable provisions of GST. Contractor has to mandatorily get registered under GST at Central and relevant State(s). Contractor shall file all the returns on timely basis and upload all the Invoices and acceptance thereof as may be required under the provisions of GST. In case, it is found that HGCL is not able to take Input Tax Credit (ITC) benefit of the taxes due to any fault of the Contractor, HGCL shall be constrained to deduct the amount from the payments to be made to the Contractor or recover the same in any other manner.

- 13.1 Bidders are required to submit a copy of the GST Registration Certificate or GST provisional certificate while submitting the bids wherever GST tax is applicable.

The responsibility of payment of GST lies with the Service Provider only. Contractor providing taxable service shall issue an Invoice, a Bill or as the case may be, a Challan which is signed, serially numbered and in accordance with rule GST Law. The invoice shall also contain the following:

- (a) Name, Address & Registration No. of such Person/ Contractor
- (b) Name & Address of the Person/ Contractor receiving Taxable Service
- (c) Description, Classification & Value of Taxable Service provided
- (d) GST Amount, if any.
- (e) HSN code of the Goods/Services.

Payments to Service Provider for claiming GST amount will be made provided the above formalities are fulfilled. Further, Employer/ HGCL may seek copies of challan and certificate from Chartered Accountant for deposit of GST collected from Employer/ HGCL.

- 13.3 In case CBIT (Central Board of Indirect Taxes and Customs) brings to the notice of Employer/ HGCL that the contractor has not remitted the amount towards GST collected from Employer/ HGCL to the government exchequer, then, that contractor may be debarred from bidding in future tenders of Employer/ HGCL for given period as per the sole discretion of Employer/ HGCL.

- 13.4 In case of statutory variation in GST during currency of the Contract, the Contractor shall submit a copy of the 'Government Notification' to evidence the date of revision. Claim for payment of GST / Statutory variation in GST, should be raised within two [02] months from the date of issue of 'Government Notification' for payment of differential (in %) GST, otherwise claim in respect of above shall not be entertained for payment of arrears. The following may also be noted: -

- a) Any increase in the rate of Non ITC based GST beyond the contractual completion period shall be to contractor's account whereas any decrease in the rate shall be passed on to the Employer/ HGCL.

13.5 HGCL will reimburse the GST to the Contractor at actuals against submission of ITC based invoices issued in accordance with GST rules. In case of any variation in the executed quantities, the amount on which the GST is applicable shall be modified in same proportion.

13.6 HGCL will reimburse the GST to the Contractor at actuals against documentary evidence subject to the ceiling amount of GST as quoted by the bidder, subject to any statutory variations. In case of any variation in the executed quantities (If directed and/ or certified by the Engineer-In-Charge) the ceiling amount on which GST is applicable will be modified on pro-rata basis.

Contractor shall ensure timely submission of correct invoice(s) with all required supporting document(s) so to enable HGCL to avail Input Tax Credit (ITC) (If applicable)

14 BID CURRENCIES:

Bidders must submit bid in the currency as mentioned Bid Data Sheet.

15 BID VALIDITY

15.1 Bids shall be kept valid for period specified in BDS from the final 'Bid Opening Date'. A Bid valid for a shorter period may be rejected by Employer as 'non-responsive'.

15.2 In exceptional circumstances, prior to expiry of the original 'Bid Validity Period', the Employer may request the Bidders to extend the 'Period of Bid Validity' for a specified additional period. The request and the responses thereto shall be made in writing or by email. A Bidder may refuse the request without forfeiture of his 'EMD, if applicable will '. A Bidder agreeing to the request will not be required or permitted to modify his Bid, but will be required to extend the validity of its 'EMD, if applicable will ' for the period of the extension and in accordance with "ITB: Clause-16" in all respects.

Note: In case of extension(s) of last due date of the Tender submission, the latest extension issued shall be considered as the final due date of Tender submission and accordingly the Tender validity should be calculated and sufficed. The validity of the Tender need to be revised by respective bidders, in case the bids are already submitted prior to the last due date of the initial Tender submission deadline.

16 EARNEST MONEY DEPOSIT (EMD)

16.1 The applicability/Non-applicability of Earnest Money Deposit (EMD) is specified under BDS.

16.2 The Bids must be accompanied with '**Earnest Money Deposit (EMD)**' in the form of either through NEFT/ RTGS transfer in the account of HGCL or '**Demand Draft**' or '**Banker's Cheque**' [in favour of **HGCL,**] or '**Bank Guarantee**' as per the format given in **Form F - 4 of the bidding documents**. Bidders shall ensure that EMD, having a validity of at least 30 (Thirty) Days beyond the validity of the bid, must accompany the Bid in the format(s) made available in the Tender Document. In case of any extension in validity of bid, the EMD shall be extended suitably. The EMD shall be submitted in Indian Rupees only.

16.3 The '**EMD**' is required to protect Employer/ HGCL against the risk of Bidder's conduct, which would warrant the EMD's forfeiture, pursuant to "ITB: Clause-16.8".

Employer/ HGCL shall not be liable to pay any Bank charges, commission or interest etc. on the amount of 'EMD'. In case 'EMD' is in the form of a 'Bank Guarantee', the same shall be from any scheduled Bank as specified in the List of Banks enclosed at Form F-22, Section-VI of Tender documents or a branch of an International Bank situated in India and registered with 'Reserve Bank of India' as Scheduled Foreign Bank. However, in case of 'Bank Guarantee' from Banks other than the Nationalized Indian Banks, the Bank must be commercial bank having net worth in excess of INR 500 Crores (Indian Rupees Five Hundred Crores Only).

16.5 Bid not accompanied with required amount of EMD shall be liable for rejection.

16.6 'Earnest Money Deposit' of all the unsuccessful Bidders disqualified at the stage of Techno- Commercial evaluation will be discharged/ returned as promptly as possible, but not later than '30 [thirty] days' of their respective disqualification.

16.7 The successful bidder's (L1 bidder) 'Earnest Money Deposit' will be discharged upon the Bidder's signing the 'Contract Agreement' and furnishing the 'Performance Security duly vetted by the Bank'.

16.8 Notwithstanding anything contained herein, the 'EMD' may also be forfeited in any of the following cases:

- (a) If a Bidder withdraws or varies his Bid during the 'Period of Bid Validity'
- (b) If a Bidder has indulged in corrupt/ fraudulent/ collusive/ coercive practice
- (c) Violates any other condition, mentioned elsewhere in the tender document including deviations or conditional bid.
- (d) In the case of a successful Bidder, if the Bidder fails to:
 - (i) acceptance of the NOA/ LOI/ LOA.
 - (ii) to furnish "Performance Security.
 - (iii) to accept 'arithmetical corrections' as per provision of the clause 30 of ITB.

- 16.9 In case EMD is in the form of 'Bank Guarantee', the same must indicate the Bid Document No and the Work for which the Bidder is quoting. This is essential to have proper correlation at a later date. The 'EMD' should be in the form provided at 'Form F-4A'.

Note: In case of extension(s) of last due date of the Tender submission, the latest extension issued shall be considered as the final due date of Tender submission and accordingly the Earnest Money Deposit validity and Bid validity should be calculated and sufficed. The validity of the submitted EMD and bid validity need to be revised by respective bidders, in case the bids are already submitted prior to the last due date of the initial Tender submission deadline.

17 VOID

18 SIGNING OF BID/TENDER DOCUMENT

- 18.1 The First and Last Pages of original tender documents including amendments, clarifications if any shall be typed or written in indelible ink [in the case of copies, photocopies are also acceptable] and shall be signed by a person or persons duly authorized to sign on behalf of the Bidder (as per POA) before uploading at online portal. The name and position held by each person signing, must be typed or printed below the signature.

19 ZERO DEVIATION AND REJECTION CRITERIA

- 19.1 **ZERO DEVIATION :** Deviation to terms and conditions of Tender Documents may lead to rejection of bid. Employer/ HGCL will accept bids based on terms & conditions of Tender Documents only. Bidder may note Employer/ HGCL will determine the substantial responsiveness of each bid to the Tender Documents pursuant to provision contained in clause 29 of ITB. For purpose of this, a substantially responsive bid is one which conforms to all terms and conditions of the Tender Documents without deviations or reservations. Employer's/ HGCL's determination of a bid's responsiveness is based on the content of the bid itself. Employer/ HGCL reserves the right to raise technical and/ or commercial query(ies), if required. The response(s) to the same shall be in writing, and no change in the price(s) or substance of the bids shall be sought, offered or permitted. The substance of the bid includes but not limited to prices, completion, scope, technical specifications, etc. Bidders are requested to not to take any deviation.

- 19.2 **REJECTION CRITERIA:** Notwithstanding the above, deviation to the following clauses of Tender document shall lead to summarily rejection of Bid:

- (a) Eligibility Criteria including General, Technical and Financial Qualifying Requirements
- (b) Firm Price

- (c) Tender Processing Fees and Earnest Money Deposit
- (d) Tender Document Fees, if applicable
- (e) Specifications & Scope of Work
- (f) Schedule of Rates (SOR)/ Price Schedule (PS)
- (g) Duration/ Period of Contract/ Completion schedule
- (h) Period of Validity of Bid
- (i) Warrantee/Guarantee/ Defect Liability Period
- (j) Arbitration/ Resolution of Dispute/ Jurisdiction of Court
- (k) Force Majeure & Applicable Laws
- (l) Any other condition specifically mentioned in the tender document elsewhere that non-compliance of the clause lead to rejection of bid

Note: Further, it is once again reminded not to mention any condition in the Bid which is contradictory to the terms and conditions of Tender document.

20 **E-PAYMENT**

HGCL has initiated payments to Contractors electronically, and to facilitate the payments electronically through '**e-banking**'. The successful bidder should give the details of his bank account as per the bank mandate form enclosed at Format F-19 in Section-VI, Sample Forms and Formats of the Tender documents.

[D] - SUBMISSION OF BIDS

21 SUBMISSION, SEALING AND MARKING OF BIDS

21.1 Bids shall be submitted through e-tender mode in the manner specified elsewhere in tender document.

21.2 Deleted

Offline documents (Specific documents only) as mentioned in clause no. 11.1 of Section - II, Instructions to Bidders (ITB) of the Tender document shall be submitted in a Sealed Covering Part. The Covering Part shall have the following Sticker

Tender Document No.	BID NOTICE No.178/CGM(T)/HGCL/DGM(Elec)/Cycle Track/2022-23/second call/ Dated:24.04.2023
Last Date of Submission	(Enter Full name and address of the Bidder)
Bids Submitted by	(Signature of the Authorized Signatory)
Authorized Signatory	(Name of the Authorized Signatory) (Stamp of the Bidder)
Bid Submitted to	Hyderabad Growth Corridor Limited(A State Government Company) Regd. Office Cum Corporate Office: D.No. 6-2-910, Visvesvaraya Bhavan, The Institution of Engineers Building, Khairatabad, Hyderabad - 500 004.

21.3 All the bids shall be addressed to the Employer at address specified in the Bid Information Sheet in Section - I, Invitation for Bids (IFB).

21.4 Bids submitted under the name of AGENT/ CONSULTANT/ REPRESENTATIVE/ RETAINER/ ASSOCIATE etc. on behalf of a bidder/ affiliate shall not be accepted.

DEADLINES FOR SUBMISSION OF BIDS

- 22.1 The bids must be submitted through e-tender mode not later than the date and time specified in the Bid Information Sheet in Section - I, Invitation for Bids (IFB).
- 22.2 The offline documents of required specific documents must be submitted through courier/ registered post/ by hand not later than the date and time specified in the Bid Information Sheet in Section - I, Invitation for Bids (IFB)/BDS
- 22.3 Employer/ HGCL may, in exceptional circumstances and at its discretion, extend the deadline for submission of Bids (clause 9 of ITB refers). In which case all rights and obligations of Employer/ HGCL and the Bidders, previously subject to the original deadline will thereafter be subject to the deadline as extended. Notice for extension of bid submission date will be uploaded on Web Portal of <http://eprocurement.telangana.gov.in> and/ or Employer's website www.hmda.gov.in.

23 LATE BIDS: DELETED

- 23.1 Any bids received after the notified date and time of closing of tenders will be treated as late bids.
- 23.2 E-tendering system shall close immediately after the deadline for submission of bid and no bids can be submitted thereafter.

Unsolicited Bids or Bids received to address other than one specifically stipulated in the tender document will not be considered for evaluation/ opening/ award if not received to the specified destination within stipulated date & time.

24 MODIFICATION AND WITHDRAWAL OF BIDS: DELETED

- 24.1 Modification and withdrawal of bids shall be as follows: -

The bidder may withdraw or modify its bid after bid submission but before the due date and time for submission as per tender document with the due written consent from the authorized signatory of the bidder.
- 24.2 The modification shall also be prepared, sealed, marked and dispatched in accordance with the provision of the clause 21 of ITB, with the outer and inner Parts additionally marked modification or withdrawal as appropriate. A withdrawal notice may also be sent by e-mail or fax but followed by a signed confirmation copy post not later than the deadline for submission of bids. No bid shall be modified/ withdrawn after the deadline for submission of bids.
- 24.3 No bid shall be allowed to be withdrawn/ modified/ substitute in the interval between the deadline for submission of bids and the expiration of the period of bid validity specified by the bidder on the Bid Form. Withdrawal/ Modification/ Substitution of a bid during this interval shall result in the forfeiture of bidder's EMD, if applicable pursuant to clause 16 of ITB and rejection of bid.

- 24.4 The latest bid submitted shall be considered for evaluation and all other bids shall be considered to be unconditionally withdrawn.
- 24.5 In case after price bid opening the lowest evaluated bidder (L1) is not awarded the job for any mistake committed by him in bidding or withdrawal of bid or modification of bid or varying in regard thereof leading to re-tendering, Employer shall forfeit EMD, if applicable paid by the bidder and such bidders shall be debarred from participation in re-tendering of the same job(s)/ item(s). Further, such bidder will be debarred for a given period as decided by Employer/ HGCL after following the due procedure.

25 EMPLOYER'S RIGHT TO ACCEPT ANY BID AND TO REJECT ANY OR ALL BIDS

Employer/ HGCL reserves the right to accept or reject any Bid, and to annul the Bidding process and reject all Bids, at any time prior to award of Contract, without thereby incurring any liability to the affected Bidder or Bidders or any obligations to inform the affected Bidder or Bidders of the ground for Employer's/ HGCL's action. However, Bidder if so desire may seek the reason (in writing) for rejection of their Bid to which Employer/ HGCL shall respond quickly.

[E] - BID OPENING AND EVALUATION

26 BID OPENING

The Technical bids will be opened online by the Chief General Manager, HGCL at the time and date as specified in the tender documents. All the Statements, documents, certificates, Demand Draft / Bank Guarantee etc., uploaded by the Tenderers will be verified and downloaded, for technical evaluation. The clarifications, particulars, if any, required from the bidders, will be obtained either online or in the conventional method by addressing the bidders. The technical bids will be evaluated against the specified parameters / criteria same as in the case of conventional tenders and the technically qualified bidders will be identified. The result of Technical bids evaluation will be displayed on the 'e' market place, which can be seen by all the Tenderers who participated in the Tenders.

The department shall carry out the technical bid evaluation solely based on the uploaded certificates/documents, DD/BG towards EMD in the e-procurement system. Those who qualify in technical will be allowed for mockup test and open the price bids of the responsive bidders those who pass the mockup test only.

27 Clarification on the Technical Bid.

- 27.1 The tender opening authority may call upon any Tenderer for clarification on the statements, documentary proof relating to the technical bid. The request for clarification and response thereto shall be in writing and it shall be only on the qualification information furnished by the Tenderer. The clarification called for from the Tenderers shall be furnished within the stipulated time, which shall not be more than a week.

27.2 The Tenderer if so desirous, shall agree in writing to furnish the clarification called for within the stipulated time and, for disqualification and rejection of his tender in the event of failure to do so.

28 Examination of technical Bids and Determination of Responsiveness

28.1 The Chief General Manager will evaluate whether each Tenderer is satisfying the eligibility criteria prescribed in the tender document and declares them as a qualified Tenderer.

Eligibility Criteria:

Single entity/ JV Consortium with members not exceeding two (2). Any agency can bid individually or can participate in only one JV Consortium

- i) The bidders should possess **Class-I (Electrical)** with A- Grade CEIG Electrical license not less than 33KV level issued by State Govt./Govt. undertakings / Public sector of Telangana State.
- ii) Copy of GST Registration.
- iii) Copy of Permanent Account Number i.e. (PAN) Card.
- iv) Proof of payment of EMD.
- v) Copy of latest Income Tax returns i.e. FY: 2021-22 submitted to IT Dept. along with proof of its submission
- vi) Liquid assets and/ or credit facilities of not less than **Rs.49.855 Crores** (credit facility/ letter of credits/ Solvency certificate certificates from banks etc.).
- vii) Satisfactorily completed as a prime contractor/JV/ authorized sub contractor in similar works of not less than **13MW** in any one year during the last 10 financial years (As per Annexure - 4).
- viii) **Minimum key parameters:**

The Minimum key quantities executed in any one financial year during last ten (10) financial years (as per Annexure-5).

SI. No	Item	unit	Qty
1	Supply installation testing & commissioning of (SITC) Solar PV Modules	MW	13
2	SITC AC & DC cables (AC- LT&HT, DC- 6Sqmm & above cable)	Km	166
3	33KV Sub-Stations with related Switch Gear	Rs.	10.0Cr

- ix) Details of existing commitments i.e. work on hand and works for which tenders are submitted (value of B: shall be furnished by the bidders).

Note: Bid capacity computed by the formula $(2AN - B)$ shall be greater than 50% of ECV.

Bid capacity: The bidder who meets the above qualification criteria and whose available bid capacity is more than the estimated contract value will be qualified for opening of price bid. The available bid capacity will be calculated as under:

Available bid capacity : $2AN - B$.

Where,

A= Value of works executed in their name in any one financial years during the last ten(10) financial years (updated to current price level at the rate of 10% per year) taking into account the work completed as well as works in progress

OR

Annual turnover in works executed in any one financial year during the last ten years /(10) financial years (updated to current price level at the rate of 10% per year) certified by chartered accountant.

N= Number of years prescribed for completion of the work for which tenders are invited (months/12).

B= value of existing commitments and ongoing works to be completed during the period of completion of works tenders are invited.

Note:

The statement showing the value of existing commitments and ongoing works as well as the stipulated period of completion remaining for each of the works listed should be issued by the Engineer-in-Charge of Govt. Dept./ undertaking not below the rank of executive Engineer or equivalent and countersigned by the rank of SE or equivalent. The statement to the above effect will be clearly enclosed to the bidding document, and the tenderer shall furnish the particulars invariably in the same format failing which the tender shall be treated as incomplete and summarily rejected.

- x) The bidders shall furnish a declaration to the effect that the required key equipment (as per the statement enclosed) is available with him or that he shall lease or hire the equipment. (as per **Annexure-8**).
- xi) The bidder should declare the availability of the experienced key personnel as per statement enclosed (**Annexure -7**) for execution of the work.
- xii) Information regarding any litigation with Government during the last five years, in which the bidder is involved (Litigation History) should be furnished. If there is no litigation history, a NIL report should be submitted. As per proforma (**Annexure-9**) enclosed.
- xiii) **Declaration on Genuineness of the certificates:** The bidder shall be required to furnish a declaration as per the statement enclosed Any incorrectness/ deviations noticed in the certificate will be viewed seriously and apart from cancelling the work (**Annexure-10**).

The application and all enclosures shall be signed by a duly authorized representative holding Power of Attorney for signing the application, and a certified copy of the Power of Attorney signed by the legally authorized representative of the firm shall accompany the application.

Note:

- a) The tenders will be opened online on the specified date by the Chief General Manager HGCL.
- b) The price-bids of such bidders who are determined to have complied with the eligibility criteria, will only be opened.

28.2 If the Technical Bid of a Tenderer is not satisfying any of the eligibility criteria it will be rejected by the Chief General Manager. However, the tender accepting authority detects any error in the evaluation of Tenders by the tender accepting authority while returning the tenders may direct the Chief General Manager, to re-evaluate the tenders.

28.3 If any alteration is made by the Tenderer in the tender documents, the conditions of the contract, the drawings, specifications or statements / formats or quantities the tender will be rejected.

28.4 The technical bid of such bidders, who are determined to have complied with the eligibility criteria, and approved by the Tender Committee of HGCL will be intimated for opening the price bid.

29 Price Bid Opening:

29.1 The price bids of qualified Tenderers whose technical bids are found satisfying the eligibility criteria shall be considered for opening in the presence of representatives of the qualified bidders.

29.2 Tenders shall be scrutinized in accordance with the conditions stipulated in the Tender document. In case of any discrepancy of non-adherence conditions the Tender accepting authority shall communicate the same which will be the binding both on the tender opening authority and the Tenderer. In case of any ambiguity, the decision taken by the Tender Accepting Authority on tenders shall be final.

29.3 Even though the bidders are free to bid for all the packages, only two works per bidder will be considered for award subjected to availability of bid capacity. The price bids of the packages will be opened for evaluation and award of work shall be as per conditions and evaluation criteria mentioned.

30 Evaluation and Comparison of Price Bids

30.1 The Chief General Manager will evaluate and compare the price bids of all the qualified Tenderers.

- i) Negotiations at any level are strictly prohibited. However, good gesture rebate, if offered by the lowest Tenderer prior to finalization of tenders may be accepted by the tender accepting authority.

- ii) Selection of Tenderer among the lowest & equally quoted tenderers will be in the following orders:
 - a) The Tenderer whose bid capacity is higher will be selected.
 - b) In case the bid capacity is also same the Tenderer whose annual turnover is more will be preferred.
 - c) Even if the criteria incidentally become the same, the turnover on similar works and thereafter machinery available for the work and then the clean track record will be considered for selection.

31. Discrepancy in Tender percentage quoted.

- 31.1** In case of any discrepancy between the overall tender percentage quoted in words and figures, the percentage quoted in words shall prevail. In case the Tenderer has quoted overall tender percentage only in words and not in figures or vice versa, such tender shall be treated as incomplete and rejected.

32. Process to be Confidential.

- 32.1** Information relating to the examination, clarification, evaluation and comparison of Tenders and recommendations for the award of a contract shall not be disclosed to Tenderers or any other persons not officially concerned with such process until the award to the successful Tenderer has been announced by the tender accepting authority. Any effort by a Tenderer to influence the processing of Tenders or award decisions may result in the rejection of his Tender.
- 33.1** No Tenderer shall contact the Chief General Manager or any authority concerned with finalization of tenders on any matter relating to its Tender from the time of the Tender opening to the time the Contract is awarded. If the Tenderer wishes to bring additional information to the notice of the Chief General Manager, it should do so in writing.
- 33.2** Before recommending / accepting the tender, the tender recommending / accepting authority shall verify the correctness of certificates submitted to meet the eligibility criteria and specifically experience. The authenticated agreements of previous works executed by the lowest Tenderer shall be called for.
- 33.3** Tenders will be finalized by the Executive Engineers / General Managers / Chief General Managers for the works costing up to Rs.2 Crores. The tenders for the works costing more than Rs.2 Crore will be referred to HGCL Tender Committee along with Technical Bid evaluation for consideration. The Committee of tenders shall scrutinize the tenders submitted by Engineer-in-Chief / Chief General Manager / Project administrators in accordance with the conditions stipulated in the Tender Document and in case any discrepancy of non-adherence to the conditions, the same shall be communicated which will be binding both on the tender

concluding authority and contractor. In case of any ambiguity the decision taken by the HGCL Tender Committee on tenders shall be final.

F. AWARD OF CONTRACT

34. Award Criteria

- 34.1 The Chief General Manager will award or recommend to the competent tender accepting authority for award of the contract to the Tenderer who is found qualified technically and in Mockup test as per the Tender conditions and whose price bid is lowest.
- 34.2 The tender accepting authority reserves the right to accept or reject any Tender or all tenders and to cancel the Tendering process, at any time prior to the award of Contract, without thereby incurring any liability to the affected Tenderer or Tenderers or any obligation to inform the affected Tenderer or Tenderers of the reasons for such action.

35. Notification of Award and Signing of Agreement.

- 35.1 The Tenderer whose Tender has been accepted will be notified of the award of the work by the Chief General Manager, prior to expiration of the Tender validity period by registered letter.

This letter (hereinafter and in the Conditions of Contract called “Letter of Acceptance”) will indicate the sum that the Government will pay the Contractor in consideration of the execution, completion, and maintenance of the Works by the Contractor as prescribed by the Contract (hereinafter and in the Contract called the “Contract Amount”).

- 35.2 When a tender is to be accepted the concerned Tenderer shall attend the office of the Chief General Manager concerned on the date fixed in the Letter of acceptance. Upon intimation being given by the Chief General Manager, of acceptance of his tender, the Tenderers shall make payment of the balance E.M.D., and additional security deposit wherever needed by way of Demand Draft or unconditional and irrevocable Bank Guarantee obtained from a **Nationalized/Schedule Bank**, and sign an agreement in the form prescribed by the department for the due fulfillment of the contract. Failure to attend the Chief General Manager’s office on the date fixed, in the written intimation, to enter into the required agreement shall entail forfeiture of the Earnest Money deposited.
- 35.3 The written agreement to be entered into between the contractor and the Government shall be the foundation of the rights and obligations of both the parties and the contract shall not be deemed to be complete until the agreement has first been signed by the contractor and then by the proper officer authorized to enter into contract on behalf of the Government.

- 35.4 The successful Tenderer has to sign an agreement within a period of 15 days from the date of receipt of communication of acceptance of his tender. On failure to do so his tender will be cancelled duly forfeiting the E.M.D., paid by him without issuing any further notice and action will be initiated for black listing the Tenderer.

36. Corrupt or Fraudulent Practices

36.1 The Government requires that the bidders / suppliers / contractors under Government financed contracts, observe the highest standard of ethics during the procurement and execution of such contracts. In pursuance of this policy, the Government

- (a) define for the purposes of the provision, the terms set forth below as follows:
- (i) “corrupt practices” means the offering, giving, receiving or soliciting of any thing of value to influence the action of a Government official in procurement process or in contract execution: and
- (ii) “fraudulent practice” means a misrepresentation of facts in order to influence a procurement process or the execution of a contract to the detriment of the Government and includes collusive practice among Tenderers (prior to or after Tender submission) designed to establish in Tender prices at artificial non-competitive levels and to deprive the Government of the benefits of free and open competition.
- (b) Will reject a proposal for award if it determines that the Tenderer recommended for award has engaged in corrupt or fraudulent practices in competing for the contract in question.
- (c) Will blacklist / or debar a firm, either indefinitely or for a stated period of time, if at any time determines that the firm has engaged in corrupt or fraudulent practices in competing for, or in executing a Government Contract.
- (d) Further more, Tenderers shall be aware of the provisions stated in the General Conditions of Contract.

37. PUBLIC PROCUREMENT POLICY FOR MICRO AND SMALL ENTERPRISES (MSEs)

37.1 As per the Public Procurement Policy for Micro and Small Enterprise (MSEs) order 2012, issued vide Gazette Notification number 503, dated 23.03.2012 by Ministry of Micro, Small and Medium Enterprise of Govt. of India, and specific to this tender, MSEs must be registered with any of the following agencies/ bodies shall be exempted from Tender Processing Fees and EMD submission, if applicable upon production of valid registration certificate.

- 37.1.1 District Industries Centre (DIC)
- 37.1.2 National Small Industries Corporation (NSIC)
- 37.1.3 Udyog Aadhaar Memorandum

MSEs participating in the tender must submit the certificate of registration with any one of the above agencies. The registration certificate issued from any of the above agencies must be valid as on close date of the tender.

The MSEs, who have applied for registration or renewal of registration with any of the above agencies/ bodies, but have not obtained the valid certificate as on close date of the tender, are not eligible for exemption/ preference.

37.2 **In case of JV/ Consortium, in order to avail the exemption to Tender Processing Fees and Earnest Money Deposit (EMD, if applicable), all the members of JV/ Consortium should be registered under any of the Categories mentioned under clause 40.1 above.**

37.3 In case the bidder is falling under above category, the bidder shall submit the documentary evidence satisfying the same.

If the bidder does not provide the above confirmation or appropriate document or any evidence, then it will be presumed that they do not qualify for any preference admissible in the tender.

38. AHR ITEMS

NOT APPLICABLE

39. RISK OF REJECTION

Any Conditional Bid will straight away run into risk of rejection.

40. INCOME TAX & CORPORATE TAX

40.1 Income tax deduction shall be made from all payments made to the contractor as per the rules and regulations in force and in accordance with the Income Tax Act prevailing from time to time

40.2 Corporate Tax liability, if any, shall be to the contractor's account.

40.3 TDS under GST as may be applicable shall be deducted as per law of Government of India in vogue.

40.4 MENTIONING OF PAN NO. IN INVOICE/ BILL

As per CBDT Notification No. 95/2015 dated 30.12.2015, mentioning of PAN no. is mandatory for procurement of goods/ services/ works/ consultancy services exceeding INR 2 Lacs per transaction.

Accordingly, supplier/ contractor/ service provider/ consultant should mention their PAN no. in their invoice/ bill for any transaction exceeding INR 2 Lacs. As provided in the notification, in case supplier/ contractor/ service provider/ consultant do not have PAN no., they have to submit declaration in Form 60 along with invoice/ bill for each transaction.

Payment of supplier/ contractor / service provider/ consultant shall be processed only after fulfilment of above requirement

===== X =====

Annexure-I

PROCEDURE FOR ACTION IN CASE CORRUPT/ FRAUDULENT/ COLLUSIVE/

COERCIVE PRACTICES

A Definitions:

- A.1 “Corrupt Practice” means the offering, giving, receiving or soliciting, directly or indirectly, anything of value to improperly influence the actions in selection process or in contract execution.

“Corrupt Practice” also includes any omission for misrepresentation that may mislead or attempt to mislead so that financial or other benefit may be obtained or an obligation avoided.

- A.2 “Fraudulent Practice” means and include any act or omission committed by an agency or with his connivance or by his agent by misrepresenting/ submitting false documents and/ or false information or concealment of facts or to deceive in order to influence a selection process or during execution of contract/ order.

- A.3 “Collusive Practice amongst bidders (prior to or after bid submission)” means a scheme or arrangement designed to establish bid prices at artificial non-competitive levels and to deprive the Employer of the benefits of free and open competition.

- A.4 “Coercive practice” means impairing or harming or threatening to impair or harm directly or indirectly, any agency or its property to influence the improper actions of an agency, obstruction of any investigation or auditing of a procurement process.

- A.5 “Vendor/ Supplier/ Contractor/ Consultant/ Bidder” is herein after referred as “Agency”

- A.6 “Competent Authority” shall mean the authority, who is competent to take final decision for Suspension of business dealing with an Agency/ (ies) and Banning of business dealings with Agency/ (ies) and shall be the “Committee” concerned.

- A.7 “Allied Agency” shall mean all the concerns within the sphere of effective influence of banned/ suspended agencies. In determining this, the following factors may be taken into consideration:

- (a) Whether the management is common;
- (b) Majority interest in the management is held by the partners or directors of banned/suspended firm.
- (c) substantial or majority shares are owned by banned/ suspended agency and by virtue of this it has a controlling voice.

“Investigating Agency” shall mean any department or unit of Employer/ HGCL investigating into the conduct of Agency/ party and shall include the Vigilance Department of the Employer/ HGCL, Central Bureau of Investigation, State Police or any other agency set up by the Central or state government having power to investigate.

B Actions against bidder(s) indulging in corrupt /fraudulent/ collusive/ coercive practice

B.1 Irregularities noticed during the evaluation of the bids:

If it is observed during bidding process/ bids evaluation stage that a bidder has indulged in corrupt/ fraudulent/ collusive/ coercive practice, the bid of such Bidder (s) shall be rejected and its Earnest Money Deposit (EMD, if applicable) shall be forfeited.

Further, such agency shall be banned for future business with Employer/ HGCL for a period specified in para B 2.2 below from the date of issue of banning order.

B.2 Irregularities noticed after award of contract

(i) During execution of contract:

If an agency, is found to have indulged in corrupt/ fraudulent/ collusive/ coercive practices, during execution of contract, the agency shall be banned for future business with Employer/ HGCL for a period specified in para B 2.2 below from the date of issue of banning order.

The concerned order(s)/ contract(s) where corrupt/ fraudulent/ collusive practices are observed, shall be suspended with immediate effect by Engineer-in-Charge (EIC)/ Employer/ HGCL whereby the supply/ work/ service and payment etc. will be suspended. The action shall be initiated for putting the agency on banning list.

After conclusion of process, the order(s)/ contract(s) where it is concluded that such irregularities have been committed shall be terminated and Contract Performance Security submitted by agency against such order(s)/ contract(s) shall also be forfeited. The amount that may have become due to the contractor on account of work already executed by him shall be payable to the contractor and this amount shall be subject to adjustment against any amounts due from the contractor under the terms of the contract.

No risk and cost provision will be enforced in such cases.

(ii) After execution of contract and during Defect liability period (DLP)/ Warranty/Guarantee Period:

If an agency is found to have indulged in corrupt/ fraudulent/ collusive/ coercive practices, after execution of contract and during DLP/ Warranty/ Guarantee Period/O&M Period, the agency shall be banned for future business with Employer/ HGCL for a period specified in para B 2.2 below from the date of issue of banning order.

Further, the Contract Performance Security submitted by agency against such order(s)/ contract(s) shall be forfeited.

(iii) After expiry of Defect liability period (DLP)/ Warranty/ Guarantee Period

If an agency is found to have indulged in corrupt/ fraudulent/ collusive/ coercive practices, after expiry of Defect liability period (DLP)/ Warranty/ Guarantee Period, the agency shall be banned for future business with Employer/ HGCL for a period specified in para B 2.2 below from the date of issue of banning order.

B.2.2 Period of Banning

Banning period shall be reckoned from the date of issuance of banning order and shall be for a period as may be decided by the Employer/ HGCL based on specific case basis. However, minimum period of ban shall be 06 (Six) months from the date of issuance of banning order.

In exceptional cases where the act of vendor/ contractor is a threat to the National Security, the banning shall be for indefinite period.

C Effect of banning on other ongoing contracts/ tenders

- C.1 If an agency is banned , such agency shall not be considered in ongoing tenders/ future tenders.
- C.2 However, if such an agency is already executing other order(s)/ contract(s) where no corrupt/fraudulent/ collusive/ coercive practice is found, the agency shall be allowed to continue till its completion without any further increase in scope except those incidental to original scope mentioned in the contract.
- C.3 If an agency is banned during tendering and irregularity is found in the case under process:
 - C.3.1 after issue of the enquiry/ bid/ tender but before opening of Technical bid, the bid submitted by the agency shall be ignored.
 - C.3.2 after opening Technical bid but before opening the Price bid, the Price bid of the agency shall not be opened and EMD, if applicable submitted by the agency shall be returned to the agency.
 - C.3.3 after opening of price bid, EMD, if applicable made by the agency shall be returned; the offer/Bid of the agency shall be ignored & will not be further evaluated. If the agency is put on banning list for fraud/ mis-appropriation of facts committed in the same tender/ other tender where errant agency emerges as the lowest (L1), then such tender shall also be cancelled and re-invited.

D. Procedure for Suspension of Bidder

D.1 Initiation of Suspension

Action for suspension business dealing with any agency/(ies) shall be initiated by C&P Department when

- (i) Vigilance Department based on the fact of the case gathered during investigation by them recommend for specific immediate action against the agency.
- (ii) Vigilance Department based on the input from Investigating agency, forward for specific immediate action against the agency.
- (iii) Nonperformance of Vendor/ Supplier/ Contractor/ Consultant leading to termination of Contract/ Order.

D.2 Suspension Procedure:

- D.2.1 The order of suspension would operate initially for a period not more than six months and is to be communicated to the agency and also to Vigilance Department. Period of suspension can be extended with the approval of the Competent Authority by one month at a time with a ceiling of six months pending a conclusive decision to put the agency on banning list.
- D.2.2 During the period of suspension, no new business dealing may be held with the agency.
- D.2.3 Period of suspension shall be accounted for in the final order passed for banning of business with the agency.
- D.2.4 The decision regarding suspension of business dealings should also be communicated to the agency.
- D.2.5 If a prima-facie, case is made out that the agency is guilty on the grounds which can result in banning of business dealings, proposal for issuance of suspension order and show cause notice shall be put up to the Competent Authority. The suspension order and show cause notice must include that (i) the agency is put on suspension list and (ii) why action should not be taken for banning the agency for future business from Employer/ HGCL.

The competent authority to approve the suspension will be same as that for according approval for banning.

D.3 Effect of Suspension of business:

Effect of suspension on other on-going/ future tenders will be as under:

- D.3.1 No enquiry/ bid/ tender shall be entertained from an agency as long as the name of agency appears in the Suspension List.
- D.3.2 If an agency is put on the Suspension List during tendering:
- D.3.3 after issue of the enquiry/ bid/ tender but before opening of Technical bid, the bid submitted by the agency shall be ignored.
- D.3.4 after opening Technical bid but before opening the Price bid, the Price bid of the agency shall not be opened and EMD, if applicable submitted by the agency shall be returned to the agency.
- D.3.5 The existing contract(s)/ order(s) under execution shall continue.

Tenders invited for procurement of goods, works and services shall have provision that the bidder shall submit a undertaking to the effect that (i) neither the bidder themselves nor their allied agency/(ies) are on banning list of Employer/ HGCL or the Ministry of New & Renewable Energy (ii) bidder is not banned by any Government Department/ Public Sector.

Annexure-II

ADDENDUM TO INSTRUCTIONS TO BIDDERS (INSTRUCTION FOR PARTICIPATION IN E-TENDER)

1. General

Special Instructions (for e-Tendering).

Submission of Online Bids is mandatory for this Tender.

This section contains detailed instructions regarding bid submission procedure under e- tendering system (e-tender portal). Bidders are required to read the following instructions in addition to various instructions mentioned elsewhere in the bid document for e-tendering. The instructions mentioned herein related to bid submission procedure shall supersede and shall prevail over the conditions enumerated elsewhere in the bid/ tender document.

Bidders who wish to participate in e-tenders must go through the 'instructions in respect of e-Tendering essentially covering security settings required for bidder's PC/ Laptop, uploading and checking the status of digital signature in the bidder's PC/ Laptop, obtaining unique login ID and password, re-setting the password, downloading of Tender document and uploading of Offer/ Bid etc.

2. About E-Tender Portal (Web)

Hyderabad Growth Corridor Limited(HGCL) has adopted a secured and user friendly e- tender system enabling bidders to Search, View, Download tender document(s) directly from the ETS Portal of M/s ElectronicTender.com (India) Pvt. Limited (<http://eprocurement.telangana.gov.in>) through ISN Electronic Tender Services Private Limited (referred as ISN-ETS) (referred hereunder as "E-Tender Portal") and also enables the bidders to participate and submit online prices directly in the e-tender portal or uploading of SOR/ Price Schedule (Price Bid) in note and attachment folder, as applicable, and

technical bids in secured and transparent manner maintaining absolute confidentiality and security throughout the tender evaluation process and award.

3. Pre-requisite for participation in bidding process

The following are the pre-requisite for participation in e-Tendering Process:

3.1 PC/ Laptop with Windows OS, Internet Explorer

Bidder must possess a PC/ Laptop with Windows 7 professional operating system and Internet Explorer 8 or 9 for hassle free bidding. Bidder is essentially required to effect the security settings as defined in the portal.

3.2 Internet Broadband Connectivity

The Bidder must have a high-speed internet connectivity (preferably Broadband) with internet explorer to access website e-Tender Portal for downloading the Tender document and uploading/ submitting the Bids.

3.3 A valid e-mail Id of the Organization/ Firm

- 3.4** E-Tendering is a new methodology for conducting Public Procurement in a transparent and secured manner. Now, the Government of India has made e-tendering mandatory. Suppliers/Vendors will be the biggest beneficiaries of this new system of procurement. For conducting electronic tendering, HGCL has decided to use the portal of M/s **Vupadhi Techno Services Pvt. Ltd.** <http://eprocurement.telangana.gov.in>. This portal is based on the world's most 'secure' and 'user friendly' software from Electronic Tender®. A portal built using Electronic Tender's software is also referred to as Electronic Tender System® (ETS).

Benefits to Suppliers are outlined on the Home-page of the portal.

Tender Bidding Methodology:

Sealed Bid System

Single Stage Two Part

4. Tender Bidding Methodology under Sealed Bid System of Single Stage Two Envelop:

4.1 Broad Outline of Activities from Bidder's Perspective:

- 4.1.1 Procure a Class III Digital Signing Certificate (DSC).
- 4.1.2 Register on Electronic Tender System® (ETS)
- 4.1.3 Create Marketing Authorities (MAs), Users and assign roles on ETS. It is mandatory to create at least one MA
- 4.1.4 View Notice Inviting Tender (NIT) on ETS
- 4.1.5 For this tender -- Assign Tender Search Code (TSC) to a MA
- 4.1.6 Download Official Copy of Tender Documents from ETS Note: Official copy of Tender Documents is distinct from downloading 'Free Copy of Tender Documents'. To participate in a tender, it is mandatory to procure official copy of Tender Documents for that tender.

4.1.7 Clarification to Tender Documents on ETS

- a) Query to HGCL (Optional)
- b) View response to queries posted by HGCL

4.1.8 Bid-Submission on ETS

4.1.9 Respond to HGCL Post-TOE queries

4.1.10 Participate in reverse auction if invited

For participating in this tender online, the following instructions are to be read carefully. These instructions are supplemented with more detailed guidelines on the relevant screens of the ETS.

4.2 Digital Certificates

For integrity of data and authenticity/ non-repudiation of electronic records, and to be compliant with IT Act 2000, it is necessary for each user to have a Digital Certificate (DC), also referred to as Digital Signature Certificate (DSC), of Class III, issued by a Certifying Authority (CA) licensed by Controller of Certifying Authorities (CCA) [refer <http://www.cca.gov.in>]

4.3 Registration

To use the Electronic Tender® portal <http://eprocurement.telangana.gov.in>, vendors need to register on the portal. Registration of each organization is to be done by one of its senior persons who will be the main person coordinating for the e-tendering activities. In ETS terminology, this person will be referred to as the Super User (SU) of that organization. For further details, please visit the website/ portal, and click on the 'Supplier Organization' link under 'Registration' (on the Home Page), and follow further instructions as given on the site. Pay Annual Registration Fee as applicable.

After successful submission of Registration details and payment of Annual Registration Fee, please contact Helpdesk (as given below), to get your registration accepted/ activated.

Important Note:

1. Interested bidders have to download official copy of the RfS/ Tender & other documents after login into the web Portal of <http://eprocurement.telangana.gov.in>). If the official copy of the documents is not downloaded from ETS Portal of within the specified period of downloading of RfS/ Tender and other documents, bidder will not be able to participate in the tender.
2. To minimize teething problems during the use of website (including the Registration process), it is recommended that the user should peruse the instructions given under 'ETS User-Guidance Centre' located on website Home Page, including instructions for timely registration on website. The instructions relating to 'Essential Computer Security Settings for Use of ETS' and 'Important Functionality Checks' should be especially taken into cognizance.

Please note that even after acceptance of your registration by the Service Provider, to respond to a tender you will also require time to complete activities related to your organization, such as creation of users, assigning roles to them, etc.

ETS Helpdesk
Vupadhi Techno Services Pvt. Ltd.

1st Floor, Ramky Grandiose,
Sy. No: 136/2 & 4, Gachibowli,
Hyderabad - 500032.
Telangana State.

Remarks 1000 hrs to 1800 hrs on all working days i.e. from Monday to Friday except Government Holidays.

5. Some Bidding Related Information for this Tender (Sealed Bid)

The bid shall be submitted by the Bidder under "Single Stage - Two Part" procedure of bidding. Under this procedure, the bid submitted by the Bidder in two Parts - First Part (also referred to as Techno - Commercial Part) and Second Part (also referred to as Price Part) shall comprise of the following documents:

5.1 Offline Documents: DELETED

Offline documents of the bid shall comprise of following documents/ programmed file- Attachments to be submitted in sealed Part, as part of First Part. **The Part shall bear {the name of Tender, the Tender No. and the words 'DO NOT OPEN BEFORE'(due date & time)}.**

Contact Persons Name: Mr. Sampath Kumar

- (a) Original Non-Refundable Tender Processing Fee as per clause no. 05 of ITB
- (b) Original Non-Refundable Cost of Tender Document, if applicable
- (c) 'Covering Letter' on Bidder's 'Letterhead' (in Original) clearly specifying the enclosed contents, as per 'Form F-0'
- (d) EMD, in original as per Clause 16 of ITB as per 'Form F-4A' or as prescribed.
- (f) Power of Attorney for authorized signatory in non-judicial stamp paper (as per 'Form F-24')
- (g) Copy of Board Resolution
- (h) The Pass-Phrase to decrypt the relevant Bid-Parts (for both Techno-Commercial and Financial) in separate sealed Parts before the start date and time of the Tender Opening Event (TOE)

Bidder shall also upload the scanned copies of all the above-mentioned original documents during online Bid Submission as a part of First Part.

"Bidder should explicitly note that no offline documents are to be submitted as a part of Second Part".

II. Online Documents/Soft Copy

Online documents/Soft copy of the bid shall comprise of following documents to be uploaded on the ISN portal <http://eprocurement.telangana.gov.in> as per provisions therein.

As part of First Part

- I. The Electronic Form of the bid for First Part (Techno-Commercial), as available on the portal, shall be duly filled.
- II. Scanned copies of all the above-mentioned original documents during online Bid Submission as a part of First Part and Bid Form for first Part
- III. Certificate of Incorporation
- IV. 'Bidder's General Information', as per 'Form F-1'.
- V. 'No Deviation Confirmation', as per 'Form F-6'
- VI. 'Bidder's Declaration regarding Banning, Liquidation etc.', as per 'Form F-7'
- VII. 'Declaration on restriction on procurement from certain countries.', as per 'Form F-8'
- VIII. 'Declaration regarding the procurement of Solar Inverters & Solar Modules from class I local suppliers, as per 'Form F-12'
- IX. 'Bidders Experience as per 'Form F-13'
- X. Disclosure for ALMM compliance as per 'Form F-14'
- XI. Format of Chartered Accountant certificate for financial capability of the bidder as per 'Form F-16'
- XII. 'E-Banking Format as per 'Form F-19'
- XIII. 'Shareholding Certificate' as per 'Form F-23'
- XIV. 'Format for Cyber Security Agreement' as per 'Form F-25'
- XV. 'Integrity Pact Format' as per 'Form F-26'
- XVI. Documents in accordance with the "Qualifying Requirements (QR)" establishing the qualification
- XVII. Document showing annual turnover for the financial years as required in Qualifying Requirements (QR) such as annual reports, profit and loss account, net worth etc. along with information as sought in enclosed Format F-16
- XVIII. Tender Document. (Only First and Last Pages of Original Tender Document duly sealed and signed/ digitally signed and all pages of amendments to Tender Documents duly sealed and signed/ digitally signed by the Authorized Signatory).

5.2 (B) As part of Second Part

- (a) The Electronic Form of the bid for Second Part (Price - Part), as available on the ISN portal, shall be duly filled. "Termed as **ELECTRONIC FORM**"
- (b) Main Price Bid of financial bid comprising of SOR-1 to 4 of the Price Schedule (available in Section - VIII, SOR), duly Completed, sealed and signed/ digitally signed shall be uploaded. "Termed as **MAIN BID**"

5.3 Special Note on Security and Transparency of Bids

Security related functionality has been rigorously implemented in ETS in a multidimensional manner. Starting with 'Acceptance of Registration by the Service Provider', provision for security has been made at various stages in Electronic Tender's software. Specifically, for Bid Submission, some security related aspects are outlined below:

As part of the Electronic Encrypted® functionality, the contents of both the 'Electronic Forms®' and the 'Main-Bid' are securely encrypted using a Pass-Phrase created by the Bidder himself. Unlike a 'password', a Pass-Phrase can be a multi-word sentence with spaces between words (e.g. I love this World). A Pass-Phrase is easier to remember, and more difficult to break. It is mandatory that a separate Pass-Phrase be created for each Bid-Part. This method of bid- encryption does not have the security and data-integrity related vulnerabilities which are inherent in e-tendering systems which use Public-Key of the specified officer of a Buyer organization for bid-encryption. Bid-encryption in ETS is such that the Bids cannot be decrypted before the Public Online Tender Opening Event (TOE), even if there is connivance between the concerned tender-opening officers of the Buyer organization and the personnel of e-tendering service provider.

CAUTION: All bidders must fill Electronic Form for each bid-part sincerely and carefully, and avoid any discrepancy between information given in the Electronic Form and the corresponding Main-Bid.

If any variation is noted between the price mentioned in the Electronic Form and the Main Bid (Refer Clause above for the definitions of Electronic Form and Main Bid), the price mentioned in the Main Bid shall prevail.

In case of any discrepancy between the values mentioned in figures and in words, the value mentioned in words will prevail.

The bidder shall make sure that the Pass-Phrase to decrypt the relevant Bid-Part is submitted into the 'Time Locked Electronic Key Box (EKB)' after the deadline of Bid Submission and before the commencement of the Online TOE of Technical Bid. The process of submission of this Pass-Phrase in the 'Time Locked Electronic Key Box' is done in a secure manner by first encrypting this Pass-Phrase with the designated keys provided by the Employer.

Additionally, the bidder shall make sure that the Pass-Phrase to decrypt the relevant Bid-Part is submitted to Employer in a sealed Part before the start date and time of the Tender Opening Event (TOE).

5.4 Other Instructions

For further instructions, the vendor should visit the home-page of the portal <http://eprocurement.telangana.gov.in>, and go to the User-Guidance Centre

The help information provided through 'ETS User-Guidance Centre' is available in three categories – Users intending to Register/ First-Time Users, Logged-in users of Buyer organizations, and Logged-in users of Supplier organizations. Various links (including links for User Manuals) are provided under each of the three categories.

Important Note: It is strongly recommended that all authorized users of Supplier organizations should thoroughly peruse the information provided under the relevant links, and take appropriate action. This will prevent hiccups, and minimize teething problems during the use of ETS.

5.5 Seven CRITICAL DO'S AND DON'TS FOR BIDDERS

Specifically, for Supplier organizations, the following 'Seven KEY INSTRUCTIONS for BIDDERS' must be assiduously adhered to:

1. Obtain individual Digital Signing Certificate (DSC or DC) of Class III well in advance of your tender submission deadline on ETS.
2. Register your organization on ETS well in advance of the important deadlines for your first tender on ETS viz 'Date and Time of Closure of Procurement of Tender Documents' and 'Last Date and Time of Receipt of Bids'. Please note that even after acceptance of your registration by the Service Provider, to respond to a tender you will also require time to complete activities related to your organization, such as creation of -- Marketing Authority (MA) [ie a department within the Supplier/ Bidder Organization responsible for responding to tenders], users for one or more such MAs, assigning roles to them, etc. It is mandatory to create at least one MA. This unique feature of creating an MA enhances security and accountability within the Supplier/ Bidder Organization.
3. Get your organization's concerned executives trained on ETS well in advance of your first tender submission deadline on ETS
4. For responding to any particular tender, the tender (ie its Tender Search Code or TSC) has to be assigned to an MA. Further, an 'Official Copy of Tender Documents' should be procured/ downloaded before the expiry of Date and Time of Closure of Procurement of Tender Documents. **Note:** Official copy of Tender Documents is distinct from downloading 'Free Copy of Tender Documents'. Official copy of Tender Documents is the equivalent of procuring physical copy of Tender Documents with official receipt in the paper-based manual tenderingsystem.
5. Submit your bids well in advance of tender submission deadline on ETS (There could be last minute problems due to internet timeout, breakdown, etc)
6. It is the responsibility of each bidder to remember and securely store the Pass-Phrase for each Bid-Part submitted by that bidder. The bidders are required to submit correct, valid and operative Pass-Phrase to decrypt either Technical Bid Part or Financial Bid Part in a separate sealed Part before due date and time of submission of bid. In the event, the bids are not opened with the pass-phrase submitted by bidder, Employer may ask for re-submission/ clarification for correct pass-phrase. If bidder fails to submit correct pass-phrase immediately as requested by Employer, the Tender Processing Fee and Tender Document Fee, if applicable, shall be forfeited and bid shall not be opened, and EMD, if applicable shall be refunded. No request on this account shall be entertained by Employer/ HGCL.

7. ETS will make your bid available for opening during the Online Public Tender Opening Event (TOE) 'ONLY IF' the status pertaining Overall Bid-Submission is 'COMPLETE'. For the purpose of record, the bidder can generate and save a copy of 'Final Submission Receipt'. This receipt can be generated from 'Bid-Submission Overview Page' only if the status pertaining overall Bid-Submission is 'COMPLETE'

NOTE:

While the first three instructions mentioned above are especially relevant to first-time users of ETS, the fourth, fifth, sixth and seventh instructions are relevant at all times.

6 Content of Bid

6.1 Un-Priced Bid

The Un-Priced Bid (i.e. Part I - Technical Bid, refer ITB of tender for details) to be uploaded using Link "TECHNICAL DOCUMENT". ***Bidders are advised to not upload duly filled Schedule of Rates (SOR)/ Price Schedule (PS) at this link.***

Before the bid is uploaded, the bid comprising of all relevant documents mentioned in the tender document should be digitally signed in accordance with the Indian IT Act 2000. If any modifications are required to be made to a document thereafter the modified documents shall be again digitally signed before uploading.

Bidders are required to upload all Tender forms and supporting documents which form part of the bid/ tender in the Collaboration Folder (C-Folder) under tender document of e-tender portal.

Uploading the documents relevant to bid before the bid submission date and time is the sole responsibility of the bidder.

The complete bid (each page) shall be continuously numbered in sequence, from start till end i.e. 1, 2, 3... n. The bid shall be uploaded along with proper index and indicating page numbers against each category of documents. A sample is suggested as under:

Sub-folder in C-Folder	Documents	Page nos. (for ref.)
1. INDEX	Index	1 to a
2. TENDER DOC	Tender Document, Corrigendum etc.	a+1 to b
3. B.E.C	BEC documents (including order, completion/ execution certificate, balance sheet, etc.)	b+1 to c
4. UN-PRICED COMMERCIAL	Un-priced Commercial offer/bid	c+1 to d
5. EMD/POA	Scanned copy of EMD/POA	d+1 to e
6. FORMATS	Formats of tender duly filed-in, signed and stamped and other	e+1 to f
7. TENDER FEE	Scanned copy of Tender Fee/ Integrity Pact (I.P.)	f+1 to f
8. OTHERS	Any other document	g+1 to n

The above shall ensure that there are no missing documents and traceability of relevant section is ensured.

Note:

- i) Bidder may save/ store the bid documents in the PC/ Laptop before submitting the bid into e-tender portal.
- ii) Bidder is required to fill up the price(s)/ rate(s) strictly in the Schedule of Rate (SOR)/Price Schedule (PS) attached with the tender.

Inadvertently, if a document is uploaded in Collaboration Folder (C Folder) by the bidders, such document can be deleted by the bidder and can be replaced by a digitally signed new/ modified document prior to due date & time.

Un-priced techno-commercial bid document should be placed in the private area earmarked in the C-folder of Tender Document in e-tender portal.

6.2 Price Bid

The Price Bid (i.e. Part II - Price Bid, refer ITB for details) to be uploaded using Link given under ISN portals & and NOT using link "TECHNICAL DOCUMENT". ***Bidders are advised not to upload any other documents and same shall be ignored.*** For detailed instructions, refer tender document and instructions as given above.

Employer/ HGCL shall not be responsible for any failure on the part of the bidder in submission of Priced Bid.

Instructions mentioned under "PRICE BID [Online]" shall be applicable in case Bidders have been asked to quote their prices on-line directly in the ISN's e-tender portal in addition to uploading of scanned copy of SOR/ PS or only the on-line price submission in the portal, as the case may be.

7 Submission of documents

Bidders are requested to upload small sized documents preferably (upto 10 MB) at a time to facilitate easy uploading into e-tender portal. Employer/ HGCL shall not be responsible in case of failure of the bidder to upload documents/ bids within specified time of tender submission.

8 Last date for submission of bids

System does not allow for submission or modification of bids beyond the deadline for bid submission. However, if the bidder for some reason intends to change the bid already entered, he may change/ revise the same on or before the last date and time of submission. The system time displayed on ISN's e-tender webpage shall be final and binding on the bidders for all purposes pertaining to various events of the subject tender and no other time shall be taken into cognizance.

Bidders are advised in their own interest to ensure that bids are uploaded in e-tender system well before the closing date and time of bid.

9 Internet connectivity

If bidders are unable to access ISN's e-tender portal or Bid Documents, the bidders may please check whether they are using proxy to connect to internet or their PC is behind any firewall and may contact their system administrator to enable connectivity. Please note that Port SSL/443 should be enabled on proxy/firewall for HTTPS connectivity. Dial-up/ Broad and internet connectivity without Proxy settings is another option.

SECTION - III

BID DATA SHEETS (BDS)

BID DATA SHEETS (BDS)

The following bid specific data for the Plant and Equipment to be procured shall amend and/ or supplement the provisions in the Section - II, Instruction to Bidders (ITB)

Sl. No.	ITB Clause Ref. No.	Bid Data Details
1.	SCOPE OF BID (ITB 1.1)	The HGCL wishes to receive bids as described in the bidding documents and submitted in following address The Hyderabad Growth Corridor Ltd., Hyderabad, Telangana Nanakramguda, Hyderabad Kind Attn: Sri.P. Sampath Kumar, Dy., GM (Elec) email: dgmelehqcl@gmail.com
2.	BIDS FROM CONSORTIUM/ JOINT VENTURE (ITB 3.0)	<u>Replace the existing clause by the following:</u> Bids from Consortium/ Joint Venture are ALLOWED
3.	NUMBER OF BIDS PER BIDDER (ITB 4.0)	<u>Replace the existing clause by the following:</u> Bidder shall submit only 'one [01] Bid' per bidder in the Bidding Process. A Bidder who submits or participates in more than 'one [01] Bid' will cause all the proposals in which the Bidder has participated to be disqualified. A Bidder

Sl. No.	ITB Clause Ref. No.	<u>Bid Data Details</u>
		<u>including its Parent, Affiliate or Ultimate Parent or any Group Company may submit a single bid.</u> <u>Note: In case a common Company/Companies directly or indirectly hold(s) more than 10% but less than 26% shareholding in more than one Bidder participating in the Tender, each of such Bidders will be required to submit the Disclosure as per the attached format. In all other cases, this disclosure is not required & standard forms & formats will be applicable.</u>
4.	TENDER PROCESSING FEE (NON-REFUNDABLE) (ITB 5.3)	<u>Replace the existing clause by the following:</u> <u>The Tender Processing Fee is exempted for MSME Vendors registered under NSIC/ Udyog Aadhaar/DIC Category only.</u>
5.	SITE VISIT (ITB 6)	<u>Add the following to the existing clause:</u> <u>6.4 The bidder shall himself obtain all the necessary information as to risks, contingencies and other circumstances which may influence or affect his offer. No consequent extra claims on any misunderstanding or otherwise shall be allowed by the Employer/HGCL.</u>
6.	GOODS & SERVICE TAX (GST) (ITB 13.0)	<u>Add the following to the existing clause:</u> <u>Tax retention clause: The supplier shall upload his Tax Invoice in the GSTN as per the provisions of the GST Act i.e., by 10th of the month subsequent to the month in which “Time of Supply” arises. In respect of orders where the entire order quantity is executed through a single Tax Invoice and if the “Tax Invoice” is not uploaded within the time limits prescribed under GST Act, the amount of tax contained in the Invoice will be withheld by HGCL till such time the invoice is uploaded in the GSTN Portal.</u> <u>In respect of orders where the entire order quantity is executed in phased manner through multiple invoices or where staggered payment is made, if the “Tax Invoice” is not uploaded within the time limits prescribed under GST Act, the tax amount will be withheld from the payment made against subsequent Invoice till such time the invoice is uploaded. The final payment is subject to compliance of all formalities under GST by the supplier.</u>

7.	BID CURRENCIES (ITB 14)	Indian Rupees (INR)
8.	BID VALIDITY (ITB 15.1)	<u>Replace the existing clause by the following:</u> The bid validity period shall be 180 (One Hundred and Eighty Days) from the date of opening of Techno-Commercial Bid (Part-I).
9.	EARNEST MONEY DEPOSIT (EMD) (ITB 16)	<u>Replace the existing clause by the following:</u> 16.7 Forfeiture of the EMD of any bidder is upon HGCL discretion and the EMD forfeited amount shall be refunded to HGCL. Notwithstanding anything contained herein, the 'EMD' may also be forfeited in any of the following cases.: (a) If a Bidder withdraws or varies his Bid during the 'Period of Bid Validity' (b) If a Bidder has indulged in corrupt/ fraudulent/ collusive/ coercive practice (c) If the Bidder modifies bids during the period of bid validity (after last date of submission). (d) Violates any other condition, mentioned elsewhere in the tender document, which may lead to forfeiture of EMD. (e) In the case of a successful Bidder, if the Bidder fails to: (i) acknowledge receipt the NOA/ LOI/ LOA within 15 (Fifteen) days from issuance of same. (ii) to furnish "Performance Security in accordance with "ITB: Clause-38" (iii) to accept 'arithmetical corrections' as per provision of the clause 30 of ITB. 16.9 MSMEs (Micro, Small and Medium Enterprises) registered under NSIC/ Udyog Aadhar/DIC Only are exempted from submission of EMD.

10.	ZERO DEVIATION & REJECTION CRITERIA (ITB 19.1 & 19.2)	Clause No 19.1, Zero deviation & Clause No 19.2, Rejection Criteria should be read in conjunction to each other & both clauses points mentioned should complement each other.
11.	DEADLINE FOR SUBMISSION OF BIDS (ITB 22.2)	Add following to the existing clause: For hard copies of bid submission purpose, the address of the Employer is: The Hyderabad Growth Corridor Ltd., Hyderabad, Telangana Nanakramguda, Hyderabad Kind Attn: Sri. P. Sampath Kumar, Dy., GM (Elec) email: dqmelehgcl@gmail.com
12.	BID OPENING (ITB 26.1)	Add following to the existing clause: The bid opening shall take place at: The Hyderabad Growth Corridor Ltd., Hyderabad, Telangana Nanakramguda, Hyderabad Kind Attn: Sri. P. Sampath Kumar, Dy., GM (Elec) email: dqmelehgcl@gmail.com
13.	AWARD (ITB 35)	Replace the existing clause by the following: Subject to "ITB: Clause-29", For Sites (Site 1,2 & 3), HGCL, Hyderabad Growth Corridor Limited (HGCL) will award the Contract Agreement (CA) on respective site basis to the successful Bidder, whose Bid has been determined to be substantially responsive and has been determined as the lowest (L1) provided that bidder is determined to be qualified to satisfactorily perform the Contract Technically.
14.	NOTIFICATION OF AWARD/ LETTER OF INTENT/ LETTER OF ALLOCATION (ITB 36.1)	Add following to the existing clause Prior to the expiry of 'Period of Bid Validity', For Sites (Site 1,2 & 3), HGCL, Hyderabad Growth Corridor Limited (HGCL) will notify the successful bidder in writing, in the form of "Notification of Award (NOA)"/ "Letter of Intent (LOI)"/ "Letter of Allocation (LOA)" on respective site basis through e-mail/ courier/ registered post, that his Bid has been accepted. The notification of award will constitute the formation of the Contract

15.	SIGNING OF CONTRACT AGREEMENT (ITB 37.1)	Add following to the existing clause State of which stamp paper is required for Contract Agreement: Telangana, The value of stamp paper will be INR 100/-
16.	NEW CLAUSE (ITB 44)	Add following new clause 44 to the ITB 44. Restrictions on procurement from a Bidder of a country which shares a land border with India: 44.1. Any Bidder (including its JV partner/ Consortium Member, wherever applicable) from a country which shares a land border with India will be eligible to bid in this tender only if bidder is registered with the Competent Authority which is the Registration Committee constituted by the Department for Promotion of Industry & Internal Trade (DPIIT) in line with the MoF OM No 6/18/2019-PPD dated 23.07.2020. Further the successful bidder shall not be allowed to subcontract supplies/services/works to any “Sub-contractor” under “Second Contract”, from a country which shares a land border with India unless such Subcontractor is registered with the competent as mentioned above. The Contractor shall not be allowed to sub-contract works to any sub-contractor/ sub-vendor from a country which shares a land border with India unless such sub-contractor is registered with the competent Authority. However, the same shall not be applicable for “First Contract i.e. Supply Contract”. However, the said requirement of registration will not apply to bidders/sub- contractors from those countries (even if sharing a land border with India) to which the Government of India has extended lines of credit or in which the Government of India is engaged in development projects. Bidders may apprise themselves of the updated lists of such countries available in the website of the Ministry of External Affairs.

ANNEXURE
TO BID DATASHEET
(BDS)
QUALIFYING
REQUIREMENTS
replacement

QUALIFYING REQUIREMENTS (QR)

Qualification of the bidder(s) will be based on their meeting the minimum eligibility criteria specified below regarding the Bidder's General Standards, Technical Experience and Financial eligibility as demonstrated by the Bidder's responses in the corresponding Bid documents. The bid can be submitted by an individual Company or by a Joint Venture/Consortium of not more than 03 (Three) companies. (Specific requirements for Joint Ventures/Consortium are given below).

1.1 GENERAL ELIGIBILITY CONDITIONS

The Bidder should be a body incorporated in India under the Companies Act, 1956 or 2013 including any amendment thereto, Government owned Enterprises who are registered and incorporated in India, Limited Liability Companies engaged in the business of Power/Infrastructure, barring those companies against whom sanction for conducting business is imposed by the Government of India or HGCL.

It's a "Domestic Competitive Bidding" and the bidder should be an Indian company registered in India and should be Group company or Holding Company or Subsidiary company of a company meeting the requirement (s) of Clause 1.2 and 1.3 below. The subsidiary of a foreign company, which is registered in India under Companies Act, 1956 or 2013 prior to the bid submission deadline, is eligible to participate.

Limited Liability Partnerships, Proprietorships, Partnerships, NGOs, Charitable Trusts, and Educational Societies are not allowed to participate in the bidding process (either individually or in Joint Venture/ Consortium).

The Bid Processing Fees is exempted for MSME bidders registered under NSIC/ DIC/ Udyog Aadhaar Category only. As the Proprietorships, Partnerships, NGOs, Charitable Trusts, Educational Societies in any form are not allowed to participate in the tendering process, hence any Proprietorships, Partnerships, NGOs, Charitable Trusts, Educational Societies MSME vendor will also not be eligible to participate in the bidding process (either individually or in Joint Venture/ Consortium).

1.2 TECHNICAL ELIGIBILITY CONDITIONS

The Bidder shall be considered meeting Technical Eligibility criteria

The bidder should have experience in Solar Projects including Design, Supply (Supply of Modules & Inverters can be in the bidder's scope in the past experience), Installation and Commissioning of Grid connected Solar PV Power Plant Capacity not less than **13 (Thirteen) MW** in any one financial year in last Ten Financial years as on last date of bid submission. However, such Grid connected Solar PV Power Plant capacity must have been in satisfactory operation for at least six (06) months prior to the last date of bid submission.

Eligibility Criteria:

Single entity/ JV Consortium with members not exceeding two (2). Any agency can bid individually or can participate in only one JV Consortium

- i) The bidders should possess **Class-I (Electrical)** with Electrical “A’ grade license issued by CEIG Electrical not less than 33KV level issued by State Govt./Govt. Undertakings/ Public Sector of Telangana State.
- ii) Copy of GST Registration.
- iii) Copy of Permanent Account Number i.e. (PAN) Card.
- iv) Proof of payment of EMD.
- v) Copy of latest Income Tax returns i.e. FY 2021-22 submitted to IT Dept. along with proof of its submission
- vi) Liquid assets and/ or credit facilities of not less than **Rs.49.855 Crores** (credit facility/ letter of credits/ Solvency certificate certificates from banks etc).
- vii) Satisfactorily completed as a prime contractor/JV/ authorized sub contractor in similar works of not less than **13MW** in any one year during the last 10 financial years (As per Annexure - 4).
- viii) **Minimum key parameters:**
The Minimum key quantities executed in any one financial year during last ten (10) financial years (as per Annexure-5).

SI. No	Item	unit	Qty
1	Supply installation testing & commissioning of (SITC) Solar PV Modules	MW	13
2	SITC AC & DC cables (AC- LT&HT, DC- 6Sqmm & above cable)	Km	166
3	33KV Sub-Stations with related Switch Gear	Rs.	10.0Cr

- ix) Details of existing commitments i.e. work on hand and works for which tenders are submitted (value of B: shall be furnished by the bidders).

Note: Bid capacity computed by the formula $(2AN - B)$ shall be greater than 50% of ECV.

Bid capacity: The bidder who meets the above qualification criteria and whose available bid capacity is more than the estimated contract value will be qualified for opening of price bid. The available bid capacity will be calculated as under:

Available bid capacity : $2AN - B$.

Where,

A= Value of works executed in their name in any one financial years during the last ten(10) financial years (updated to current price level at the rate of 10% per year) taking into account the work completed as well as works in progress

OR

Annual turnover in works executed in any one financial year during the last ten years /(10) financial years (updated to current price level at the rate of 10% per year) certified by chartered accountant.

N= Number of years prescribed for completion of the work for which tenders are invited (months/12).

B= value of existing commitments and ongoing works to be completed during the period of completion of works tenders are invited.

Note:

The statement showing the value of existing commitments and ongoing works as well as the stipulated period of completion remaining for each of the works listed should be issued by the Engineer-in-Charge of Govt. Dept./ undertaking not below the rank of executive Engineer or equivalent and countersigned by the rank of SE or equivalent. The statement to the above effect will be clearly enclosed to the bidding document, and the tenderer shall furnish the particulars invariably in the same format failing which the tender shall be treated as incomplete and summarily rejected.

- x) The bidders shall furnish a declaration to the effect that the required key equipment (as per the statement enclosed) is available with him or that he shall lease or hire the equipment. (as per Annexure-8).
- xi) The bidder should declare the availability of the experienced key personnel as per statement enclosed (Annexure -7) for execution of the work.
- xii) Information regarding any litigation with Government during the last five years, in which the bidder is involved (Litigation History) should be furnished. If there is no litigation history, a NIL report should be submitted. As per proforma (Annexure-9) enclosed.
- xiii) **Declaration on Genuineness of the certificates:** The bidder shall be required to furnish a declaration as per the statement enclosed Any incorrectness/ deviations noticed in the certificate will be viewed seriously and apart from cancelling the work (Annexure-10).

The application and all enclosures shall be signed by a duly authorized representative holding Power of Attorney for signing the application, and a certified copy of the Power of Attorney signed by the legally authorized representative of the firm shall accompany the application.

Note:

- c) The tenders will be opened online on the specified date by the Chief General Manager HGCL.
- d) The price-bids of such bidders who are determined to have complied with the eligibility criteria, will only be opened.

1.3 FINANCIAL ELIGIBILITY CONDITIONS

The Annual Turnover of the bidder in any one financial year of the last ten financial years should be **INR 50,00,00,000/- (Indian rupees Fifty crores only)**. ***The turnover shall mean Revenue from Operations as incorporated in the profit & loss account excluding other income, e.g. sale of fixed assets.*** This must be the individual Company's turnover and not that of any group of Companies. A summarized sheet of average turnover, certified by a practicing Chartered Accountant/Statutory Auditor should be compulsorily enclosed along with corresponding annual accounts.

AND

The net worth for the last financial year should be positive. "Net Worth" of the Bidder shall be calculated as per the Companies Act, 2013.

- 1.3.1 **For participation as Bidder** : In case the bidder is a subsidiary of a holding company, financial eligibility criteria referred to in clause 1.3 above shall be of that subsidiary company only (i.e. excluding its holding company).
- 1.3.2 The Bidder will provide a copy each of audited annual report of previous three financial years for ascertaining their turnover and Net Worth for the purpose of verification.
- 1.3.3 The derivation of "Net Worth" of the Bidder shall be calculated as per Company Act 2013.
- 1.3.4 Other income (as per the Companies Act, 2013 including amendment/clarifications), shall not be considered for arriving at annual turnover.
- 1.3.5 A scanned copy of Certificate of Incorporation of the Bidder shall be furnished in the bid (through online mode).
- 1.3.6 Bidders shall furnish documentary evidence as per the prescribed format (online as well as offline), duly certified by the Authorized Signatory and the Statutory Auditor / Practicing Chartered Accountant of the Bidding Company in support of their financial eligibility.
The Bidder shall furnish the following documentary evidences along with the Bid in support of meeting of above-mentioned Financial Eligibility Criteria:
 - (i) "Details of Financial capability of Bidder" as per format F-16 duly signed and stamped by a Chartered Accountant (format attached).
 - (ii) Audited financial results i.e. Annual Report including Audited Balance Sheet and Profit & Loss Account Statement for immediate three preceding financial years to meet the above Financial Eligibility Criteria. In case of tenders having the submission deadline up to 30th September of the relevant financial year and audited financial results of immediate 3 preceding financial years being not available, the

bidder has an option to submit the audited financial results of three years immediately prior to relevant financial year. In case the bid submission deadline is after 30th September of the relevant financial year, bidder has to compulsorily submit the audited financial results of immediately preceding three financial years.

1.3.7. For participation as Bidder :

In case the bidder is a subsidiary of a holding company, financial eligibility criteria referred to in the clause above, shall be of that subsidiary company only (i.e. excluding its holding company). A job executed by a Bidder for its own plant/ projects cannot be considered as experience for the purpose of meeting the Eligibility Conditions of the tender. Also, the jobs executed for Subsidiary/ Fellow subsidiary/ Holding company will not be considered as experience for the purpose of meeting Eligibility Conditions.

1.4 JOINT VENTURE & CONSORTIUM CONDITIONS:

In case the bid is submitted by a Joint venture (JV) or Consortium of two or more companies as partners, they must meet the following requirements: -

- 1.4.1 The Lead partner/member of the JV/Consortium shall meet individually/Jointly 100% of Technical Eligibility Conditions given at para 1.2 above.
- 1.4.2 There can be a maximum of 02 (Two) partners/members in a JV/Consortium. The Lead partner of the JV/Consortium shall meet individually not less than 50% of minimum Financial Eligibility Conditions (MAAT & Working Capital) given at para 1.3 above. However, all the JV/Consortium partners/members must meet collectively 100% Financial Eligibility Conditions given at para 1.3 above. Herein, apart from the Lead partner, a minimum of 25% Financial eligibility should be met individually by all other partners/members of the JV/Consortium.
- 1.4.3 In the case of a joint venture/Consortium, all members shall be jointly and severally liable for the execution of the entire Contract in accordance with the Contract terms. The JV/Consortium shall nominate the Lead partner/Member of the JV/Consortium who shall have the authority to conduct all business for and on behalf of any and all the members of the JV/Consortium during the Bidding process and, in the event the JV/Consortium is awarded the Contract, during contract execution.
- 1.4.4 A firm that is a Bidder (either individually or as a JV/Consortium member) shall not participate as a Bidder or as JV/Consortium member in more than one Bid. Such participation shall result in the disqualification of all Bids in which the firm is involved. However, this does not limit the participation of a Bidder as subcontractor in another Bid or of a firm as a subcontractor in more than one Bid.

- 1.4.5 The Bid Security & Performance Security (In case of award) of a JV/Consortium shall be in the name of the JV/Consortium that submits the Bid. If the JV/Consortium has not been legally constituted into a legally enforceable JV /Consortium at the time of bidding, the Bid Security shall be in the names of all the members of the JV/Consortium or in the name of **“only lead bidder”**. The Bid Security shall be executed in the names of all the members of the JV/Consortium.
- 1.4.6 Similarly, in case of the award, the Performance Security, to be submitted by a JV/Consortium shall be in the name of the JV/Consortium that has been awarded the NOA/Contract Agreement. If the JV/Consortium has not been legally constituted into a legally enforceable JV/Consortium, the Performance Security shall be in the names of all the partners/members of the JV/Consortium or in the name of **“only lead bidder”**. The Performance Security Bank Guarantee shall be executed in the names of all the partners/members of the JV/Consortium or in the name of the lead bidder.
- 1.4.7 In the case that the Bidder is a JV/Consortium, the Bid shall be signed by the authorized representative of the Lead partner/member of the JV/Consortium on behalf of the JV/Consortium, and so as to be legally binding on all the partners/members as evidenced by a power of attorney signed by their legally authorized representatives.
- 1.4.8 The joint venture/Consortium agreement should indicate precisely the responsibility of all partners/members of JV/Consortium in respect of planning, design, manufacturing, supply, installation, commissioning and training. All members of JV /Consortium should have active participation in execution during the currency of the Contract. The composition or the constitution of the JV/Consortium shall not be varied/modified subsequently without prior approval of the Employer/HGCL.
- 1.4.9** The Joint Venture/Consortium must collectively satisfy the Criteria of clauses 1.2 & 1.3 above (Routes I & II) for which purpose, the relevant figure of average annual turnover and liquid assets/credit facilities for each of the partners of the JV/Consortium shall be added together to arrive at total eligibility of the Joint Venture/Consortium. The net worth of each Partner of JV/Consortium should be positive.
- 1.4.10 A Joint Venture (JV), may or may not be incorporated as a Registered Company.
- 1.4.11 A Joint Venture (JV), if incorporated as a Registered Company, is required to submit Bid Securing Declaration and Performance Security (In case of award) in the name of Joint Venture only.
- 1.4.12 A JV is required to submit Form of Undertaking by the JV Partners (F - 17) and Form of Power of Attorney (F - 17A) as per prescribed formats enclosed in Section - VI (Sample Forms and Formats). A Consortium is required to submit Form of Undertaking by the Consortium Partners (F - 18) and Form of Power of Attorney (F - 18A) as per prescribed formats enclosed in Section - VI (Sample Forms and Formats).

1.4.13 JV/ Consortium is also required to declare detailed scope of work to be executed by each partner/member of JV.

1.4.14 The Employer may assess the capacity and capability of the bidder, to ascertain that the bidder can successfully execute the scope of work covered under the package within stipulated completion period. This assessment shall inter-alia include (i) document verification, (ii) bidder's facilities visit, (iii) details of works executed, works in hand, anticipated in future & the balance capacity available for the present scope of work, (iv) details of plant and machinery, testing facilities, design capabilities, manpower and financial resources, (v) details of quality systems in place, (vi) past experience and performance, (vii) customer feedback, (viii) banker's feedback etc.

1.4.15 In order to avail the benefits of exemptions in JV/ Consortium, all the partners of JV/ Consortium must be MSME Vendors/ Developers registered under NSIC/ Udyog Aadhaar/DIC Category only.

Employer/HGCL reserves the right to waive minor deviations if they do not materially affect the capability of the Bidder to perform the contract.

SECTION - IV

GENERAL CONDITION S OF CONTRACT (GCC)

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Preamble

This Section (Section - IV) of the Bidding Documents [named as General Conditions of Contract (GCC)] provides the performance of the Contractor, payments under the contract or matters affecting the risks, rights and obligations of the parties under the contract. This Section contains provisions that are to be used unchanged unless Section - V [named as Special Conditions of Contract (SCC)] states otherwise as any changes in GCC or any complementary information that may be needed has been shown in SCC. If there is a conflict between the provisions of Section - IV & Section - V, the provisions of Section - V shall prevail.

Bidders may note that the respective rights of the Employer/ HGCL and Bidders/ Contractors shall be governed by this Tender Documents and Contracts (to be) signed between the Employer/ HGCL and the Contractor for the respective package(s). The provisions of this Tender Documents shall always prevail over any other documents in case of contradiction.

Further in all matters arising out of the provisions of this Section - IV and the Section - V of the Tender Documents, the laws of the Union of India shall be the governing laws and courts of the State of the Project HGCL shall have exclusive jurisdiction.

[A] DEFINITIONS

1. Definition of Terms:

1.1 In this **TENDER** (as here-in-after defined) the following words and expressions shall have the meanings hereby assigned to them except where the context otherwise required.

1.1.1 **ADJUDICATOR** means the person or persons named as such in the SCC to make a decision on or to settle any dispute or difference between the HGCL and the Contractor.

1.1.2 **AFFILIATE** shall mean a company that either directly or indirectly

- a. controls or
- b. is controlled by or
- c. is under common control with a Bidding Company

"Control" means HGCLship, directly or indirectly, of more than 50% (fifty percent) of the voting shares of such Company or right to appoint majority Directors.

1.1.3 **AFFECTED PARTY** means HGCL or the Contractor whose performance has been affected by an event of Force Majeure

1.1.4 **APPLICABLE LAW** means any statute, law, regulation, ordinance, notification, rule, regulation, judgment, order, decree, bye-law, approval, directive, guideline, policy, requirement or other governmental restriction or any similar form of decision of, or determination by, or any interpretation or administration having the force of law in the Republic of India and the State Government, by any Government Authority or instrumentality thereof, whether in effect as of the date of this Contract or thereafter.

1.1.5 **APPROVED** shall mean approved in writing including subsequent written confirmation of previous verbal approval and "APPROVAL" means approval in writing including as aforesaid.

1.1.6 **APPOINTING AUTHORITY** for the purpose of arbitration shall be the any person so designated by the HGCL.

1.1.7 **ARBITRATOR** means the person or persons appointed by agreement between the HGCL and the Contractor to make a decision on or to settle any dispute or difference between the HGCL and the Contractor referred to him or her by the parties.

1.1.8 **B.I.S.** means specifications of Bureau of Indian Standards (BIS).

1.1.9 **BID** means the Techno Commercial proposal together with Price Bid submitted by the Bidder along with all documents/ credentials/ attachments/ annexure etc., in response to the Tender, in accordance with the terms and conditions hereof.

1.1.10 **BIDDER** means Bidding Company submitting the Bid. Any reference to the Bidder includes Bidding Company including its successors, executors and permitted assigns as the context may require.

1.1.11 **CEA** means the Central Electricity Authority.

- 1.1.12 **CHARTERED ACCOUNTANT** means a person practicing in India or a firm whereof all the partners practicing in India as a Chartered Accountant(s) within the meaning of the Chartered Accountants Act, 1949.
- 1.1.13 **COLLABORATOR** means the firms/ corporations who has provided technological support to the manufacturer for the detailed Design/establishing production line for the specific Equipment.
- 1.1.14 **COMPANY** means a body incorporated in India under the Indian Companies Act, 1956 or Companies Act, 2013 including any amendment thereto.
- 1.1.15 **CONTRACT** shall mean the Agreement between the HGCL and the Contractor for the execution of the works including therein all Contract documents.
- 1.1.16 **CONTRACTOR** means the person or the persons, Company or Corporation whose Tender has been accepted by the Employer and includes the Contractor's legal representatives his/ her successors and permitted assigns.
- 1.1.17 **CONTRACT DOCUMENTS** mean collectively the Tender Documents, Designs, Drawings, Specification, Schedule of Quantities and Rates, Notification of Award/ Letter of Intent/ Letter of Acceptance and agreed variations if any, and such other documents constituting the Tender and acceptance thereof.
- 1.1.18 **CONTRACTOR'S EQUIPMENT** means all plant, facilities, equipment, machinery, tools, apparatus, appliances or things of every kind required in or for installation, completion and maintenance of Facilities that are to be provided by the Contractor, but does not include Plant and Equipment, or other things intended to form or forming part of the Facilities.
- 1.1.19 **CONTRACT PRICE/CONTRACT VALUE/PROJECT VALUE** means the firm value of the final quoted price, as a result of e- RA (if applicable), by the successful bidder specified in its financial proposal as the sum of individual contract value of Supply & Services (Supply of goods, F&I, Design, installation, Civil Works etc) as mentioned under the different work heads specified in the financial proposal including all the applicable taxes/ Goods & Service taxes.
- 1.1.20 **CONTRACTOR'S REPRESENTATIVE** means any person nominated by the Contractor and approved by the Employer to perform the duties delegated by the Contractor.
- 1.1.21 **CONSULTANT** means Techno-Commercial experts who are the consulting engineer to the Employer/ HGCL for this project.
- 1.1.22 **COMMISSIONING** means a project shall be considered commissioned if all equipment as per rated capacity, mentioned under the scope of contract, has been installed and energy has flown into grid.
- 1.1.23 **COMPLETION OF FACILITIES** means that the Facilities (or a specific part thereof where specific parts are specified in the SCC) have been completed operationally and structurally and put in a tight and clean condition, and that all work in respect of Pre-commissioning of the Facilities or such specific part thereof has been completed; and Commissioning along with Operational Acceptance has been attained as per Technical Specifications.

- 1.1.24 **CHANGE ORDER** means an order given in writing by the Engineer-in-Charge/Project Manager to effect additions to or deletion from and alteration in the works.
- 1.1.25 **DAY** means a day of 24 hours from midnight to midnight irrespective of the number of hours worked in that day.
- 1.1.26 **DEFECT LIABILITY PERIOD** in relation to scope of work means 12 (Twelve) months from the date of Operational acceptance during which the Contractor stands responsible for rectifying all defects/rejection that may appear in the works executed by the Contractor in pursuance of the Contract and includes warranties against Manufacturing/ Fabrication/ Erection/ Construction defects covering all materials plants, equipment, components and the like supplied by the Contractor, works executed against workmanship defects.
- 1.1.27 **DEVELOPER** means an entity who has either executed or got executed the work/ project as HGCL of solar projects.
- 1.1.28 **DRAWINGS** shall include maps, plans and tracings, supporting documents, specifications or prints or sketches thereof with any modifications approved in writing by the Engineer-in-Charge/Project Manager and such other drawing as may, from time to time, be furnished by the Contractor.
- 1.1.29 **EARNEST MONEY DEPOSIT (EMD)** means the unconditional and irrevocable Tender Security in the form of Demand Draft/ Banker's Cheque/ Bank Guarantee to be submitted along with the Bid by the Bidder as prescribed in the Tender document.
- 1.1.30 **EMPLOYER** means the Company/ Corporation/ Government Entity, named in the **BDS/ SCC**, who is responsible for getting the Project/Facilities implemented. The Employer may be HGCL himself or an agency appointed by the HGCL and shall include the legal successors or permitted assigns of the Employer.
- 1.1.31 **ENGINEER-IN-CHARGE (EIC)/PROJECT MANAGER** shall mean the person designated from time to time by the Employer and shall include those who are expressly authorized by him to act for and on his behalf for operation of this Contract.
- 1.1.32 **EFFECTIVE DATE** means the date of issuance of Notification of Award/ Letter of Intent/Letter of Award (LOA) from which the Time for Completion shall be determined or any other specific date as provided in the Notice to Proceed (NTP), in case the NTP is applicable.
- 1.1.33 **FACILITIES** mean the Plant and Equipment to be supplied and installed, as well as all the installation services including all infrastructure as mentioned in scope of works to be carried out by the Contractor under the Contract.
- 1.1.34 **FINAL ACCEPTANCE/FINAL COMPLETION** means acceptance of Facilities by the Employer at the end of one year from the date of Operational Acceptance and upon demonstration of minimum annual parameters as specified in the technical specifications and completion of works under the punch list which certifies the Contractor's fulfilment of the Contract in respect of Functional and Plant Performance Guarantees of the Facilities.

- 1.1.35 **GUARANTEE TEST(S)** means the test(s) specified in the Technical Specifications to be carried out to ascertain whether the Facilities or a specified part thereof is able to attain the Functional Guarantees specified in the Technical Specifications during/ after successful Commissioning followed by Trial - Operation.
- 1.1.36 **GOODS & SERVICE TAX (GST)** means taxes or cess levied under the Central Goods and Services Tax Act, Integrated Goods and Services Tax Act, Goods and Services Tax (Compensation to States) Act and various State/Union Territory Goods and Services Tax Laws and applicable cesses, if any under the laws in force (hereinafter referred to as relevant GST Laws) w.e.f. 01.07.2017, which shall be fully complied with by Bidders..
- 1.1.37 **IEC** means specifications of International Electro-Technical Commission.
- 1.1.38 **INSTALLATION SERVICES** means all those services ancillary to the supply of the Plant and Equipment for the Facilities, to be provided by the Contractor under the Contract; e.g., transportation and provision of marine or other similar insurance (s), inspection, expediting, site preparation works (including the provision and use of Contractor's Equipment and the supply of all structural and construction materials required), installation including civil and allied works etc., testing, pre-commissioning, commissioning, operations, maintenance, the provision of operations and maintenance manuals, training of Employer's Personnel etc..
- 1.1.39 **MWp** means Mega-Watt Peak.
- 1.1.40 **MWh** means Mega-Watt Hour.
- 1.1.41 **kWh** shall mean Kilo-Watt-hour.
- 1.1.42 **LANGUAGE FOR DRAWINGS AND INSTRUCTION** All the drawings, titles, notes, instruction, dimensions, etc. shall be in English Language only.
- 1.1.43 **MNRE** means Ministry of New and Renewable Energy, Government of India.
- 1.1.44 **MOBILIZATION** shall mean establishment of sufficiently adequate infrastructure by the Contractor at Site comprising of construction equipment, aids, tools tackles including setting of site offices with facilities such as power, water, communication etc. establishing manpower organization comprising of Resident Engineers, Supervising Personnel and an adequate strength of skilled, semi-skilled and un-skilled workers, who with the so established infrastructure shall be in a position to commence execution of work at site(s), in accordance with the agreed Time Schedule of Completion of Work. Mobilization shall be considered to have been achieved, if the Contractor is able to establish infrastructure as per Time Schedule, where so warranted in accordance with agreed schedule of work implementation to the satisfaction of Engineer-in-Charge/Project Manager.
- 1.1.45 **NET-WORTH** shall have same meaning as defined in Company Act 2013 and Amendment, if any.

- 1.1.46 **NOTICE IN WRITING OR WRITTEN NOTICE** shall mean a notice in written, typed or printed characters sent (unless delivered personally or otherwise proved to have been received by the addressee) by Email/registered post to the latest known private or business address or registered office of the addressee and shall be deemed to have been received in the ordinary course of post it would have been delivered.
- 1.1.47 **NOTIFICATION OF AWARD (NOA)/LETTER OF INTENT (LOI)/LETTER OF AWARD (LOA)** means the official notice issued vide Letter/ E-mail by the Employer/HGCL notifying the Contractor that his bid has been awarded.
- 1.1.48 **NTP** means Notice to Proceed subsequent to the placement of LOA/NOA/LOI. NTP will be issued by the HGCL within a maximum time frame of 90 days from the date of LOA/NOA/LOI, failing which the LOA/NOA/LOI shall stand withdrawn without creating any further liability on either party. In such case EMD/PBG, as applicable submitted by the contractor shall be released.
- 1.1.49 **OPERATIONAL ACCEPTANCE** means the acceptance of the Plant Facilities (or any part of the Facilities where the Contract provides for acceptance of the Facilities in parts) by the Employer/HGCL, which certifies the Contractor's fulfilment of the Contract in respect of meeting Plant Functional and Performance Guarantees of the Facilities and completion of works.
- 1.1.50 **O & M** means comprehensive Operation & Maintenance of Commissioned Project/ Work/ Facilities under the contract.
- 1.1.51 **HGCL** means the Company/ Corporation/ Government Entity, named in the **BDS/ SCC**, who has decided to set up the Facilities under his HGCLship at his designated location and shall include the legal successors or permitted assigns of the HGCL.
- 1.1.52 **PARENT COMPANY** means a company that holds more than Fifty Percent (50%) of the paid-up equity capital directly or indirectly in the Bidding Company as the case may be.
- 1.1.53 **PLANT AND EQUIPMENT** means permanent plant, equipment, machinery, apparatus, articles and things of all kinds to be provided and incorporated in the Facilities by the Contractor under the Contract (including the spare parts to be supplied by the Contractor) but does not include Contractor's Equipment.
- 1.1.54 **PRE-COMMISSIONING** means the testing, checking and other requirements specified in the Technical Specifications that are to be carried out by the Contractor in preparation for Commissioning.
- 1.1.55 **SCC** means the Special Conditions of Contract.
- 1.1.56 **HGCL** means Hyderabad Growth Corridor Limited, Hyderabad
- 1.1.57 **SITE** means the land and other places upon which the Facilities are to be installed, and such other land or places as specified in the SCC of the Contract as forming part of the Site.

- 1.1.58 **SPECIFICATION** shall mean all directions the various Technical Specifications, provisions attached and referred to the Tender Documents which pertain to the method and manner of performing the work or works to the quantities and qualities of the work or works and the materials to be furnished under the Contract for the work or works, as may be amplified or modified by the Employer or Engineer-in-Charge/Project Manager during the performance of Contract in order to provide the unforeseen conditions or in the best interests of the work or works. It shall also include the latest edition of relevant Standard Specifications including all addenda/ corrigenda published before entering into Contract.
- 1.1.59 **SUB-CONTRACTOR** including vendors, means any person to whom execution of any part of the Facilities, including preparation of any design or supply of any Plant and Equipment, is sub- contracted directly or indirectly by the Contractor, and includes its legal successors or permitted assigns.
- 1.1.60 **TEMPORARY WORKS** shall mean all temporary works of every kind required in or about the execution, completion or maintenance of site works.
- 1.1.61 **TENDER/TENDER DOCUMENT/ BIDDING DOCUMENT** means the entire set of documents vide which Employer invite bids for Projects/ Works/ Facilities that are submitted within a finite deadline by the Bidder.
- 1.1.62 **TIME FOR COMPLETION** means the time within which Completion of the Facilities is to be attained in accordance with the specifications, as a whole (or of a part of the Facilities where a separate Time for Completion of such part has been prescribed) as specified in the SCC of the tender document
- 1.1.63 **WEEK** means a period of any consecutive seven days.
- 1.1.64 **WORKING DAY** means any day which is not declared to be holiday or rest day by the Employer/ HGCL.

[B] GENERAL INFORMATION

2. General Information

2.1.a Location of Site

The proposed location of Project site(s) is/ are defined in the Scope of work under Technical specifications & also defined under the Special conditions of contract (SCC).

2.1.b Access by Road

Contractor, if necessary, shall build other temporary access roads to the actual site of construction for his own work at his own cost. The Contractor shall be required to permit the use of the roads so constructed by him for vehicles of any other parties who may be engaged on the project site. The Contractor shall also facilitate the construction of any permanent roads should the construction thereof starts while he is engaged on this work. He shall make allowance in his Tender for any inconvenience he anticipates on such account. Non-availability of access roads, railway siding and railway wagons for the use of the Contractor shall in no case condone any delay in the execution of work nor be the cause for any claim for compensation against the Employer/ HGCL.

2.2 Scope of Work

The scope of work is defined in the Section - VII, Scope of Work and Technical Specifications (TS) of the Tender document. In addition, the Contractor shall provide all necessary materials, equipment, labour etc. for the execution and maintenance of the work till completion unless otherwise mentioned in the Tender Document.

2.3 Construction Water Supply

Contractor will have to make his own arrangements for supply of water to his labour camps and for works. The water quality should be suitable for use in civil construction work. All pumping installations, pipe network and distribution system will have to be carried out by the Contractor at his own risk and cost. Alternatively, the Employer/ HGCL at his discretion may endeavor to provide water to the Contractor at the Employer's/ HGCL's source of supply provided the Contractor makes his own arrangement for the water meter which shall be in custody of the Employer/ HGCL and other pipe networks from source of supply, so as not to interfere with the layout and progress of the other construction works. In such case, a separate accounting shall be maintained for billing purpose on monthly basis. However, the HGCL does not guarantee the supply of water and this does not relieve the Contractor of his responsibility in making his own arrangement and for the timely completion of the various works as stipulated.

2.4 Construction Power Supply

2.4.1 Contractor has to arrange for the construction power supply of their own. However, subject to availability, Employer/ HGCL may provide access to the nearest available point in his location for supply power at only one point, from where the Contractor will make his own arrangement for temporary distribution through a temporary energy meter (sealed by Employer/ HGCL). All the works will be done as per the applicable regulations with information to the Engineer-in-Charge/Project Manager. The temporary line will be removed forthwith after the completion of work or if there is any hindrance caused to the other works due to the alignment of these lines, the Contractor will re-route or remove the temporary lines at his own cost. The Contractor at his own cost will also provide suitable electric meters, fuses, switches, etc. for purposes of payment to the Employer/ HGCL which should be in the custody and control of the Employer/ HGCL. The cost of power supply shall be payable to the Employer/ HGCL by the Contractor. The Employer/ HGCL shall not, however, guarantee the supply of electricity nor have any liability in respect thereof. No claim for compensation for any failure or short supply of electricity will be admissible.

2.4.2 It shall be the responsibility of the Contractor to provide and maintain the complete installation on the load side of the supply with due regard to safety requirement at site. All cabling, equipment, installations etc. shall comply in all respects with the latest statutory requirements and safety provisions i.e., as per the Central/ State Electricity Acts and Rules etc. The Contractor will ensure that his equipment and Electrical Wiring etc., are installed, modified, maintained by a licensed Electrician/ Supervisor.

2.4.3 At all times, IEA regulations shall be followed failing which the Employer/ HGCL has a right to disconnect the power supply without any reference to the Contractor. No claim shall be entertained for such disconnection. Power supply will be reconnected only after production of fresh certificate from authorized

electrical supervisors.

- 2.4.4** The Employer/ HGCL is not liable for any loss or damage to the Contractor's equipment as a result of variation in voltage or frequency or interruption in power supply or other loss to the Contractor arising therefrom.
- 2.4.5** The Contractor will have to provide and install his own lights and power meters which will be governed as per Central/ State /UT Government Electricity Rules.
- 2.4.6** In case of damage of any of the Employer's/ HGCL's equipment on account of fault, intentional or unintentional on the part of the Contractor, the Employer/ HGCL reserves the right to recover the cost of such damage from the Contractor's bill.

2.5 Land for Contractor's Field Office, Godown and Workshop

The Employer/ HGCL will, at his own discretion and convenience and for the duration of the execution of the work make available near the site, land for construction of Contractor's Temporary Field Office, godowns, workshops and assembly yard required for the execution of the Contract. The Contractor shall at his own cost construct all these temporary buildings and provide suitable water supply and sanitary arrangement. On completion of the works undertaken by the Contractor, he shall remove all temporary works erected by him and have the Site cleaned as directed by Engineer-in-Charge/Project Manager. If the Contractor shall fail to comply with these requirements, the Employer/HGCL may, at the expenses of the Contractor remove such surplus and rubbish materials and dispose off the same as he deems fit and get the site cleared as aforesaid; and Contractor shall forthwith pay the amount of all expenses so incurred and shall have no claim in respect of any such surplus materials disposed off as aforesaid. But the Employer/ HGCL reserves the right to ask the Contractor any time during the pendency of the Contract to vacate the land by giving 07 (Seven) day notice on security reasons or on national interest or otherwise. The Contractor shall put up temporary structures as required by them for their office, fabrication shop and construction stores only in the area allocated to them on the project site by the Employer/ HGCL or his authorized representative. No tea stalls/ canteens should be put up or allowed to be put up by any Contractor in the allotted land or complex area without written permission of the Employer/ HGCL. No unauthorized buildings, constructions or structures should be put up by the Contractor anywhere on the project site. For uninterrupted fabrication work, the Contractor shall put up temporary covered structures at his cost within Area in the location allocated to them in the project site by the Employer/ HGCL or his authorized representative. No person except for authorized watchman shall be allowed to stay in the plant area/ Contractor's area after completion of the day's job without prior written intimation to Engineer-in-Charge/Project Manager.

2.6 Land for Residential Accommodation

No Land shall be made available for residential accommodation for staff and labour of Contractor.

[C] ADDITIONAL GENERAL INSTRUCTIONS TO BIDDERS

3. Documents

- 3.1 **Corrections and Erasures:** All correction(s) and alteration(s) in the entries of Tender document shall be signed in full by the bidder with date. No erasure or over writing is permissible.
- 3.2 **Witness:** Witness and sureties shall be persons of status and property and their names, occupation and address shall be stated below their signature.
- 3.3 **Details of Experience:** The bidder should furnish, along with his Tender, details of previous experience in having successfully completed in the recent past works of this nature, together with the names of Employers/ HGCLs, location of sites and value of Contract, date of commencement and completion of work, delays if any, reasons of delay and other details along with documentary evidence(s).
- 3.4 **Liability of Government of India:** It is expressly understood and agreed by and between bidder or/ Contractor and Employer/ HGCL that, Employer/ HGCL is entering into this agreement solely on its own behalf. In particular, it is expressly understood and agreed that the Government of India has no liabilities, obligations or rights hereunder. It is expressly understood and agreed that Employer/ HGCL is an independent legal entity with power and authority to enter into Contracts solely on its own behalf under the applicable Laws of India and general principles of Contract Law. It is further understood and agreed that the Government of India is not and shall not be liable for any acts, omissions, commissions, breaches or other wrongs arising out of the Contract. Accordingly, bidder/ Contractor hereby expressly waives, releases and foregoes any and all actions or claims, including cross claims, impleader claims or counter claims against the Government of India arising out of this Contract and covenants not to sue to Government of India as to any manner, claim, cause of action or thing whatsoever arising of or under this agreement.

4. Transfer of Tender Documents

- 4.1 Transfer of Tender Documents purchased by one intending bidder to another is not permissible.

5. Right of Employer/HGCL to Accept or Reject Tender

- 5.1 The right to accept the Tender will rest with the Employer/ HGCL. The Employer/ HGCL, however, does not bind himself to accept the lowest Tender, and reserves to itself the authority to reject any or all the Tenders received without assigning any reason whatsoever. At the option of the Employer/HGCL, the work for which the Tender had been invited, may be awarded to one Contractor or split between more than one bidders, in which case the award will be made for only that part of the work, in respect of which the bid has been accepted. The quoted rates should hold good for such eventualities. Tenders in which any of the particulars and prescribed information are missing or are incomplete in any respect and/ or the prescribed conditions are not fulfilled are liable to be rejected. The Tender containing uncalled for remarks or any additional conditions are liable to be rejected. Canvassing in connection with Tenders is strictly prohibited and

Tenders submitted by the Tenderers who resort to canvassing will be liable to rejection.

6. Time Schedule & Progress Reporting

- 6.1 The work shall be executed strictly as per the Time Schedule specified in Section - V, Special Conditions of Contract (SCC). The period of construction given in Time Schedule includes the time required for mobilization as well as testing, rectifications if any, retesting and completion in all respects to the entire satisfaction of the Engineer-in- Charge.
- 6.2 A joint program of execution of the work will be prepared by the Contractor based on priority requirement of this project & submitted to the Engineer in charge. This program will take into account the time of completion mentioned above and the time allowed for the priority works by the Engineer-in- Charge.
- 6.3 Monthly/ Weekly construction program will be drawn up by the Engineer-in-Charge jointly with the Contractor, based on availability of work fronts and the joint construction program as per above. The Contractor shall scrupulously adhere to these targets/ programs by deploying adequate personnel, construction tools and tackles and he shall also supply himself all materials of his scope of supply in good time to achieve the targets/ programs. In all matters concerning the extent of targets set out in the weekly and monthly programs and the degree of achievements the decision of the Engineer-in-Charge will be final and binding on the Contractor.
- 6.4 The Contractor shall monitor progress of all the activities specified in the work schedule referred in GCC above and submit the progress report to the Project Manager as per the Contract Co-ordination procedure.
- 6.6 If at any time the Contractor's actual progress falls behind the scheduled program, or it becomes apparent that it will so fall behind, the Contractor shall, at the request of the Employer or the Project Manager/ EIC, prepare and submit to the Project Manager/ EIC a revised program, taking into account the prevailing circumstances, and shall notify the Project Manager/ EIC, of the steps being taken to expedite progress so as to attain Completion of the Facilities within the Time for Completion. If any extension thereof entitled under GCC, or any extended period as may otherwise be agreed upon between the Employer and the Contractor, Contractor shall submit the revised plan for completion of Facility accordingly.
- 6.7 Maintenance of Records of Progress Review Meeting

The Contractor shall be required to attend all site progress review meetings organized by the MD/ HGCL or his authorized representative. The deliberations in the meetings shall inter-alia include the program, progress of work (including details of manpower, material, tools and plants deployed by the Contractor vis-à-vis agreed schedule), inputs to be provided by Employer, delays, if any and recovery program, specific hindrances to work and work instructions by Employer. The minutes of the meetings shall be recorded with the 'Project Manager/ EIC' or his authorized representative.

7. Conflict of Interest and Bidder's Responsibility

- 7.1 A bidder shall not have a Conflict of Interest. All bidders found to have Conflict of Interest shall be disqualified. A bidder may be considered to have a Conflict of Interest with one or more parties in this bidding process, if:

- (a) They have a controlling partner in common; or
- (b) They receive or have received any direct or indirect subsidy from any of them; or
- (c) They have the same legal representative for the purposes of this bid; or
- (d) They have a relationship with each other, directly or through common third parties, that puts them in a position to have access to information about or influence on the bid of another bidder, or influence the decision of the Employer/ HGCL regarding this bidding process; or
- (e) A bidder submits more than one bid in this bidding process, either individually [including bid submitted as an agent/ authorized representative on behalf of one or more bidders or through license - licensor route, wherever permitted as per the provision of the Qualification Requirement for the bidders in the Annexure to Bid Data Sheet (BDS)] or as a partner in a Joint Venture/ Consortium, except for alternative offers permitted under Tender. This will result in the disqualification of all such bids; or
- (f) A bidder or any of its Affiliates participated as a consultant in the preparation of the Design or Technical Specification or Detailed Project Report (DPR) of the Plant and Installation of services*/ goods and related services** that are the subject of the bid; or
- (g) A bidder or any of its Affiliate has been hired (or is proposed to be hired) by the Employer/ HGCL as a Project Manager for the Contract

Unquote:

* Applicable for Supply & Installation Contracts

** Applicable for Supply & Supply cum Supervision of Installation Contracts

- 7.2 The intending bidder shall be deemed to have visited the Site and familiarized himself before submitting the Tender. Non-familiarity with the site conditions will not be considered a reason either for extra claims or for not carrying out the works in strict conformity with the Drawings and Specifications or for any delay in performance.

8. Retired Government or Company Officers

- 8.1 No Engineer of Gazetted rank or other Gazetted Officer employed in Engineering or Administrative duties in an Engineering Department of the States/ UT/ Central Government or of the Employer/ HGCL is allowed to work as a Contractor for a period of two years after his retirement from Government Service, or from the employment of the Employer/ HGCL. The Contract, if awarded, is liable to be cancelled if either the Contractor or any of his employees is found at any time to be such a person, who has not obtained the permission of the State/ Central Government before submission of Tender, or engagement in the Contractor's service as the case may be.

9. Representatives and Field Management & Controlling

- 9.1 Project Manager / Engineer- In –Charge (EIC):

If the Project Manager/ EIC is not named in the Contract, then within seven (7) days of the Effective Date, the Employer shall appoint and notify the Contractor in writing of the name of the Project Manager/ EIC. The Employer may from time to time appoint some other person as the Project Manager/ EIC in place of the person previously so appointed and shall give a notice of the name of such other person to the Contractor without delay. The Employer shall take reasonable care, unless unavoidable to see that no such appointment is made at such a time or in such a manner as to impede the progress of work on the Facilities. The Project Manager/EIC shall represent and act for the Employer at all times during the currency of the Contract.

All notices, instructions, information and other communications given by the Contractor to the Employer under the Contract shall be given to the Project Manager/ EIC, except as herein otherwise provided.

9.2 Contractor's Representative & Construction Manager

If the Contractor's Representative is not named in the Contract, then within seven (07) days of the Effective Date, the Contractor shall appoint the Contractor's Representative and shall request the Employer in writing. If the Employer objects to the appointment within seven (07) days giving the reason therefor, then the Contractor shall appoint a replacement within seven (07) days of such objection, and the foregoing provisions of this GCC shall apply thereto.

9.3 The Contractor's Representative shall represent and act for the Contractor at all times during the tenure of the Contract and shall give to the Project Manager/ EIC all the Contractor's notices, instructions, information and all other communications under the Contract.

9.4 All notices, instructions, information and all other communications given by the HGCL/ Employer or the Project Manager/ EIC to the Contractor under the Contract shall be given to the Contractor's Representative or, in its absence, its deputy, except as herein otherwise provided.

9.5 The Contractor shall not revoke the appointment of the Contractor's Representative without the Employer consent, which shall not be unreasonably withheld. If the Employer consents thereto, the Contractor shall appoint some other person as the Contractor's Representative, pursuant to the procedure set out in GCC Clauses.

9.6 The Contractor's Representative may, subject to the approval of the Employer (which shall not be unreasonably withheld), at any time delegate to any person any of the powers, functions and authorities vested in him or her. Any such delegation may be revoked at any time. Any such delegation or revocation shall be subject to a prior notice and shall specify the powers, functions and authorities thereby delegated or revoked. No such delegation or revocation shall take effect unless and until a copy thereof has been delivered to the Project Manager/EIC.

9.7 Any act or exercise by any person of powers, functions and authorities so delegated to him or her in accordance with this GCC Clause shall be deemed to be an act or exercise by the Contractor's Representative.

9.8 Notwithstanding anything stated in GCC Clause above, for the purpose of execution of contract, the Employer and the Contractor shall finalize and agree to a Contract Co-ordination Procedure and all the communication under the Contract shall be in accordance with such Contract Co-ordination Procedure.

9.9 From the commencement of installation of the Facilities at the Site until Final Acceptance, the Contractor's Representative shall appoint a suitable person as the construction manager (hereinafter referred to as "the Construction Manager"). The Construction Manager shall supervise all work done at the Site by the Contractor and shall be present at the Site throughout normal working hours except when on leave, sick or absent for reasons connected with the proper Performance of the Contract. Whenever the Construction Manager is absent from the Site, a suitable person shall be appointed to act as his or her deputy.

9.10 The Employer may object to any Contractor's representative or person employed by the Contractor in the execution of the Contract who, in the reasonable opinion of the Employer, may behave inappropriately, may be incompetent or negligent, or may commit a serious breach of the Site regulations and safety.

- 9.11 If any representative or person employed by the Contractor is removed in accordance with GCC Clause above, the Contractor shall, where required, promptly appoint a replacement. The Engineer-in-Charge/Project Manager may also authorize his representatives to assist in performing his duties and functions.

9.13 Hindrance Register

The Contractor may also maintain a Hindrance Register where reasons along with documentary evidence for delay/ fault may be recorded from time to time and at the time of occurrence of the hindrance and get it duly certified by the Project Manager or his authorized representative.

10. Note to Schedule of Rates

- 10.1 The Schedule of Rates should be read in conjunction with Section - VIII of the Tender documents.
- 10.2 The bidder shall be deemed to have studied the Drawings (if any), Specifications and details of work to be done within Time Schedule and to have acquainted himself of the condition prevailing at site.
- 10.3 Rates must be filled in the Schedule of Rates of original Tender Documents. If quoted in separate typed sheets no variation in item description or specification shall be accepted.

11. Policy for Tenders under Consideration

- 11.1 Only Those Tenders which are complete in all respects and are strictly in accordance with the Terms and Conditions and Technical Specifications of Tender Document, shall be considered for evaluation. Such Tenders shall be deemed to be under consideration immediately after opening of Tender and until such time an official intimation of acceptance/ rejection of Tender is made by Employer to the Bidder.
- 11.2 Zero Deviation: Bidders to note that this is a Zero Deviation Tender. Employer will appreciate submission of bids based on the terms and conditions in the enclosed General Conditions of Contract (GCC), Special Conditions of Contract (SCC), Instructions to Bidders (ITB), Scope of Work, technical specifications etc. to avoid wastage of time and money in seeking clarifications on technical/ commercial aspects of the bids. Bidder may note that no technical and commercial clarifications will be sought for after the receipt of the bids. In case of any deviation/ nonconformity observed in the bid, it will be liable for rejection.

12. Clarification of Tender Document

- 12.1 Verbal clarification and information given by Employer or its representatives shall not in any way be binding on Employer.

13. Local Conditions

- 13.1 The site is being offered to the bidders for Project on **“as is where is”** basis assuming that bidders have acquainted themselves appropriately with all the local site conditions & no plea will be entertained before/after the award of contract on this ground. It will be imperative on each bidder to acquaint himself of all local site conditions and factors which may have any effect on the execution of work covered under the Tender Document. In their own interest, the bidder(s) is/ are requested to familiarize themselves with the Indian Income Tax Act 1961, Indian Companies Act 1956, and Indian Customs Act 1962, GST Act and other related Acts and Laws and Regulations of India with their latest amendments, as applicable. Employer shall not entertain any requests for clarifications from the bidder regarding such local conditions.

- 13.2 It must be understood and agreed that such factors have properly been investigated and considered while submitting the Tender. Any claim for financial or Forex or any other adjustments to Value of Contract, on lack of clarity of such factors shall not be entertained.

14. Extension of Time for Commissioning

- 14.1 The time for Project commissioning as specified in the SCC is firm & final binding till the final commissioning of the Plant facilities. The Time(s) for Commissioning specified in the SCC shall be extended if the Contractor is delayed or impeded in the Performance of any of its obligations under the Contract due to delay in fulfilment of obligations by the HGCL or by the reason of any of occurrence of Force Majeure as provided in the Tender for Force Majeure.
- 14.2 The Contractor shall at all times use its reasonable efforts to minimize any delay in the Performance of its obligations under the Contract.

[D] GENERAL OBLIGATIONS

15.1 Priority of Contract Documents

Several documents forming the Contract are to be taken as mutually explanatory of one another, but in case of ambiguities or discrepancies the same shall be explained and adjusted by the Engineer-in-Charge/Project Manager who shall thereupon issue to the Contractor instructions thereon and in such event, unless otherwise provided in the Contract, the priority of the documents forming the Contract shall be as follows:

- a) Contract Agreement and the appendices, along with the amendments, if any thereto
- b) Notification of award (NoA)/Letter of Intent (LOI) /Letter of Award (LOA)/Notice to proceed (NTP)
- c) Special Conditions of Contract
- d) General Conditions of Contract
- e) BDS & ITB
- f) Special Technical Conditions on Technical Specifications
- g) Technical Specifications and Drawings
- h) Price Schedules submitted by the Contractor
- i) Other completed Bidding forms submitted with the Bid
- j) Any other documents forming part of the Employer's Requirements

In the event of any ambiguity or conflict between the Contract Documents listed above, the order of precedence shall be the order in which the Contract Documents are listed above. For eg. Contract agreement have precedence over NOA.

15.2 Headings and Marginal Notes

All headings and marginal notes to the clauses of these General Conditions of Contract or to the Specifications or to any other Tender Document are solely for the purpose of giving a concise indication and not a summary of the contents thereof, and they shall never be deemed to be part thereof or be used in the interpretation or construction thereof of the Contract.

15.3 Singular and Plural

In Contract Documents unless otherwise stated specifically, the singular shall include the plural and vice versa wherever the context so requires.

15.4 Interpretation

Words implying 'Persons' shall include relevant Corporate Companies/ Registered Associations/ Body of Individuals/ Firm of Partnership' as the case may be.

16 Special Conditions of Contract (SCC)

- 16.1** Special Conditions of Contract shall be read in conjunction with the General Conditions of Contract, Specification of Work, Drawings and any other documents forming part of this Contract wherever the context so requires.
- 16.2** Notwithstanding the sub-division of the documents into these separate sections and volumes every part of each shall be deemed to be supplementary to and complementary of every other part and shall be read with and into the Contract so far as it may be practicable to do so.
- 16.3** Where any portion of the General Condition of Contract is repugnant to or at variance with any provisions of the Special Conditions of Contract, unless a different intention appears the provisions of the Special Conditions of Contract shall be deemed to over-ride the provisions of the General Conditions of Contract and shall to the extent of such repugnancy, or variations, prevail.
- 16.4** Wherever it is mentioned in the specifications that the Contractor shall perform certain Work or provide certain facilities, it is understood that the Contractor shall do so at his cost and the Value of Contract shall be deemed to have included cost of such performance and provisions, so mentioned.
- 16.5** The materials, design and workmanship shall satisfy the relevant Indian Standards, the Job Specifications contained herein and Codes referred to. Where the job specification stipulate requirements in addition to those contained in the standard codes and specifications, these additional requirements shall also be satisfied.

17 Contractor to obtain his own Information

- 17.1** The Contractor in fixing his cost shall for all purpose whatsoever reason may be, deemed to have himself independently obtained all necessary information for the purpose of preparing his tender and his tender as accepted shall be deemed to have taken into account all contingencies as may arise due to such information or lack of same. The correctness of the details, given in the Tender Document to help the Contractor to make up the tender is not guaranteed. The Contractor shall be deemed to have examined the Contract Documents, to have generally obtained his own information in all matters whatsoever that might affect the carrying out of the works at the scheduled rates and to have satisfied himself to the sufficiency of his offer.

Any error in description of quantity or omission therefrom shall not vitiate the Contract or release the Contractor from executing the work comprised in the Contract according to Drawings and Specifications at the scheduled rates. He is deemed to have known the scope, nature and magnitude of the Works and the requirements of materials and labour involved etc., and as to what all works he has to complete in accordance with the Contract documents whatever be the defects, omissions or errors that may be found in the Documents. The Contractor shall be deemed to have visited surroundings, to have satisfied himself to the nature of all existing structures, if any, and also as to the nature and the conditions of the Railways, Roads, Bridges and Culverts, means of transport and communication, whether by land, water or air, and as to possible interruptions thereto and the access and egress from the site, to have made enquiries, examined and satisfied himself as to the sites for obtaining sand, stones, bricks and other materials, the sites for disposal of surplus materials, the available accommodation as to whatever required, depots and such other buildings as may be necessary for executing and completing the works, to have made local independent enquiries as to the sub-soil, subsoil water and variations thereof, storms, prevailing winds, climatic conditions and all other similar matters effecting these works.

He is deemed to have acquainted himself as to his liability of payment of Government Taxes, duties and other charges, levies etc. Any neglect or omission or failure on the part of the Contractor in obtaining necessary and reliable information upon the foregoing or any other matters affecting the Contract shall not relieve him from any risks or liabilities or the entire responsibility from completion of the works at the scheduled rates and times in strict accordance with the Contract. It is, therefore, expected that should the Contractor have any doubt as to the meaning of any portion of the tender Document he shall set forth the particulars thereof in writing to Employer before bid submission.

The Employer may provide such clarification as may be necessary in writing to Contract, such clarifications as provided by Employer shall form part of tender Documents. No verbal agreement or inference from conversation with any effect or employee of the Employer either before, during or after the execution of the Contract Agreement shall in any way affect or modify and of the terms or obligations herein contained. Any change in layout due to site conditions or technological requirement shall be binding on the Contractor and no extra claim on this account shall be entertained.

18 Time of Performance

18.1 Time for Mobilization

The work covered by this Tender shall be commenced immediately upon issuance of the NOA/ LOI/LOA/NTP (whichever is later) and be completed on or before the dates as mentioned in the Time Schedule of Completion of Work under Section - V, Special Conditions of Contract (SCC). The Contractor should bear in mind that time is the essence of this contract agreement. Request for revision of construction time after tenders are submitted will not receive any consideration. The mobilization period is included within the overall Completion Schedule, not over and above the completion time to any additional work or any other reasons.

18.2 Time Schedule of Construction

- 18.2.1 The general time schedule of construction is given in the Section - V, Special Conditions of Contract (SCC) of the Tender Documents. Contractor should prepare a detailed monthly or weekly construction program. The Work shall be executed strictly as per the Time Schedule given in the Tender Documents. The period of construction given includes the time required for mobilization testing, rectifications, if any, retesting and completion in all respects in accordance with Contract Document.
- 18.2.2 The Contractor shall submit a detailed Bar chart in MS Projects consisting of adequate number of activities covering various key phases of the Work such as Design, Procurement, Manufacturing, Shipment and Field Erection activities. This network shall also indicate the intermediate milestones and interface facilities to be provided by the Employer, if any and the dates by which such facilities are needed.

19. Force Majeure

- 19.1 A 'Force Majeure' means any event or circumstance or combination of events those stated below that wholly or partly prevents or unavoidably delays an Affected Party in the performance of its obligations under this Agreement, but only if and to the extent that such events or circumstances are not within the reasonable control, directly or indirectly, of the Affected Party and could not have been avoided if the Affected Party had taken reasonable care or complied with Prudent Utility Practices:
- Act of God, including, but not limited to lightning, fire not caused by Contractors' negligence and explosion (to the extent originating from a source external to the site), earthquake, volcanic eruption, landslide, unprecedented flood, cyclone, typhoon or tornado;

- Any act of war (whether declared or undeclared), invasion, armed conflict or act of foreign enemy, blockade, embargo, revolution, riot, insurrection, terrorist or military action, quarantine;
- Radioactive contamination or ionizing radiation originating from a source in India or resulting from another Force Majeure Event mentioned above.

19.2 Force Majeure Exclusions

Force Majeure shall not include (i) any event or circumstance which is within the reasonable control of the Parties and (ii) the following conditions, except to the extent that they are consequences of an event of Force Majeure:

- Unavailability, late delivery, or changes in cost of the plant, machinery, equipment, materials, spare parts or consumables for the Power Project;
- Delay in the performance of any Contractor, sub-Contractor or their agents;
- Non-performance resulting from normal wear and tear typically experienced in power generation materials and equipment;
- Strikes at the facilities of the Contractor / Affected Party;
- Insufficiency of finances or funds or the agreement becoming onerous to perform; and
- Non-performance caused by, or connected with, the Affected Party's:
 - o Negligent or intentional acts, errors or omissions;
 - o Failure to comply with an Indian Law; or
 - o Breach of, or default under this Contract Agreement.
- Normal rainy seasons and monsoon
- Any Transport strikes not directly affecting the delivery of goods from manufacturer to site

19.3 In the event of either party being rendered unable by Force Majeure to perform any obligation required to be performed by them under this Contract, relative obligation of the party affected by such Force Majeure shall be treated as suspended during the period which the Force Majeure clause lasts.

19.4 Upon occurrence of such causes, the party alleging that it has been rendered unable as aforesaid, thereby, shall notify the other party in writing by registered notice within 7 (Seven) Days of the alleged beginning thereof giving full particulars and satisfactory evidence in support of its claim. Further, within 14 (Fourteen) days, the Contractor will furnish a detailed Contingency Plan to overcome the effects of the incident and bring the project on its schedule after cessation of the effect of Force Majeure.

19.5 The Affected Party shall give notice to the other Party of (i) the cessation of the relevant event of Force Majeure; and (ii) the cessation of the effects of such event of Force Majeure on the performance of its rights or obligations under this Agreement, as soon as practicable after becoming aware of each of these cessations.

19.6 Time for Performance of the relative obligation suspended by the force majeure shall stand extended by the period for which such Force Majeure clause lasts.

19.7 If works are suspended by Force Majeure conditions lasting for more than two months, the Employer/HGCL shall have the option of cancelling this Contract in whole or part thereof, at its discretion.

19.8 The Contractor will not be entitled to claim any compensation for Force Majeure conditions and shall take appropriate steps to insure its men and materials utilized by it under the Contract.

20. Liquidated Damages (LD)

- a. If for any reason, which does not entitle the contractor to an extension of item, the rate of progress of works, or any section is at any time, in the opinion of the Chief General Manager too slow to ensure completion by the prescribed time or extended time for completion Chief General Manager shall so notify the contractor in writing and the contractor shall there upon take such steps as are necessary and the Chief General Manager may approve to expedite progress so as to complete the works or such section by the prescribed time or extended time. The contractor shall not be entitled to any additional payment for taking such steps. If as a result of any notice given by the Chief General Manager under this clause the contractor shall seek the Chief General Managers permission to do any work at night or on Sundays, if locally recognized as days or rest, or their locally recognized equivalent, such permission shall not be unreasonably refused.
- b. If the contractor fails to complete whole of the works or any part thereof or section of the works within the stipulated periods of individual mile stones (including any bonafide extensions allowed by the competent authority without levying liquidated damages), the Chief General Manager may without prejudice to any other method of recovery will deduct one tenth of one percent of contract value per calendar day or part of the day for the period of delays subject to a maximum of 10% of the contract value not as a penalty from any monies in his hands due or which may become due to the contractor. The payment or deductions of such damages shall not relieve the contractor from his obligation to complete the works, or from any other of his obligations and liabilities under the contract.
- c. The liquidated damages for the whole of the work subject to force majeure:

Mile stone No.	Period from the date of signing the agreement	Minimum percentage of work to be completed (Cumulative)	Liquidated damages amount in Rupees per day
For the Milestone-1	End of 2 months	Completion of SITC of Stretch-1 ie., from Nanakramguda to TSPA Junction	Rs 50,000/-
For the Milestone-2	End of 3 months	SITC of two stretches ie., Completion of 100% of work	Rs 1,00,000/-
Incentive	Incentive for early completion of whole work	0.5% of initial contract price per month not exceeding 1.5% of contract price.	

The maximum amount of liquidated damages for the whole of the works is ten percent of final contract price.

If the progress is made good such that the total work is completed within the final milestone, the intermediate liquidated damages levied for milestones 1 &2 shall be reviewed by the Chief General Manager, HGCL and waive off the same.

21. Rights of the HGCL to forfeit Contract Performance Security

- 21.1 Whenever any claim against the Contractor for the payment of a sum of money arises out of or under the Contract, the Contractor shall pay to the HGCL on demand any balance remaining due to the HGCL & HGCL shall be entitled to recover such sum by appropriating in part or whole from the Contract Performance Security of the Contractor. In the event of the security being insufficient or if no security has been taken from the Contractor, then the balance or the total sum recoverable, as the case may be shall be deducted from any sum then due or which at any time thereafter may become due to the Contractor.

22. Failure by the Contractor to comply with the provisions of the Contract

22.1 If the Contractor refuses or fails to execute the Work or any separate part thereof with such diligence as will ensure its completion within the time specified in the Contract/O&M Contract or extension thereof or fails to perform any of his obligation under the Contract/O&M Contract or in any manner commits a breach of any of the provisions of the Contract/O&M Contract it shall be open to the Employer/ HGCL at its option by written notice to the Contractor:

a) To determine the event in which the Contract/O&M Contract shall stand terminated and shall cease to be in force and effect on and from the date appointed by the HGCL on that behalf, whereupon the Contractor shall stop forthwith any of the Contractor's work then in progress, except such work as the HGCL may, in writing, require to be done to safeguard any property or work, or installations from damage.

22.2 In such events of Clause 22.1(a) above.

a) The whole or part of the Contract Performance Security furnished by the Contractor is liable to be forfeited without prejudice to the right of the HGCL to recover from the Contractor the excess cost referred to in the Clause aforesaid, the HGCL shall also have the right of taking possession and utilizing in completing the works or any part thereof, such as materials, equipment and plants available at work site belonging to the Contractor as may be necessary and the Contractor shall not be entitled for any compensation for use or damage to such materials, equipment and plant.

22.3 Before determining the Contract as per Clause 22.1(a) provided in the judgement of the HGCL, the default or defaults committed by the Contractor is/ are curable and can be cured by the Contractor if an opportunity given to him, then the HGCL may issue Notice in writing calling the Contractor to cure the default within such time specified in the Notice.

22.4 The HGCL shall also have the right to proceed or take action as per 22.1(a) above, in the event that the Contractor becomes bankrupt, insolvent, compounds with his creditors, assigns the Contract in favor of his creditors or any other person or persons, or being a company or a corporation goes into voluntary liquidation, provided that in the said events it shall be necessary for the HGCL to give prior notice to the Contractor. Refer clause 25 below for further details on the "termination" of contract pertaining to various contractual situations.

23. Contractor remains liable to pay compensation if

23.1 In any case in which any of the powers conferred upon the HGCL by Clause 22.0 thereof shall have become action not taken under clause 22 exercisable and the same had not been exercised, the non-exercise thereof shall not constitute a waiver of any of the conditions hereof and such powers shall notwithstanding be exercisable in the event of any further case of default by the Contractor for which by any clause or clauses hereof he is declared liable to pay compensation amounting to the whole of his Contract Performance Security, and the liability of the Contractor for past and future compensation shall remain unaffected.

24. Contract Performance Security

Against the work Contract and Operation & maintenance contract of the project, within 07 (seven) days from the issuance of the Notification of Award/ Letter of Intent/Letter of Award (LOA) from HGCL, the successful bidder shall furnish an unconditional and irrevocable Contract Performance Security in accordance with Special Conditions of the Contract. The Contract Performance Security shall be in the form of either Banker's Cheque or Demand Draft or Bank Guarantee and shall be in the currency of the Contract and will be issued in the name of the HGCL as specified in the SCC. The Contract Performance Security shall be for an amount equal to specified in Special Conditions of Contract (SCC) towards faithful performance of the contractual obligations, performance of equipment and shall cover entire Contract. The validity of Contract Performance Security shall be in conjunction with the provisions mentioned under Section - V, Special Conditions of Contract (SCC).

Bank Guarantee towards Contract Performance Security shall be from any scheduled bank as specified in the List of Banks enclosed at Form F-22, Section - VI, Sample Forms and Formats of Tender documents or a branch of an International Bank situated in India and registered with Reserve Bank of India as scheduled foreign bank in case of Indian bidder. However, in case of Bank Guarantees from Banks other than the Nationalized Indian banks, the Bank must be a commercial Bank having net worth in excess of INR 500 Crores and a declaration to this effect should be made by such commercial bank either in the Bank Guarantee itself or separately on its letterhead. This Bank Guarantee shall be valid for a period in conjunction with the provisions mentioned under Section - V, Special Conditions of Contract (SCC).

The Contract Performance Security may also be submitted in the form of 'crossed payee account only' Demand Draft/ Banker's Cheque in favor of as mentioned in the BDS/SCC.

In case of default or failure of the successful bidder to comply with the requirements of this article shall constitute sufficient grounds for the cancellation of the award and forfeiture of the EMD, if applicable.

In case of default or failure of the Contractor to comply with the requirements of any of the Obligations covered under this Tender Document and/ or Contract Agreement shall constitute sufficient grounds for forfeiture of the Contract Performance Security.

The Contract Performance Security has to cover the entire contract value including extra works/ services also. As long as the Contract Performance Security submitted at the time of award takes care of the extra works/ services executed and total executed value are within the awarded contract price, there is no need for additional Contract Performance Security. As soon as the total executed value exceeds the ceiling of awarded contract price by more than 0.5%, the Contractor shall furnish additional Contract Performance Security on proportionate basis of the percentage as defined in the Special Conditions of Contract (SCC) for the additional amount in excess to the original contract value.

Further, any delay beyond 30 (Thirty) days shall attract interest @ 1.25% per month on the total Contract Performance Security amount, calculated on pro-rata basis accordingly. HGCL at its sole discretion may cancel the NOA/ LOI/LOA & forfeit 100% of EMD if applicable, in case Contract Performance Security is not submitted within 45 (Forty-five) days from issuance of NOA/ LOI/LOA. However, total project completion period shall remain same. Part Security shall not be accepted. The Zero Date shall be counted from the date of LOA or NTP whichever is later.

If the Contractor/ Sub-Contractor or their employees or the Contractor's agents and representatives shall damage, break, deface or destroy any property belonging to the Employer or others during the execution of the Contract, the same shall be made good by the Contractor at his own expenses and in default thereof, the Engineer-in-Charge may cause the same to be made good by other agencies and recover expenses from the Contractor (for which the certificate of the Engineer-in-Charge shall be final).

All compensation or other sums of money payable by the Contractor to the HGCL under terms of this Contract may be deducted from or paid by the encashment of a sufficient part of his Contract Performance Security or from any sums which may be due or may become due to the Contractor by the HGCL of any account whatsoever and in the event of his Contract Performance Security being reduced by reasons of any such deductions or sale of aforesaid, the Contractor shall within 10 (Ten) days thereafter make good in cash, bank drafts as aforesaid any sum or sums which may have been deducted from or realized by encashment of his Contract Performance Security, or any part thereof. No interest shall be payable by the HGCL for sum deposited as Contract Performance Security.

25. Termination of contract

25.1 Termination for HGCL's Convenience

25.1.1 The HGCL may at any time terminate the Contract for any reason by giving the Contractor a notice of termination that refers to this GCC Sub-Clause 25.1.

25.1.2 Upon receipt of the notice of termination under GCC Sub-Clause 25.1.1, the Contractor shall either immediately or upon the date specified in the notice of termination

- (a) cease all further work, except for such work as the HGCL may specify in the notice of termination for the sole purpose of protecting that part of the Facilities already executed, or any work required to leave the Site in a clean and safe condition
- (b) terminate all subcontracts, except those to be assigned to the HGCL pursuant to paragraph (d) (ii) below
- (c) remove all Contractor's Equipment from the Site, repatriate the Contractor's and its Subcontractors' personnel from the Site, remove from the Site any wreckage, rubbish and debris of any kind, and leave the whole of the Site in a clean and safe condition, and
- (d) subject to the payment specified in GCC Sub-Clause 25.1.3,
 - (i) deliver to the HGCL the parts of the Facilities executed by the Contractor up to the date of termination
 - (ii) to the extent legally possible, assign to the HGCL all right, title and benefit of the Contractor to the Facilities and to the Plant as of the date of termination, and, as may be required by the HGCL, in any subcontracts concluded between the Contractor and its Subcontractors; and
 - (iii) deliver to the HGCL all non-proprietary drawings, specifications and other documents prepared by the Contractor or its Subcontractors as at the date of termination in connection with the Facilities.

25.1.3 In the event of termination of the Contract under GCC Sub-Clause 25.1.1, the HGCL shall pay to the Contractor the following amounts:

- (a) the Contract Price, properly attributable to the parts of the Facilities executed by the Contractor as of the date of termination
- (b) the costs reasonably incurred by the Contractor in the removal of the Contractor's Equipment from the Site and in the repatriation of the Contractor's and its Subcontractors' personnel
- (c) any amounts to be paid by the Contractor to its Subcontractors in connection with the termination of any subcontracts, including any cancellation charges
- (d) costs incurred by the Contractor in protecting the Facilities and leaving the Site in a clean and safe condition pursuant to paragraph (a) of GCC Sub-Clause 25.1.2
- (e) the cost of satisfying all other obligations, commitments and claims that the Contractor may in good faith have undertaken with third Parties in connection with the Contract and that are not covered by paragraphs (a) through (d) above.

25.2 Termination for Contractor's Default

25.2.1 Neither the Employer nor the Contractor may assign the Contract or any part thereof, or anyright, benefit, obligation, or interest therein or thereunder, to any third party without the expressprior written consent of the other Party, which consent shall not be unreasonably withheld, except that the Contractor shall be entitled to assign any monies due and payable to it or that may become due and payable to it under the Contract, either absolutely or by way of charge.

25.2.2 The HGCL, without prejudice to any other rights or remedies it may possess, may terminate the Contract forthwith in the following circumstances by giving a notice of termination and its reasons therefor to the Contractor, referring to this GCC Sub-Clause 25.2:

- (a) if the Contractor becomes bankrupt or insolvent, has a receiving order issued against it, compounds with its creditors, or, if the Contractor is a corporation, a resolution is passed or order is made for its winding up, other than a voluntary liquidation for the purposes of amalgamation or reconstruction, a receiver is appointed over any part of its undertaking or assets, or if the Contractor takes or suffers any other analogous action in consequenceof debt
- (b) if the Contractor assigns or transfers the Contract or any right or interest therein in violationof the provision of GCC Clause 25.2.1.
- (c) if the Contractor, in the judgment of the HGCL has engaged in Fraud and Corruption in competing for or in executing the Contract.

25.2.3 If the Contractor

- (a) has abandoned or repudiated the Contract
- (b) has without valid reason failed to commence work on the Facilities promptly or has suspended the progress of Contract performance for more than twenty-eight (28) days after receiving a written instruction from the HGCL to proceed
- (c) persistently fails to execute the Contract in accordance with the Contract or persistently neglects to carry out its obligations under the Contract without just cause
- (d) refuses or is unable to provide sufficient materials, services or labor to execute and complete the Facilities in the manner specified in the program at rates of progress that give reasonable assurance to the HGCL that the Contractor can attain Completion of theFacilities by the Time for Completion as extended,

then the HGCL may, without prejudice to any other rights it may possess under the Contract, give a notice to the Contractor stating the nature of the default and requiring the Contractor to remedy the same. If the Contractor fails to remedy or to take steps to remedy the same within fourteen (14) days of its receipt of such notice, then the HGCL may terminate the Contract forthwith by giving a notice of termination to the Contractor that refers to this GCC Sub-Clause25.2.

25.2.4 Upon receipt of the notice of termination under GCC Sub-Clauses 25.2.2 or 25.2.3, the Contractor shall, either immediately or upon such date as is specified in the notice of termination,

- (a) cease all further work, except for such work as the HGCL may specify in the notice of termination for the sole purpose of protecting that part of the Facilities already executed,or any work required to leave the Site in a clean and safe condition

- (b) terminate all subcontracts, except those to be assigned to the HGCL pursuant to paragraph (d) below
- (c) deliver to the HGCL the parts of the Facilities executed by the Contractor up to the date of termination
- (d) to the extent legally possible, assign to the HGCL all right, title and benefit of the Contractor to the Facilities and to the Plant as of the date of termination, and, as may be required by the HGCL, in any subcontracts concluded between the Contractor and its Subcontractors
- (e) deliver to the HGCL all drawings, specifications and other documents prepared by the Contractor or its Subcontractors as of the date of termination in connection with the Facilities.

25.2.5 The HGCL may enter upon the Site, expel the Contractor, and complete the Facilities itself or by employing any third Party. The HGCL may, to the exclusion of any right of the Contractor, take over and use with the payment of a fair rental rate to the Contractor, with all the maintenance costs to the account of the HGCL and with an indemnification by the HGCL for all liability including damage or injury to persons arising out of the HGCL's use of such equipment, any Contractor's Equipment owned by the Contractor and on the Site in connection

with the Facilities for such reasonable period as the HGCL considers expedient for the supply and installation of the Facilities.

Upon completion of the Facilities or at such earlier date as the HGCL thinks appropriate, the HGCL shall give notice to the Contractor that such Contractor's Equipment will be returned to the Contractor at or near the Site and shall return such Contractor's Equipment to the Contractor in accordance with such notice. The Contractor shall thereafter without delay and at its cost remove or arrange removal of the same from the Site.

25.2.6 Subject to GCC Sub-Clause 25.2.7, the Contractor shall be entitled to be paid the Contract Price attributable to the Facilities executed as of the date of termination, the value of any unused or partially used Plant on the Site, and the costs, if any, incurred in protecting the Facilities and in leaving the Site in a clean and safe condition pursuant to paragraph (a) of GCC Sub-Clause

25.2.4. Any sums due the HGCL from the Contractor accruing prior to the date of termination shall be deducted from the amount to be paid to the Contractor under this Contract.

25.2.7 If the HGCL completes the Facilities, the cost of completing the Facilities by the HGCL shall be determined.

If the sum that the Contractor is entitled to be paid, pursuant to GCC Sub-Clause 25.2.6, plus the reasonable costs incurred by the HGCL in completing the Facilities, exceeds the Contract Price, the Contractor shall be liable for such excess.

If such excess is greater than the sums due the Contractor under GCC Sub-Clause 25.2.6, the Contractor shall pay the balance to the HGCL, and if such excess is less than the sums due the Contractor under GCC Sub-Clause 25.2.6, the HGCL shall pay the balance to the Contractor.

The HGCL and the Contractor shall agree, in writing, on the computation described above and the manner in which any sums shall be paid.

25.3 Termination by the Contractor

25.3.1 If

- (a) the HGCL has failed to pay the Contractor any sum due under the Contract within the specified period, has failed to approve any invoice or supporting documents, or commits a substantial breach of the Contract, the Contractor may give a notice to the HGCL that requires payment of such sum, with interest, requires approval of such invoice or supporting documents, or specifies the breach and requires the HGCL to remedy the same, as the case may be. If the HGCL fails to pay such sum together with such interest, fails to approve such invoice or supporting documents or give its reasons for withholding such approval, fails to remedy the breach or take steps to remedy the breach within fourteen (14) days after receipt of the Contractor's notice, or
- (b) the Contractor is unable to carry out any of its obligations under the Contract for any reason attributable to the HGCL, including but not limited to the HGCL's failure to provide possession of or access to the Site or other areas or failure to obtain any governmental permit necessary for the execution and/or completion of the Facilities,

then the Contractor may give a notice to the HGCL thereof, and if the HGCL has failed to pay the outstanding sum, to approve the invoice or supporting documents, to give its reasons for withholding such approval, or to remedy the breach within twenty-eight (28) days of such notice, or if the Contractor is still unable to carry out any of its obligations under the Contract for any reason attributable to the HGCL within twenty-eight (28) days of the said notice, the Contractor may by a further notice to the HGCL referring to this GCC Sub-Clause 25.3.1, forthwith terminate the Contract.

25.3.2 The Contractor may terminate the Contract forthwith by giving a notice to the HGCL to that effect, referring to this GCC Sub-Clause 25.3.2, if the HGCL becomes bankrupt or insolvent, has a receiving order issued against it, compounds with its creditors, or, being a corporation, if a resolution is passed or order is made for its winding up (other than a voluntary liquidation for the purposes of amalgamation or reconstruction), a receiver is appointed over any part of its undertaking or assets, or if the HGCL takes or suffers any other analogous action in consequence of debt.

25.3.3 If the Contract is terminated under GCC Sub-Clauses 25.3.1 or 25.3.2, then the Contractor shall immediately

- (a) cease all further work, except for such work as may be necessary for the purpose of protecting that part of the Facilities already executed, or any work required to leave the Site in a clean and safe condition
- (b) terminate all subcontracts, except those to be assigned to the HGCL pursuant to paragraph (d) (ii)
- (c) remove all Contractor's Equipment from the Site and repatriate the Contractor's and its Subcontractors' personnel from the Site, and
- (d) subject to the payment specified in GCC Sub-Clause 25.3.4,
 - (i) deliver to the HGCL the parts of the Facilities executed by the Contractor up to the date of termination

- (ii) to the extent legally possible, assign to the HGCL all right, title and benefit of the Contractor to the Facilities and to the Plant as of the date of termination, and, as may be required by the HGCL, in any subcontracts concluded between the Contractor and its Subcontractors, and
- (iii) deliver to the HGCL all drawings, specifications and other documents prepared by the Contractor or its Subcontractors as of the date of termination in connection with the Facilities.

25.3.4 If the Contract is terminated under GCC Sub-Clauses 25.3.1 or 25.3.2, the HGCL shall pay to the Contractor all payments specified in GCC Sub-Clause 25.1.3, and reasonable compensation for all loss, except for loss of profit, or damage sustained by the Contractor arising out of, in connection with or in consequence of such termination.

25.3.5 Termination by the Contractor pursuant to this GCC Sub-Clause 25.3 is without prejudice to any other rights or remedies of the Contractor that may be exercised in lieu of or in addition to rights conferred by GCC Sub-Clause 25.3.

25.4 In this GCC Clause 25, the expression "Facilities executed" shall include all work executed, Installation Services provided, and all Plant acquired, or subject to a legally binding obligation to purchase, by the Contractor and used or intended to be used for the purpose of the Facilities, up to and including the date of termination.

25.4 In this GCC Clause 25, in calculating any monies due from the HGCL to the Contractor, account shall be taken of any sum previously paid by the HGCL to the Contractor under the Contract, including any advance payment.

26. Members of the Employer/ HGCL not individually Liable

26.1 No Director, or official or employee of the Employer/ HGCL shall in any way be personally bound or liable for the acts or obligations under the Contract or answerable for any default or omission in the observance or performance of any of the acts, matters or things which are herein contained.

27. Employer/HGCL not bound by Personal Representations

27.1 The Contractor shall not be entitled to any increase on the price or any other right or claim whatsoever by reason of any representation, explanation statement or alleged representation, promise or guarantees given or alleged to have been given to him by any person.

28. Contractor's office at Site & Vehicle requirement for Mobility at site

28.1 The Contractor shall provide and maintain an office at the site for the accommodation of his agent and staff (With meeting room) and such office shall be open at all reasonable hours to receive instructions, notice or other communications. The contractor shall make provisions and arrangements at site locations for temporary office (Portacabin) including all basic amenities having electricity, furniture, Air Conditioning, Sanitation, toilet facilities, etc. at his own cost on behalf & purpose of the Employer and his staff and shall get the clearance of local authorities for setting up/construction of such facilities. Bidders are required to consider all such temporary office construction cost into his account, while bidding. These facilities shall be provided till the final handing over of the project to the HGCL. The contractor shall ensure that the area is kept clean and sanitary conditions are maintained as laid down by the local authorities controlling the area.

Further contractor shall also provide & ensure availability of required Vehicle (Preferably SUV Segment Vehicles requiring Off Roding) with driver and fuel/Maintenance/Consumables etc for the purpose of Mobility in and around the site for official purposes for the Employer & its staff. The vehicle as mentioned is to be provided during project construction and O&M period and inspection by HGCL as per requirement, failing which HGCL shall have full right for alternate arrangement at the risk & cost of the contractor

29. Contractor's Subordinate Staff and their Conduct

- 29.1 The Contractor, on or after award of the Work shall name and depute a qualified Engineer having sufficient experience in carrying out work of similar nature, to whom the equipment, materials, if any, shall be issued and instructions for works given. The Contractor shall also provide sufficient and qualified staff to superintend the execution of the Work, competent sub-agents, foremen and leadinghands including those specially qualified by previous experience to supervise the types of works contained in the Contract in such manner as will ensure work of the best quality, expeditious working. Whenever in the opinion of the Engineer-in-Charge/Project Manager additional properly qualified supervisory staff is considered necessary, they shall be employed by the Contractor without additional charge on accounts thereof. The Contractor shall ensure that Sub-Contractors, if any, shall provide competent and efficient supervision, over the work entrusted to them.
- 29.2 If and whenever any of the Contractor's or Sub-Contractor's agents, sub-agents, assistants, foremen, or other employees be guilty of any misconduct or be incompetent or insufficiently qualified or negligent in the performance of their duties, it is undesirable for administrative or any other reason for such person or persons to be employed in the works, the Contractor, shall at once remove such person or persons from employment thereon. Any person or persons so removed from the works shall not again be employed in connection with the Works. Any person so removed from the Work shall be immediately replaced at the expense of the Contractor by a qualified and competent substitute. Should the Contractor be requested to repatriate any person removed from the works he shall do so and shall bear all costs in connection herewith.
- 29.3 The Contractor shall be responsible for the proper behavior of all the staff, foremen, workmen, and others, and shall exercise a proper degree of control over them and in particular and without prejudice to the said generality, the Contractor shall be bound to prohibit and prevent any employees from trespassing or acting in any way detrimental or prejudicial to the interest of the community or of the properties or occupiers of land and properties in the neighborhood and in the event of such employee so trespassing, the Contractor shall be responsible therefore and relieve the HGCL of all consequent claims or actions for damages or injury or any other grounds whatsoever. The Contractor shall be liable for any liability to HGCL on account of deployment of Contractor's staff etc. or incidental or arising out of the execution of Contract. The Contractor shall be liable for all acts or omissions on the part of his staff, Foremen and Workmen and others in his employment, including misfeasance or negligence of whatever kind in the course of their work or during their employment, which are connected directly or indirectly with the Contract.
- 29.4 If and when required by the HGCL and Contractor's personnel entering upon the HGCL's premises shall be properly identified by badges or gate passes which must be worn at all times on HGCL's premises. Contractor may be required to obtain daily entry passes for his Staff/ Employees to work within operating areas. These being safety requirements, no relaxations on this account shall be given to Contractor.
- 29.5 Contractor shall at all times provide Employer/ HGCL access to site and office during construction/ O&M periods and also provide them with any data/ information sought for.

30. Sub-letting of Works

- 30.1 Sub-contracting other than for labour contract/engagement of labour, shall be permitted with the information to the Employer. However, sub-contracting for 100% of the contract on back to back basis shall not be permitted. Any part of the Contract nor any share or interest therein shall in any manner or degree be transferred, assigned or sublet by the Contractor directly or indirectly to any person, firm or corporation whatsoever without the consent in writing, of the Employer except as provided for in the succeeding Clause.

i) Sub-Contracts for Temporary Works etc.

The Employer may give written consent to Sub-Contract for the execution of any part not exceeding 50% of the Work at the site on written request by the original contractor, being entered in to by Contractor provided each individual Sub-contract is informed to the Engineer-in-Charge/Project Manager.

ii) List of Sub-Contractors to be supplied

The Contractor shall furnish to the Engineer-in-Charge/Project Manager list of all Sub-Contractors or other persons or firms engaged by the Contractor and working at the Site during the previous month with particulars of the general nature of the Subcontract or works done by them.

iii) Contractor's Liability not Limited by Sub-Contractors

Notwithstanding any sub-letting with such approval as aforesaid and notwithstanding that the Engineer-in-Charge/Project Manager shall have received copies of any Subcontracts, the Contractor shall be and shall remain solely responsible for the quality, proper and expeditious execution of the Contract in all respects as if such sub-letting or Subcontracting had not taken place, and as if such work had been done directly by the Contractor. The Contractor shall bear all responsibility for any act or omission on the part of sub-Contractors in regard to work to be performed under the Contract.

iv) No Remedy for Action Taken under this Clause

No action taken by the HGCL under the clause shall relieve the Contractor of any of his liabilities under the Contract or give rise to any right or compensation, extension of time or otherwise failing which the HGCL shall have the right to remove such Sub-Contractor(s) from the site.

vi) Termination/ Cancellation of Contract

HGCL is nowhere liable for the communication, acts and deeds and performance of the sub-Contractor as engaged by the principal Contractor. Principal Contractor solely is responsible and liable for the entire execution of project and performance of contract.

Subject to poor performance and prolonged delay of the project on account of inefficient sub-letting of the project work, HGCL may take a final decision to terminate the contract of the principal Contractor which will be binding and non-revertible and henceforth no plea in this regard shall be entertained.

vii) Subject to the provisions of the Contract, the Contractor shall be solely responsible for the manner in which the Contract is performed. All employees, representatives or Sub-Contractors engaged by the Contractor in connection with the Performance of the Contract shall be under the complete control of the Contractor and shall not be deemed to be employees of the HGCL. Nothing contained in the Contract or in any subcontract awarded by the Contractor shall be construed to create any contractual relationship between any such employees, representatives or Sub-Contractors and the HGCL.

viii) Under no circumstances the sub-Contractor shall claim or shall put any binding to the HGCL and at all times the sub-Contractor must be managed by the Contractor. The HGCL shall not be responsible for any claims at any time by the Contractor in relation to the sub-Contractor.

ix) No relaxation, forbearance, delay or indulgence by either party in enforcing any of the terms and conditions of the Contract or the granting of time by either party to the other shall prejudice, affect or restrict the rights of that party under the Contract, nor shall any waiver by either party of any breach of Contract operate as waiver of any subsequent or continuing breach of Contract.

x) Any waiver of a party's rights, powers or remedies under the Contract must be in writing, must be dated and signed by an authorized representative of the party granting such waiver, and must specify the right and the extent to which it is being waived.

xi) The Contractor shall be solely responsible for ensuring that his sub-Contractors fulfil and comply with the statutory requirements of Labour and other Laws.

31. Power of Entry

31.1 If the Contractor shall not commence the Work in the manner previously described in the Contract documents or if he shall at any time in the opinion of the Engineer-in-Charge/Project Manager:

- i) fail to carry out the Work in conformity with the Contract documents, or
- ii) fail to carry out the Work in accordance with the Time Schedule, or
- iii) substantially suspend the Work for a period of minimum 14 (Fourteen) days without authority from the Engineer-in-Charge/Project Manager, or
- iv) fail to supply sufficient or suitable construction plant, temporary works, labour, materials or things, or
- v) Commit, suffer, or permit any other breach of any of the provisions of the Contract on his part to be performed or observed or persist in any of the above-mentioned breaches of the Contract for 14 (Fourteen) days, after notice in writing shall have been given to the Contractor by the Engineer-in-Charge/Project Manager requiring such breach to be remedied, or
- vi) if the Contractor abandon the Work, or
- vii) If the Contractor during the continuance of the Contract shall become bankrupt, make any arrangement or composition with his creditors or go into liquidation whether compulsory or voluntary not being merely a voluntary liquidation for the purpose of amalgamation or reconstruction

then in any such case, the HGCL shall have the power to enter upon site and take possession thereof and of the materials, temporary Work, construction plant, and stock thereon, and to revoke the Contractor's license to use the same, and to complete the Work by his agents, other Contractors or workmen or to sublet the same upon any terms and to such other person, firm or corporation as the HGCL in his absolute discretion may think proper to employ and for the purpose aforesaid to use or authorize the use of any materials, temporary work, Construction Plant, and stock as aforesaid, without making payment or allowance to the Contractor for the said materials other than such as may be certified in writing by the Engineer-in-Charge/Project Manager to be reasonable, and without making any payment or allowance to the Contractor for the use of the temporary said works, construction plant and stock or being liable for any loss or damage thereto, and if the HGCL shall by reason of his taking possession of the Work or of the Work being completed by other Contractor (due account being taken of any such extra work or works which may or be omitted) then the amount of such excess shall be deducted from any money which may be due for work done by the Contractor under the Contract and not paid for.

Any deficiency shall forthwith be made good and paid to the HGCL by the Contractor and the HGCL shall have power to sell in such manner and for such price as he may think fit all or any of the construction plant, materials etc. constructed by or belonging to Contractor and to recoup and retain the said deficiency or any part thereof out of proceeds of the sale.

32. Contractor's responsibility

32.1 Contractor's responsibility

32.1.1 The Contractor shall grade/level the land identified for development of the mentioned Power Plant along with the design, procure, manufacture (including associated purchases and/or subcontracting), install, commission and complete the Facilities, carry out the Guarantee tests with due care and diligence in accordance with the Contract along with interconnecting transmission system including Right of Way for Transmission Line and the comprehensive O&M of the complete facilities for the period as defined under the tender document. It is Contractor's responsibility to coordinate with state/central agencies in order to get any

permission whatsoever, required for successful development & operation of Plant till its desired life.

- 32.1.2 The Contractor shall acquire, on behalf of HGCL, in the HGCL's name, all permits, approvals and/or licenses from all local, state or national government authorities or public service undertakings in the Country/State where the Site is located that are necessary for the setting up of the Plant & operation of Plant till its desired life as mentioned under the Contract, including, but not limited to, entry permits for all imported HGCL's/Employer's Equipment (if any). In this regard, any document required from HGCL/ Employer shall be intimated at least 10 days prior to submission. Contractor has to ensure safe keeping of the documents and diligent use. It is the responsibility of the Contractor to safe keep and return all the original approvals, permits, licenses, certificates and other relevant document generated as a result of the setting up of project and comprehensive O&M process to the HGCL.
- 32.1.3 In the matter of connectivity of Plant to DISCOM's substation, the HGCL will take the necessary connectivity permission, however, all the other permissions and clearances as deemed required by the State Agency/DISCOM for Bay allocation, technical/regulatory compliance for interconnection, ROW etc are to be taken by the Contractor. Statutory fees pertaining to such shall be reimbursed by the HGCL on production of the documentary evidence/Demand note over and above the contract value. Further, Contractor shall also facilitate HGCL in getting the required permissions/agreements as required for the energy accounting by State agencies/DISCOM.
- 32.1.4 The Contractor shall acquire in its name all permits, approvals and/or licenses from all local, state or national government authorities or public service undertakings in the country where the Site is located that are necessary for the Performance of the Contract, including, but not limited to, the right of way for the access to site and for erection of transmission lines as applicable, visas for the Contractor's and Sub-Contractor's personnel and entry permits for all imported Contractor's Equipment. The Contractor shall acquire all other permits, approvals and/or licenses that are not the responsibility of the HGCL and that are necessary for the Performance of the Contract.
- 32.1.5 Contractor shall also seek for any exemption applicable for the project as per the orders released from GOI time to time in appropriate Formats including all the required attachments. In this regard, Contractor shall be responsible to take all necessary certificates as a proof of exemptions on behalf of HGCL. However, all the documents required from HGCL, as needed for the process, will be provided by HGCL. The demand of such documents shall be made to the HGCL in at least 10 days advance.
- 32.1.6 The Contractor shall comply with all laws in force at the place, where the Facilities are installed and where the Installation Services are carried out. The laws will include all national, provincial, municipal or other laws that affect the Performance of the Contract and binding upon the Contractor. The Contractor shall indemnify and hold harmless the Employer/HGCL from and against any and all liabilities, damages, claims, fines, penalties and expenses of whatever nature arising or resulting from the violation of such laws by the Contractor or its personnel, including the SubContractors and their personnel.
- 32.1.7 Any plant, material, spares & spares inventory and services that will be incorporated in or be required for the facilities.
- 32.1.8 Unless otherwise specified in the Contract or agreed upon by the HGCL and the Contractor, the Contractor shall provide/ deploy sufficient, properly qualified operating and maintenance personnel; shall supply and make available all raw materials, spares, tools and tackles and other materials and facilities; and shall perform all work and services of whatsoever nature, to properly carry out Pre- commissioning, Commissioning and Guarantee Tests, all in accordance with the provisions of "Scope of Works and Supply by the HGCL/Employer" to the Contract Agreement at or before the time specified in the program furnished by the Contractor and in the manner thereupon specified or as otherwise agreed upon by the HGCL/ Employer and the Contractor.

33. Other Agencies at Site

- 33.1 The Contractor shall have to execute the Work in such place and conditions where other agencies may also be engaged for other works such as site grading, filling, and levelling, electrical and mechanical engineering works, etc. No claim shall be entertained due to Work being executed in the above circumstances. The Contractor shall allow such agencies to use the facilities like roads, etc constructed by the Contractor in order that they are able to carry out their respective scope of works unhindered

34. Notice

34.1 To the Contractor

Any notice hereunder may be served on the Contractor or his duly authorized representative at the job site or may be served.

Any communication sent shall be confirmed within two (2) days after receipt. Any communication sent by facsimile or e-mail shall be deemed to have been delivered on date of its dispatch and personal delivery deemed to have been delivered on date of delivery. Either party may change its postal, facsimile or e-mail address or addresses for receipt of such notices by ten (10) days' notice to the other party in writing.

34.2 To the Employer/HGCL

Any notice to be given to the Employer/HGCL under the terms of the Contractor shall be served by sending the same by mail to or delivering the same at the offices of Employer/ HGCL at the mentioned address in the Tender document.

35. Right of Various Interests

- 35.1 i) The HGCL reserves the right to distribute the work between more than one agency(ies). The Contractor shall cooperate and afford other agency(ies) reasonable opportunity for access to the Work for the carriage and storage of materials and execution of their works.

ii) Wherever the work being done by any department of the Employer/ HGCL or by other agency(ies) employed by the Employer/ HGCL is contingent upon Work covered by this Contract, the respective rights of the various interests involved shall be determined jointly to secure the completion of the various portions of the work in general harmony.

36. Patents and Royalties

- 36.1 The copyright in all drawings, documents and other materials containing data and information furnished to the HGCL/ Employer by the Contractor herein shall remain vested in the Contractor or, if they are furnished to the HGCL/ Employer directly or through the Contractor by any third party, including suppliers of materials, the copyright in such materials shall remain vested in such third party. The HGCL/ Employer shall however be free to reproduce all drawings, documents, specification and other material furnished to the HGCL/ Employer for the purpose of the contract including, if required, for operation and maintenance of the facilities.

The Contractor shall indemnify the HGCL/ Employer against third party claims of infringement of patent, trademark or industrial design rights arising from use of goods or any part thereof in India.

The Contractor, if licensed under any patent covering equipment, machinery, materials or compositions of matter to be used or supplied or methods and process to be practiced or employed in the performance of this Contract, agrees to pay all royalties and license fees which may be due with respect thereto. If any equipment, machinery, materials, composition of matters, be used or supplied or methods and processes to be practiced or employed in the performance of this Contract, is covered by a patent under which the Contractor is not licensed then the Contractor before supplying or using the equipment, machinery materials, composition method or processes shall obtain such licenses and pay such royalties and

license fees as may be necessary for performance of this Contract. In the event the Contractor fails to pay any such royalty or obtain any such license, any suit for infringement of such patents which is brought against the Contractor or the Employer/ HGCL as a result such failure will be defended by the Contractor at his own expense and the Contractor will pay any damages and costs awarded in such suit. The Contractor shall promptly notify the Employer/ HGCL if the Contractor has acquired the knowledge of any plant under which a suit for infringement could be reasonably brought because of the use by the HGCL of any equipment, machinery, materials, process, methods to be supplied hereunder. The Contractor agrees to and does hereby grant to Employer/ HGCL, together with the right to extend the same to any of the subsidiaries of the Employer/ HGCL as irrevocable, royalty free license to use in any country, any invention made by the Contractor or his employee in or as result of the performance of the Work under the Contract.

- 36.2 All charges on account of royalty, tollage, rent or any other levy on materials obtained for the work or temporary work or part thereof (excluding materials provided by the Employer/ HGCL, if any) shall be borne by the Contractor.
- 36.3 The Contractor shall not sell or otherwise dispose of or remove except for the purpose of this Contract, the sand, stone, clay, ballast, earth, rock or other substances, or materials obtained from any excavation made for the purpose of the Work or any building or produce upon the site at the time of delivery of the possession thereof, but all such substances, materials, buildings and produce shall be the property of the HGCL provided that the Contractor use the same for the purpose of the work.
- 36.4 The copyright in all drawings, documents and other materials containing data and information furnished to the Employer/HGCL by the Contractor herein shall remain vested in the Contractor or, if they are furnished to the Employer/HGCL directly or through the Contractor by any third party, including suppliers of materials, the copyright in such materials shall remain vested in such third party. The Employer/HGCL shall however be free to reproduce all drawings, documents, specification and other material furnished to the Employer/HGCL for the purpose of the contract including, if required, for operation and maintenance of the facilities.
- 36.5 The Employer/HGCL and the Contractor shall keep confidential and shall not, without the written consent of the other party hereto, divulge to any third party any documents, data or other information furnished directly or indirectly by the other party hereto in connection with the Contract, whether such information has been furnished prior to, during or following termination of the Contract. Notwithstanding the above, the Contractor may furnish to its Sub-Contractor(s) such documents, data and other information it receives from the Employer/HGCL to the extent required for the Sub- Contractor(s) to perform its work under the Contract, in which event the Contractor shall obtain from such Sub-Contractor(s) an undertaking of confidentiality similar to that imposed on the Contractor under this clause.
- 36.6 The Employer/HGCL shall not use such documents, data and other information received from the Contractor for any purpose other than the operation and maintenance of the Facilities. Similarly, the Contractor shall not use such documents, data and other information received from the Employer/HGCL for any purpose other than the design, procurement of Plant and Equipment, construction or such other work and services as are required for the Performance of the Contract.
 - The obligation of a party under GCC Sub-Clauses 36.5 and 36.6 above, however, shall not apply to that information which
 - Now or hereafter becomes available in the public domain through no fault of that party
 - Can be proven to have been possessed by that party at the time of disclosure and which was not previously obtained, directly or indirectly, from the other party hereto.
 - Otherwise lawfully becomes available to that party from a third party that has no obligation of confidentiality.

- The above provisions of this Clause shall not in any way modify any undertaking of

confidentiality given by either of the parties hereto prior to the date of the Contract in respect of the Facilities or any part thereof.

- The provisions of this clause shall survive termination, for whatever reason, of the Contract.

37. Liens

- 37.1 If, at any time there should be evidence or any lien or claim for which the HGCL might have become liable and which is chargeable to the Contractor, the HGCL shall have the right to retain out of any payment then due or thereafter to become due an amount sufficient to completely indemnify the HGCL against such lien or claim and if such lien or claim be valid, the HGCL may pay and discharge the same and deduct the amount so paid from any money which may be or may become due and payable to the Contractor. If any lien or claim remain unsettled after all payments are made, the Contractor shall refund or pay to the HGCL all money that the latter may be compelled to pay in discharging such lien or claim including all costs and reasonable expenses. HGCL reserves the right to do the same.
- 37.2 The HGCL shall have lien on all materials, equipment including those brought by the Contractor for the purpose of erection, testing and commissioning of the Work.
- 37.3 The final payment shall not become due until the Contractor delivers the complete release or waiver of all liens arising or which may arise out of his agreement or receipt in full or certification by the Contractor that all invoices for labour, materials, services have been paid in lien thereof and if required in any case an affidavit that so far as the Contractor has knowledge or information the releases and receipts include all the labour and material for which a lien could be filled.
- 37.4 Contractor will indemnify and hold the Employer/ HGCL harmless, for a period of 02 (Two) years after the issue of Final Acceptance from all liens and other encumbrances against the Employer/ HGCL on account of debts or claims alleged to be due from the Contractor or his Sub-Contractor to any person including Sub-Contractor and on behalf of Employer/ HGCL will defend at his own expense, any claim or litigation brought against the Employer/ HGCL or the Contractor in connection therewith. Contractor shall defend or contest at his own expense any fresh claim or litigation by any person including his Sub-Contractor, till its satisfactory settlement even after the expiry of 02 (Two) years from the date of issue of Final Acceptance.

38. Delays by Employer/ HGCL or his Authorized representatives

- 38.1 In case the Contractor's performance is delayed due to any act or omission on the part of the Employer/ HGCL or his authorized representatives, then the Contractor may be given due extension of time for the completion of the Work after proper due diligence by Employer/ HGCL, to the extent such omission on the part of the Employer/ HGCL has caused delay in the Contractor's performance of his Work.
- 38.2 No adjustment in Contract Price shall be allowed for reasons of such delays and extensions granted except as provided in Tender Document, where the Employer/ HGCL reserves the right to seek indulgence of Contractor to maintain the agreed Time Schedule of Completion. In such an event the Contractor shall be obliged for working by Contractor's personnel for additional time beyond stipulated working hours as also Sundays and Holidays and achieve the completion date/interim targets.

39. Payment if the Contract is terminated

- 39.1 If the Contract shall be terminated as per the provisions of the Tender/ Contract, the Contractor shall be paid by the HGCL in so far as such amounts or items shall not have already been covered by payments of amounts made to the Contractor for the Work executed prior to the date of termination at the cost provided for in the Contract and in addition to the following:

- a) The amount payable in respect of any preliminary items, so far as the Work or service

comprised therein has been carried out or performed and an appropriate portion as certified by Engineer-in-Charge of any such items or service comprised in which has been partially carried out or performed.

b) Any other expenses which the Contractor has expended for performing the Work under the Contract subject to being duly recommended by Engineer-in-Charge/Project Manager and approved by Employer/ HGCL for payment, based on documentary evidence of his having incurred such expenses.

39.2 The Contractor will be further required to transfer the title, Guarantee/Warranty of the Material/Products/Spares of the works & Projects including but not limited to PV Modules, Power Conditioning Units (PCU)/ Inverters, Transformers, Batteries (If applicable) etc and provide the following in the manner and as directed by the Employer/ HGCL.

a) Any and all completed works.

b) Such partially completed Work including drawings, information and Contract rights as the Contractor has specially performed, produced or acquired for the performance of the Contractor.

40. No Waiver of Rights

40.1 Neither the inspection by the Employer/ HGCL or any of their officials, Employees or representatives nor any order by the Employer/ HGCL for payment of money or any payment for or acceptance of the whole or any part of the Work by the Employer/ HGCL nor any extension of time, nor any possession taken by HGCL shall operate as a waiver of any provision of the Contract, or of any power herein reserved to the Employer/ HGCL, or any right to damages herein provided, nor shall any waiver of any breach in the Contract be held to be a waiver of any other subsequent breach.

41. Certificate not to Affect Right of Employer and Liability of Contractor

41.1 No interim payment certificate(s) issued by the Employer/ HGCL, nor any sum paid on account by the Employer/ HGCL, nor any extension of time for execution of the work granted by Employer/ HGCL shall affect or prejudice the rights of the Employer/ HGCL against the Contractor or relieve the Contractor of his obligations for the due performance of the Contract, or be interpreted as approval of the Work done or of the equipment supplied and no certificate shall create liability for the Employer/ HGCL to pay for alterations, amendments, variations or additional works not ordered, in writing, by Employer/ HGCL or discharge the liability of the Contractor for the payment of damages whether due, ascertained, or certified or not or any sum against the payment of which he is bound to indemnify the Employer/ HGCL.

42. Language and Measures

42.1 All documents pertaining to the Contract including Specifications, Schedules, Notices, Correspondence, Operating and Maintenance Instructions, Drawings or any other writing shall be written in English language only. The SI System of measurement shall be used in the Contract unless otherwise specified. Any literature/standard required for the execution of the project work will be provided by the Contractor in the English language only.

43. Guarantee Tests & Operational Acceptance and Transfer of Title

43.1 Functional Guarantees

43.1.1 The Contractor guarantees that during the Guarantee Test, the Facilities and all parts thereof shall attain the Functional Guarantees specified under Technical Specifications, subject to and upon the conditions therein specified. Respective compensation in case of the Non-achievement of the same is mentioned in the SCC of the tender document.

43.1.2 If, for reasons attributable to the Contractor, the guaranteed level of the Functional Guarantees

specified under Technical Specifications are not met either in whole or in part, the Contractor shall, within a mutually agreed time, at its cost and expense make such changes, modifications and/ or additions to the Plant or any part thereof as may be necessary to meet such Guarantees. The Contractor shall notify the Employer/HGCL upon completion of the necessary changes, modifications and/or additions, and shall seek the Employer's/HGCL's consent to repeat the Guarantee Test. If the level of the specified Functional Guarantee parameters, as demonstrated even during repeat of the Guarantee Test(s), are outside the acceptable shortfall limit, the Employer/HGCL may at its option, either

- Reject the Equipment and advise immediate replacement to suit the provisions of Technical Specification without any additional cost or;
- Reject the Equipment and recover the payments already made, or;
- Terminate the Contract and recover the payments already made, or;
- Accept the equipment after levy of liquidated damages in accordance with the provisions specified.

43.2 Plant Performance Guarantee Test

The Plant Performance Guarantee (as mentioned in TS) Test shall be conducted by the Contractor after Commissioning of the Facilities to ascertain whether the Facilities or the relevant part(s) can attain the Functional Guarantees specified in the Contract Documents. The Contractor's and Project Manager's advisory personnel shall attend the Guarantee Test. The detailed procedure for Performance Guarantee Test shall be carried out as per procedure laid down in Technical Specifications.

43.3 Operational Acceptance

43.3.1 Operational Acceptance shall occur in respect of the Facilities when:

- a) The Plant Performance Guarantee in accordance with the procedure specified in "Technical Specifications" has been successfully completed and the Functional Guarantees are met.
- b) Completion of the Facilities have been achieved as per Technical Specifications

43.3.2 The milestone payment linked with successful Operation acceptance shall be released subjected to following:

- a) All "As- Built" Drawings and documents are submitted.
- b) Detailed Engineering Document with detailed specification, schematic drawing, circuit drawing, cable routing plans and test results, manuals for all deliverable items, Operation, Maintenance & Safety Instruction Manual and other information about the project are submitted
- c) Bill of material of the installed Facility is submitted.
- d) Inventory of recommended and mandatory spares including special tools and tackles at project Site are provided
- e) All the required approvals and NOC's as required, are submitted
- f) List of deviation from the approved drawings with reason for deviation is submitted
- g) List of punch points, duly signed, is provided.
- h) Settlement of liquidity damages against delay and performance (Liquidity Damages)
- i) Certificates of final levels as set out for various works
- j) Certificates of tests performed for various Works.
- k) Material appropriation, Statement for the materials issued by the HGCL, if applicable for the Work and list of surplus materials returned to the HGCL's store duly supported by necessary documents.
- l) Warranty certificates for each equipment are handed over to HGCL' and 'Statutory approvals/permits/ NOC are handed over to HGCL'
- m) Supply of all mandatory and recommend spares

- 43.3.3 At any time after the events set out in GCC Sub- Clause 43.3.1 have occurred, the Contractor may give a notice to the Project Manager/EIC requesting the issue of an Operational Acceptance Certificate in the form acceptable to the Employer in respect of the Facilities or the part thereof specified in such notice as at the date of such notice.
- 43.3.4 The HGCL shall, after consultation with the Project Manager/EIC, and within thirty (30) days after receipt of the Contractor's notice, issue an Operational Acceptance.
- 43.3.5 If within thirty (30) days after receipt of the Contractor's notice, the Project Manager fails to issue the Operational Acceptance or fails to inform the Contractor in writing of the justifiable reasons why the HGCL has not issued the Operational Acceptance, the Facilities shall be deemed to have been accepted as at the date of the Contractor's said notice.
- 43.3.6 The start date of the Comprehensive Operation and Maintenance shall be reckoned from the date mentioned in the Operational Acceptance Certificate.

43.4 Final Acceptance

43.4.1 Final Acceptance shall occur in respect of the Facilities when:

- a) The plant has achieved the Operational acceptance; and
- b) Handing over – Taking over of Plant should have been completed; and
- c) Successful demonstration of the performance guarantees
- d) Contractor has provided the list of recommended spares with detailed specification, source and price for further procurement; and
- e) The Contractor has paid the liquidated damages, if any, as specified in SCC thereto;
- f) Account reconciliation and NCR/ Punch list closure.

43.4.2 At any time after the events set out in GCC Sub – Clause 43.4.1 have occurred, the Contractor may give a notice to the Project Manager/EIC requesting the issue of Final Acceptance in the form acceptable to the Employer in respect of the Facilities or the part thereof specified in such notice as at the date of such notice.

43.4.3 The Employer shall, after consultation with the Project Manager/EIC, and within thirty (30) days after receipt of the Contractor's notice, issue Final Acceptance.

43.4.4 If within thirty (30) days after receipt of the Contractor's notice, the HGCL fails to issue the Final Acceptance or fails to inform the Contractor in writing of the justifiable reasons why the HGCL has not issued the Final Acceptance, the Facilities shall be deemed to have been accepted as at the date of the Contractor's said notice.

43.4.5 The O&M contract period may further be extended for a period as per mutually agreed terms and conditions.

43.4.6 In case the HGCL wishes to extend the O&M period beyond the agreed period under this contract, he shall intimate Contractor at least 6 months prior to the completion period. The Contractor may accept the offer as per the terms and conditions to be mutually agreed with the HGCL.

44. Release of Confidential Information

44.1 The Contractor shall not communicate or use in advertising, publicity, sales releases or in any other medium, photographs, or other reproduction of the Work under this Contract or description of the site dimensions, quantity, quality or other information, concerning the Work unless prior written permission has been obtained from the Employer/ HGCL.

44.2 The HGCL/ Employer and the Contractor shall keep confidential and shall not, without the written consent of the other party hereto, divulge to any third party any documents, data or other

information furnished directly or indirectly by the other party hereto in connection with the Contract, whether such information has been furnished prior to, during or following termination of the Contract. Notwithstanding the above, the Contractor may furnish to its Sub Contractor(s) such documents, data and other information it receives from the Employer/HGCL to the extent required for the Sub Contractor(s) to perform its work under the Contract, in which event the Contractor shall obtain from such Sub Contractor(s) an undertaking of confidentiality similar to that imposed on the Contractor.

- 44.3 The HGCL/ Employer shall not use such documents, data and other information received from the Contractor for any purpose other than the operation and maintenance of the Facilities. Similarly, the Contractor shall not use such documents, data and other information received from the HGCL/ Employer for any purpose other than the design, procurement of Plant and Equipment, construction or such other work and services as are required for the Performance of the Contract.
- 44.4 The obligation of a party above, however, shall not apply to that information which
- Now or hereafter enters the public domain through no fault of that party
 - Can be proven to have been possessed by that party at the time of disclosure and which was not previously obtained, directly or indirectly, from the other party hereto.
 - Otherwise lawfully becomes available to that party from a third party that has no obligation of confidentiality.
- 44.5 The above provisions of this GCC shall not in any way modify any undertaking of confidentiality given by either of the parties hereto prior to the date of the Contract in respect of the Facilities or any part thereof.
- 44.6 The provisions of this GCC Clause 44 shall survive termination, for whatever reason, of the Contract.

45. Operation & Maintenance

- 45.1 The Operation and Maintenance shall be comprehensive. The maintenance service provided shall ensure project functioning of the Solar PV system as a whole and Power Evacuation System to the extent covered in the Contract. All preventive / routine maintenance and breakdown / corrective maintenance required for ensuring maximum uptime shall have to be provided. Accordingly, the Comprehensive Operation & Maintenance shall have two distinct components as described below:

45.2 Preventive / Routine Maintenance:

This shall be done by the Contractor regularly and shall include activities such as cleaning and checking the health of the Solar PV system, cleaning of module surface, tightening of all electrical connections, and any other activity including the associated civil works, as mentioned in TS, wear and tear that may be required for proper functioning of the Solar PV system as a whole. Necessary maintenance activities, Preventive and Routine for Transformers and associated switch gears and transmission line also shall be included.

45.3 Breakdown / Corrective maintenance:

Whenever a fault occurs, the Contractor has to attend to rectify the fault & the fault must be rectified within the 72 hours from the time of occurrence of fault. The Contractor must maintain all the records pertaining to all such faults and necessary measures taken.

The date of Comprehensive Operation & Maintenance Contract period shall begin from the date of Operational acceptance. However, operation of the Power Plant means operation of system as per TS and workmanship in order to keep the project trouble free covering the O&M period. The Contractor must demonstrate the committed CUF at the end of every year in accordance with commitment made in line with the Performance guarantees.

45.4 Serviceability Level Agreement (SLA)

- 45.4.1 Contractor shall make efforts to maintain 100 % serviceability of complete Plant including all other associated infrastructure developed by the Contractor during execution of project as its scope of work & the respective report of the same shall be submitted to the HGCL.
- 45.4.2 Contractor shall maintain a Complaint log book, which shall include the timing of logging of complaint including unique Complaint number, time of closure of complaint & it's Root Cause Analysis.
- 45.4.3 Contractor will be responsible for maintaining the Insurance Policy for the complete Plant and Facilities during the Erection period and O&M period also. He shall maintain seamless insurance cover during construction and O&M phases. Copy of policies shall be given to the owner. However, damages occurred due to accidents shall be rectified by the contractor and the same will be re-imbursed by HGCL for that quantity after successful re-installation/rectification.
- 45.4.4 Such rectification work carried out by HGCL doesn't exempts/relieves Contractor from its responsibility towards subsequent operation, maintenance, repair & replacement of such component/ infrastructure of the Plant or meeting the performance parameters of the Plant.
- 45.4.5 O&M Routine & Manpower: Contractor shall provide Preventive / Routine Maintenance schedule based on Original Equipment manufacturer and good engineering practices. The team deployed for the O&M must have the sufficient experience of executing the similar tasks.

However, Contractor shall engage additional manpower as and when need arise.

- 45.5 Bidder is requested to provide the list of all the spares required to maintain the facility for O&M period. Contractor agrees to supply such spare parts, as recommended or otherwise required for the effective and hassle-free operation and maintenance of the Facilities. However, the Contractor, with its previous experience, is to provide a list of spares including specifications, supplier details and indicative price, as recommended by him and OEM. The Contractor shall keep and maintain the inventory of such spares for the hassle-free operation during the complete O&M period without additional cost to HGCL. Also, at the end of penultimate year of the O&M contract, Contractor shall supply a list of all recommended spares as per the operational requirement of the plant and with reference to the mean time between failures (MTBF), along with detailed specifications, supplier details and tentative cost for future purchase. The price of such spare parts shall include the breakup of taxes and duties as applicable towards purchase and supply of spare parts. HGCL, at its discretion, will purchase the spare as required for future operation. However, the Contractor shall replenish the mandatory spares at his cost prior to the completion of the O&M period.

46. Completion of Contract

- 46.1 Unless otherwise terminated under the provisions of any other relevant clause, this Contract shall be deemed to have been completed at the expiration of the Period of Liability/ Validity of the Contract as provided under Section - V, Special Conditions of Contract (SCC).

47. Pre - Commissioning & Commissioning

- 47.1 As soon as installation of the Facilities has, in the opinion of the Contractor, been completed as specified in the Technical Specifications, excluding items not materially affecting the operation or safety of the Facilities, the Contractor shall so notify the Project Manager/ EIC in writing to witness the pre- commissioning of the facility.
- 47.2 If the Project Manager/EIC is satisfied that the Facilities have reached Completion, the Project Manager/EIC shall, within seven (7) days after receipt of the Contractor's notice, arrange to witness the pre – commissioning of the Facilities.

- 47.3 If the Project Manager/EIC notifies the Contractor of any defects and/or deficiencies, the Contractor shall then correct such defects and/or deficiencies and shall repeat the procedure described in GCCSub- Clause 47.1.
- 47.4 If the Project Manager/EIC is still not so satisfied, then it shall notify the Contractor in writing of any defects and/or deficiencies within seven (7) days after receipt of the Contractor's repeat notice, and the above procedure shall be repeated.
- 47.5 As soon as all works in respect of Pre-commissioning are completed and, in the opinion of the Contractor, the Facilities are ready for Commissioning, the Contractor shall so notify the Project Manager in writing. The Contractor shall commence Commissioning of the facilities as per the GCCSub – Clause 47.6.
- 47.6 Commissioning of the Facilities shall be completed by the Contractor as per procedures detailed in the Technical Specifications and in the presence of the Project Manager/ EIC and HGCL.
- 47.7 If the Project Manager/EIC fails to inform the Contractor of any defects and/or deficiencies within fourteen (14) days after receipt of the Contractor's notice under GCC Sub-Clause 47.1 or within seven (7) days after receipt of the Contractor's repeated notice under GCC Sub-Clause 47.3, then the Facilities shall be deemed to have taken up the date of the Contractor's notice or repeated notice, accepted for commissioning, as the case may be.
- 47.8 As soon as possible after Commissioning, the Contractor shall complete all outstanding minor items so that the Facilities are fully in accordance with the requirements of the Contract, failing which the Employer will undertake such completion and deduct the costs thereof from any monies owing to the Contractor.
- 47.9 As soon as possible after Commissioning, the Contractor shall make the facility ready for the performance test (PR test) and inform the Project Manager/EIC at least 7 (seven) days prior to the start of the performance test as per the procedure mentioned in the Technical Specifications.
- 47.10 Upon successful Operational Acceptance of the Facilities as per GCC sub clause 43.3, the Contractor shall be responsible for the care and custody of the Facilities, together with the risk of loss or damage thereto, and shall thereafter take over the Facilities or the relevant part thereof for the agreed duration of comprehensive operation and maintenance as stipulated and mutually agreed terms and conditions.

[E] PERFORMANCE OF WORK

48. Execution of Work

- 48.1 All the Works shall be executed in strict conformity with the provisions of the Contract Documents and with such explanatory detailed drawings, specification and instructions as may be furnished from time to time by the Contractor whether mentioned in the Contract or not. The Contractor shall be responsible for ensuring that works throughout are executed in the most substantial, proper and workmanlike manner with the quality of material and workmanship in strict accordance with the Specifications. The Contractor shall provide all necessary materials, equipment, labour etc. for execution and maintenance of Work till completion unless otherwise mentioned in the Contract.
- 48.2 All materials shall be brand new & of the best quality and workmanship capable of satisfactory operation under the operating and climatic conditions as has been specified. Unless otherwise specified, they shall conform in all respect to the latest edition of the relevant IS

codes specification wherever Indian specifications apply or IEC codes or equivalent internationally accepted standard.

48.3 The Contractor shall supply & deliver all equipment and materials for installation at site. The Contractor shall arrange for transportation, loading & unloading, local sifting, EAR insurance and safe storage of materials at project site at his own cost & risk.

48.4 If the Contractor offers equipment manufactured in accordance with other international well recognized standards (mentioned above), he shall, in that case, supply a copy in English of the Standard Specification adopted and shall clearly mention in what respect such standard specification differs from Indian Standard Specifications. The Plant, equipment, and materials offered by the Contractor should comply with one consistent set of Standards only to make the system compatible and work in harmony as far as possible, except if mentioned otherwise.

49. Void

50. Work in Monsoon and Dewatering

50.1 Unless otherwise specified elsewhere in the tender, the execution of the Work may entail working in the monsoon also. The Contractor must maintain a minimum labour force as may be required for the job and plan and execute the construction and erection according to the prescribed schedule. No extra rate will be considered for such work in monsoon.

50.2 During monsoon and other period, it shall be the responsibility of the Contractor to keep the construction work site free from water logging at his own cost.

51. Change in Laws and Regulations

51.1 If, after the date seven (7) days prior to the date of Bid submission, in the country where the Site is located, any law, regulation, ordinance, order or by-law having the force of law is enacted, promulgated, abrogated or changed (which shall be deemed to include any change in interpretation or application by the competent authorities) that subsequently affects the costs and expenses of the Contractor and/or the Time for Completion, the Contract Price shall be correspondingly increased or decreased, and/or the Time for Completion shall be reasonably adjusted to the extent that the Contractor has thereby been affected in the Performance of any of its obligations under the Contract. Contractor shall promptly and within 15 days of such enactment coming into force, forward relevant supporting documents to HGCL.

However, these adjustments would be restricted to direct transactions between the Employer and Contractor and Bought out items (dispatched directly from sub vendor's works to Site). These adjustments shall not be applicable on procurement of raw materials, intermediary components and intermediary services etc. by the Contractor.

Notwithstanding the foregoing, such additional or reduced costs shall not be separately paid or credited if the same has already been accounted for in the price adjustment provisions where applicable.

However, any increase in the rate of these taxes, duties and levies beyond the contractual completion period shall be to Contractor's account and any decrease shall be passed on to HGCL.

The term Change in Law shall refer to the occurrence of any of the following events pertaining to this project only after the date seven (7) days prior to the date of Bid submission, including

(i) the enactment of any new law; or (ii) an amendment, modification or repeal of an existing law; or (iii) any change in the rates of any Taxes including any duties and cess or introduction of any new tax made applicable for setting up the project.

However, Change in Law shall not include (i) any change in taxes on corporate income or (ii) any change in any withholding tax on income or dividends distributed to the shareholders of the Contractor, or (iii) any change on account of regulatory measures by the Appropriate Commission.

52. General Conditions for Construction and Erection Work

- 52.1 Overtime work is permitted in cases of need and the HGCL will not compensate the same. Shift working at 2 or 3 shifts per day may become necessary and the Contractor should take this aspect into consideration for formulating his rates. No extra claims will be entertained by the HGCL on this account.
- 52.2 The Contractor must arrange for the placement of workers in such a way that the delayed completion of the Work or any part thereof for any reason whatsoever will not affect their proper employment. The HGCL will not entertain any claim for idle time payment whatsoever.
- 52.3 The Contractor shall submit to the HGCL/ Employer progress reports at regular intervals regarding the state and progress of Work. The details and format of the report will be mutually agreed after the award of Contract. The Contractor shall provide display boards showing progress and labour strengths at worksite. Updated project schedule in MS Projects shall also be furnished by Contractor as per agreed interval.

53. Design and Engineering

- 53.1 The Work covered under this Contract having to be executed by the Contractor on a lump-sum firm price quoted by him, the HGCL will not accept any proposals for changes in Value of Contract or extension in time on account of any such changes which may arise to the Contractor's scope of Work as a result of detailed Engineering and thereafter during the execution of Work.

53.2 Specifications and Drawings

- 53.2.1 The Contractor shall execute the basic and detailed design and engineering work in compliance with the provisions of the Contract, or where not so specified, in accordance with good and sound engineering practice.
- 53.2.1 The Contractor shall be responsible for any discrepancies, errors or omissions in the specifications, drawings and other technical documents that it has prepared, whether such specifications, drawings and other documents have been approved by the Project Manager/ EIC or not, provided that such discrepancies, errors or omissions are not because of inaccurate information furnished in writing to the Contractor by or on behalf of the HGCL/ Employer.

53.3 Codes and Standards

- 53.3.1 Wherever references are made in the Contract to codes and standards in accordance with which the Contract shall be executed, the edition or the revised version of such codes and standards current at the date of bid submission shall apply unless otherwise specified.

53.3.2 Approval / Review of Technical Documents by Project Manager

The Contractor shall prepare list of documents and drawings i.e. Master drawing list (MDL) as per technical specifications and furnish to the Project Manager/EIC for review & Approval of

the same within 14 days from the Zero date.

- 53.3.3 Within ten (10) working days after receipt by the Project Manager of any document requiring the Project Manager's approval, the Project Manager shall either return one copy thereof to the Contractor with its approval endorsed thereon or shall notify the Contractor in writing of its disapproval thereof and the reasons therefor and the modifications that the Project Manager proposes.
- 53.3.4 The Project Manager shall not disapprove any document, except on the grounds that the document does not comply with some specified provision of the Contract or that it is contrary to good engineering practice.
- 53.3.5 If the Project Manager disapproves the document, the Contractor shall modify the document and resubmit it for the Project Manager's approval. If the Project Manager approves the document subject to modification(s), the Contractor shall make the required modification(s), and upon resubmission with the required modifications the document shall be approved.
- 53.3.6 The procedure for submission of the documents by the Contractor and their approval by the Project Manager shall be as per the Contract Co-ordination procedure.
- 53.3.7 If any dispute or difference occurs between the Employer and the Contractor in connection with or arising out of the disapproval by the Project Manager of any document and/or any modification(s) thereto that cannot be settled between the parties within a reasonable period, then such dispute or difference may be settled in accordance with GCC Clause (Settlement of Dispute) hereof. If such dispute or difference is referred as per GCC clause, the Project Manager shall give instructions as to whether and if so, how, Performance of the Contract is to proceed. The Contractor shall proceed with the Contract in accordance with the Project Manager's instructions, provided that if the Arbitration upholds the Contractor's view on the dispute, then the Contractor shall be reimbursed by the HGCL for any additional costs incurred by reason of such instructions and shall be relieved of such responsibility or liability in connection with the dispute and the execution of the instructions as the Arbitration shall decide, and the Time for Completion shall be extended accordingly.
- 53.3.8 The Project Manager's approval, with or without modification of the document furnished by the Contractor, shall not relieve the Contractor of any responsibility or liability imposed upon it by any provisions of the Contract except to the extent that any subsequent failure results from modifications required by the Project Manager.
- 53.3.9 The Contractor shall not depart from any approved document unless the Contractor has first submitted to the Project Manager an amended document and obtained the Project Manager's approval thereof, pursuant to the provisions of this GCC Clause.
- 53.3.10 If the Project Manager requests any change in any already approved document and/or in any document based thereon, generally shall be taken care by the Contractor if the change is not causing any major financial impact.

54. Drawings to be supplied by the HGCL/Employer

- 54.1 The drawings attached with tender, if any, are only for the general guidance to the Contractor to enable him to visualize the type of work contemplated and scope of work involved. The Contractor will be deemed to have studied the Drawings and formed an idea about the Work involved.
- 54.2 The Contractor shall be deemed to have gone through the Drawings supplied to him thoroughly and carefully and in conjunction with all other connected drawings and bring to the notice of the Engineer-in-Charge/ Project Manager discrepancies, if any, therein before actually carrying out

the Work.

- 54.3 Copies of all detailed working drawings relating to the Work shall be kept at the Contractor's office on the site and shall be made available to the Engineer-in-Charge/Project Manager at any time during the Contract. The drawings and other documents issued by the Employer/ HGCL shall be returned to the Employer/ HGCL on completion of the Work.

55. Drawings to be supplied by the Contractor

- 55.1 The drawings/ data which are to be furnished by the Contractor shall be furnished within the specified time.
- 55.2 Where approval/ review of drawings before manufacture/ construction/ fabrication has been specified, it shall be Contractor's responsibility to have these drawings prepared as per the TS and get it approved before proceeding with manufacture/ construction/ fabrication as the case may be. Any change that may have become necessary in these drawings during the execution of the work shall have to be carried out by the Contractor at no extra cost. All as built drawings shall bear the certification stamp duly signed by both the Contractor and Engineer-in-Charge/Project Manager. The Contractor shall incorporate any modifications and/ or corrections as highlighted/notified and submit the drawings for approval practically within the next 7 working days.
- 55.3 The Drawings submitted by the Contractor shall be reviewed by the Engineer-in-Charge/Project Manager as far as practicable within 10 (Ten) working days. The Contractor shall incorporate any modifications and/ or corrections as highlighted/notified and submit the drawings for approval. Any delays arising out of failure by the Contractor to rectify the drawing in good time shall not alter the Contract Completion Time.
- 55.4 All GA & GFC drawings shall be provided in soft as well as Hard form in appropriate format/size to Employer for review & approval.

All as built drawings showing all corrections, adjustments & deviations, if any, etc shall be furnished by the Contractor in 04 (Four) Hard Copies & a soft copy for record purpose to the Employer/ HGCL immediately after the operational acceptance.

56. Setting out Works

- 56.1 The Contractor shall provide, fix and be responsible for the maintenance of all stakes, templates, level marks, profiles and other similar things and shall take all necessary precautions to prevent their removal or disturbance and shall be responsible for the consequence of such removal or disturbance should the same take place and for their efficient and timely reinstatement. The Contractor shall also be responsible for the maintenance of all existing survey marks, boundary marks, distance marks and center line marks, either existing or supplied and fixed by the Contractor. Contractor shall carry out Geotech investigation at site at his own cost and design the foundations accordingly. A copy of the investigation report shall also be forwarded to Employer and HGCL
- 56.2 Before beginning the Works, the Contractor shall at his own cost, provide all necessary reference and level posts, pegs, bamboos, flags, ranging rods, strings and other materials for proper layout of the works in accordance with the schemes for bearing marks as required. The center, longitudinal or face lines and cross lines shall be marked by means of small masonry pillars. Each pillar shall have distinct mark at the center to enable theodolite to be set over it.
- 56.3 Pillars bearing geodetic marks located at the sites of units of Works under construction should be protected and fenced by the Contractor.

- 56.4 On completion of Work, the Contractor must submit the geodetic documents according to which the Work was carried out.

57. Geological Discoveries

All fossils, coins, articles of value or antiquity and structures and other remains or things of geological or archaeological interest discovered on the site where the services are performed, be deemed to be the absolute property of the HGCL. The Contractor shall take reasonable precautions to prevent the personnel or any other persons from removing or damaging any such article or thing and shall immediately upon the discovery thereof and, before removal, acquaint the HGCL of such discovery and carry out, at the expense of the HGCL, the HGCL's orders as to the disposal of the same.

58. Materials to be supplied by Contractor

58.1 Plant and Equipment

The Contractor shall procure and transport all the Plant and Equipment in an expeditious and orderly manner to the Site to achieve completion of activities as per schedule to enable commissioning of the Project by the scheduled commissioning date. Contractor shall deliver supplies at site in accordance to its erection sequence. HGCL may hold payment against supplies in case same is delivered more than 03 (Three) months before its erection requirement (except for imported items)

58.2 Transportation

The Contractor shall ensure that all the plant and equipment required to complete the Facility at site, are procured and dispatched. The Contractor shall at its own risk and expense transport all the Plant and Equipment and the Contractor's Equipment to the Site by the mode of transport that the Contractor judges most suitable under all the circumstances. Contractor shall be responsible to assess in advance suitability of access roads, bridges, culverts, etc for transportation of ODC shipments, if any and arrange to transport them accordingly.

58.3 Packing and Marking

The Contractor shall be responsible for securely protecting and packing the plant & equipment as per prescribed standards in force to withstand the journey and ensuring safety of materials and also arrival of materials at destination in original condition and good for contemplated use. Packing case size & weight shall take into consideration the remoteness of the goods final destination and absence of heavy material handling facilities at all points in transit.

Packing lists of materials shall be provided in each package to facilitate checking up of the contents at the destination.

In order to import any items, associated with the Solar PV Power Project, from abroad or from any other state in India, Contractor shall have to arrange any clearance, permission, if required at his own risk, from any Government (Government of State & Government of India) or any Government (Government of State & Government of India) controlled organization for transportation of materials from manufacturing shop to delivery at Site. Contractor shall take necessary insurances to ensure safe transit & consequential risks. All packing material is the property of the HGCL and shall be immediately taken into the safe storage.

58.4 Storage of Equipment

The plant and equipment thus procured under the scope of the contract must be kept in safe custody till put under operation, essentially free from water contact. All the spares, as required for the trouble-free comprehensive O&M of Plant, must be kept under secure storage during O&M

period. Contractor has to ensure the appropriate and proper storage arrangement prior to the arrival of the equipment including containers, temporary structures, sheds, platforms etc at its own cost.

The Contractor shall procure and provide within the Value of Contract the whole of the materials required for the construction including steels, cement and other building materials, tools, tackles, construction plant and equipment for the completion and maintenance of the Work except the materials which will be issued by the HGCL and shall make his own arrangement for procuring such materials and for the transport thereof. The HGCL may give necessary recommendation to the respective authority if so desired by the Contractor but assumes no further responsibility of any nature. The HGCL will insist on the procurement of materials which bear ISI stamp and/ or which are supplied by reputed suppliers.

- 58.5 The Contractor shall properly store all materials either issued to him or brought by him to the Site to prevent damages due to rain, wind, direct exposure to sun, etc. as also from theft, pilferage, etc. for proper and speedy execution of his works. The Contractor shall maintain sufficient stocks of all materials required by him including commissioning spares.

59. Stores Supplied by the HGCL

- 59.1 If the Specification of the Work provides for the use of any material of special description to be supplied from the HGCL's stores or it is required that the Contractor shall use certain stores to be provided by the HGCL, such materials and stores, there for as hereinafter mentioned being so far as practicable for the convenience of the Contractor, but not so as in any way to control the meaning or effect of the Contract, the Contractor shall be bound to purchase and shall be supplied such materials and stores as are from time to time required to be used by him for the purpose of the Contract only. After the completion of the Work, however, the Contractor has to account for the full quantity of materials supplied to him as per relevant clauses in this document.
- 59.2 All materials so supplied to the Contractor shall remain the absolute property of the HGCL and shall not be removed on any account from the Site of the Work and shall be at all times open for inspection to the Engineer-in-Charge/Project Manager. Any such materials remaining unused at the time of the completion or termination of the Contract shall be returned to the HGCL's stores or at a place as directed by the Engineer-in-Charge/Project Manager in perfectly good condition at Contractor's cost.
- 59.3 HGCL, at his sole discretion and upon request from Contractor, may provide appropriate space for storage outside the site on chargeable basis mutually agreeable to both parties. However, the transportation of equipment from store to site shall be arranged by the Contractor at his cost and risk

60. Conditions for Issue of Materials

- 60.1 i) Materials specified as to be issued by the HGCL will be supplied to the Contractor by the HGCL from his stores. It shall be responsibility of the Contractor to take delivery of the materials and arrange for its loading, transport and unloading at the Site of Work at his own cost. The materials shall be issued between the working hours and as per the rules of the HGCL as framed from time to time. Once the material is issued or taken over by the Contractor, then the same will be his own liability to store, use & maintain.
- ii) The Contractor shall bear all incidental charges for the storage and safe custody of materials at site after these have been issued to him.
- iii) Materials specified as to be issued by the HGCL shall be issued in standard sizes as obtained from the manufacturers.
- iv) The Contractor shall construct suitable godowns at the Site of Work for storing the materials safe against damage by rain, dampness, fire, theft etc. He shall also employ necessary watch and ward along with proper illumination establishment for the purpose.

v) It shall be duty of the Contractor to inspect the materials supplied to him at the time of taking delivery and satisfy himself that they are in good condition. After the materials have been delivered by the HGCL, it shall be the responsibility of the Contractor to keep them in good condition and if the materials are damaged or lost, at any time, they shall be repaired and/or replaced by him at his own cost.

vi) The HGCL shall not be liable for delay in supply or non-supply of any materials which the HGCL has undertaken to supply where such failure or delay is due to natural calamities, act of enemies, transport and procurement difficulties and any circumstances beyond the control of the HGCL. In no case, the Contractor shall be entitled to claim any compensation or loss suffered by him on this account.

vii) It shall be responsibility of the Contractor to arrange in time all materials required for the Work other than those to be supplied by the HGCL. If, however, in the opinion of the Engineer-in-Charge/Project Manager the execution of the Work is likely to be delayed due to the Contractor's inability to make arrangements for supply of materials which normally he has to arrange for, the Engineer-in-Charge/Project Manager shall have the right at his own discretion to issue such materials, if available with the HGCL or procure the materials from the market or as elsewhere and the Contractor will be bound to take such materials at the rates decided by the Engineer-in-Charge/Project Manager. This, however, does not in any way absolve the Contractor from responsibility of making arrangements for the supply of such materials in part or in full, should such a situation occur nor shall such action by HGCL constitute a reason for the delay in the execution of the Work.

viii) The Contractor shall, if desired by the Engineer-in-Charge/Project Manager, be required to execute an Indemnity Bond in the prescribed form for safe custody and accounting of all materials issued by the HGCL.

ix) The Contractor shall furnish to the Engineer-in-Charge/Project Manager sufficiently in advance a statement showing his requirement of the quantities of the materials to be supplied by the HGCL and the time when the same will be required by him for the works, so as to enable the Engineer-in-Charge/Project Manager to make necessary arrangements for procurement and supply of the material.

x) Account of the materials issued by the HGCL shall be maintained by Contractor indicating the daily receipt, consumption and balance in hand. This account shall be maintained in a manner prescribed by the Engineer-in-Charge/Project Manager along with all connected papers viz. requisitions, issues, etc., and shall be always available for inspection in the Contractor's office at Site.

xi) The Contractor should see that only the required quantities of materials are got issued. The Contractor shall not be entitled to cartage and incidental charges for returning the surplus materials, if any, to the stores wherefrom they were issued.

xii) Materials/ Equipment(s) supplied by Employer/ HGCL shall not be utilized for any purpose(s) than issued for.

61. Material Procured with Assistance of HGCL/ Return of Surplus

Notwithstanding anything contained to the contrary in any or all the clauses of this Contract where any materials for the execution of the Contract are procured with the assistance of the HGCL either by issue from HGCL's stock or purchases made under order or permits or licenses issued by Government, the Contractor shall hold the said materials as trustee for the HGCL and use such materials economically and solely for the purpose of the Contract and not dispose them off without the permission of the HGCL and return, if required by the Engineer-in-Charge/Project Manager, shall determine having due regard to the condition of the materials.

62. Materials obtained from Dismantling

If the Contractor in the course of execution of the Work is called upon to dismantle any part for reasons other than those stipulated in Clauses 67 and 70 hereunder, the materials obtained in the Work of dismantling etc., will be considered as the HGCL's property and will be disposed

off to the best advantage of the HGCL.

63. Articles of Value Found

All gold, silver and other minerals of any description and all precious stones, coins, treasure relics, antiquities and other similar things which shall be found in, under or upon the Site, shall be the property of the HGCL and the Contractor shall duly preserve the same and shall from time to time deliver the same to such person or persons indicated by the HGCL.

64. Discrepancies between Instructions

Should any discrepancy occur between the various instructions furnished to the Contractor, his representative or staff or any doubt arises as to the meaning of any such instructions or should there be any misunderstanding between the Contractor's staff and the Engineer-in-Charge/Project Manager's staff, the Contractor shall refer the matter immediately in writing to the Engineer-in-Charge/Project Manager whose decision thereon shall be final and conclusive and no claim for losses alleged to have been caused by such discrepancies between instructions, doubts, or misunderstanding shall in any event be admissible.

65. Action where no Specification is issued

In case of any class of Work for which there is no Specification supplied by the Employer/HGCL as mentioned in the Tender Documents such Work shall be carried out in accordance with Indian Standard Specifications and if the Indian Standard Specifications do not cover the same, the Work should be carried out as per standard Engineering Practice subject to the approval of the Engineer-in-Charge/Project Manager.

66. Inspection of Works

- 66.1 The Engineer-in-Charge/Project Manager will have full power and authority to inspect the Work at any time wherever in progress either on the Site or at the Contractor's premises/ workshops wherever situated, premises/ workshops of any person, firm or corporation where Work in connection with the Contract may be in hand or where materials are being or are to be supplied, and the Contractor shall afford or procure for the Engineer-in-Charge/Project Manager every facility and assistance to carry out such inspection. The Contractor shall, at all time during the usual working hours and at all other time at which reasonable notice of the intention of the Engineer-in-Charge/Project Manager or his representative to visit the Work shall have been given to the Contractor, either himself be present or receive orders and instructions, or have a responsible representative duly accredited in writing, present for the purpose. Orders given to the Contractor's representative shall be considered to have the same force as if they had been given to the Contractor himself. The Contractor shall give not less than 15 (Fifteen) day notice in writing to the

Engineer-in-Charge/Project Manager for carrying out the inspection, Domestic or Overseas) so as to reach out for the inspection by the concerned officials and/or measurement of any work in order that the same may be inspected and measured. Employer/HGCL at his own discretion may or may not attend the scheduled inspection calls as arranged by the contractor on account of pre-occupation and other site exigencies.

In the event of breach of above the same shall be uncovered at Contractor's expense for carrying out such measurement or inspection.

- 66.2 The Contractor is to provide at all time during the progress of the Work and the maintenance period, proper means of access with ladders, gangways etc. to move and adopt as directed for inspection or measurements of the Work by the Engineer-in-Charge/Project Manager.
- 66.3 The Contractor shall make available to the Engineer-in-Charge/Project Manager free of cost all necessary instruments and assistance in checking or setting out of Work and in the checking of any Work made by the Contractor for the purpose of setting out and taking measurements of

Work.

67. Tests for Quality of Work

- 67.1 All workmanship shall be of the respective kinds described in the Contract Documents and in accordance with the instructions of the Engineer-in-Charge/Project Manager and shall be subjected from time to time to such test as the Engineer-in-Charge/Project Manager may direct at the place of manufacture or fabrication or on the site or at all or any such places.

The cost of inspection/ pre-dispatch inspection/ in-stage inspection (Exclusive of Employer/HGCLs representatives TA/DA) shall be borne by Contractor. Such pre-dispatch inspection(s) at the manufacturer's facility shall be carried out in the presence of the Employer/HGCL or their authorized representatives, for such items as is specified by the Engineer-in-Charge/Project Manager. Travel expenses for the representatives of Employer/HGCL for such inspections shall be borne by the Employer/HGCL respectively.

However, in case re-inspection is necessitated on account of non-acceptance of item(s) due to failure on Factory Acceptance Test(s), the cost of associated travel and accommodation for the revisit shall be borne by the Contractor. A minimum of 07 (Days) notice shall be given by the Contractor for witnessing such inspection at the works

The Contractor shall provide assistance, instruments, labour and materials as are normally required for examining, measuring and testing any workmanship as may be selected and required by the Engineer-in-Charge/Project Manager and keep all stage inspection/ material TC readily available for the Inspector.

- 67.2 All the tests that will be necessary in connection with the execution of the Work as decided by the Engineer-in-Charge/Project Manager shall be carried out at the field-testing laboratory of the HGCL by paying the charges as decided by the HGCL from time to time. In case of non-availability of testing facility with the HGCL, the required test shall be carried out at the cost of Contractor at Government or any other accredited testing laboratory.
- 67.3 If any tests are required to be carried out in conjunction with the Work or materials or workmanship not supplied by the Contractor, such tests shall be carried out by the Contractor and cost of such tests shall be reimbursed by the HGCL.
- 67.4 The PV modules/ inverters/ cables and other Balance of system equipment deployed in the solar PV power Plant shall have valid test certificates for their qualification as per above specified IEC/ IS Standards by one of the NABL Accredited /Govt approved Test Centers in India. In case of module types/ equipment for which such Test facilities may not exist in India, test certificates from reputed ILAC Member body accredited Labs abroad (with proof of accreditation) will be acceptable.

68. Samples for Approval

In case of requirement, the Contractor shall furnish to the Engineer-in-Charge/Project Manager for approval, when requested or if required by the specifications, adequate samples of all materials and finished to be used in the Work. Such samples shall be submitted before the Work is commenced and in ample time to permit tests and examinations thereof. All materials furnished and finishes applied in actual Work shall be fully equal to the approved samples.

69. Action and Compensation in case of Poor/Non-compliant Work

If, against documentary proofs, is established that any work has been executed with unsound, imperfect or unskilled workmanship, or with materials of any inferior description, or that any

materials or articles provided by the Contractor for the execution of the Work are unsound, or of a quality inferior to that contracted for, or otherwise not in accordance with the Contract, the Contractor shall on demand in writing from the Engineer-in-Charge/Project Manager or his authorized representative specifying the Work, materials or articles complained of notwithstanding that the same may have been inadvertently passed, certified and paid for, forthwith rectify or remove and reconstruct the Work so specified and provide other proper and suitable materials or articles at his own cost. In the case of any such failure the Engineer-in-Charge/Project Manager may on expiry of notice period rectify or remove and re-execute the Work or remove and replace with others, the materials or articles complained of to as the case may be at the risk and expense in all respects of the Contractor. The decision of the Engineer-in-Charge/Project Manager as to any question arising under this clause shall be final and conclusive. No additional time for Project completion shall be granted for undertaking such replacement/ rectification works by Contractor.

70. Suspension of Works

i) Subject to the provisions of sub-para (ii) of this clause, the Contractor shall, if ordered in writing by the Engineer-in-Charge/Project Manager, or his representative, temporarily suspend the Works or any part thereof for such written order, proceed with the Work therein ordered to be suspended until, he shall have received a written order to proceed therewith. The Contractor shall not be entitled to claim compensation for any loss or damage sustained by him by reason of temporary suspension of the Works aforesaid. However, suitable time extension may be considered at the sole discretion of the HGCL.

ii) In case of suspensions of entire Work, ordered in writing by Engineer-in-Charge/Project Manager, for a period of more than 03 (Three) months, the Contractor shall have the option to terminate the Contract.

71. HGCL may do Part of Work

Upon failure of the Contractor to comply with any instructions given in accordance with the provisions of this Contract the HGCL has the alternative right, instead of assuming charge of entire Work, to place additional labour force, tools, equipment and materials on such parts of the Work, as the HGCL may designate or also engage another Contractor to carry out the Work. In such cases, the HGCL shall deduct from the amount which otherwise might become due to the Contractor, the cost of such work and material with 110% (Hundred & Ten Percent) of the actual cost of works and materials.

72. Possession prior to Completion

The Engineer-in-Charge/Project Manager shall have the right to take possession of or use any completed or partially completed Work or part of the Work. Such possession or use shall not be deemed to be an acceptance of any work completed in accordance with the Contract Agreement. If such prior possession or use by the Engineer-in-Charge/Project Manager delays the progress of Work, equitable adjustment in the time of completion will be made and the Contract Agreement shall be deemed to be modified accordingly.

73. Defects Liability Period

[24 (Twenty Four) Months Period of Liability from the date of Operational Acceptance]

73.1 The Contractor must warrant that the Facilities shall be free from defects in the design, engineering, materials and workmanship of the Plant and Equipment supplied and of the work executed.

73.2 If it shall appear to the Project Manager that any supplies have been executed with unsound, imperfect or unskilled workmanship, or with materials of any inferior description, or that any materials or articles provided by the Contractor for the execution of Contractor are unsound or otherwise not in accordance with the Contract, the Contractor shall on demand in writing

inform the Project Manager or its authorized representative specifying the item, materials or articles complained of, notwithstanding that the same may have been inadvertently passed, certified and paid for. The Contractor shall forthwith rectify or remove and replace that item so specified and provide other proper and suitable materials or articles at its own charge and cost, and in the event of failure to do so within a period to be specified by the Project Manager in its demand aforesaid, the Project Manager may on expiry of notice period rectify or remove and re-execute the time or remove and replace with others, the materials or articles complained of as the case may be at the risk and expense in all respects of the Contractor. The decisions of the Project Manager in this regard shall be final and binding.

- 73.3 The Contractor shall also be undertaking the operation and maintenance of the Facility and consequently shall be required to rectify any defects that emerge during the operation of the Facilities for the entire term of this Contract.
- 73.4 The Defect Liability Period shall be of twelve (12) months from the date of Operation Acceptance, during which the Contractor must repair any defect identified by the Project Manager / EIC after commissioning of the Plant. All the expenses to repair the defects shall be borne by the Contractor and no additional cost charged to the HGCL.
- 73.5 If during the Defect Liability Period any defect should be found in the design, engineering, materials and workmanship of the Plant and Equipment supplied or of the work executed by the Contractor, the Contractor shall promptly, in consultation and agreement with the HGCL regarding appropriate remedying of the defects, and at its cost, repair, replace or otherwise make good (as the Contractor shall, at its discretion, determine) such defect as well as any damage to the Facilities caused by such defect.
- 73.6 Furthermore, without prejudice to the generality of the foregoing, it is clarified that the Contractor shall also be responsible for the repair, replacement or making good of any defect, or of any damage to the Facilities arising out of or resulting from any of the following causes:
- Improper operation or maintenance of the Facilities by the Contractor during operation and maintenance of the Facility; and
 - Operation of the Facilities outside specifications of the Facilities.
- 73.7 The Contractor may, with the consent of the HGCL, remove any Plant and Equipment or any part of the Facilities that are defective from the Site, if the nature of the defect and/or any damage to the Facilities caused by the defect is such that repairs cannot be expeditiously carried out at the Site.
- 73.8 If the repair, replacement or making good is of such a nature that it may affect the efficiency of the Facilities or any part thereof, the HGCL may give to the Contractor a notice requiring that tests of the defective part of the Facilities shall be made by the Contractor immediately upon completion of such remedial work, whereupon the Contractor shall carry out such tests.
- 73.9 If such part fails the tests, the Contractor shall carry out further repair, replacement or making good (as the case may be) until that part of the Facilities passes such tests. The tests, in character, shall in any case be not inferior to what has already been agreed upon by the HGCL and the Contractor for the original equipment/part of the Facilities.
- 73.10 If the Contractor fails to commence the work necessary to remedy such defect or any damage to the Facilities caused by such defect within a reasonable time (which shall in no event be considered to be less than seven (7) days), the HGCL may, following a notice to the Contractor, proceed to do such work, and the costs incurred by the HGCL in connection therewith shall be paid to the HGCL by the Contractor or may be deducted by the HGCL from any monies due to

the Contractor or claimed under the Performance Guarantee, without prejudice to other rights, which the HGCL may have against the Contractor in respect of such defects.

- 73.11 If the Facilities or any part thereof cannot be used by reason of such defect and/or making good of such defect, the Defect Liability Period of the Facilities or such part, as the case may be, shall be extended by a period equal to the period during which the Facilities or such part cannot be used by the HGCL because of any of the aforesaid reasons. Upon correction of the defects in the Facilities or any part thereof by repair/replacement, such repair/replacement shall have the defect liability period of twelve (12) months from such replacement.
- 73.12 In addition, the Contractor shall also provide an extended warranty for any such component of the Facilities and for the period of time. Such obligation shall be in addition to the defect liability specified.
- 73.13 Latent defect liability: Notwithstanding, the defect liability period of 12 months above, the plant shall carry a latent defect liability of 5 years from date of operational acceptance towards any design/ manufacturing defects in the equipment supplied by the Contractor
- 73.14 The Contractor's liability under this contract for any reason, what so ever, shall be limited to the total Contract Price (Including GST etc)

73.15 Limitation of Liability

Notwithstanding anything contrary contained herein, the aggregate total liability of Contractor under the Agreement or otherwise shall be limited to **100% of Agreement/ Contract Value** except in case of Patent Infringement liability. However, neither party shall be liable to the other party for any indirect and consequential damages, loss of profits or loss of production.

73.16 Guarantee/ Warranty

As enumerated in Clause no. 73 (Defect Liability Period) of GCC, the overall DLP should stand valid for a period of 12 (Twelve) Months from the date of Operational Acceptance. However, Contractor needs to ensure following Guarantees/ Warranties to the best possible extent for the successful execution of the Contract. Subsequently, necessary Guarantee/ Warranty Certificate shall be produced by the Contractor prior to Operational Acceptance of the Facility.

Guarantee/ Warranty as defined under Section VII, Scope of work and Technical specifications will prevail in this clause. Contractor should guarantee the Plant facility for the workmanship for a period of 05 (Five) years from the date of Operational Acceptance.

73.16.1 The Contractor must ensure that the goods supplied under the Contract are new, unused and of most recent or current models and incorporate all recent improvements in design and materials unless provided otherwise in the Contract.

73.16.2 The Contractor shall transfer the warranty/ guarantees of the equipment as such from the OEM/ supplier in the name of the HGCL. The period of the warranty / guarantee for each equipment shall be as per the "Technical Specifications"

73.16.3 During the period of Warranty / Guarantee the Contractor shall remain liable to replace any defective parts, that becomes defective in the plant, of its own manufacture or that of its sub- Contractors, under the conditions provided for by the Contract under and arising solely from faulty design, materials or workmanship & faulty operation/repair, provided such defective parts are not repairable at Site. After replacement, the Contractor is allowed to take back the defective parts to its works at his expenses.

73.16.4 At the end of warranty/ guarantee period, the Contractor's liability shall cease subjected to fulfillment of its liability under GCC Clause 73.16 (Defect Liability Period). In respect of goods not mentioned for the warranty/ guarantee in "Technical Specifications", the HGCL shall be entitled to the benefit of such guarantee given to the Contractor by the original Supplier or manufacturer of such goods eg Performance guarantee for 25 yrs for modules, etc.

73.16.5 During the Comprehensive Operation & Maintenance period, the Contractor shall be responsible for any defects in the work due to faulty workmanship or due to use of sub-standard materials in the work. Any defects in the work during the guarantee period shall therefore, be rectified by the Contractor without any extra cost to the HGCL within a reasonable time as may be considered from the date of receipt of such intimation from the HGCL failing which the HGCL reserves the right to take up rectification work at the risk and cost of the Contractor.

74. Care of Works

74.0 From the commencement to completion of the Work & till the completion of O&M period (If applicable), the Contractor shall take full responsibility for the care for all works including all temporary works and in case any damages, loss or injury shall happen to the Work or to any part thereof or to any temporary works from any cause whatsoever, shall at his own cost repair and make good the same so that at completion the Work shall be in good order and in conformity in every respects with the requirement of the Contract and the Engineer-in-Charge's instructions. Also damage to external property of Third Parties.

74.1 Defects Prior to Taking Over

If at any time, before the Work is taken over, the Engineer-in-Charge/Project Manager shall:

a) Decide that any works done or materials used by the Contractor or by any Sub-Contractor is defective or not in accordance with the Contract, or that the works or any portion thereof are defective, or do not fulfill the requirements of Contract (all such matters being hereinafter, called 'Defects' in this clause), and

b) As soon as reasonably practicable, gives to the Contractor notice in writing of the said decision, specifying particulars of the defects alleged to exist or to have occurred, then the Contractor shall at his own expenses and with all speed make good the defects so specified.

In case, Contractor shall fail to do so, the HGCL may take, at the cost of the Contractor, such steps as may take in all circumstances, be reasonable to make good such defects. The expenditure so incurred by the HGCL will be recovered from the amount due to the Contractor. The decision taken

by the HGCL in this regard towards the amount to be recovered from the Contractor will be final and binding on the Contractor. As soon as the Work has been completed in accordance with the Contract (except in minor respects that do not affect their use for the purpose for which they are intended and have passed the tests on completion, the HGCL shall be deemed to have taken over the Work on the date so certified.

74.2 Defects after Taking Over (If applicable)

In order that the Contractor could obtain a Completion he shall make good, with all possible speed, any defect arising from the defective materials supplied by the Contractor or workmanship or any act or omission of the Contractor or that may have been noticed or developed, after the works has been taken over, the period allowed for carrying out such Work will be normally 01 (One) Month. If any defect be not remedied within a reasonable time, the HGCL may proceed to do the Work at Contractor's risk and expense and deduct from the final

bill such amount as may be decided by the HGCL. If by reason of any default on the part of the Contractor a Completion has not been obtained in respect of any portion of the Work within 01 (One) Month after the date fixed by the Contract for the completion of the Work, the HGCL shall be at liberty to use the Work or any portion thereof in respect of which a completion has not been obtained, provided that the Work or the portion thereof so used as aforesaid shall be afforded reasonable opportunity for completing these works for the Completion .

75. Guarantee/ Transfer of Guarantee

For the major Material/Products/Spares of the works & Projects including but not limited to PV Modules, Power Conditioning Units (PCU)/ Inverters, Transformers, Batteries (If applicable) etc the Contractor shall invariably engage OEMs/Sub-Contractors who are specialists in the field and OEM's/OPM's/firms of repute and such a OEM/OPM/Sub-Contractor shall furnish guarantees/warranties for their workmanship to the HGCL directly in the name of HGCL only without any deviation. The Contractor shall give the guarantee/warranty to the HGCL directly For other minor Material/Products/Spares also.

In case of the contract termination/cancellation and wherein the title of Guarantee/Warranty for the major Material/Products/Spares of the works & Projects including but not limited to PV Modules, Power Conditioning Units (PCU)/ Inverters, Transformers, Batteries (If applicable) is in the name of the Contractor, then all such Guarantee/Warranties for all such major products/material/spares will have to be mandatorily transferred in the name of HGCL as soon as the contract is terminated/cancelled & no plea/deviation from the Contractor side will be entertained in this regard.

For works like water-proofing, acid and alkali resisting materials, pre-construction soil treatment against termite or any other specialized works etc. the Contractor shall invariably engage Sub- Contractors who are specialists in the field and firms of repute and such a Sub-Contractor shall furnish guarantees for their workmanship to the HGCL, through the Contractor. In case such a Sub- Contractor is not prepared to furnish a guarantee to the HGCL, the Contractor shall give that guarantee to the HGCL directly.

76. Installation and Training of Employer's/ HGCL's Personnel

76.1 Tools & Tackles

The Contractor shall provide technically suitable tools and tackles for installation & erection of Plant & Machineries conforming to relevant BIS safety and technical standards for proper execution of work. The HGCL, in no way, shall be responsible for supply of any tools and tackles for implementation of the work and also to carry out operation & maintenance activities.

76.2 Setting up/Supervision/Labor

76.2.1 Bench Mark:

The Contractor shall be responsible for the true and proper setting-up of the Facilities in relation to bench marks, reference marks which are mutually agreed upon by the Contractor and HGCL.

If, at any time during the progress of installation of the Facilities, any error shall appear in the position, level or alignment of the Facilities, the Contractor shall forthwith notify the Project Manager of such error and, at its own expense, immediately rectify such error.

76.2.2 Contractor's Supervision:

The Contractor shall give or provide all necessary supervision during the installation of the Facilities, and the Construction Manager or its deputy shall be constantly on the Site to provide full-time supervision of the installation. The Contractor shall provide and employ only technical personnel who are skilled and experienced in their respective fields and supervisory staff who

are competent to adequately supervise the work at hand.

76.2.3 Labour:

The Contractor shall provide and employ on Site in the installation of the Facilities such skilled, semi- skilled and unskilled labor as is necessary for proper and timely execution of the Contract. The Contractor is encouraged to use local labor that has the necessary skills.

Unless otherwise provided in the Contract, the Contractor shall be responsible for the recruitment, transportation, accommodation, sanitation, first aid facility and catering of all labor, local or expatriate, required for the execution of the Contract and for all payments in connection therewith.

The Contractor shall be responsible for obtaining all necessary permit(s) and/or visa(s) from the appropriate authorities for the engagement and entry of all labour and personnel to be employed by Contractor on the Site including that of his sub-Contractors.

The Contractor shall at all times during the progress of the Contract use its best endeavors to prevent any unlawful, riotous or disorderly conduct or behavior by or amongst its employees and the labour of its SubContractors.

The Contractor shall, in all dealings with its labour and the labour of its SubContractors currently employed on or connected with the Contract, pay due regard to all recognized festivals, official holidays, religious or other customs and all local laws and regulations pertaining to the employment of labor.

The Contractor shall keep the HGCL indemnify, during construction as well as during O&M period, in respect of compliance with the statutory provisions in respect to the labor employed at site.

Upon completion of the construction activities/ O&M activities, the Contractor shall obtain no – objection certificate (NOC) from local/ statutory bodies in respect to the fulfillment of all compliance and submit a copy to the HGCL prior to the final settlement

76.3 Contractor's Equipment

- 76.3.1 All equipment brought by the Contractor onto the Site shall be deemed to be intended to be used exclusively for the execution of the Contract. The Contractor shall not remove the same from the Site without the Project Manager's consent that such Contractor's Equipment is no longer required for the execution of the Contract.

All the necessary approvals with due taxes, insurance and license, as required for the use of equipment at site, are to be taken by the Contractor.

The equipment should be in a good operating condition for safe use at site. The operator shall be competent to operate. It is advised to keep adequate spares, consumables, etc to reduce the breakdown time.

- 76.3.2 Unless otherwise specified in the Contract, upon completion of the Facilities, the Contractor shall remove from the Site all Equipment brought by the Contractor onto the Site.

76.4 Site Regulations and Safety

Contractor shall submit the EHS policy for the site to the Project Manager/EIC within 14

(fourteen) days from effective date and shall abide by the rules and regulations of the EHS policy.

The Contractor shall have to provide necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident, which may cause damage to any equipment / material or injury to workmen. The HGCL shall not be liable for any such accidents during the performance of the contract.

The Contractor, if required, will provide necessary safety training to workmen. Also, Contractor shall engage sufficient security guards to protect Facility from any theft and unauthorized access to site during the entire construction and O&M periods.

76.5 Site Clearance

76.5.1 Site Clearance in Course of Performance

In the course of carrying out the Contract, the Contractor shall keep the Site reasonably free from all unnecessary obstruction, store or remove any surplus materials, clear away any wreckage, packaging material, rubbish & debris and temporary installations from the Site, and remove any Contractor's Equipment no longer required for execution of the Contract with due approval of the HGCL.

76.5.2 Site Clearance after Completion

After Completion of all parts of the Facilities, the Contractor shall clear away and remove all wreckage, packaging material, rubbish & debris and temporary works & installations of any kind from the Site with due approval of the HGCL and shall leave the Site and Facilities clean and safe.

76.5.3 Disposal of Scrap

The term 'Scrap' shall refer to scrap/ waste/ remnants arising out of the unpacking of equipment, construction debris, breakage of modules, fabrication of structural steel work and piping work at the project site in the course of execution of the contract and shall also include any wastage of cables during the termination process while installing the cables.

The Contractor shall with the agreement of the HGCL promptly remove from the site any 'Scrap' generated during Performance of any activities at site in pursuance of the Contract.

The disposal of such Scrap shall vest with the Contractor for the items supplied by the Contractor and issued by HGCL under this contract for installation and construction without any additional cost to the HGCL. The removal of scrap shall be subject to the due approval of HGCL & Contractor producing the necessary clearance from the relevant authorities, if required by the law, in respect of disposal of the scrap. The liability for the payment of the applicable GST shall be that of the Contractor.

The Contractor shall also indemnify to keep the HGCL harmless from any act of omission or negligence on the part of the Contractor in following the statutory requirements with regard to removal/disposal of scrap. The undertaking shall be furnished by Contractor as per Format enclosed in the Section VI of Forms & Formats. Further, in case the laws require the HGCL to take prior permission of the relevant Authorities before handing over the scrap to the Contractor, the same shall be obtained by the Contractor on behalf of the HGCL.

In case, the scrap is generated against the free issue material supplied by the HGCL, the Contractor shall maintain a separate record of same and intimate HGCL before its disposal thereafter. After due approval from HGCL, the scrap should be disposed of and its value shall be remitted to the HGCL.

76.5.4 Watch & Ward and Lighting

The Contractor shall provide and maintain at its own expense all lighting, fencing, watch and ward wherever necessary for the proper execution and the protection of the Facilities, or for the safety of the HGCLs and occupiers of adjacent property and for the safety of the public during the entire construction and O&M period.

76.6 Training

The period and the nature of training for the individual personnel shall be agreed upon mutually between the Contractor and the HGCL. These personnel shall be given special training at the shops, where the equipment will be manufactured and/ or in their collaborator's works and where possible, in any other plant where equipment manufactured by the Contractor or his collaborators is under installation or test to enable those personnel to become familiar with the equipment being furnished by the Contractor. HGCL shall bear the cost of Boarding, Lodging & Travel only for the said personnels.

The Contractor undertakes to provide training to Personnel selected and sent by the HGCL at the works of the Contractor without any cost to the HGCL.

77. Replacement of Defective Parts and Materials

- 77.1 If during the progress of the Work, HGCL shall decide and inform in writing to the Contractor, that the Contractor has manufactured any plant or part of the plant unsound or imperfect or has furnished plant inferior to the quality specified, the Contractor on receiving details of such defects or deficiencies shall at his own expenses within 07 (Seven) days of his receiving the notice, or otherwise within such time as may be reasonably necessary for making it good, proceed to alter, re-construct or remove such work and furnish fresh equipment's up to the standards of the specifications.

In case the Contractor fails to do so, HGCL may on giving the Contractor 07 (Seven) day notice in writing of his intentions to do so, proceed to remove the portion of the Work so complained of and at the cost of Contractor's, perform all such works or furnish all such equipment's provided that nothing in the clause shall be deemed to deprive the HGCL of or affect any rights under the Contract, the HGCL may otherwise have in respect of such defects and deficiencies.

78. Indemnity

- 78.1 If any action is brought before a Court, Tribunal or any other Authority against the Employer/ HGCL or an officer or agent of the Employer/ HGCL, for the failure, omission or neglect on the part of the Contractor to perform any acts, matters, covenants or things under the Contract, or damage or injury or death caused by the alleged omission or negligence on the part of the Contractor, his agents, representatives or his Sub-Contractor's, or in connection with any claim based on lawful demands of Sub-Contractor's workmen suppliers or employees, the Contractor, shall in such cases indemnify and keep the Employer/ HGCL and/ or their representatives harmless from all losses, damages, expenses or decrees arising out of such action.

79. Construction Aids, Equipment, Tools & Tackles

- 79.1 Contractor shall be solely responsible for making available for executing the Work, all requisite Construction Equipments, Special Aids, Barges, Cranes and the like, all Tools, Tackles and Testing Equipment and Appliances, including imports & insurance of such equipment etc. as required. In case of import of the same the rates applicable for levying of Custom Duty on such

Equipment, Tools & Tackles and the duty drawback applicable thereon shall be ascertained by the Contractor from the concerned authorities of Government of India. It shall be clearly understood that HGCL shall not in any way be responsible for arranging to obtain Custom Clearance and/ or payment of any duties and/ or duty draw backs, license etc. for such equipment's so imported by the Contractor and the Contractor shall be fully responsible for Goods and Service Tax (GST) and documentation with regard to the same. Bidder in his own interest may contact, for any clarifications in the matter, concerned agencies/ Dept./ Ministries of Govt. of India. All clarifications so obtained and interpretations thereof shall be solely the responsibility of the Contractor.

[F] CERTIFICATES AND PAYMENTS

80. Schedule of Rates and Payments

80.1 Contractor's Remuneration : DELETED

For the billing procedure will follow as per the prescribed payment terms as defined in Special Conditions of Contract (SCC). The price to be paid by the HGCL to Contractor for the whole of the Work to be done and for the performance of all the obligations undertaken by the Contractor under the Contract Documents shall be ascertained by the application of the respective Schedule of Rates (the inclusive nature of which is more particularly defined by way of application but not of limitation, with the succeeding Clause of this clause) and payment to be made accordingly for the Work actually executed and approved by the Engineer-in-Charge/Project Manager. The sum so ascertained shall (excepting only as and to the extent expressly provided herein) constitute the sole and inclusive remuneration of the Contractor under the Contract and no further or other payment whatsoever shall be or become due or payable to the Contractor under the Contract.

80.2 Schedule of Rates to be Inclusive

The prices quoted by the Contractor shall remain firm till the contract period and shall not be subject to escalation. Schedule of Rates shall be deemed to include and cover all costs, expenses and liabilities of every description and all risks of every kind to be taken in executing, completing and handing over the Work to the HGCL by the Contractor. The Contractor shall be deemed to have known the nature, scope, magnitude and the extent of the Work and materials required though the

Contract Document may not fully and precisely furnish them. Bidders in the Schedule of Rates should cover all costs as he may consider necessary to cover the cost of any works and materials as may be reasonable and necessary to complete the Work. Generality of this present provision shall not be deemed to cut down or limit in any way because in certain cases it may and in other cases it may not be expressly stated that the Contractor shall do or perform a work or supply articles or perform services at his own cost or without addition of payment or without extra charge or words to the same effect or that it may be stated or not stated that the same are included in and covered by the Schedule of Rates.

80.3 Schedule of Rates to Cover Construction Equipments, Materials, Labour etc.

Without in any way limiting the provisions of the preceding Clause the Schedule of Rates shall be deemed to include and cover the cost of all construction equipment, temporary Work (except as provided for herein), pumps, materials, labour, insurance, fuel, consumables, stores and appliances to be supplied by the Contractor and all other matters in connection with each item in the Schedule of Rates and the execution of the Work or any portion thereof finished, complete in every respect and maintained as shown or described in the Contract Documents

or as may be ordered in writing during the continuance of the Contract.

80.4 Schedule of Rates to Cover Royalties, Rents and Claims

The Schedule of Rates (i.e., Contract Value) shall be deemed to include and cover the cost of all Royalties and Fees for the articles and processes, protected by letters, patent or otherwise incorporated in or used in connection with the Work, also all Royalties, Rents and other payments in connection with obtaining materials of whatsoever kind for the Work and shall include an Indemnity to the Employer/ HGCL which the Contractor hereby gives against all actions, proceedings, claims, damages, costs and expenses arising from the incorporation in or use on the Work of any such articles, processes or materials, other municipal or local Board Charges, if levied on materials, equipment or machineries to be brought to site for use on Work shall be borne by the Contractor.

80.5 Schedule of Rates to Cover GST/Applicable taxes

No exemption or reduction of Duties, Goods & Service Tax (GST) on Works Contract quay or any port dues, transport charges, stamp duties or Central or State Government or Local Body or Municipal Taxes or from or of any other body, whatsoever, will be granted or obtained, all of which expenses shall be deemed to be included in and covered by the Schedule of Rates. The Contractor shall also obtain and pay for all permits or other privileges necessary to complete the Work.

80.6 Schedule of Rates to Cover Risks of Delay

The Schedule of Rates shall be deemed to include and cover the risk of all possibilities of delay and interference with the Contractor's conduct of Work which occur from any causes including orders of the Employer/ HGCL in the exercise of his power and on account of extension of time granted due to various reasons and for all other possible or probable causes of delay.

80.7 Schedule of Rates Cannot be altered

(a) For Engineering, Procurement and Commissioning (EPC) Contracts or Lumpsum Turnkey (LSTK) Contracts, the total Project/ Contract Value stands to be fixed inclusive of entire items, Materials, Spares, Consumables, Services, Erection and all quoted and unquoted items/ Services in the Bill of Quantity (BOQ) of the Tender/ Contract. Contract Value of such EPC Contracts comprises of all the related costs required for successful execution of the work. The final payment outlay or total cost of the project will be limited to the total value of the EPC Contract and O & M Contract. Any kind of variations related to Total Contract Value shall be to Contractor's account. The payment will be made according to the Work carried out, for which purpose an item wise, or work wise Schedule of Rates shall be furnished, suitable for evaluating the value of Work done and preparing running account bill.

(b) If applicable, For Item Rate Contracts, no alteration will be allowed in the Schedule of Rates by reason of works or any part of them being modified, altered, extended, diminished or committed. The Schedule of Rates are fully inclusive of rates which have been fixed by the Contractor and agreed to by the Employer/HGCL and cannot be altered.

Based on the mechanism of Tender as described in the Special Conditions of Contract (SCC), the methodologies described above shall prevail. However, payment for any additional work which is not covered in the Schedule of Rates, shall only be released on issuance of change order.

Cost Control

81. Bill of Quantities:

The Bill Quantities shall contain items for the construction work to be done by the Contractor.

The Contractor is paid for the quantity of the work done at the estimate rate in the Bill of Quantities for each item plus or minus Tender percentage.

82. Changes in the Quantities:

82.1 The contractor is bound to execute all supplemental works that are found essential, incidental and inevitable during execution of main work.

82.2 The payment of rates for such supplemental items of work will be regulated as under; Supplemental items directly deducible from similar items in the original agreement.

- The rates shall be derived by adding to or subtracting from the agreement rate of such similar item the cost of the difference in the quantity of materials labour between the new items and similar items in the agreement worked out with reference to the Standard Schedule of Rates adopted in the sanctioned estimate with which the tenders are accepted plus or minus over all tender percentage.
- ii. (a) Similar items but the rates of which cannot be directly deduced from the original agreement.

(b) Purely new items which do not correspond to any item in the agreement.
- iii. The rates of all such items shall be Estimated Rates plus or minus overall Tender premium.

83. Extra Items:

83.1 Extra items of work shall not vitiate the contract. The contractor shall be bound to execute extra items of work as directed by the Engineer - in – Charge. The rates for extra items shall be worked out by the Deputy General Managers per the conditions of the Contract and the same are binding on the Contractor.

83.2 The contractor shall before the 15th day of each month, submit in writing to the Deputy General Manager a statement of extra items if any that they have executed during the preceding month failing which the contractor shall not be entitled to claim any.

83.3 Entrustment of additional items:

- Where ever additional items not contingent on the main work and outside the scope of original agreement are to be entrusted to the original contractor dispensing with bids and if the value of such items exceeds the limits up to which the officer is empowered to entrust works initially to contractor without calling for tenders, approval of **next higher authority** shall be obtained. Entrustment of such items on nomination shall be at rates not exceeding the estimated rates.
- Entrustment of the additional items contingent on the main work will be authorised by the officers up to the monetary limits up to which they themselves are competent to accept items in the original agreement so long as the total amounts up to which they are competent to accept in an original agreement rates for such items shall be worked out in accordance with the procedure (I) For all items of work in excess of the quantities shown in the Bill of Quantities of the Tenders, the rate payable for such items shall be estimate rates for the items (+) or (-) over all tender percentage accepted by the competent authority.

- o Entrustment of either the additional or supplemental items shall be subject to the provisions of the agreement entered into by a Competent Authority after the tender is accepted. The Chief General Manager being the higher authority who entered into the agreement approves the rate for the items / variation in quantity in the current agreement. The items shall not be ordered by an officer on his own responsibility if the revised estimate or deviation statement providing for the same requires the sanction of higher authority.

Note: It may be noted that the term Estimate Rate used above means the rate in the sanctioned estimate with which the tenders are accepted, or if no such rates is available in the estimate, the rate derived will be with reference to the Standard Schedule of Rates adopted in the sanctioned estimate with which tenders are accepted.

84. Cash flow forecasts:

84.1 When the program is updated, the contractor is to provide the Engineer-in-charge with an updated cash flow forecast.

85. Payment Certificates:

- The Contractor shall submit to the Engineer - in - Charge monthly statements of the estimated value of the work completed less the cumulative amount certified previously.
- The Engineer - in - Charge shall check the Contractor's monthly statement within 14 days.
- The value of work executed shall be determined by the Engineer - in - Charge.
- The value of work executed shall comprise the value of the quantities of the items in the Bill of Quantities completed.
- The Engineer - in - Charge may exclude any item certified in a previous certificate or reduce the proportion of any item previously certified in any certificate in the light of later information.

86. Payments:

- a. Payment for the work done by the contractor will be made for the finished work based on the measurements recorded in measurement books by any officer of the department not lower in rank than a Assistant Engineer and check measured by any officer not lower in rank than a Deputy Executive Engineer. The measurement shall be recorded at various stages of the work done and also after work is completed. The contractor shall be present at the time of recording of each set of measurement and their check measurement and accept them then and there so as to avoid disputes at a later stage. If the contractor is not available at the work spot at the time of recording measurements or check measurements the particulars of measurements shall be signed by the authorized agent of contractor based on which the contractor shall accept the set of measurements without any further dispute. If for any reason the contractor's authorized agent is also not available at site when the department decides to suspend the work recording of measurements in the absence of the contractor or his authorized representative the department shall not entertain any claim from the contractor for any loss incurred by him on this account. The Contractor shall however note that the Department cannot indefinitely wait for recording the measurement due to the absence of the Contractor and his authorized agent and check measure them even in the absence of the contractor.
- b. For all material.
 - i. 60% of the agreement rate for each item will be paid for all material as per site requirement after testing at factory and conveyance to site as per specifications and on production
 - ii. 20% of the agreement rate after laying, fixing, and erection & 20% agreement rate after commissioning and put in use.

c. Payments and Certificates:

- i. Payments shall be adjusted for recovery of advance payments, liquidated damages in terms of tender conditions and security deposit for the due fulfillment of the contract. Payment will be made to the Contractor under the certificate duly recording the measurements in the Measurement Books, issued at reasonably frequent intervals by the Engineer - in - Charge, and payment will be the sum equal to 100% of the value of work done as so certified by Engineer-in-charge.
- ii. In case of over payments or wrong payment if any made to the contractor due to wrong interpretation of the provisions of the contract, APSS or Contract conditions etc., such unauthorized payment will be deducted in the subsequent bills or final bill for the work or from the bills under any other contracts with the HGCL or at any time there after from the deposits available with the HGCL.
- iii. Any recovery or recoveries advised by the Government Department either state or central, due to non-fulfillment of any contract entered into with them by the contractor shall be recovered from any bill or deposits of the contractor.
- iv. No claim shall be entertained, if the same is not represented in writing to the Engineer-in-Charge within 15 days of its occurrence.
- v. The contractor is not eligible for any compensation for inevitable delay in handing over the site or for any other reason. In such case, suitable extensions of time will be granted after considering the merits of the case.
- vi. **Wherever the audit parties of A.G point out that the contractor is unintendly benefited, then the employer is empowered to recover the same amount from the Contractor and it is binding on the contractor**

d. Intermediate Payments:

- i. For intermediate Stage of work, only part rates as fixed by the Engineer - in - Charge will be paid.
- ii. Part rates shall be worked out for the work done portion based on the actual operations involved keeping in view the value of the balance work to be done, to avoid unintended benefit to the Contractor in initial Stage.
 - a. 70% of the agreement rate for each item will be paid for all material as per site requirement after testing at factory and conveyance to site as per specifications and on production
 - b. 15% of the agreement rate after laying, fixing, and erection & 15% agreement rate after commissioning and put in use.
- iii. Full rate shall be paid when the work is completed to the full profile as noted in the drawings.
- iv. For earthwork in cutting, 10% of the quantity will be with held for intermediate payments and the same will be released after completing the work to the profiles as per drawings and disposal of the spoil material at the specified places and handing over the balance useful stone. For this purpose a length of 25 mts. will be taken as a Unit.

- v. For earth work, embankment formation work, 10% of the quantity will be withheld for intermediate payments and the same will be released after completing the bund to the profiles as per drawings including trimming of side slopes and all other works contingent to the bund profile. For this purpose, 25mts of length will be taken as a Unit.
- vi. For the structure works either with masonry or concrete where the height of structure is more than three meters, the quantities executed in the lower level will be withheld at the rate of one percent for every three meters height, if the balance height of the structure work is more than three meters in being over the executed level and the same will be released only after the entire work is completed as certified by the Engineer-in-Charge.
- vii. For C.M. & C.D. works and for lining works, spread over more than 2 Km. In length 5 percent of the concrete and Masonry quantities will be withheld and the same will be released after completion of all C.M. & C.D. works and lining for the entire length certified by the Engineer-in-Charge.
- viii. Where payment is intended for aggregates by Bill of Quantities item based on stock measurements, 10% of the quantity measured will be withheld. No payment or advance will be made for unfixed materials when the rates are for finished work in site.

87. Interest on Money due to the Contractor:

- No omission by the Deputy General Manager or the sub-divisional officer to pay the amount due upon certificates shall vitiate or make void the contract, nor shall the contractor be entitled to interest upon any guarantee fund or payments in arrear, nor upon any balance which may, on the final settlement of his accounts, found to be due to him.

88. Certificate of Completion of works:

- a. Certificate of Completion of works:
 - i. When the whole of the work has been completed and has satisfactorily passed any final test that may be prescribed by the Contract, the Contractor may give a notice to that effect to the Engineer-in-Charge accompanied by an undertaking to carry out any rectification work during the period of maintenance, such notice and undertaking shall be in writing and shall be deemed to be request by the Contractor for the Engineer-in-Charge to issue a Certificate of completion in respect of the Works. The Engineer-in-Charge shall, within twenty one days of the date of delivery of such notice either issue to the Contractor, a certificate of completion stating the date on which, in his opinion, the works were completed in accordance with the Contract or give instructions in writing to the Contractor specifying all the Works which, in the Engineer-in-Charge's opinion, required to be done by the Contractor before the issue of such Certificate. The Engineer-in-Charge shall also notify the Contractor of any defects in the Works affecting completion that may appear after such instructions and before completion of the Works specified there in. The Contractor shall be entitled to receive such Certificate of the Completion within twenty one days of completion to the satisfaction of the Engineer-in-Charge of the Works so specified and making good of any defects so notified.
 - ii. Similarly, the Contractor may request and the Engineer-in-Charge shall issue a Certificate of Completion in respect of:
 - b. Any section of the permanent works in respect of which a separate time for completion is provided in the Contract, and
 - c. Any substantial part of the permanent works which has been both completed to the satisfaction of the Engineer-in-Charge and occupied or used by the department.

- iii. If any part of the Permanent Works shall have been completed and shall have satisfactorily passed any final test that may be prescribed by the Contract, the Engineer-in-Charge may issue such certificate, and the Contractor shall be deemed to have undertaken to complete any outstanding work in that part of the Works during the period of Maintenance.

89. Procedure for Measurement and Billing

89.1 Billing Procedure

Following procedures shall be adopted for billing of works executed by the Contractor.

- i. If applicable, For Item Rate Contracts, all measurements shall be recorded in sextuplicate on standard measurement sheets in duly approved formats for scrutiny and passing by Employer/HGCL. Employer/ HGCL shall scrutinize and check the measurements recorded on the sheets and shall certify correctness of the same on the measurement sheets.
- ii. Engineer-in-Charge/Project Manager shall pass the bills after carrying out the comprehensive checks in accordance with the terms and conditions of the Contracts, within 21 (Twenty-one) days of submission of the bills along with all necessary enclosures and documents, complete in all respects and send the same to the HGCL to effect payment to the Contractor as per the defined payment terms.
- iii. HGCL shall make all endeavor to make payments of undisputed amount of the bills submitted based on the joint measurements within 30 (Thirty) days from the date of certification by the Engineer-in-Charge/Project Manager. Any disputed claims/amounts will be mutually settled and paid accordingly.
- iv. Measurements shall be recorded as per the methods of measurement spelt out in Contract Document.

89.2 Secured Advance on Material

Unless otherwise provided in the SCC of the tender, no 'Secured Advance' on security of materials brought to site for execution of contracted items(s) shall be paid to the Contractor whatsoever.

89.3 Dispute in Mode of Measurement

In case of any dispute as to the mode of measurement not covered by the Contract to be adopted for any item of Work, mode of measurement as per latest Indian Standard Specifications shall be followed.

89.4 Rounding-Off of Amounts

In calculating the amount of each item due to the Contractor in every certificate prepared for payment, sum of less than 50 paise shall be omitted and the total amount on each certificate shall be rounded off to the nearest rupees, i.e., sum of less than 50 paise shall be omitted and sums of 50 paise and more up to one rupee shall be reckoned as one rupee.

90. BOQ in Tender

- 90.1 The payment against BOQ item shall be made only on completion of that item (Excluding Milestones linked payment structure) as per the provision of the Contract after certification by

91. Running Account Payments to be regarded as Advance

- 91.1 All running account payments shall be regarded as payment by way of advance against the final payment only and not as payments for Work actually done and completed and shall not preclude the requiring of bad, unsound and imperfect or unskilled work to be removed and taken away and reconstructed or re-erected or be considered as an admission of the due performance of the Contract, or any part thereof, in this respect, or of the occurring of any claim by the Contractor, nor shall it conclude, determine or affect in any way the powers of the Employer/ HGCL under these conditions or any of them as to the final settlement and adjustment of the accounts or otherwise, or in any other way vary or affect the Contract.

The final bill shall be submitted by the Contractor within 01 (One) Month of the date of the final acceptance of the Work; otherwise, the Engineer-in-Charge/Project Manager's measurement and of total amount payable for the Work accordingly shall be final and binding on all parties

92. Notice of Claims for Additional Payments

- 92.1 Should the Contractor consider that he is entitled to any extra payment for any extra/ additional Works or Material change in original Specifications carried out by him in respect of Work he shall forthwith give notice in writing to the Engineer-in-Charge/Project Manager that he claims extra payment. Such notice shall be given to the Engineer-in-Charge/Project Manager upon which Contractor bases such claims and such notice shall contain full particulars of the nature of such claim with full details of amount claimed. Irrespective of any provision in the Contract to the contrary, the Contractor must intimate his intention to lodge claim on the HGCL within 10 (Ten) days of the commencement of happening of the event and quantify the claim within 30 (Thirty) days, failing which the Contractor will lose his right to claim any compensation /reimbursement/ damages etc. or refer the matter to arbitration. Separate bills shall be raised for the extra claim.

- 92.2 Engineer-in-Charge/Project Manager shall review such claims within a reasonable period of time and cause to discharge these in a manner considered appropriate after due deliberations thereon. However, Contractor shall be obliged to carry on with the Work during the period in which his claims are under consideration by the HGCL, irrespective of the outcome of such claims, where additional payments for Works considered extra are justifiable in accordance with the Contract provisions,

HGCL shall arrange to release the same in the same manner as for normal Work payments. Such of the extra works so admitted by HGCL shall be governed by all the terms, conditions, stipulations and specifications as are applicable for the Contract. Settlement of all extra claims shall be taken up after Project commissioning.

93. Payment of Contractor's Bill

- 93.1 Payment due to the Contractor shall be made by the HGCL either by e-Banking or by Account Payee Cheque forwarding the same to registered office or the notified office of the Contractor. In no case, will HGCL be responsible if the cheque is mislaid or misappropriated by unauthorized person/ persons. In all cases, the Contractor shall present his bill duly pre-receipted on proper revenue stamp & payment shall be made in Indian Currency only.
- 93.2 In general payment of final bill shall be made to Contractor within 60 days of the submission of bill on joint measurements, after completion of all the obligations under the Contract against the final completion.

94. Receipt for Payment

- 94.1 Receipt for payment made on account of work when executed by a company, must be signed by a person holding due power of attorney in this respect on behalf of the Contractor, except when the Contractors are described in their tender as a limited company in which case the receipts must be signed in the name of the company by one of its principal officers or by some other person having authority to give effectual receipt for the company.

95. Handing Over – Taking Over

- 95.1 Subsequent to Operational Acceptance of the Facilities by the Employer and within 15 (Fifteen) days of the commencement of the O&M period, the Contractor shall furnish an Indemnity Bond/undertaking as per “Sample Forms and Formats” which is to be executed by the Contractor for the plant handed over by HGCL for performance of its O&M Contract (Entire Solar Photo Voltaic Plant).

The Facility shall be taken over by the HGCL upon successful Operational Acceptance in accordance with GCC Clause 43.3 (Operational Acceptance).

Immediately after taking over of complete facilities (s), the Facilities will be handed over to the Contractor for Comprehensive Operation & Maintenance for a period as mentioned in the Contract document.

96. Final Decision and Final Acceptance

- 96.1 Upon expiry of the period of liability & the Works have been duly maintained by the Contractor during monsoon or such period as hereinbefore provided in Clause 73 & 74 and that the Contractor has in all respect duly made-up any subsidence and performed all his obligations under the Contract, the HGCL give a final acceptance to that effect and the Contractor shall not be considered to have fulfilled the whole of his obligations under Contract until Final acceptance shall have been given by the HGCL notwithstanding any previous entry upon the Work and taking possession, working or using of the same or any part thereof by the HGCL.

97. Certificate and Payments on Evidence of Completion

- 97.1 Except the Final Acceptance, no other payments on general account shall be taken to be an admission by the HGCL of the due performance of the Contract or any part thereof or of occupancy or validity of any claim by the Contractor.

98. Deductions from the Contract Price

- 98.1 All costs, damages or expenses which HGCL may have paid or incurred, which under the provisions of the Contract, the Contractor is liable/ will be liable, will be claimed by the HGCL. All such claims shall be billed by the HGCL to the Contractor regularly as and when they fall due. Such claims shall be paid by the Contractor within 15 (Fifteen) days of the receipt of the corresponding bills and if not paid by the Contractor within the said period, the HGCL may, then, deduct the amount from any moneys due including Contract Performance Security or becoming due to the Contractor under the Contract or may be recovered by actions of law or otherwise, if the Contractor fails to satisfy the HGCL of such claims.

[G] TAXES AND INSURANCE

99. Taxes & Duties – Goods & Service Tax

- 99.1 Except as otherwise specifically provided in the Contract, the Contractor shall bear and pay all taxes, duties, levies and charges assessed on the Contractor, its Sub-contractor or their employees by all municipal, state or national government authorities in connection with the Facilities in and outside of the country where the Site is located.
- 99.2 The HGCL shall bear and pay/reimburse to the Contractor Goods and Services Tax (GST)

applicable on: (a) Plant and Equipment (including Type Test Charges) and Mandatory Spares to be supplied from within the Employer's country to be incorporated in the Facilities, by the law of country where the site is located, (b) local transportation & insurance, other local costs incidental to delivery of plant & equipment including mandatory and (c) Installation Services including Erection, Civil & Allied Works and other services. However, all other taxes, duties & levies as may be applicable on goods and services specified in under the contract and on the materials used for civil construction works and erection & commissioning shall be to the contractor's account and no separate claim in this regard will be entertained by the Employer. Notwithstanding anything to contrary contained in the Contract, the Contractor's right to payment under the Contract is subject to issuance of valid tax invoice, payment of applicable GST to the credit of appropriate Government and submission of valid particulars of tax invoice under GST returns in accordance with GST Law. The Contractor shall issue tax invoices, file appropriate returns, and deposit the applicable GST to the account of appropriate government within the time limit prescribed under the GST Law. In the event of any default, Contractor shall be liable to pay any penalty/demand raised on HGCL due to default by Contractor, and the same shall be recovered/Contractor shall make good the loss. The Contractor shall be responsible for the issuance of e-way bill and other compliances relating to e-way bill as per GST law. The HGCL will deduct GST at source at the applicable rates in case transactions under the contract are liable to GST deduction at source as per the prevailing provisions of GST Law.

99.3 If any tax exemptions, reductions, allowances or privileges are available to the Contractor in the country where the Site is located, the HGCL shall use its best endeavors to enable the Contractor to benefit from any such tax savings to the maximum allowable extent.

99.4 The Contractor agrees to and does hereby accept full and exclusive liability for the payment of any and GST/all Taxes. now in force or hereafter imposed, increased, modified, from time to time in respect of Works and materials and all contributions and taxes for unemployment compensation, insurance and old age pensions, PF, ESI etc or annuities now in force or hereafter imposed by any Central or State Government authorities which are imposed with respect to or covered by the wages, salaries or other compensations paid to the persons employed by the Contractor and the Contractor shall be responsible for the compliance of all Sub-Contractors, with all applicable Central, State, Municipal and local law and regulation and requirement of any Central, State or local Government agency or authority.

Contractor further agrees to defend, indemnify and hold Employer/ HGCL harmless from any liability or penalty which may be imposed by the Central, State or Local authorities by reason or any violation by Contractor or Sub-Contractor of such laws, suits or proceedings that may be brought against the Employer/ HGCL arising under, growing out of, or by reason of the work provided for by this Contract, by third parties, or by Central or State Government authority or any administrative sub- division thereof. Tax deductions will be made as per the rules and regulations in force in accordance with acts prevailing from time to time.

99.5 Bidder should quote all-inclusive prices including the liability of GST (in line with the given SOR Format) whether on the works contract as a whole or in respect of bought out components used by the Contractor in execution of the Contract. HGCL/Employer shall not be responsible for any such liability of the Contractor in respect of this Contract.

99.6 Goods & Service Tax (GST) [applicable for both Centre and state] and other levies [if any] payable by the Contractor under the Contract, or for any other cause, shall be included in the rates/ prices and the total bid-price submitted by the Bidder. Applicable rate of GST shall be indicated in Agreed SOR formats.

100. Income Tax

As per Indian Income Tax Act & Rules, HGCL is required to deduct Income Tax at source from all the payments to be made to Nonresident/ Foreign Contractor. For this purpose, the Contractor shall be required to either furnish (i) the certificate from Indian Tax Authority or (ii) Ruling from "the Authority for Advance Ruling (AAR)" determining the applicable rate of Income tax in India before release of first payment. The Contractor will be required to submit

PAN details to the Project Manager before the submission of the first bill. If the HGCL orders any spare at a later date a, all applicable additional taxes & duties, if any, not included in the original price shall be to the account of HGCL. Notwithstanding anything to contrary contained in the agreement/Purchase Order, Contractor/Supplier's right to payment under the contract/agreement/Purchase Order is subject to issuance of valid tax invoice, payment of applicable GST to the credit of appropriate Government and submission of a valid particulars of tax invoice under GST returns in accordance with GST Act. Unless expressly stated otherwise, a common mechanism for reconciliation of input credit mismatch, to be followed by both HGCL and Contractor/Supplier, shall be mutually agreed so that both parties follow the same procedure for disclosing the transactions in their respective returns. Notwithstanding anything in the agreement/ contract, penalty / damages shall be recovered in case the Contractor/Supplier makes a default in following the agreed procedure. Contractor/Supplier shall issue tax invoices, file appropriate returns, and deposit the applicable GST to the account of appropriate government within the time limit prescribed under the GST Law. In the event of any default, Contractor/Supplier shall be liable to pay the amount which may be imposed on HGCL due to such default. Contractor/Supplier should comply with the provisions of e-way bill notified by appropriate authorities from time to time. The existing provisions regarding road permit will continue till such time if applicable. HGCL will deduct GST at source at the applicable rates in case transactions under the contract are liable to GST deduction at source as per the prevailing provisions of GST Law.

101. Void

102. Insurance

To the extent specified in the Contract Agreement, the Contractor shall at its expense take out and maintain in effect, or cause to be taken out and maintained in effect, during the performance of the Contract, the insurances set forth below in the sums and with the deductibles and other conditions. The identity of the insurers and the form of the policies shall be subject to the approval of the HGCL, who should not unreasonably withhold such approval.

- 102.1 During the Contract period including O&M period, i.e., during Construction & O&M period, all insurance related expenses shall be borne by the Contractor. The goods supplied under the Contract shall be fully insured against the loss or damage incidental to manufacture or acquisition, transportation, storage and delivery in such a manner that HGCL shall not incur any financial loss, as long as the plant continues to remain under the custody of the Contractor.
- 102.2 In case of any loss or damage or pilferage or theft or fire accident or combination of the said incident etc. under the coverage of insurance, the Contractor shall lodge the claim as per rules of insurance. Any FIR required to be lodged to local Police Station shall be the responsibility of the Contractor.
- 102.3 The Contractor shall arrange to supply/ rectify/ recover the materials without waiting for settlement of the insurance claim and even if the claim is unsettled for timely completion of the project. The final financial settlement with the insurance company shall rest up on the HGCL.
- 102.4 In case of any delay of the project attributable to the Contractor, the Contractor himself in consultation with HGCL/Employer shall take the extension of insurance. Any financial implications shall be borne by the Contractor.
- 102.5 The Contractor should arrange for providing insurance coverage to its workmen under Workmen's Compensation Act or similar Rules and Acts as applicable during execution of work for covering risk against any mishap to its workmen. The Contractor shall also undertake a Third-Party Insurance and shall at all times keep HGCL indemnified against any Third-Party claims and shall arrange to settle them at the earliest. The HGCL/ Employer will not be liable for any such loss or mishap.
- 102.6 All other insurance like – transit insurance (Marine/ Cargo/ others as applicable), Construction

All Risk, Erection All Risk, workmen compensation, fire, third party liability, insurance against theft, Contractor's Equipments, machinery breakdown policy, business interruption insurance, Property damage Insurance & Environmental risk insurance as required during the Construction and O&M period of the Plant shall be in the contractor's scope & shall borne by the Contractor.

- 102.7 HGCL shall be named as co – insured under all insurance policies taken out by the Contractor, except for the workmen compensation, third party liability and HGCL's liability insurances. All insurers' rights of subrogation against such co – insured for losses or claims arising out of the performance of the contract shall be waived under such policies.
- 102.8 All the insurance cover taken for the construction and O&M period shall be seamless in nature & preferably taken from the same insurance company.
- 102.9 The insurance is to be suitably taken for the activity/ act which is required to cover all the risks associated to the activity / act. The Contractor shall be responsible to take suitable insurance till the completion of the O&M contract and indemnify the Employer/HGCL from all associated risks whatsoever.
- 102.10 The Contractor shall be responsible to take suitable insurance(s) and claim management during and till the completion of the O&M contract and indemnify the HGCL from all associated risks whatsoever.

Various Types of Insurance to be taken by Contractor during Construction & O&M period:

102.11 Employees State Insurance (ESI) Act

The Contractor agrees to and does hereby accept full and exclusive liability for the compliance with all obligations imposed by the Employee State Insurance Act 1948 and the Contractor further agrees to defend, indemnify and hold Employer/ HGCL harmless for any liability or penalty which may be imposed by the Central, State or Local authority by reason of any asserted violation by Contractor or Sub-Contractor of the Employees' State Insurance Act, 1948, and also from all claims, suits or proceeding that may be brought against the Employer/ HGCL arising under, growing out of or by reasons of the work provided for by this Contractor, by third parties or by Central or State Government authority or any political sub- division thereof.

The Contractor agrees to fill in with the Employee's State Insurance Corporation, the Declaration Forms, and all forms which may be required in respect of the Contractor's or Sub-Contractor's employees, who are employed in the Work provided for or those covered by ESI from time to time under the Agreement. The Contractor shall deduct and secure the agreement of the Sub-Contractor to deduct the Employee's contribution as per the first schedule of the Employee's State Insurance Act from wages and affix the Employees Contribution Card at wages payment intervals.

The Contractor shall remit and secure the agreement of Sub-Contractor to remit to the Employee's State Insurance Corporation Account, the Employee's contribution as required by the Act. The Contractor agrees to maintain all cards and Records as required under the Act in respect of employees and payments and the Contractor shall secure the agreement of the Sub-Contractor to maintain such records. Any expenses incurred for the contributions, making contributions or maintaining records shall be to the Contractor's or Sub-Contractor's account.

102.12 Workmen Compensation and Employer's/HGCL's Liability Insurance

Insurance shall be effected for all the Contractor's employees engaged in the performance of this Contract. If any of the work is sublet, the Contractor shall require the Sub-Contractor to

provide workman's Compensation and HGCL's liability insurance for the latter's employees if such Employees are not covered under the Contractor's Insurance.

102.13 Accident or Injury to Workmen

The Employer/ HGCL shall not be liable for or in respect of any damages or compensation payable at law in respect or in consequence of any accident or injury to any workman or other person in the Employment of the Contractor or any Sub-Contractor and the Contractor shall indemnify and keep indemnified the Employer/ HGCL against all such damages and compensation (save and except and aforesaid) and against all claims, demands, proceedings, costs, charges and expenses, whatsoever in respect or in relation thereto.

102.14 Transit/Cargo Insurance

In respect of all items to be transported by the Contractor to the Site of Work and any consequential risks, the cost of transit insurance shall be borne by the Contractor and the quoted price shall be inclusive of this cost.

Covering loss or damage occurring, while in transit from the supplier's or manufacturer's works or stores until arrival at the Site, to the Facilities (including spare parts therefor) and to the construction equipment to be provided by the Contractor or its Subcontractors.

Amount	Deductible Limits	Parties Insured	From	To
110% of the Ex-works value of supply	Nil	Contractor & HGCL	Ware House	Ware House + 60 Days

102.15 Comprehensive Automobile Insurance

This insurance shall be in such a form as to protect the Contractor against all claims for injuries, disability, disease and death to members of public including Employer's/ HGCL's men and damage to the property of others arising from the use of motor vehicles during on or off the site operations, irrespective of the Employer ship of such vehicles.

102.16 Installation All Risks Insurance

Covering physical loss or damage to the Facilities at the Site, occurring prior to Completion of the Facilities, with extended maintenance coverage for the Contractor's liability in respect of any loss or damage occurring during the Defect Liability Period while the Contractor is on the Site for the purpose of performing its obligations during the Defect Liability Period.

Amount	Deductible Limits	Parties Insured	From	To
110% of the Ex-works value of supply	Nil	Contractor, Sub contractor & HGCL	Receipt at site	Upto Defect Liability period

102.17 Comprehensive General Liability Insurance

102.17.1 This insurance shall protect the Contractor against all claims arising from injuries, disabilities, disease or death of member of public or damage to property of others due to any act or omission on the part of the Contractor, his agents, his

employees, his representatives and Sub-Contractor's or from riots, strikes and civil commotion.

102.17.2 Contractor shall take suitable Group Personal Accident Insurance Cover for taking care of injury, damage or any other risks in respect of his Engineers and other Supervisory staff who are not covered under Employees State Insurance Act.

102.17.3 The policy shall cover third party liability. The third party (liability shall cover the loss/ disablement of human life (person not belonging to the Contractor) and also cover the risk of damage to others materials/ equipment/ properties during construction, erection and commissioning at site & during O&M period.

The value of third-party liability for compensation for loss of human life or partial/ full disablement shall be of required statutory value but not less than INR 02 (Two) Lakhs per death, INR 1.5 (One and Half) Lakhs per full disablement and INR 1 (One) Lakh per partial disablement and shall nevertheless cover such compensation as may be awarded by Court by Law in India and cover for damage to others equipment/ property as approved by the Purchaser. However, third party risk shall be maximum to INR 10 (Ten) Lakhs for death.

102.17.4 The Contractor shall also arrange suitable insurance to cover damage, loss, accidents, risks etc., in respect of all his plant, equipment and machinery, erection tools & tackles and all other temporary attachments brought by him at site to execute the work.

102.17.5 The Contractor shall take out insurance policy in the joint name of HGCL and Contractor from one or more nationalized insurance company from any branch office at Project site.

102.17.6 Any such insurance requirements as are hereby established as the minimum policies and coverages which Contractor must secure and keep in force must be complied with, Contractor shall at all times be free to obtain additional or increased coverages at Contractor's sole expenses.

102.18 The Contractor shall also arrange suitable insurance to cover following during the O&M Period:

102.18.1 **Machinery Breakdown:** Electrical & or machinery breakdown of any machinery or other equipment resulting in costly repairs or even replacement of the solar panel.

102.18.2 **Business Interruption:** Cover for period of operational downtime i.e., covering the cash flow of the solar business as a result of an insured peril, for example fire or storm damage, machinery breakdown or equipment failure.

102.18.3 **Property Damage:** The insurance should cover material damage due to external causes such as fire, theft, vandalism, sabotage, hail damage, snow load, lightning strike, overload, operational mistakes, clumsiness, negligence & theft.

102.18.4 **Employers Liability:** Provides cover against the risk of accident from usual workplace risks such as working at height & manual handling during construction & O&M period..

102.18.5 **Environmental Risk Insurance:** Environmental damage coverage indemnifies solar system HGCLs of the risk of either environmental damage done by their development or pre-existing damage on the development site.

102.19 ANY OTHER INSURANCE REQUIRED UNDER LAW OR REGULATIONS OR BY HGCL:

Contractor shall also carry and maintain any and all other insurance(s) which he may be

required under any law or regulation from time to time without any extra cost to HGCL. He shall also carry and maintain any other insurance which may be required by the HGCL.

Amount	Deductible Limits	Parties Insured	From	To
To be indicated by the Contractor	Nil	Contractor, Sub contractor & HGCL	Receipt at site	Upto Defect Liability period

103 Damage to Property or to any Person or any Third Party

103.1 Contractor shall be responsible for making good to any loss or any damage to structures and properties belonging to the HGCL or being executed or procured or being procured by the HGCL or of other agencies within in the premises of all the work of the HGCL, if such loss or damage is due to fault and/ or the negligence or willful acts or omission of the Contractor, his employees, agents, representatives or Sub-Contractors.

103.2 The Contractor shall take sufficient care in moving his plants, equipment and materials from one place to another so that they do not cause any damage to any person or to the property of the Employer/ HGCL or any third party including overhead and underground cables and in the event of any damage resulting to the property of the HGCL or of a third party during the movement of the aforesaid plant, equipment or materials the cost of such damages including eventual loss of production, operation or services in any plant or establishment as estimated by the Employer/ HGCL or ascertained or demanded by the third party shall be borne by the Contractor. Third party liability risk shall be INR 1 (One) Lakh for single accident and limited to INR 10 (Ten) Lakhs.

103.3 The Contractor shall indemnify and keep the Employer/ HGCL harmless of all claims for damages to property other than Employer's/ HGCL's property arising under or by reason of this agreement, if such claims result from the fault and/ or negligence or willful acts or omission of the Contractor, his employees, agents, representative of Sub-Contractor.

[H] LABOUR LAWS

104 Labour Laws & Indemnity Bond

104.1 No labour below the age of 18 (Eighteen) Years shall be employed on the Work.

104.2 The Contractor shall at his expense comply with all labour laws and keep the Employer/ HGCL indemnified in respect thereof.

104.3 The Contractor shall employ labour in sufficient numbers either directly or through Sub- Contractor's to maintain the required rate of progress and of quality to ensure workmanship of the degree specified in the Contract.

104.4 The Contractor shall indemnify the Employer/ HGCL against any payments to be made under and for the observance of the provisions of the aforesaid labour compliances without prejudice to his right to obtain indemnity from his Sub-Contractor's.

104.5 The Contractor shall also indemnify to keep the HGCL/ Employer harmless from any act of omission or negligence on the part of the Labour Laws compliance in following the statutory requirements with regard to Labour laws. Against the signing of the contract, The Indemnity Bond shall be furnished by Contractor as per Format enclosed under Forms and Formats for the labour law compliance.

104.6 Upon completion of the construction activities/ O&M activities, the Contractor shall obtain no – objection certificate (NOC) from local/ statutory bodies in respect to the fulfillment of all compliance related to labour law and submit a copy to the Employer/HGCL prior to the final settlement

105 Void

106 Contractor to Indemnify the Employer/ HGCL

106.1 The Contractor shall indemnify the Employer/ HGCL and every member, office and employee of the Employer/ HGCL, also the Engineer-in-Charge/Project Manager and his staff against all actions, proceedings, claims, demands, costs and expenses whatsoever arising out of or in connection with the matters referred to in Clause 95 and elsewhere and all actions, proceedings, claims, demands, costs and expenses which may be made against the Employer/ HGCL for or in respect of or arising out of any failure by the Contractor in the performance of his obligations under the Contract Document.

The Employer/ HGCL shall not be liable for or in respect of or arising out of any failure by the Contractor in the performance of his obligations under the Contract Document. The Employer/ HGCL shall not be liable for or in respect of any demand or compensation payable by law in respect or in consequence of any accident or injury to any workmen or other person. In the employment of the Contractor or his Sub-Contractor the Contractor shall indemnify and keep indemnified the Employer/ HGCL against all such damages and compensations and against all claims, damages, proceedings, costs, charges and expenses whatsoever in respect thereof or in relation thereto.

106.2 Payment of Claims and Damages

Should the Employer/ HGCL have to pay any money in respect of such claims or demands as aforesaid the amount so paid and the costs incurred by the Employer/ HGCL shall be charged to and paid by the Contractor and the Contractor shall not be at liberty to dispute or question the right of the Employer/ HGCL to make such payments notwithstanding the same, may have been made without the consent or authority or in law or otherwise to the contrary.

106.3 In every case in which by virtue of the provisions of Section 12, Sub-section (i) of workmen's compensation Act, 1923 or other applicable provision of Workmen Compensation Act or any other Act, the Employer/ HGCL is obliged to pay compensation to a workman employed by the Contractor in execution of the Work, the Employer/ HGCL will recover from the Contractor the amount of the compensation so paid, and without prejudice to the rights of Employer/ HGCL under Section 12, Sub-section (2) of the said Act, Employer/ HGCL shall be at liberty to recover such amount or any part thereof by deducting it from the Contract Performance Security or from any sum due to the Contractor whether under this Contract or otherwise. The Employer/ HGCL shall not be bound to contest any claim made under Section 12, Sub-section (i) of the said act, except on the written request of the Contractor and upon his giving to the Employer/ HGCL full security for all costs for which the Employer/ HGCL might become liable in consequence of contesting such claim.

107 Health and Sanitary Arrangements for Workers

107.1 In respect of all labour directly or indirectly employed in the Works for the performance of the Contractor's part of this agreement, the Contractor shall comply with or cause to be complied with all the rules and regulations of the local sanitary and other authorities or as framed by the HGCL from time to time for the protection of health and sanitary arrangements for all workers.

107.2 The Contractor shall provide in the labour colony all amenities such as electricity, water and other sanitary and health arrangements. The Contractor shall also provide necessary surface transportation to the place of work and back to the colony for their personnel accommodated in the labour colony.

107.2.1 APPLICABLE LAWS & SETTLEMENT OF DISPUTES

108 Arbitration

Normally all disputes should be settled by negotiations between the HGCL and the concerned parties.

In case any dispute / difference is not settled through negotiations, the respective parties can seek remedy through Arbitration only by invoking the same within 120 days of raising of dispute. No disputes shall be referred to civil courts other than through Arbitration.

The issues/ disputes which cannot be mutually resolved through negotiations within the time stipulated above, all such disputes shall be referred to arbitration by Sole Arbitrator.

HGCL shall suggest a panel of three independent and distinguished persons to the Contractor/ Supplier (as the case may be) to select any one among them to act as the Sole Arbitrator. In the event of failure of the other parties to select the Sole Arbitrator within 30 (Thirty) Days from the receipt of the communication suggesting the panel of arbitrators, the right of selection of the sole arbitrator by the other party shall stand forfeited and HGCL shall have discretion to proceed with the appointment of the Sole Arbitrator. The decision of HGCL on the appointment of the sole arbitrator shall be final and binding on the parties. The award of sole arbitrator shall be final and

binding on the parties and unless directed/ awarded otherwise by the sole arbitrator, the cost of arbitration proceedings shall be shared equally by the parties.

The Arbitration proceedings shall be in English language and venue shall be the State of the HGCL as specified in the BDS/SCC. Subject to the above, the provisions of (Indian) Arbitration & Conciliation Act 1996 and the Rules framed there under shall be applicable. All matter relating to this contract are subject to the exclusive jurisdiction of the court situated in the state of the HGCL as specified in the BDS/SCC.

Above mentioned Arbitration clause will be applicable for the Disputes where the amount of claim is less than or equal to INR 1Crore (Indian Rupees One Crore only). For the cases of disputes where the amount of claim is more than INR 1 Crore (Indian Rupees One Crore), such disputes will be settled through commercial Courts established under the Commercial Courts, Commercial Division and Commercial Appellate divisions of High Courts act, 2015. Before going to the commercial court for settlement, such disputes be first referred to conciliation for settlement and in case of failure in conciliation, disputes be referred to the commercial courts for adjudication. In cases of funded packages, the aforesaid changes shall be implemented after concurrence of the funding agency.

FOR THE SETTLEMENT OF DISPUTES BETWEEN GOVERNMENT DEPARTMENT AND ANOTHER AND ONE GOVERNMENT DEPARTMENT AND PUBLIC ENTERPRISE AND ONE PUBLIC ENTERPRISE AND ANOTHER THE ARBITRATION SHALL BE AS FOLLOWS:

In the event of any dispute or difference between the parties hereto, such dispute or difference shall be resolved amicably by mutual consultation or through the good offices of empowered agencies of the Government.

If such resolution is not possible, then, the unresolved dispute or difference shall be referred to arbitration of an arbitrator to be nominated by Secretary, Department of Legal Affairs ("Law Secretary") in terms of the Office Memorandum No.55/3/1/75-CF, dated the 19th December 1975 issued by the Cabinet Secretariat (Department of Cabinet Affairs), as modified from time to time.

The Arbitration Act 1940 (10 of 1940) shall not be applicable to the arbitration under this clause. The award of the Arbitrator shall be binding upon parties to the dispute. Provided, however, any party aggrieved by such award may make a further reference for setting aside or revision of the award to Law Secretary whose decision shall bind the parties finally and conclusively.

109 Jurisdiction

The Contract shall be governed by and constructed according to the laws in force in INDIA. The Contractor hereby submits to the jurisdiction of the Courts situated at “**State of the HGCL of the Project as defined in the BDS/SCC**” for the purposes of disputes, actions and proceedings arising out of the Contract, the courts at “**State of the HGCL of the Project**” only will have the jurisdiction to hear and decide such disputed, actions and proceedings.

[J] SAFETY CODES

110 General

Contractor shall adhere to safe construction practice and guard against hazardous, and unsafe working conditions and shall comply with Safety rules as set forth herein.

Any hazardous material used during construction or used as part of the plant has to be taken back by the supplier for recycling or dumping purpose after its operating / working life, so that it may not affect the environment or any living being. Contractor have to comply with State Pollution Board regulation.

111 Safety Regulations

i) In respect of all labour, directly employed in the Work for the performance of Contractor's part of this agreement, the Contractor shall at his own expense arrange for all the safety provisions as per safety codes of C.P.W.D., Indian Standards Institution. The Electricity Act, The Mines Act and such other acts as applicable.

ii) The Contractor shall observe and abide by all fire and safety regulations of the HGCL. Before starting construction, Contractor shall consult with HGCL's safety Engineers or Engineer-in-Charge/Project Manager and must make good to the satisfaction of the Employer/ HGCL any loss or damage due to fire to any portion of the work done or to be done under this agreement or to any of the Employer's/HGCL's existing property.

112 First Aid and Industrial Injuries

i) Contractor shall maintain first aid facilities for its employees and those of its Sub-Contractor.

ii) Contractor shall make arrangements for Ambulance Service, on requirement basis and for the treatment of industrial injuries. Names of those providing these services shall be furnished to Employer/ HGCL prior to start of construction and their telephone numbers shall be prominently posted in Contractor's Site Office.

ii) All critical industrial injuries shall be reported promptly to Employer/ HGCL, and a copy of Contractor's report covering each personal injury requiring the attention of a physician shall be furnished to the Employer/ HGCL.

113 General Rules

Site is a Nonsmoking zone area. Hence, smoking within the battery area is strictly prohibited.

114 Contractor's Barricades

i) Contractor shall erect and maintain barricades required in connection with his operation to guard or protect: -

- a) Excavations
- b) Hoisting Areas.
- c) Areas adjudged hazardous by Contractor's or Employer's/ HGCL's inspectors.
- d) Employer's/ HGCL's existing property subject to damage by Contractor's Operations.
- e) Rail Road unloading spots

ii) Contractor's employees and those of his Sub-Contractor's shall become acquainted with Employer's/ HGCL's barricading practice and shall respect the provisions thereof.

iii) Barricades and hazardous areas adjacent to, but not located in normal routes of travel shall be marked by red flasher lanterns at nights.

115 Working at Height

i) Scaffolding or staging more than 4 meters above the ground or floor, swing suspended from an overhead support or erected with stationary support shall have a guard rail properly attached, bolted, braced and otherwise retarded at least one meter high above the floor or platform of such scaffolding or staging and extending along the entire length of the outside and ends thereof with only such openings as may be necessary for the delivery of materials. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure.

ii) Working platform, gangway and stairway should be so constructed that they should not sag unduly or unequally and if the height of platform of the gangway or the stairway is more than 4 meters above the ground level or floor level, they should be closely boarded, should have adequate width and should be suitably fastened as in ii) above.

iii) Every opening in the floor of a building or in a working platform shall be provided with suitable means to prevent the fall of persons or materials by providing suitable fencing or railing whose minimum heights shall be 1 meter.

iv) While working at the substation and for construction of transmission line Towers and their subsequent maintenance, experienced manpower should be deployed with appropriate protection Equipments, such as insulating gloves, fall arrestor etc.

116 Excavation and Trenching

All trenches 1.5 Meters or more in depth, shall at all times be supplied with at least one ladder for each 50 Meters length or fraction thereof. Ladder shall be extended from bottom of the trenches to at least 1 meter above the surface of the ground. The sides of the trenches which are 1.5 Meters in depth shall be stepped back to give suitable slope or securely held by timber bracing, so as to avoid the danger of sides to collapse. The excavated materials shall not be placed within 1.5 Meters of the edge of the trench or half of the trench width whichever is more. Cutting shall be done from top to bottom. Under no circumstances undermining or undercutting shall be done.

117 Demolition/ General Safety

i) Before any demolition work is commenced and also during the progress of the demolition work

a) All roads and open areas adjacent to the work site shall either be closed or suitably protected.

b) No electric cable or apparatus which is liable to be a source of danger shall remain electrically charged.

c) All practical steps shall be taken to prevent danger to persons employed from risk of fire or explosion or flooding. No floor, roof or other part of the building shall be so overloaded with debris or materials as to render it unsafe.

- ii)** All necessary personal safety equipment as considered adequate should be kept available for the use of the persons employed on the Site and maintained in condition suitable for immediate use, and the Contractor shall take adequate steps to ensure proper use of equipment by those concerned.
- a)** Workers employed on mixing asphaltic materials, cement and lime mortars shall be provided with protective footwear and protective gloves.
- b)** Those engaged in white washing and mixing or stacking of cement bags or any material which are injurious to the eyes be provided with protective goggles.
- c)** Those engaged in welding and cutting works shall be provided with protective face & eye shield, hand gloves, etc.
- d)** Stone breakers shall be provided with protective goggles and protective clothing and seated at sufficiently safe intervals.
- e)** When workers are employed in sewers and manholes, which are in use, the CONTRACTOR shall ensure that the manhole covers are opened and are ventilated at least for an hour before the workers are allowed to get into the manholes, and the manholes so opened shall be cordoned off with suitable railing and provided with warning signals or board to prevent accident to the public.
- f)** The CONTRACTOR shall not employ men below the age of 18 years and women on the work of painting with products containing lead in any form. Wherever men above the age of 18 years are employed on the work of lead painting, the following precautions should be taken.
- 1)** No paint containing lead or lead product shall be used except in the form of paste or ready made paint.
 - 2)** Suitable face masks should be supplied for use by the workers when paint is applied in the form of spray or a surface having lead paint dry rubbed and scrapped.
 - 3)** Overalls shall be supplied by the Contractor to the workmen and adequate facilities shall be provided to enable the working painters to wash them during and on cessation of work.
- iii)** When the work is done near any place where there is risk of drowning, all necessary safety equipment should be provided and kept ready for use and all necessary steps taken for prompt rescue of any person in danger and adequate provision should be made for prompt first aid treatment of all injuries likely to be sustained during the course of the work.
- iv)** Use of hoisting machines and tackles including their attachments, anchorage and supports shall conform to the following standards or conditions:
- a)** These shall be of good mechanical construction, sound materials and adequate strength and free from latent defect and shall be kept in good working order.
- b)** Every rope used in hoisting or lowering materials or as means of suspension shall be of durable quality and adequate strength and free from patent defects.
- c)** Every crane driver or hoisting appliance operator shall be properly qualified and no person under the age of 21 years should be in charge of any hoisting machine including any scaffolding, winch or give signals to the operator.
- d)** In case of every hoisting machine and of every chain ring hook, shackle, swivel, and pulley block used in hoisting or lowering or as means of suspension, the safe working load shall be ascertained by adequate means. Every hoisting machine and all gears referred to above shall be plainly marked with the safe working load of the conditions under which it is applicable and the same shall be clearly indicated. No part of any machine or any gear referred to above in this paragraph shall be loaded beyond safe working load except for the purpose of testing.

e) As regards Contractor's machines, the Contractor shall notify the safe working load of the machine to the Engineer-in-Charge/Project Manager whenever he brings any machinery to Site of Work and get it verified by the Engineer concerned.

v) Motors, gears, transmission lines, electric wiring and other dangerous parts of hoisting appliances should be provided with efficient safeguards. Hoisting appliances should be provided with such means as to reduce to minimum the accidental descent of the load, adequate precautions should be taken to reduce the minimum risk of any part or parts of a suspended load becoming accidentally

displaced. When workers are employed on electrical installations which are already energized, insulating mats, wearing apparel, such as gloves, sleeves, and boots as may be necessary should be provided. The workers shall not wear any rings, watches and carry keys or other materials which are good conductors of electricity.

vi) All scaffolds, ladders and other safety devices mentioned or described herein shall be maintained in safe conditions and no scaffolds, ladder or equipment shall be altered or removed while it is in use. Adequate washing facilities should be provided at or near places of work.

vii) These safety provisions should be brought to the notice of all concerned by displaying on a notice board at a prominent place at the work-spot. The person responsible for compliance of the safety code shall be named therein by the Contractor.

viii) To ensure effective enforcement of the rules and regulations relating to safety precautions, the arrangements made by the Contractor shall be open to inspection by the Engineer-in-Charge/Project Manager or safety Engineer of the HGCL.

ix) Notwithstanding the above clauses there is nothing in these to exempt the Contractor for the operations of any other Act or rules in force in the Republic of India. The work throughout including any temporary works shall be carried out in such a manner as not to interfere in any way whatsoever with the traffic on any roads or footpath at the site or in the vicinity thereto or any existing works whether the property of the Administration or of a third party.

In addition to the above, the Contractor shall abide by the safety code provision as per C.P.W.D. Safety code and Indian Standard Safety Code from time to time.

118 Care in Handling Inflammable Gas

The Contractor has to ensure all precautionary measures and exercise utmost care in handling the inflammable gas cylinder/ inflammable liquids/ paints etc. as required under the law and/ or as advised by the fire Authorities of the HGCL or Administration.

119 Temporary Combustible Structures

Temporary combustible structures will not be built near or around work site.

120 Precautions against Fire

The Contractor will have to provide portable Fire Extinguishers, Fire Buckets and drums at worksite as per specifications & standards. They will have to ensure all precautionary measures and exercise utmost care in handling the inflammable gas cylinders/ inflammable liquid/ paints etc. as advised by Engineer-in-Charge/Project Manager. Temporary combustible structure will not be built near or around the work-site.

121 Explosives

Explosives shall not be stored or used on the Work or on the Site by the Contractor without the permission of the Engineer-in-Charge/Project Manager in writing and then only in the manner and to the extent to which such permission is given. When explosives are required for the Work they will be stored in a special magazine to be provided at the cost of the Contractor in accordance with the Explosives Rules. The Contractor shall obtain the necessary license for the storage and the use of explosives and all operations in which or for which explosives are employed shall be at sole risk and responsibility of the Contractor and the Contractor shall indemnify the Employer/HGCL against any loss or damage resulting directly or indirectly therefrom. Only licensed persons shall be engaged for handling and working with explosives.

122 Preservation of Place

The Contractor shall take requisite precautions and use his best endeavors to prevent any riotous or unlawful behavior by or amongst his worker and others employed on the works and for the preservation of peace and protection of the inhabitants and security of property in the neighborhood of the Work. In the event of the HGCL requiring the maintenance of a Special Police Force at or in the vicinity of the site during the tenure of works, the expenses thereof shall be borne by the Contractor and if paid by the HGCL shall be recoverable from the Contractor.

123 Outbreak of Infectious Diseases

- 115.1 The Contractor shall remove from his camp such labour and their facilities who refuse protective inoculation and vaccination when called upon to do. Should Cholera, Plague or other infectious diseases break out the Contractor shall burn the huts, beddings, clothes and other belongings or used by the infected parties and promptly erect new huts on healthy sites as failing which the work may be done by the HGCL and the cost thereof recovered from the Contractor.

124 Use of Intoxicants

- 116.1 The unauthorized sale of spirits or other intoxicants, beverages upon the work in any of the buildings, encampments or tenements owned, occupied by or within the control of the Contractor or any of his employee is forbidden and the Contractor shall exercise his influence and authority to the utmost extent to secure strict compliance with this condition.

In addition to the above, the Contractor shall abide by the safety code provision as per C.P.W.D. safety code, Indian Standard Code & OHSAS 18001 framed from time to time.

SECTION - V

SPECIAL CONDITIONS OF CONTRACT (SCC)

SPECIAL CONDITIONS OF CONTRACT (SCC)

The following Special Conditions of Contract (SCC) shall supplement the General Conditions of Contract (GCC). Wherever there is a conflict, the provisions herein shall prevail over those in the GCC. The corresponding Clause number(s) of the GCC is/ are indicated in parentheses.

SCC No.	GCC Clause Ref. No. (If Applicable)	Details/ Description/ Special Conditions
1	(GCC clause 1)	The Hyderabad Growth Corridor Ltd., Hyderabad, Telangana Nanakramguda, Hyderabad Kind Attn: Sri.P. Sampath Kumar, Dy., GM (Elec) email: dgmelehqcl@gmail.com <u>Replace the existing clause by following:</u> 1.1.49 Operational Acceptance means completion of all the required works / activities to achieve the rated generation and successful demonstration of Guaranteed Performance Ratio (PR) for 30 days. <u>Add following to the existing clause:</u> 1.1.65 Synchronization means connection of Solar plant output to grid, after the approval of TS TRANSCO & TS NPDCL.
2.		The Engineer-in-Charge: “Shall be intimated to the successful bidder”
3.		The Time for Commissioning for the cumulative capacity of 13 MW Solar PV Power Plant at 02 different sites along with ORR as per this tender document in total shall be 3(Three) Months from the Date of the Notification of Award NOA/LOA/LOI. Further Contractor is also to provide Operation & Maintenance Contract of Solar Photo Voltaic Plant for a period of 5 (Five) years from the date of Operational Acceptance of the Plant.

SCC No.	Stage	Reference from D
3.1	Issue of NOA/ LOA/ LOI (as applicable)	Zero Date (D)
3.2	Project Synchronization / Date of successful commissioning	<u>D + 2 Months</u>
3.3	Operational Acceptance of plant	<u>D + 2.5 Months</u>
3.4	Final Acceptance of plant	<u>D + 3 Months</u>

	The contractor shall submit the L2/L3 schedule for the project clearly indicating the timelines for submission and approval of drawings, procurement, and transportation of materials to site, their installation at site, commissioning, PR test etc.	
4	Location of Site (GCC clause 2.1.a)	Add following to the existing clause regarding the Site Visit: 1. The proposed solar powered plants are to set up at various areas of HGCL. 2. Prospective Bidders are advised to visit the site to study the actual conditions and go through the plans/ drawings connected to the present scope of work including power evacuation system, including ground conditions, availability of water etc and get acquainted with the same before attending Pre-bid meeting. For seeking visit of site or any clarifications bidders may contact the D,General Manager (Ele) (e-mail: dgmelehgcl@gmail.com). 3. The Bidder at the Bidder's own responsibility, cost and risk shall inspect and examine the site and its surrounding, and shall satisfy themselves fully before submitting bids as to the form and nature of the site, the geological conditions decisive for the success of the project, the means of access to the site, the loading and unloading facilities etc. In general, the Bidders shall themselves obtain all necessary information as to risks,

		contingencies, and other circumstances susceptible to influence or affect their bids. 4. Although certain information is provided in Scope of work of this tender document, it should be checked by the Bidders, any neglect or failure to obtain or confirm such information will not relieve the Bidders from any liability or responsibilities to carry out the works according to the contract. HGCL will assist the Bidders in obtaining the data required but will not assume responsibility either for the data obtained or for their completeness. 5. Bidders shall acquaint themselves on their own responsibility with laws and regulations in India under which the work is to be performed including those which may influence, in general or in detail, design, supply, transportation, erection, operation of the equipment and requirement of manpower. Any failure or neglect to do so will not absolve the potential Contractor from his contractual obligation. 6. It is specially emphasized that it shall be the responsibility of the Bidders to have themselves familiarized with the prevailing conditions and that no claim relating thereto for additional payment or adjustment of a Contract price will be acceptable after the submission of their Bid. 7. It shall be deemed that the tenderer has visited the site/area and got fully acquainted with the working conditions and other prevalent conditions and fluctuations thereto whether he visits the site/area or not and has taken all the factors into account while quoting his rates. 8. Prospective Bidders are advised to carefully read the Tender documents along with Annexures, understand them in the proper perspective and then fill the Technical Bid Format, Commercial Bid Format and Price Bid Format (SOR). 9. In connection with the site visit, the bidder shall submit an Undertaking along with the bid. 10. Cutting of trees, bushes and vegetation from site (including permission from local authorities) and disposal thereof. The site should be maintained in vegetation free condition during erection stage (at any point of time) i.e., up to start of O&M Contract.
5	Construction Water Supply (GCC clause 2.3)	<u>Add following to the existing clause:</u> The Contractor shall make own arrangements from the nearest supply source as per HGCL terms and conditions, If HGCL providing the water supply for construction & O&M activity. Source of Water and Power supply will be provided by HGCL on free of cost
6	Construction PowerSupply	<u>Add following to the existing clause:</u>
	(GCC clause 2.4)	The Contractor shall make own arrangements from the nearest supply source as per HGCL terms and conditions, If HGCL providing the Power supply for construction activity

7	Construction Power Supply (GCC clause 2.4.3)	<u>Replace the existing clause by following:</u> At all times, CEA regulations shall be followed failing which the Employer/ HGCL has a right to disconnect the power supply without any reference to the Contractor. No claim shall be entertained for such disconnection. Power supply will be reconnected only after production of fresh certificate from authorized electrical supervisors.
8	Land for Contractor's Field Office, Godown and Workshop (GCC clause 2.5)	<u>Add following to the existing clause:</u> The Employer/ HGCL will, at his own discretion and convenience and for the duration of the execution of the work make available near the site on chargeable basis as per HGCL terms and conditions for construction of Contractor's Temporary Field Office, godowns, workshops and assembly yard required for the execution of the Contract.
9	Land for Residential Accommodation (GCC clause 2.6)	<u>Replace the existing clause by following:</u> Contractor has to arrange the accommodation for their staff/labour. However, subject to availability, Employer/HGCL may provide land for temporary construction or quarters or guest house on chargeable basis, as per the company rules. The Employer/HGCL shall not, however, guarantee the provision of land/accommodation.
10	Liquidated Damages (GCC clause 20)	Subject to Force Majeure Clause, if the contractor fails to comply with the timelines for completion, the application of LD will be as per Clause: 20 of Section-IV of GCC.
11	Contract Performance Security (Section-IV GCC clause 24)	<u>Replace the existing clause by following:</u> Against EPC/ LSTK & O&M Contract of the project, within 30 (Thirty) days from the issuance of the Notification of Award/ Letter of Intent/Letter of Award, the successful bidder shall furnish unconditional and irrevocable individual Bank Guarantees for respective sites (Site 1,2 & 3) issued by any Public-Sector Banks (SBI & Associates and Nationalized banks) or Private Bank incorporated in India having net worth of ₹ 5,000.00 Crores and above for the previous Financial Year in the fashion as mentioned below. However, where value of BG is Rs.5 lakhs and above, the BG should be obtained from the bank branches situated at Hyderabad / or operative and payable at bank branches situated at Hyderabad The Contract Performance Security shall be in the form of Bank Guarantee only and shall be in the currency of the Contract and will be issued individually for all the sites (Site 1&2) in the name of the HGCL as “Hyderabad Growth Corridor Limited, Telangana State”

		The Contract Performance Security against this Contract need to be furnished as mentioned below: 1. First Stage (Supply and Services): The value of the Contract Performance Security shall be 10% (Ten percent) of the Contract Value (i.e., total sum of the Supply & Service Contract). This Performance security will be valid for a total period of 66Months (3 Months Project commissioning period) + prescribed O & M Period, i.e., 60 Months + 03 Months additional) from the date of its issuance. The successful bidder can submit Contract Performance Security with initial validity of one year and the same may be extended every year till completion of the total 66months period. 2. The Contract Performance Security shall be towards faithful performance of the contractual obligations and performance of equipment. 3. In case of any default or failure of the Contractor to comply with the requirements of any of the Obligations covered under this Tender Document and/ or Contract Agreement shall constitute sufficient grounds for forfeiture of the entire Contract Performance Security, in such cases, the liability on account of GST will be borne by the contractor. 4. Further, any delay beyond 30 (Thirty) days shall attract delay charges @ 1.25% per month on the total Contract Performance Security amount, calculated on pro-rata basis accordingly. However, total project completion period shall remain same. Part Security shall not be accepted. Further, HGCL at its sole discretion may cancel the Contract Agreement/ NOA & forfeit 100% of EMD inclusive of GST, in case Contract Performance Security is not submitted within 45 (Forty-Five) days from issuance of NOA/LOA/LOI. Contract Performance Security submitted shall be released to the Contractor without any interest not later than 75 (Seventy-Five) days after the successful completion of the complete O&M period (5 Years) subject to the approval and acceptance of the O&M period deliverables.
12.	Termination for HGCL's Convenience (GCC sub-clause 25.1.3 (e))	<u>Replace the existing clause by following:</u> Void.

13	Termination for Contractor's Default (GCC clause 25.2)	<u>Add following to the existing clause:</u> 25.2.3 Termination of Contract for non-performance and subsequently putting the contractor on Banning List: In case of termination of Contract herein set forth except under conditions of Force Majeure and termination after expiry of contract, the Contractor shall be put under Banning List [i.e. neither any enquiry will be issued to the party by Employer/ HGCL against any type of tender nor their offer will be considered by Employer/ HGCL against any ongoing tender(s) where contract between Employer/ HGCL and that particular Contractor (as a bidder) has not been finalized] for a period as decided by the Employer/ HGCL to such Contractor.																																													
14	Termination by the Contractor (GCC clause 25.3)	<u>Replace the existing clause by following:</u> Void.																																													
15	Contractor's responsibility (GCC clause 32.1.3)	<u>Replace the existing clause by following:</u> In the matter of connectivity of Plant to DISCOM's substation, the HGCL will take the necessary connectivity permission, however, all the other permissions and clearances as deemed required by the State Agency/DISCOM for Bay allocation, technical/regulatory compliance for interconnection, ROW etc. are to be taken by the Contractor. All the required fees including statutory fees, Supervision charges etc. shall be paid / born by the contractor which will be reimbursed by HGCL after submission of necessary documents.																																													
16	Design and Engineering (GCC clause 53)	<u>Add following to the existing clause:</u> 53.4 The timelines for submission of major drawings are as follows: <table border="1"> <thead> <tr> <th>S. No.</th><th>Document / Drawing</th><th>Timeline for submission</th></tr> </thead> <tbody> <tr> <td></td><td>Issue of LoA</td><td>Zero Date (D)</td></tr> <tr> <td>1</td><td>Topographical Survey</td><td>D + 1 weeks</td></tr> <tr> <td>2</td><td>PV Modules</td><td>D + 3 weeks</td></tr> <tr> <td>3</td><td>Power Conditioning Unit</td><td>D + 3 weeks</td></tr> <tr> <td>4</td><td>DC SLD</td><td>D + 3 weeks</td></tr> <tr> <td>5</td><td>AC SLD</td><td>D + 3 weeks</td></tr> <tr> <td>6</td><td>Geotechnical Investigation Report</td><td>D + 4 weeks</td></tr> <tr> <td>7</td><td>Cable Laying Design Report</td><td>D + 4 weeks</td></tr> <tr> <td>8</td><td>Module fixing on Structure</td><td>D + 6 weeks</td></tr> <tr> <td>9</td><td>Transformer</td><td>D + 8 weeks</td></tr> <tr> <td>10</td><td>HT Switchgear Panel</td><td>D + 8 weeks</td></tr> <tr> <td>11</td><td>Array Layout</td><td>D + 10 weeks</td></tr> <tr> <td>12</td><td>Cables and Transmission Line</td><td>D + 12 weeks</td></tr> <tr> <td>13</td><td>SCADA</td><td>D + 12 weeks</td></tr> </tbody> </table>	S. No.	Document / Drawing	Timeline for submission		Issue of LoA	Zero Date (D)	1	Topographical Survey	D + 1 weeks	2	PV Modules	D + 3 weeks	3	Power Conditioning Unit	D + 3 weeks	4	DC SLD	D + 3 weeks	5	AC SLD	D + 3 weeks	6	Geotechnical Investigation Report	D + 4 weeks	7	Cable Laying Design Report	D + 4 weeks	8	Module fixing on Structure	D + 6 weeks	9	Transformer	D + 8 weeks	10	HT Switchgear Panel	D + 8 weeks	11	Array Layout	D + 10 weeks	12	Cables and Transmission Line	D + 12 weeks	13	SCADA	D + 12 weeks
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17	Conditions for Issue of Materials (GCC clause 60)	No material will be issued to the contractor by the HGCL, HGCL for this Project.
18	Training (GCC clause 76.6)	The contractor shall provide training (free of cost) to the personnel of HGCL/Nominated person of HGCL for 50 (Fifty) man-days at his works/or at the mutually agreed designated place/ or at site for erection, testing, commissioning and O&M of the Project. Expenses towards travel, lodging, and boarding for the personnel shall be borne by HGCL. Such training may include Class Room & hands on experience etc as mutually agreed.
19	Taxes & Duties - Goods & Service Tax (GCC clause 91)	<u>Add the following to the existing clause:</u> Exemption of GST or any other taxes and duties (if applicable) on Items Manufactured for Grid Connected Solar Photo Voltaic Power Generation Projects As per extant guidelines of Department of Revenue, Ministry of Finance, Govt. of India, the benefits of exemption of GST or any other taxes and duties (if applicable on items manufactured for grid connected Solar Photo Voltaic Power Generation Projects) may be available for the items to be supplied under the contract, Bidder may appraise itself of the relevant policies and quote accordingly. The Employer shall issue the requisite certificate (if required) as specified in the relevant policy of Govt. of India. However, if the certificate is required to be issued by any department/ ministry of Government of India or State Government where the Project is located other than Employer, the Bidder shall itself be responsible for obtaining such certificate from the concerned department/ministry. In such a case, the Employer may issue recommendatory letter to the bidder. The Bidder shall be solely responsible for obtaining the benefits of exemption on excise duty GST or any other taxes and duties (if applicable) on the items to be supplied under the contract from the Govt. of India. In case of failure of the bidder to receive the benefits partly or fully from Govt. of India or in case of delay in receipt of such benefits, the Employer shall neither be responsible nor liable in this regard in any manner whatsoever.

SECTION - VI

SAMPLE FORMS, FORMATS & ANNEXURES

Preamble

This Section (Section - VI) of the Bidding Documents [named as Sample Forms and Procedures (FP)] provides proforma to be used by the bidders at the time of their bid preparation and by the Contractor subsequent to the award of Contract.

The Bidder shall complete, sign and submit with its bid the relevant FORMS to be used unamended, in accordance with the requirements included in the Bidding Documents.

The Bidder shall provide the EMD, if applicable in the form included hereafter acceptable to the Employer, pursuant to the provisions in the instructions to Bidders.

The Performance Security (ies) and Bank Guarantee for Advance Payment forms should not be completed by the bidders at the time of their bid preparation. Only the successful Bidder will be required to provide the Performance Security(ies) and Bank Guarantee for Advance Payment, according to one of the forms indicated herein acceptable to the Employer and pursuant to the provisions of the General and Special Conditions of Contract, respectively.

Depending on specific facts and circumstances related to the Bid/ Tender and the contract, the text of the Forms herein may need to be modified to some extent. The Employer reserves the right to make such modifications in conformity with such specific facts and circumstances and rectify and consequent discrepancies, if any. However, modifications, if any, to the text of the Forms that may be required in the opinion of the Bidder/ Contractor shall be effected only if the same is approved by the Employer. The Employer's decision in this regard shall be final and binding.

<u>LIST OF FORMS & FORMAT</u>	
Form No.	Description
F-0	COVERING LETTER
F-1	BIDDER'S GENERAL INFORMATION
F-2	FORMAT FOR INDEMNITY BOND TO BE EXECUTED BY THE CONTRACTOR FOR THE REMOVAL / DISPOSAL OF SCRAP/DISPOSAL OF SURPLUS MATERIAL – IF APPLICABLE
F-3	INDEMNITY BOND TO BE EXECUTED BY THE CONTRACTOR FOR THE PLANT HANDED OVER FOR PERFORMANCE OF ITS O&M CONTRACT (ENTIRE SOLAR PHOTO VOLTAIC PLANT) – DURING HANDOVER
F-4A	PROFORMA OF "BANK GUARANTEE" FOR "EARNEST MONEY DEPOSIT (EMD)"
F-5	INDEMNITY BOND
F-6	NO DEVIATION CONFIRMATION
F-7	DECLARATION REGARDING BANNING AND LIQUIDATION, COURT RECEIVERSHIP ETC.
F-8	DECLARATION ON RESTRICTION ON PROCUREMENT FROM CERTAIN COUNTRIES: MoF OM No 6/18/2019-PPD dated 23.07.2020
F-9	PROFORMA OF "BANK GUARANTEE" FOR "CONTRACT PERFORMANCE SECURITY"
F-12	DECLARATION REGARDING THE PROCUREMENT OF SOLAR INVERTERS & SOLAR MODULES FROM CLASS I LOCAL SUPPLIERS
F-13	BIDDER'S EXPERIENCE
F-14	DISCLOSURE FOR ALMM COMPLIANCE
F-15	DECLARATION REGARDING THE FULFILLMENT OF DOMESTIC CONTENT REQUIREMENT CLAUSE
F-16	FORMAT OF CHARTERED ACCOUNTANT CERTIFICATE FOR FINANCIAL CAPABILITY OF THE BIDDER
F-17	FORMAT FOR JOINT VENTURE AGREEMENT
F-17A	FORMAT FOR POWER OF ATTORNEY FOR JOINT VENTURE AGREEMENT
F-18	FORMAT FOR CONSORTIUM AGREEMENT
F-18A	FORMAT FOR POWER OF ATTORNEY FOR CONSORTIUM AGREEMENT
F-19	E-BANKING FORMAT
F-20	PROFORMA OF "BANK GUARANTEE" FOR "ADVANCE PAYMENT"

F-22	LIST OF BANKS
F-23	SHAREHOLDING CERTIFICATE
F-24	POWER OF ATTORNEY FOR BIDDING COMPANY
F-25	FORMAT FOR CYBER SECURITY AGREEMENT
F-26	INTEGRITY PACT FORMAT

COVERING LETTER

(The Covering Letter should be submitted on the Letter Head of the Bidding Company)

Ref.No. _____

Date: _____

From: _____ *(Insert name and address of Bidding Company)*

Tel.#: _____

Fax#: _____

E-mail address#To _____

Hyderabad Growth Corridor Limited(A State
Government Company)Nanakramguda,
Hyderabad

Sub: Bid for "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different stretches including 5 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India"

Dear Sir / Madam,

1. We, the undersigned.... [insert name of the 'Bidder'] having read, examined and understood in detail the tender Document for Bid for "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different stretches including 5 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India" hereby submit our Bid comprising of Techno Commercial Bid and Price Bid.

"We confirm that neither we nor any of our Parent Company/ Affiliate / Ultimate Parent Companyhas submitted Bid other than this Bid directly or indirectly in response to the aforesaid tender."

Note: In case a common Company/Companies directly or indirectly hold(s) more than 10% but less than 26% shareholding in more than one Bidder participating in the Tender, each of such Bidders will be required to submit the Disclosure as per the attached format. In such a case, the above-mentioned highlighted declaration/statement will be modified suitably by the bidder. In all other cases, this disclosure is not required & standard covering letter & other forms & formatswill be applicable.

2. We give our unconditional acceptance to the tender, dated and tender documents attached thereto, issued by HGCL, as amended. As a token of our acceptance to the tender documents, the same have been initialled by us and enclosed to the Bid. We shall ensure that we execute such tender documents as per the provisions of thetender and provisions of such tender documents shall be binding on us.

3. Bid Capacity

We have bid for the Total capacity of 13 MW (AC).

4. Bid Processing Fees

We have enclosed a Bid Processing Fees of INR..... (*Insert Amount*), in the form of Demand Draft/ Banker's Cheque no..... (*Insert reference of the DD/ Banker's Cheque*) dated..... (*Insert date of DD/ banker's cheque*) from..... (*Insert name of Bank providing DD/ banker's cheque*) and valid up to and including in terms of Clause of this tender.

5. Earnest Money Deposit

We have enclosed an Earnest Money Deposit of INR..... (*Insert Amount*), in the form of bank guarantee no..... (*Insert reference of the bank guarantee*) dated(*Insert date of bank guarantee*) as per Format F-4 from(*Insert name of Bank providing BG*) and valid up to and including in terms of Clause of this tender. The offered quantum of power by us is __MWp. (*Insert total capacity offered*).

6. We have submitted our Price Bid strictly as per this tender, without any deviations, conditions and without mentioning any assumptions or notes for the Price Bid in the said format(s).

7. In case we are a Successful Bidder, we shall furnish a declaration at the time of commissioning of the Project to the effect that neither we have availed nor we shall avail in future any Incentive other than received from HGCL for implementation of the project.

8. Acceptance

We hereby unconditionally and irrevocably agree and accept that the decision made by Hyderabad Growth Corridor Limited in respect of any matter regarding or arising out of the tender shall be binding on us. We hereby expressly waive any and all claims in respect of Bid process.

We confirm that there are no litigations or disputes against us, which materially affect our ability to fulfil our obligations with regard to execution of projects of capacity offered by us.

9. Familiarity with Relevant Indian Laws & Regulations

We confirm that we have studied the provisions of the relevant Indian laws and regulations as required to enable us to submit this Bid and execute the tender documents, in the event of our selection as Successful Bidder. We further undertake and agree that all such factors as mentioned in tender have been fully examined and considered while submitting the Bid.

10. Contact Person

Details of the contact person are furnished as under:

Name	:	
Designation	:	
Company	:	
Address	:	
Phone Nos.	:	
Fax Nos.	:	
E-mail address	:	

11. We are enclosing herewith the Part-I (Covering Letter, Processing Fees, EMD (If applicable), Bid Securing Declaration etc through Offline and Online, Techno-Commercial documents through online as per clause no. 11.1 of Section - II, ITB) and Part II (Price Bids) (through online) containing duly signed formats, each one duly sealed separately, in one original as desired by you in the tender for your consideration as per clause no. 11.0 of Section - II, ITB.

It is confirmed that our Bid is consistent with all the requirements of submission as stated in the tender and subsequent communications from HGCL. The information submitted in our Bid is complete, strictly as per the requirements stipulated in the tender and is correct to the best of our knowledge and understanding. We would be solely responsible for any errors or omissions in our Bid. We confirm that all the terms and conditions of our Bid are valid for acceptance for a period as specified in BDS from the date of opening of "Techno-Commercial/ Un-priced Bid". We confirm that we have not taken any deviation so as to be deemed non-responsive.

Dated the _____ day of _____, 20....

Thanking you,

We remain, Yours

faithfully,

Name, Designation and Signature of Authorized Person in whose name Power of Attorney/ Board Resolution/ Declaration. Copy of Power of Attorney/ Board Resolution/ Declaration should be enclosed along with Covering Letter.

BIDDER'S GENERAL INFORMATION**(To be submitted on the Letter Head of the Bidding Company)**

Sr. No.	Description	Remarks
1	Name of the Bidder <i>(Incase of JV/ Consortium, name of all the members to be indicated and lead member to be identified)</i>	
2	Status of the Firm	
3	Mailing Address of Registered Office	
4	Mailing Address of Operation Office	
5	E-mail	
6	Web site	
7	Authorized Contact Person(s) with Name, Designation, Address and Mobile Phone No., E-mail address / Fax No. to whom all references shall be made	
8	Year of Incorporation	
9	Number of Years in Operation	
10	ISO Certification Yes/No	
11	Name of the Banker	
12	Branch Details of Bank	
13	Type of Account with Account Number	
14	IFSC Code	
15	Permanent Account Number (PAN) of the Bidder	<i>(Copy of PAN Card to be enclosed)</i>
16	Whether the Vendor is registered/ Likely to be registered under GST	Yes or No : If Yes, then customer will be treated as registered customer & he will have to provide further details as stated on the below left hand side
17	GST ID (Proof to be submitted – GST No acknowledgement OR Email from GoI)	
18	GSTN Address	
19	PF Registration Number with Details	<i>(Copy of Registration to be enclosed)</i>
20	ESI Registration Number with Details	<i>(Copy of Registration to be enclosed)</i>

21	Have the Bidder/ Company ever been debarred by any Govt. Dept./ Undertaking for undertaking any work	Yes/No <i>(If answer is YES, please provide details)</i>
22	Reference of any document information attached by the Bidder other than specified in the tender.	
23	Whether the Bidder wishes to form a Project Company for execution of work	Yes/No
24	Bidding company is listed in India	Yes/No
25	Whether company is MSME as on the bidding date	Yes/No

(Signature of Authorized Signatory)

With Stamp

**Indemnity Bond to be executed by The Contractor for The Removal / Disposal of
Scrap/Disposal of Surplus Material**

(TO BE EXECUTED ON STAMP PAPER OF APPROPRIATE VALUE)

INDEMNITY BOND

This INDEMNITY BOND executed this day of 20..... by(Name of Company)....., a Company registered under the Companies Act, 1956/2013 Concern and having its registered office(s) at(Office Address)....., hereinafter called the Indemnifier(s)/ Contractor(s) (which expression shall, unless excluded by or repugnant to the context, be deemed to mean and include its successors, administrators and executors).

IN FAVOUR OF

M/s , having its registered office at.....(hereinafter referred to as “.....”)

1. M/s..... has awarded the Contractor(s), contract for execution of work (“Scope of Work”) as mentioned in the NOA/contract agreement no..... dated , entered into between M/s and Contractor(s), relating to(Name & Address of Project/Station)... (hereinafter called ‘the Project’).
2. The Indemnifier(s) for the purpose of execution of its Scope of Work had from time to time procured and stored(Details of Material)... at the Project Site.
3. After completion of the Scope of Work by Indemnifier(s), it has been identified that scrap (Details of Scrap Material & its Quantity).....and/or surplus (Details of Surplus Material & its Quantity)belonging to Indemnifier(s) is lying at the said Project Site.
4. Now, the scrap (Details of Scrap Material & its Quantity)... and/or surplus (Details of Surplus Material & its Quantity)..... belonging to the Indemnifier(s), requires to be removed by Indemnifier(s) from the Project Site.

NOW THEREFORE THIS INDEMNITY BOND WITNESSETH AS UNDER:

1. That Indemnifier(s) by way of this indemnity requests M/s..... to issue approval in favour of Indemnifier(s) for removal of scrap(Details of Scrap Material & its Quantity)... and/or surplus(Details of Surplus Material & its Quantity).....belonging to Indemnifier(s), from the project.
2. That the Indemnifier(s) shall ensure clearing of its scrap (Details of Scrap Material & its Quantity).....and/or surplus (Details of Surplus Material & its Quantity).....by itself, as aforesaid.

3. That Indemnifier(s) in consideration of the premises above, for itself and its respective, executors, administrators and assigns, jointly and severally agree and undertake from time to time and at all times hereafter to indemnify M/s..... and keep M/s..... indemnified from and against all claims, demands, actions, liabilities and expenses which may be made or taken against or incurred by M/s..... by reason of the issue of necessary approval by M/s..... and permitting Indemnifier(s) to remove scrap(Details of Scrap Material & its Quantity)... and/or surplus(Details of Surplus Material & its Quantity).....belonging to Indemnifier(s), from the project.

4. That Indemnifier(s) undertakes to indemnify and keep M/s... harmless from any act of omission or negligence on the part of the Contractor in following the statutory requirements with regard to removal/disposal of scrap and surplus belonging to Indemnifier(s), from the Project Site aforesaid, by the Indemnifier(s). Further, in case the laws require M/s..... to take prior permission of the relevant Authorities before handing over the scrap and/or surplus to the Indemnifier, the same shall be obtained by the Indemnifier on behalf of M/s.....

IN WITNESS WHEREOF, the Indemnifier(s), through its authorized representative, has executed these presents on the Day, Month and Year first mentioned above at

.....(Name of the Place).....

Witness:

Indemnifier

1.

2.

(Authorised Signatory)

INDEMNITY BOND TO BE EXECUTED BY THE CONTRACTOR FOR THE PLANT HANDED OVER FOR PERFORMANCE OF ITS O&M CONTRACT (ENTIRE SOLAR PHOTO VOLTAIC PLANT)

(TO BE EXECUTED ON STAMP PAPER OF APPROPRIATE VALUE)

INDEMNITY BOND

THIS INDEMNITY BOND IS made this..... day of 20.....by a Company registered under the Companies Act, 1956/2013 concern having its Registered Office at..... (hereinafter called as "Contractor" or "Obligor" which expression shall include its successors) in favour of M/s....., a Company incorporated under the Companies Act, 1956 having its Registered Office at and its Project at(hereinafter called "M/s....." which expression shall include its successors and assigns) :

WHEREAS M/s..... has awarded to the Contractor a Contract for vide its Letter of NOA/Contract Agreement No..... dated and its Amendment No..... (Applicable when amendments have been issued) (hereinafter called the "Contract") in terms of which M/s.... is required to hand over various Equipment and facilities provided under Supply & Service Contract , herein after called "Solar Photo Voltaic Plant" to the Contractor for execution of the Contract.

AND WHEREAS, The Contractor is required to execute an Indemnity Bond in favour of M/s... for the Solar Photo Voltaic Plant handed over to it by M/s..... for the purpose of Performance of the Contract/O&M portion of the Contract.

NOW, THEREFORE, this Indemnify Bond witnesseth as follows:

1. That in consideration of Solar Photo Voltaic Plant as mentioned in the Contract, Valued at Rs.....#..... (Rupees.....) handed over to the Contractor for the purpose of Performance of the Contract, the Contractor hereby undertakes to indemnify and shall keep M/s... indemnified, for the full value of the Solar Photo Voltaic Plant. The Contractor hereby acknowledges actual receipt of the Solar Photo Voltaic Plant as detailed in the Schedule appended hereto. The Contractor shall hold such Solar Photo Voltaic Plant in trust as a "Trustee" for and on behalf of M/s.....

2. That the Contractor is obliged and shall remain absolutely responsible for the safe O&M/protection and custody of the Solar Photo Voltaic Project against all risks whatsoever till completion of O&M Contract in accordance with the terms of the Contract and is taken over by M/s..... The Contractor undertakes to keep M/s..... harmless against any loss or damage that may be caused to the Solar Photo Voltaic Plant.

3. The Contractor undertakes that the Solar Photo Voltaic Plant shall be used exclusively for the Performance/execution of the Contract strictly in accordance with its terms and conditions and no part of the Solar Photo Voltaic Plant shall be utilised for any other work or purpose whatsoever. It is clearly understood by the Contractor that non-observance of the obligations under this Indemnify Bond by the Contractor shall inter-alia constitute a criminal breach of trust on the part of the Contractor for all intents and purposes including legal/penal consequences.

4. That M/s.... is and shall remain the exclusive HGCL of the Solar Photo Voltaic Plant free from all encumbrances, charges or liens of any kind, whatsoever. The Solar Photo Voltaic Plant shall at all times be open to inspection and checking by Engineer-in-Charge/Engineer or other employees /agents authorised by him in this regard. Further, M/s... shall always be free at all times to take possession of the Solar Photo Voltaic Plant in whatever form the Solar Photo Voltaic Plant may be, if in its opinion, the Solar Photo Voltaic Plant are likely to be endangered, mis-utilised or converted to uses other than those specified in the Contract, by any acts of omission or commission on the part of the Contractor or any other person or on account of any reason whatsoever and the Contractor binds itself and undertakes to comply with the directions of demand of M/s... to return the Solar Photo Voltaic Plant without any demur or reservation.

5. That this Indemnify Bond is irrevocable. If at any time any loss or damage occurs to the Solar Photo Voltaic Plant or the same or any part thereof is mis- utilised in any manner whatsoever, then the Contractor hereby agrees that the decision of the Engineer-in-Charge/Engineer of M/s.... as to assessment of loss or damage to the Solar Photo Voltaic Plant shall be final and binding on the Contractor. The Contractor binds itself and undertakes to replace the lost and/or damaged Solar Photo Voltaic Plant at its own cost and / or shall pay the amount of loss to M/s.... without any demur, reservation or protest. This is without prejudice to any other right or remedy that may be available to M/s.... against the Contractor under the Contract and under this Indemnify Bond.

6. NOW THE CONDITION of this Bond is that if the Contractor shall duly and punctually comply with the terms of and conditions of this Bond to the satisfaction of M/s..., THEN, the above Bond shall be void, but otherwise, it shall remain in full force and virtue.

IN WITNESS WHEREOF, the Contractor has hereunto set its hand through its authorised representative under the common seal of the Company, the day, month and year first above mentioned

SCHEDULE

Particulars of the Equipment / Facilities handed-over	Quantity	Value	Other details, (if any)	Signature of Attorney in token of receipt

WITNESS

For and on behalf of

M/s.

I. 1. Signature Name

2. Name Signature

3. Address Designation

Authorised representative*

II.

1. Signature

2. Name Common Seal
(In case of Company)

3. Address

* Indemnity Bonds are to be executed by the authorised persons and (i) In case of contracting Company under common seal of the Company of (ii) having the power of attorney issued under common seal of the company with authority to execute Indemnity Bonds, (iii) In case (ii) the original Power of Attorney if it is specifically for our contract or a Photostat copy of the Power of Attorney if it is a General Power of Attorney and such documents should be attached to Indemnity Bond.

The value shall be sum of Supply and Erection Contract value.

**PROFORMA OF BANK GUARANTEE FOR EARNEST
MONEY DEPOSIT (EMD) :**

(To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place of execution.)

Ref. _____ _Bank Guarantee No. _____

Date: _____

In consideration of the ----[Insert name of the Bidder] (hereinafter referred to as 'Bidder') submitting the response to NIT inter alia for selection of the Project in response to the NIT No. _____ dated issued by Hyderabad Growth Corridor Limited (hereinafter referred to as HGCL) and HGCL considering such response to the NIT of[insert the name of the Bidder] as per the terms of the NIT, the _____ [insert name & address of bank] hereby agrees unequivocally, irrevocably and unconditionally to pay to HGCL at [Insert Name of the Place from the address of HGCL] forthwith on demand in writing from HGCL or any Officer authorized by it in this behalf, any amount upto and not exceeding Rupees -----[Insert amount] only, on behalf of M/s.

_____ [Insert name of the Bidder] .

This guarantee shall be valid and binding on this Bank up to and including _____ [insert date of validity in accordance with NIT] and shall not be terminable by notice or any change in the constitution of the Bank or the term of contract or by any other reasons whatsoever and our liability hereunder shall not be impaired or discharged by any extension of time or variations or alternations made, given, or agreed with or without our knowledge or consent, by or between parties to the respective agreement.

Our liability under this Guarantee is restricted to Rs. _____ (Rs. _____ only). Our Guarantee shall remain in force until _____ [insert date of validity in accordance with NIT]. HGCL shall be entitled to invoke this Guarantee till _____ [Insert date which is 30 days after the date in the preceding sentence].

The Guarantor Bank hereby agrees and acknowledges that the HGCL shall have a right to invoke this BANK GUARANTEE in part or in full, as it may deem fit.

The Guarantor Bank hereby expressly agrees that it shall not require any proof in addition to the written demand by HGCL, made in any format, raised at the above-mentioned address of the Guarantor Bank, in order to make the said payment to HGCL.

The Guarantor Bank shall make payment hereunder on first demand without restriction or conditions and notwithstanding any objection by ----- [Insert name of the Bidder] and/or any other person. The Guarantor Bank shall not require HGCL to justify the invocation of this BANK GUARANTEE, nor shall the Guarantor Bank have any recourse against HGCL in respect of any payment made

hereunder.

This BANK GUARANTEE shall be interpreted in accordance with the laws of India and the courts at Hyderabad shall have exclusive jurisdiction.

The Guarantor Bank represents that this BANK GUARANTEE has been established in such form and with such content that it is fully enforceable in accordance with its terms as against the Guarantor Bank in the manner provided herein.

This BANK GUARANTEE shall not be affected in any manner by reason of merger, amalgamation, restructuring or any other change in the constitution of the Guarantor Bank.

This BANK GUARANTEE shall be a primary obligation of the Guarantor Bank and accordingly HGCL shall not be obliged before enforcing this BANK GUARANTEE to take any action in any court or arbitral proceedings against the Bidder, to make any claim against or any demand on the Bidder or to give any notice to the Bidder or to enforce any security held by HGCL or to exercise, levy or enforce any distress, diligence or other process against the Bidder.

Notwithstanding anything contained hereinabove, our liability under this Guarantee is restricted to Rs. _____ (Rs. _____ only) and it shall remain in force until _____ [Date to be inserted on the basis of this NIT] with an additional claim period of thirty (30) days thereafter. We are liable to pay the guaranteed amount or any part thereof under this Bank Guarantee only if HGCL serves upon us a written claim or demand.

Signature _____

Name _____

Power of Attorney No. _____

For

_____ [Insert Name of the Bank] _____

Banker's Stamp and Full Address.

Dated this _____ day of __, 20__

Bank Contact Details & E Mail ID is to be provided

(TO BE EXECUTED ON STAMP PAPER OF APPROPRIATE VALUE)

INDEMNITY BOND

THIS INDEMNITY BOND IS made this..... day of 20..... by **M/s.**, a Company registered under the Companies Act, 1956/ having its Registered Office at(hereinafter called as "Contractor" or "Obligor" which expression shall include its successors) in favour of M/s, a Company incorporated under the Companies Act, 1956/2013 having its Registered Office at and their authorised representatives/Employer i.e., Hyderabad Growth Corridor Limited(HGCL) and its Project athereinafter called "....." which expression shall include its successors and assigns) :

WHEREAS/ HGCL has awarded to the Contractor a Contract vide its NOA NoDated & Contract Agreement No.....dated.....executed on (hereinafter called the "Contract").

NOW, THEREFORE, this Indemnify Bond witnesseth as follows:

1. That in consideration of Solar Photo Voltaic Plant as mentioned in the Contract, total valued at Rs. (Rupeesonly), valued as a summation of the referred contracts and handed over to the Contractor for the purpose of Performance of the Contract (s), the Contractor hereby undertakes to indemnify and shall keep M/sand their authorized representatives/Employer i.e., HGCL, indemnified, for the full value of the Solar Photo Voltaic Plant during the tenure of this contract or its extension if agreed to.
2. Contractor confirm that they are aware of all the contractual obligations regarding payment to all personnel and labours including that of the sub – vendors & sub –contractors of Contractor, involved in this project., as well as various contractual, statutory and non – statutory obligations as per legal requirements and have complied with the obligations to the best of information and knowledge.
3. Contractor also confirms, that they are in compliance of all the provisions and requirements, including but not limited to, the following acts/ laws/ provisions and other applicable statutory and non-statutory regulations:

- a. Contract Labor (Regulation & Abolition) Act 1970
 - b. Wages Act 1936
 - c. Minimum Wages Act 1948
 - d. Employer's Liability Act 1938
 - e. Workmen's Compensation Act 1923
 - f. Industrial Dispute Act 1947
 - g. Maturity Benefit Act 1961
 - h. Mines Act 1952
 - i. Employees State Insurance Act 1948.
4. Contractor, hereby confirm, that in the event any contractual or statutory obligation is found to be deficient in compliance or found non-complied, Contractor will indemnify M/s...../ HGCL and take necessary actions to comply with the same within the permissible time period.
 5. Contractor also confirm, should there be any charges or penalty, of any kind imposed on M/s.... or its authorized representatives i.e., HGCL, for non-compliance of legal or statutory requirements by Contractor during its tenure of the contract, Contractor agrees to indemnify M/s.....and HGCL against all damages, libel or suit, in full for the same on raising such a demand by M/s..... / HGCL and promises to pay the same within 30 days.
 6. We do further undertake that this above stated warranty is inclusive of materials and labour.
 7. That the Contractor is obliged and shall remain absolutely responsible for the safe O&M/ protection and custody of the Solar Photo Voltaic Project against all risks whatsoever till completion of O&M Contract in accordance with the terms of the Contract. The Contractor undertakes to keep M/s... ..and HGCL harmless against any loss or damage that may be caused to the Solar Photo Voltaic Plant.

IN WITNESS WHEREOF, the Contractor has hereunto set its hand through its authorised representative under the common seal of the Company, the day, month and year first above mentioned

For _____ Engineers Limited

(_____)

Designation: Director

Name:

Date:

(Authorized Representative of _____)

NO DEVIATION CONFIRMATION
(To be submitted on the Letter Head of the Bidding Company)

Ref.No. _____

Date: _____

From: _____ *(Insert name and address of Bidding Company)*

Tel.#: _____

Fax#: _____

E-mail address# _____

To

Hyderabad Growth Corridor Limited(A
State Government Company)

Sub: Bid for the "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 05 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India".

Dear Sir / Madam,

We understand that any 'deviation/ exception' in any form may result in rejection of bid. We, therefore, certify that we have not taken any 'exception/ deviation' anywhere in the bid and we agree that if any 'deviation/ exception' is mentioned or noticed, our bid may be rejected.

Place:

Date:

[Signature of Authorized Signatory of Bidder]

Name:

Designation:

Seal:

**DECLARATION REGARDING BANNING, LIQUIDATION,
COURT RECEIVERSHIP ETC.**

(To be submitted on the Letter Head of the Bidding Company)

Ref.No. _____

Date: _____

From: _____ (Insert name and address of Bidding Company)

Tel.#: _____

Fax#: _____

E-mail address# _____

To

Hyderabad Growth Corridor Limited(A
State Government Company)

Sub: Bid for the "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 05 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India".

Dear Sir / Madam,

We hereby confirm that we are not on Banning List by Employer/ HGCL or Public Sector Project Management Consultant due to "poor performance" or "corrupt and fraudulent practices" or any other reason or banned by Government department/ Public Sector on due date of submission of bid.

Further, we confirm that neither we nor our allied agency/(ies) (as defined in the Procedure for Action in case of Corrupt/ Fraudulent/ Collusive/ Coercive Practices) are on banning list of Employer/ HGCL or the Ministry of New & Renewable Energy.

We also confirm that we are not under any liquidation, court receivership or similar proceedings or 'bankruptcy'.

In case it comes to the notice of Employer/ HGCL that we have given wrong declaration in this regard, the same shall be dealt as 'fraudulent practices' and action shall be initiated as per the Procedure for action in case of Corrupt/Fraudulent/Collusive/Coercive Practices.

Further, we also confirm that in case there is any change in status of the declaration prior to award of contract, the same will be promptly informed to Employer/ HGCL by us.

Place:

Date:

[Signature of Authorized Signatory of Bidder]

Name:

Designation:

Seal:

DECLARATION

RESTRICTION ON PROCUREMENT FROM CERTAIN COUNTRIES:

MoF OM No 6/18/2019-PPD dated 23.07.2020

(This should be submitted on the Letter Head of the Bidding Company/ Each Member of Consortium)

Ref.No. _____

Date: _____

From: _____ (Insert name and address of Bidding Company/ Lead Member of Consortium)

Tel.#: Fax#:

E-mail address#

To

HGCL

Sub: Response to the Tender No datedfor the tender for
.....
.....

Dear Sir/ Madam,

This is with reference to attached order No. OM no. 6/18/2019-PPD dated 23rd July 2020 issued by Department of Expenditure, MoF, Govt of India.

Bidders are required to submit the relevant Declaration 1 And/or 2. Kindly provide the relevant declaration(s) pertaining to you.

We are hereby submitting the following declaration in this regard:

Any Bidder (including its JV partner/Consortium Member, wherever applicable) from a country which shares a land border with India will be eligible to bid in this tender only if bidder is registered with the Competent Authority which is the Registration Committee constituted by the Department for Promotion of Industry & Internal Trade (DPIIT) in line with the MoF OM No 6/18/2019-PPD dated 23.07.2020. Further the successful bidder shall not be allowed to subcontract supplies/services/works to any "Sub-contractor" under "Second Contract" i.e Service Contract, from a country which shares a land border with India unless such Subcontractor is registered with the competent as mentioned above.

The Contractor shall not be allowed to sub-contract works to any sub-contractor/ sub-vendor from a country which shares a land border with India unless such sub-contractor is registered with the competent Authority. However, the same shall not be applicable for "First Contract i.e. Supply Contract".

However, the said requirement of registration will not apply to bidders/sub- contractors from those countries (even if sharing a land border with India) to which the Government of India has extended lines of credit or in which the Government of India is engaged in development projects. Bidders may apprise themselves of the updated lists of such countries available in the website of the Ministry of External Affairs.

Declaration 1:

Certification for Tenders

"I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India; I certify that this bidder is not from such a country or, if from such a country, has been registered with the Competent Authority. I hereby certify that this bidder fulfils all requirements in this regard and is eligible to be considered. Where applicable, evidence of valid registration by the Competent Authority shall be attached]."

And

Declaration 2:

Certification for Tenders for Works involving possibility of sub-contracting (Applicable for Second Contract" i.e Service Contract)

"I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India and on sub-contracting to contractors from such countries; I certify that this bidder is not from such a country or, if from such a country, has been registered with the Competent Authority and will not sub-contract any work to a contractor from such countries unless such contractor is registered with the Competent Authority. I hereby certify that this bidder fulfils all requirements in this regard and is eligible to be considered. [Where applicable, evidence of valid registration by the Competent Authority shall be attached]."

We further declare that the above statement is true & correct. We are aware that if at any stage it is found to be incorrect, our response to the tender will be rejected.

Dated the _____ day of _____, 20....

Thanking you,

We remain,
Yours faithfully,

Name, Designation, Seal and Signature of Authorized Person in whose name Power of Attorney/ Board Resolution/ Declaration.

**FORMAT FOR PERFORMANCE SECURITY BANK
GAURANTEE**

(To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place of execution.)

Ref. _____ Date: _____

Bank Guarantee No. _____

In consideration of the [Insert name of the Bidder] (hereinafter referred to as 'Bidder') submitting the response to NIT inter alia for selection of the Project in response to the NIT No. _____ dated issued by Hyderabad Growth Corridor Limited (hereinafter referred to as HGCL) and HGCL considering such response to the NIT of [insert the name of the Bidder] as per the terms of the NIT, the _____ [insert name & address of bank] hereby agrees unequivocally, irrevocably and unconditionally to pay to HGCL at [Insert Name of the Place from the address of HGCL] forthwith on demand in writing from HGCL or any Officer authorized by it in this behalf, any amount upto and not exceeding Rupees ----- [Insert amount] only, on behalf of M/s.

_____ [Insert name of the Bidder] .

This guarantee shall be valid and binding on this Bank up to and including _____ [insert date of validity in accordance with NIT] and shall not be terminable by notice or any change in the constitution of the Bank or the term of contract or by any other reasons whatsoever and our liability hereunder shall not be impaired or discharged by any extension of time or variations or alternations made, given, or agreed with or without our knowledge or consent, by or between parties to the respective agreement.

Our liability under this Guarantee is restricted to Rs. _____ (Rs. _____ only). Our Guarantee shall remain in force until _____ [insert date of validity in accordance with NIT]. HGCL shall be entitled to invoke this Guarantee till _____ [Insert date which is 30 days after the date in the preceding sentence].

The Guarantor Bank hereby agrees and acknowledges that the HGCL shall have a right to invoke this BANK GUARANTEE in part or in full, as it may deem fit.

The Guarantor Bank hereby expressly agrees that it shall not require any proof in addition to the written demand by HGCL, made in any format, raised at the above-mentioned address of the Guarantor Bank, in order to make the said payment to HGCL.

The Guarantor Bank shall make payment hereunder on first demand without restriction or conditions and notwithstanding any objection by ----- [Insert name of the Bidder] and/or any other person.

The Guarantor Bank shall not require HGCL to justify the invocation of this BANK GUARANTEE, nor shall the Guarantor Bank have any recourse against HGCL in respect of any payment made hereunder.

This BANK GUARANTEE shall be interpreted in accordance with the laws of India and the courts at Hyderabad shall have exclusive jurisdiction.

The Guarantor Bank represents that this BANK GUARANTEE has been established in such form and with such content that it is fully enforceable in accordance with its terms as against the Guarantor Bank in the manner provided herein.

This BANK GUARANTEE shall not be affected in any manner by reason of merger, amalgamation, restructuring or any other change in the constitution of the Guarantor Bank.

This BANK GUARANTEE shall be a primary obligation of the Guarantor Bank and accordingly HGCL shall not be obliged before enforcing this BANK GUARANTEE to take any action in any court or arbitral proceedings against the Bidder, to make any claim against or any demand on the Bidder or to give any notice to the Bidder or to enforce any security held by HGCL or to exercise, levy or enforce any distress, diligence or other process against the Bidder.

Notwithstanding anything contained hereinabove, our liability under this Guarantee is restricted to Rs. _____ (Rs. _____ only) and it shall remain in force until _____ [Date to be inserted on the basis of this NIT] with an additional claim period of thirty (30) days thereafter. We are liable to pay the guaranteed amount or any part thereof under this Bank Guarantee only if HGCL serves upon us a written claim or demand.

Signature _____

Name _____

Power of Attorney No. _____

For

_____[Insert Name of the Bank]____

Banker's Stamp and Full Address.

Dated this _____ day of __, 20__

Bank Contact Details & E Mail ID is to be provide

**DECLARATION REGARDING THE MANDATORY PROCUREMENT
OF SOLAR MODULES &
INVERTERS FROM CLASS I LOCAL SUPPLIERS**

(To be submitted on the Letter Head of the Statuary Auditor/Cost Auditor of the company (In case of companies) or from a practicing cost accountant or practicing chartered accountant (In respect of suppliers other than companies))

Reference: Ministry of New & Renewable Energy (MNRE) Order No 283/22/2019-GRID SOLAR dated 23rd Sep 2020 & any amendments thereof, for the Public Procurement (Preference to Make in India) to provide for Purchase Preference (linked with local content) in respect of Renewable Energy (RE) Sector.

Reference 2: Department for Promotion of Industry and Internal Trade (DPIIT) Notification No. P- 45021/2/2017-PP (BE-II) dated 4th June, 2020.

Ref.No. _____

Date:

To
Hyderabad Growth Corridor Limited(A
State Government Company)

Sub: Bid for “

Dear Sir / Madam,

We hereby confirm that in line with the mandate of *Ministry of New & Renewable Energy (MNRE) Order No 283/22/2019-GRID SOLAR dated 23rd Sep 2020 for the Public Procurement (Preference to Make in India) & any amendments thereof, to provide for Purchase Preference (linked with local content) in respect of Renewable Energy (RE) Sector & Department for Promotion of Industry and Internal Trade (DPIIT) Notification No. P-45021/2/2017-PP (BE-II) dated 4th June, 2020*, we hereby declare that:

- a) The procurement of **Solar Modules** for the subject tender will be done from Class I Local suppliers only
- b) The procurement of **Solar Inverters** for the subject tender will be done from Class I Local suppliers only.

- c) The percentage of Local content in the procurement of mentioned Solar Modules is..... % (In words)
- d) The percentage of Local content in the procurement of mentioned Solar Inverters is..... % (In words).

A Class-I local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, has local content equal to or more than 50%, as defined under the above said Orders.

False declaration will be in the breach of the code of integrity under Rule 175(1)(i)(h) of the General Financial rules for which a bidder or its successors can be debarred for up to two years as per rule 151 (iii) of the general Financial rules along with such other actions as may be permissible under the law.

Also, In case it comes to the notice of Employer/ HGCL that we have given wrong declaration in this regard, the same shall be dealt as 'wrong declaration under fraudulent practices' and action shall be initiated as per the Procedure for action in case of Corrupt/Fraudulent/Collusive/Coercive Practices.

(Signature and stamp (on each page) of Authorized Signatory of Bidding Company.

Name:

Date:

Place:

Signature and stamp (on each page) of Statuary Auditor/Cost Auditor/ practicing cost accountant or practicing chartered accountant of Bidding Company.

Name:

Date:

Place:

Form F-13**BIDDER'S EXPERIENCE****(To be submitted on the Letter Head of the Bidding Company)**

Ref.No. _____

Date: _____

From: _____ (Insert name and address of Bidding Company)

Tel.#:

Fax#:

E-mail address#

To

Hyderabad Growth Corridor Limited(A
State Government Company)

Sub: Bid for "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 05 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India".

Dear Sir / Madam,

Sl. No	Description of the Services	LOA /WO No. and date	Full Address & phone nos. of Client. Name, designation and address of Engineer/ Officer-in-Charge (for cases other than purchase)	Postal	Value of Contract/ Order (Specify Currency Amount)	Date of Commencement of Services	Scheduled Completion Time (Months)	Date of Actual Completion	Reasons for delay in execution, if any
(1)	(2)	(3)	(5)		(6)	(7)	(8)	(9)	(10)

Place:

Date:

[Signature of Authorized Signatory of Bidder]

Name:

Designation:

Seal

Format for Disclosure for ALMM compliance

[On the letter head of Bidding Company/ Lead Member in a Bidding JV/Consortium]

Disclosure

Ref.No. _____

Date: _____

From: _____ (Insert name and address of Bidding Company/ Lead Member of JV/Consortium)

To

Hyderabad Growth Corridor Limited(A

State Government Company)

Sub: Response to Tender No.dated for the “Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 05 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India”

Dear Sir/ Madam,

We declare that we are fully aware of the binding provisions of the ALMM Order and the Lists(s) thereunder, while quoting the for the subject tender of “Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 5years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India”.

We further understand that the List-I (Solar PV Modules) of ALMM Order, Annexure-I of the OM, issued by MNRE on 10th March, 2021 will be updated by MNRE from time to time. We also understand that the Modules to be procured for this project, shall be from the List-I of the ALMM Order applicable on the date of invoicing of such modules.

We also further understand and accept that we shall be liable for penal action, including but not limited to blacklisting and invocation of Performance Bank Guarantee, if we are found not complying with the provisions of ALMM Order, including those mentioned above.

Dated the _____ day of _____, 20....

Thanking you,

We remain,

Yours faithfully,

Name, Designation, Seal and Signature of Authorized Signatory

**DECLARATION REGARDING THE FULFILLMENT OF
DOMESTIC CONTENT REQUIREMENT CLAUSE**

Ref.No. _____ To
Hyderabad Growth Corridor Limited(A State Government Company)

Date: _____

Sub: Bid for "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 05 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India".

Dear Sir / Madam,

We, M/s..... (name of the bidder) hereby certify and declare the following:

The cells and modules to be used in the subject project shall be manufactured domestically and comply with the Domestic Content Requirements as laid down by Ministry of New and Renewable Energy and incorporated in the Guidelines and the tender documents under the CPSU Scheme.

The Cells & modules shall be manufactured indigenously, in factories registered in India. The manufacturing process shall be carried out in India from the stage of finished cells till modules and semi-finished/blue cells shall not be used in this process.

The above information is complete and accurate in all material respects, and there is no material information omitted from this Certificate that makes the information contained herein misleading or inaccurate.

In case of any non-compliance in respect of meeting the DCR requirement as laid down by Ministry of New and Renewable Energy, we shall be fully responsible for payment of any penalty and / or legal action.

We shall maintain, and present upon request, to the Inspection committee of MNRE/HGCL, any documentation necessary in support of this certificate. Further, we shall not bar the inspection committee of MNRE/HGCL from visiting our factories/installations in relation to the agreement between our Company and HGCL / MNRE or their representative.

In case of any discrepancies found upon the inspection conducted by the committee, it may be dealt as per the decisions of MNRE/HGCL.

Signature of the Authorized Person

Name:

Date:

Place:

FORMAT OF CHARTERED ACCOUNTANT
CERTIFICATE FOR FINANCIAL
CAPABILITY OF THE BIDDER

(To be submitted on the Letter Head of the Chartered Accountant)

Ref.No. _____ Date: _____

To

Hyderabad Growth
Corridor Limited(A
State Government
Company)

Sub: Bid for "Tender for Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of 13 MW (AC) Solar PV Power Plant at different sites including 05 years plant O&M at Hyderabad Growth Corridor Limited, Telangana, India".

Dear Sir / Madam,

We have verified the Annual Accounts and other relevant records of M/s... Name
of the bidder) and certify the following

Further, we certify that the Financially Evaluated Entity (ies) had an Annual Turnover

A. ANNUAL TURNOVER OF LAST 10 YEARS:

Year	Amount (Currency)
Year 1:	
Year 2:	
Year 3:	
Year 4:	
Year 5:	
Year 6:	
Year 7:	
Year 8:	
Year 9:	
Year 10:	

And

Net worth (strike out whichever is not applicable) of INR..... Crore computed as per instructions provided
in this tender based on unconsolidated audited annual accounts as per last FY.

B. FINANCIAL DATA FOR LAST AUDITED FINANCIAL YEAR:

Description	Year_____
	Amount (Currency)
1. Current Assets	
2. Current Liabilities	
3. Working Capital (Current Assets- Current liabilities)	
4. Net Worth (As mentioned under Annexure to BDS)	

Yours faithfully

(Signature and stamp (on each page) of Authorized Signatory of Bidding Company.

Name:
.....
.....

Date:
.....
.....

Place:
.....
.....

Signature and stamp (on each page) of Chartered Accountant/Statutory Auditors of Bidding Company.

Name:
.....
.....

Date:
.....
.....

Place:

.....
.....

Notes:

Audited consolidated annual accounts of the Bidder may also be used for the purpose of financial criteria provided the Bidder has at least 50% equity in each company whose accounts are merged in the audited consolidated accounts and provided further that the financial capability of such companies (of which accounts are being merged in the consolidated accounts) shall not be considered again for the purpose of evaluation of the Bid.

FORMAT FOR JOINT VENTURE AGREEMENT

***(To be stamped in accordance with Stamp Act, the Non Judicial Stamp Paper of
Appropriate Value should be in the name of the Joint Venture)***

CONSORTIUM AGREEMENT AMONGST M/s & M/s for Bid
Specification No. OF Hyderabad Growth Corridor Limited

This consortium agreement executed on this day Month of Two thousand
..... amongst M/S..... a company incorporated under the laws of
.....(country) and having its registered office at (herein after called
the "Lead Company" which expression shall include its successors, executors and permitted
assigns) and M/S....., a company incorporated under the laws of
.....(country) and having its registered office at (herein after called
the Constituent Company which expression shall include its successors, executors and permitted
assigns), for the purpose of making a bid and entering into a contract (In case of award) against
the specification no. " for(Name of
the work) for HYDERABAD GROWTH CORRIDOR LIMITED HGCL Office Building, 2nd
floor, Khajaguda Road, Adjacent to ORR, Nanakramguda, Hyderabad-500104 (hereinafter called
the "Owner"). The leader of the consortium shall abide by the general terms and conditions
specified by the Purchaser. WHEREAS the Owner invited bids as per the above mentioned
specifications for the design, manufacture, supply and erection, testing and commissioning of
Equipment's/Materials stipulated in the bidding documents under subject package
.....(Name of the work)

AND WHEREAS Clause 2, Section-ITB and BDS (documents establishing the Qualification of
Bidder) & Qualification Criteria in Annexure to BDS forming part of the Tender Documents, inter-
alia stipulates that an Undertaking of two or more qualified manufacturers as partners, meeting the
requirements of Qualification Criteria in Annexure to BDS, as applicable may bid, provided, the
Consortium fulfills all other requirements under Qualification Criteria in Annexure to BDS and in
such a case, the Bid Forms shall be signed by all the partners so as to legally bind all the Partners
of the Consortium, who will be jointly and severally liable to perform the Contract and all
obligations hereunder.

The above clause further states that the consortium agreement shall be attached to the bid and the
contract performance guarantee will be as per the format enclosed with the bidding document
without any restriction or liability for either party.

AND WHEREAS the bid has been submitted to the Owner vide proposal No..... dated
..... by Lead Company based on the Constituent agreement between all the consortium
under these presents and the bid in accordance with the requirements of Specifications
(Qualification Requirements).

NOW THIS INDENTURE WITNESSTH AS UNDER

In consideration of the above premises and agreements all the partners to this consortium members do hereby now agree as follows:

1. In Consideration of the award of the Contract by the Owner to the Consortium members, we , the members to the consortium agreement do hereby agree that Ms.....shall act as Lead Company and further declare and confirm that we shall jointly and severally be bound up to the Owner for the successful performance of the contract and shall be fully responsible for the design, manufacture, supply and successful performance of the equipment's in accordance with the Contract.
2. In case of any breach of the said contract by the Lead company or other constituent of the Consortium agreement, the Lead company do hereby agree to be fully responsible for the successful performance of the contract and to carry out all the obligations and responsibilities under the contract in accordance with the requirements of the Contract.
3. Further, if the Owner suffers any loss or damage on account of any breach in the contract or any shortfall in the performance of the equipment in meeting the performance guaranteed as per the specification in terms of the contract, the Lead company of these presents undertake to promptly make good such loss or damages caused to the Owner, on its demand without any demur. It shall not be necessary or obligatory for the Owner to proceed against Lead Partner to these presents before proceeding against or dealing with the other Partner (s).
4. The Financial liability of the members of the consortium agreement to the Owner, with respect to any of the claims arising out of performance or non-performance of the obligations set forth in the said agreement, read in conjunction with the relevant conditions of the contract shall, however, not be limited in any way so as to restrict or limit the liabilities of any of the members of the agreement.
5. It is expressly understood and agreed between the members to this consortium agreement that the responsibilities and obligations of each of the partners shall be as delineated hereunder (To be incorporated suitably by the member to this agreement). However, each member of the consortium will be severally and jointly responsible for the faithful performance of the contract. It is further agreed by the member that the above sharing of responsibilities and obligations shall not in any way be a limitation of joint and several responsibilities of the members under this contract.

SL NO	Name of the Lead Bidder / Consortium Partner	Scope of work / participation
01		
02		

We further declare ourselves bound to execute our scope of work/participation as stated above.

6. This Consortium agreement shall be constructed and interpreted in accordance with the laws of India and the courts of Hyderabad shall have the exclusive jurisdiction in all matters arising there under.
7. In case of an award of a Contract, we the Consortium members do hereby agree that we shall be jointly & severally responsible for the works respectively executed and Lead company shall furnishing a contract performance security from a bank in favor of the PURCHASER in the forms acceptable to purchaser for value of 5% of the Contract price in the current / Currencies of the contract or Five Percent(5%) of the updated estimate cost if the Contract Price is less than the updated estimate cost of the tender.
8. It is further agreed that the Consortium agreement shall be irrevocable and shall form an integral part of the contract and shall continue to be enforceable till the owner discharges the same. It shall be effective from the date first mentioned above for all purposes and intents.

IN WITNESS WHEREOF the partners to the Consortium have through their authorized representatives executed these presents and affixed Common seals of their companies, on the day, month and year first mentioned above.

For Lead Company	For Constituent 1
Name:	Name:
Designation:	Designation:
Common Seal of the Company	Common Seal of the Company

WITNESS

1)

Signature.....
Name.....
Designation.....

2)

Signature.....
Name.....
Designation.....

FORMAT FOR POWER OF ATTORNEY
FOR JOINT VENTURE
AGREEMENT

*(To be stamped in accordance with Stamp Act, the Non-Judicial Stamp Paper of
Appropriate Value should be in the name of the Joint
Venture)*

KNOW ALL MEN BY THESE PRESENTS THAT WE, the Partners whose details are given hereunder

..... have formed a Joint Venture under the laws of

..... and having our Registered Office(s)/ Head Office(s) at

..... (hereinafter called the 'Joint Venture' which expression shall unless repugnant to the context or meaning thereof, include its successors, administrators and assigns) acting through M/s

..... being the Partner in-charge do hereby constitute, nominate and appoint M/s..... a Company incorporated under the

laws of and having its Registered/ Head Office at

..... as our duly constituted lawful Attorney (hereinafter called "Attorney" or "Authorised Representative" or "Partner In-charge") to exercise all or any of the powers for and on behalf of the Joint Venture in regard to Tender No..... Package the bids for which

have been invited by HGCL, (hereinafter called the 'Employer') to undertake the following acts:

- i) To submit proposal and participate in the aforesaid Bid Specification of the Employer on behalf of the "Joint Venture".
- ii) To negotiate with the Employer the terms and conditions for award of the Contract pursuant to the aforesaid Bid and to sign the Contract with the Employer for and on behalf of the "Joint Venture".
- iii) To do any other act or submit any document related to the above.
- iv) To receive, accept and execute the Contract for and on behalf of the "Joint Venture".

It is clearly understood that the Partner In-charge (Lead Partner) shall ensure performance of the Contract(s) and if one or more Partner fail to perform their respective portions of the Contract(s), the same shall be deemed to be a default by all the Partners.

It is expressly understood that this Power of Attorney shall remain valid binding

and irrevocable till completion of the Defect Liability Period in terms of the Contract.

The Joint Venture hereby agrees and undertakes to ratify and confirm all the whatsoever the said Attorney/ Authorised Representatives/ Partner in-charge quotes in the bid, negotiates and signs the Contract with the Employer and/or proposes to act on behalf of the Joint Venture by virtue of this Power of Attorney and the same shall bind the Joint Venture as if done by itself.

IN WITNESS, THEREOF the Partners Constituting the Joint Venture as aforesaid have executed these presents on this day of under the Common Seal(s) of their Companies.

for and on behalf of the

Partners of Joint Venture

.....

.....

.....

The Common Seal of the above Partners of the Joint Venture:

The Common Seal has been affixed there unto in the presence of:

WITNESS

1. Signature.....

Name

Designation

Occupation

2. Signature.....

Name

Designation

Occupation

Note:

1. For the purpose of executing the Agreement, the non-judicial stamp papers of appropriate value shall be purchased in the name of Joint Venture.

2. The Agreement shall be signed on all the pages by the authorised representatives of each of the partners and should invariably be witnessed.

FORMAT FOR CONSORTIUM AGREEMENT

*(To be stamped in accordance with Stamp Act, the Non-Judicial Stamp Paper of
Appropriate Value should be in the name of the Consortium)*

THIS JOINT DEED OF UNDERTAKING executed on this..... day of.....Two
Thousandand..... by a company incorporated under the laws
No.1" which expression shall include its successors & executors) and M/s.....a
companyincorporated under the laws of and having its
Registered Office at
..... (hereinafter called the "Party No.2" which expression shall include its
successors & executors) and M/s.. a Company incorporated under the
laws of
and having its Registered Office at (hereinafter called the "Party No.3" which
expression shall include its successors & executors) for the purpose of making a bid and
entering into a contract [hereinafter called the "Contract" {in case of award}] against the Tender
No..... for (insert name of the package alongwith project name) of
HGCL, a Company incorporated under the Companies Act of 2013 having its registered office at
(hereinafter called the "Employer").

WHEREAS the Party No.1, Party No.2 and Party No.3 have entered into an Agreement
dated.....

AND WHEREAS the Employer invited bids as per the above-mentioned Specification for the
design, manufacture, supply, erection, testing and commissioning including O & M of Equipment/
Materials stipulated in the Tender Documents under (insert name of the package
alongwith project name)
.....

AND WHEREAS Clause 2, Section-ITB and BDS (documents establishing the Qualification of
Bidder) & Qualification Criteria in Annexure to BDS forming part of the Tender Documents, inter-
alia stipulates that an Undertaking of two or more qualified manufacturers as partners, meeting
the requirements of Qualification Criteria in Annexure to BDS, as applicable may bid, provided,
the Consortium fulfills all other requirements under Qualification Criteria in Annexure to BDS and
in such a case, the Bid Forms shall be signed by all the partners so as to legally bind all the
Partners of the Consortium, who will be jointly and severally liable to perform the Contract and all
obligations hereunder.

The above clause further states that this Undertaking shall be attached to the bid and the
Contract Performance Security will be as per the format F-9 enclosed with the Tender
Documents without any restrictions or liability for either party.

AND WHEREAS the bid is being submitted to the Employer vide proposal No..... dated
.....
by Party No.1 based on this Undertaking between all the parties; under these presents and the
bid in accordance with the requirements of Clause 2, Section-ITB and BDS (documents
establishing the Qualification of Bidder) & Qualification Criteria in Annexure to BDS, has been
signed by all the parties.

NOW THIS UNDERTAKING WITNESSETH AS UNDER:

In consideration of the above premises and agreements all the parties of this Deed of Undertaking do hereby declare and undertake:

1. In requirement of the award of the Contract by the Employer to the Consortium Partners, we, the Parties do hereby undertake that M/s... the Party No.1, shall act as Lead Partner and further declare and confirm that we the parties to the Consortium shall jointly and severally be bound unto the Employer for the successful performance of the Contract and shall be fully responsible for the design, manufacture, supply and successful performance of the equipment in accordance with the Contract:
2. In case of any breach or default of the said Contract by any of the parties to the Consortium, the party(ies) do hereby undertake to be fully responsible for the successful performance of the Contract and to carry out all the obligations and responsibilities under the Contract in accordance with the requirements of the Contract.
3. Further, if the Employer suffers any loss or damage on account of any breach in the Contract or any shortfall in the performance of the equipment in meeting the performances guaranteed as per the specification in terms of the Contract, the Party(ies) of these presents undertake to promptly make good such loss or damages caused to the Employer, on its demand without any demur. It shall not be necessary or obligatory for the Employer to proceed against Lead Partner to these presents before proceeding against or dealing with the other Party(ies), the Employer can proceed against any of the parties who shall be jointly and severally liable for the performance and all other liabilities/ obligations under the Contract to the Employer.
4. The financial liability of the Parties of this Deed of Undertaking to the Employer, with respect to any of the claims rising out of the performance or non-performance of the obligations set forth in this Deed of Undertaking, read in conjunction with the relevant conditions of the Contract shall, however not be limited in any way so as to restrict or limit the liabilities or obligations of any of the Parties of this Deed of Undertaking.
5. It is expressly understood and agreed between the Parties to this Undertaking that the responsibilities and obligations of each of the Parties shall be as delineated in Appendix - I *(to be suitably appended by the Parties along with this Undertaking in its bid)* to this Deed of Undertaking. It is further undertaken by the parties that the above sharing of responsibilities and obligations shall not in any way be a limitation of joint and several responsibilities of the Parties under the Contract.
6. It is also understood that this Undertaking is provided for the purposes of undertaking joint and several liabilities of the partners to the Consortium for submission of the bid and performance of the Contract and that this Undertaking shall not be deemed to give rise to any additional liabilities or obligations, in any manner or any law, on any of the Parties to this Undertaking or on the Consortium, other than the express provisions of the Contract.
7. This Undertaking shall be construed and interpreted in accordance with the provisions of the Contract.
8. In case of an award of a Contract, we the parties to this Deed of Undertaking do hereby agree that we shall be jointly and severally responsible for furnishing a

Contract Performance Security from a bank in favour of the Employer in the currency/ currencies of the Contract.

9. It is further agreed that this Deed of Undertaking shall be irrevocable and shall form an integral part of the bid and shall continue to be enforceable till the Employer discharges the same or upon the completion of the Contract in accordance with its provisions, whichever is earlier. It shall be effective from the date first mentioned above for all purposes and intents.

IN WITNESS, WHEREOF, the Parties to this Deed of Undertaking have through their authorized representatives executed these presents and affixed Common Seals of their companies, on the day, month and year first mentioned above.

Common Seal of
has been affixed in my/ our
presence pursuant to Board of
Director's Resolution dated

For Lead Partner (Party No.-1)
For and on behalf of M/s
.....

Name

Designation

Signature

(Signature of the authorized
representative)

WITNESS :

I.

II.

Common Seal of
has been affixed in my/our
presence pursuant to Board of
Director's Resolution dated

For Party No.-2
For and on behalf of
M/s.....

Name

(Signature of the authorized
representative)

Designation

Signature

WITNESS :

I.

II.

Common Seal of
has been affixed in my/ our
presence pursuant to Board of
Director's Resolution dated

For Party No.-3
For and on behalf of M/s.
.....

Name

Designation

Signature

(Signature of the authorized
representative)

WITNESS :

I.

II.

Note:

1. For the purpose of executing the Joint Deed of Undertaking, the non-judicial stamp papers of appropriate value shall be purchased in the name of Consortium.
2. The Undertaking shall be signed on all the pages by the authorized representatives of each of the partners and should invariably be witnessed.

**FORMAT FOR POWER OF ATTORNEY OF
CONSORTIUM AGREEMENT**

*(To be stamped in accordance with Stamp Act, the Non-Judicial Stamp Paper of
Appropriate Value should be in the name of the Consortium)*

KNOW ALL MEN BY THESE PRESENTS THAT WE, the Partners whose details are given hereunder

..... have formed a Consortium under the laws of

..... and having our Registered Office(s)/ Head Office(s) at

..... (hereinafter called the 'Consortium' which expression shall unless repugnant to the context or meaning thereof, include its successors, administrators and assigns) acting through M/s

..... being the Partner in-charge do hereby constitute, nominate and appoint M/s..... a Company incorporated under the laws of

..... and having its Registered/ Head Office at

..... as our duly constituted lawful Attorney (hereinafter called "Attorney" or "Authorised Representative" or "Partner In-charge") to exercise all or any of the powers for and on behalf of the Consortium in regard to Tender No..... Package..... the bids for which have

been invited by HGCL, (hereinafter called the 'Employer') to undertake the following acts:

- i) To submit proposal and participate in the aforesaid Bid Specification of the Employer on behalf of the "Consortium".
- ii) To negotiate with the Employer the terms and conditions for award of the Contract pursuant to the aforesaid Bid and to sign the Contract with the Employer for and on behalf of the "Consortium".
- iii) To do any other act or submit any document related to the above.
- iv) To receive, accept and execute the Contract for and on behalf of the "Consortium".

It is clearly understood that the Partner In-charge (Lead Partner) shall ensure performance of the Contract(s) and if one or more Partner fail to perform their respective portions of the Contract(s), the same shall be deemed to be a default by all the Partners.

It is expressly understood that this Power of Attorney shall remain valid binding and irrevocable till completion of the Defect Liability Period in terms of the Contract.

The Consortium hereby agrees and undertakes to ratify and confirm all the whatsoever the said Attorney/ Authorized Representatives/ Partner in-charge

quotes in the bid, negotiates and signs the Contract with the Employer and/or proposes to act on behalf of the Consortium by virtue of this Power of Attorney and the same shall bind the Consortium as if done by itself.

IN WITNESS, THEREOF the Partners Constituting the Consortium as aforesaid have executed these presents on this day of under the Common Seal(s) of their Companies.

for and on behalf of the Partners of Consortium

.....
.....
.....

The Common Seal of the above Partners of the Consortium:

The Common Seal has been affixed there unto in the presence of:

WITNESS

1. Signature.....
Name
Designation
Occupation
2. Signature.....
Name
Designation
Occupation

Note:

1. For the purpose of executing the Agreement, the non-judicial stamp papers of appropriate value shall be purchased in the name of Consortium.
2. The Agreement shall be signed on all the pages by the authorized representatives of each of the partners and should invariably be witnessed.

e-BANKING FORMAT

(To be submitted on the Letter Head of the Bidder)

1. Bidder/ Customer
Name :
2. Bidder/ Customer
Code :
3. Bidder/ Customer
Address
:
4. Bidder/ Customer
E-mail ID
5. Particulars of Bank Account
 - a) Name of Bank :
 - b) Name of Branch :
 - c) Branch Code :
 - d) Address :
 - e) Telephone Number :
 - f) Type of Account :
 - g) Account Number :
 - h) RTGS IFSC Code :
 - i) NEFT IFSC Code :
- j) 9 digit MICR code :

I/ We hereby authorize Hyderabad Growth Corridor Limited to release any amount due to me/ us in the bank account as mentioned above. I/ We hereby declare that the particulars given above are correct and complete. If the transaction is delayed or lost because of incomplete or incorrect information, we would not hold the Hyderabad Growth Corridor Limited responsible.

(Signature of Vendor/ Customer)

BANK CERTIFICATE

We certify that ----- has an Account no ----- with us and we confirm that the details given above are correct as per our records.

Bank stamp

Date
officer of bank)

(Signature of authorized

**PROFORMA OF BANK GUARANTEE FOR
ADVANCEPAYMENT**

*(To be stamped in accordance with Stamp Act, the Non-Judicial Stamp Paper of
Appropriate Value should be in the name of the issuing Bank)*

Bank Guarantee No.:

.....

Date:

.....

.....

NOA/ Contract No.....

..... *[Name of Contract]*

To:

Hyderabad Growth Corridor
Limited(A State Government
Company)

Dear Sir / Madam,

We refer to the Contract ("the Contract") signed on*(insert date of the Contract)*.....between you
and M/s *(Name of Contractor)*

(or)

vide notification of award issued on *(insert date of the notification of award)* by you to
M/s

..... *(Name of Contractor)* having its Principal place of business at*(Address of Contractor)* and Registered Office at *(Registered address of Contractor)*

..... ("the Contractor") concerning

.....

(Indicate brief scope of work) for the complete execution of the *(insert name of Package alongwith name of the Project)*

Whereas, in accordance with the terms of the said Contract, the HGCL has agreed to pay or
cause to be paid to the Contractor an interest bearing Advance Payment against furnishing of an
irrevocable bank guarantee for an amount of *(Amount in figures and words)*which is equal
to 110% of the amount of Advance Payment.

By this letter, we, the undersigned, *(insert name & address of the issuing bank)*, a
Bank (which expression shall include its successors, administrators, executors and assigns)
organized under the laws of and having its Registered/ Head Office at

..... (insert address of registered office of the bank) do hereby irrevocably guarantee repayment of.....(Amount of the bank guarantee in figures and words)..... upon the first demand of the Employer without cavil or argument in the event that the Contractor fails to commence or fulfill its obligations under the terms of the said Contract for reasons whatsoever.

Provided always that the Bank's obligation shall be limited to the amount of this Bank guarantee or an amount equal to the outstanding balance of the Advance Payment and the accrued interest on the Advance Payment, taking into account such amounts, which have been repaid by the Contractor from time to time in accordance with the terms of payment of the said Contract, as certified by you.

This Guarantee shall remain in full force from the date upon which the said Advance Payment is received by the Contractor up to sixty (60) days beyond the date on which the entire Advance Payment along with the accrued interest if any due thereon has been fully adjusted in terms of the Contract i.e., up to sixty (60) days beyond the date of Completion of the Facilities under the Contract. This Guarantee may be extended from time to time, as may be desired by M/s Hyderabad Growth Corridor Limited on whose behalf this Guarantee has been issued.

Any claims to be made under this Guarantee must be received by the Bank during its period of validity, i.e. up to sixty (60) days beyond the date of Completion of the Facilities by the Employer i.e. up to and inclusive of..(dd/mm/yy).

Notwithstanding anything contained herein:

1. Our liability under this Bank Guarantee shall not exceed_____ (value in figures)_____
[_____(value in words) _____].
2. This Bank Guarantee shall be valid upto _____(validity date)_____.
3. We are liable to pay the guaranteed amount or any part thereof under this Bank Guarantee only & only if we receive a written claim or demand on or before _____ (validity date)_____.

For and on behalf of the Bank

[Signature of the authorised
signatory(ies)] Signature_____

Name_____

Designation_____

POA Number_____

Contact Number(s): Tel. _____ Mobile _____

Fax Number_____

email _____

Common Seal of the Bank_____W

itness:Signature_____

Name_____

Address_____

Contact Number(s): Tel.Mobile_____

email _____

Note:

1. For the purpose of executing the Bank Guarantee, the non-judicial stamp papers of appropriatevalue shall be purchased in the name of Bank who issues the 'Bank Guarantee'.
2. The Bank Guarantee shall be signed on all the pages by the Bank Authorities indicating their POAnos. and should invariably be witnessed.

LIST OF BANKS*(For Reference Purpose)*

SBI AND ASSOCIATES	OTHER PUBLIC SECTOR BANKS
1. State Bank of India	1. IDBI Bank Limited
2. State Bank of Indore	FOREIGN BANKS
	1. Bank of America NA
	2. Bank of Tokyo Mitsubishi UFJ Ltd.
	3. BNP Paribas
	4. Calyon Bank
	5. Citi Bank N.A.
NATIONALISED BANKS	6. Deutsche Bank A.G
1. Allahabad Bank	7. The HongKong and Shanghai Banking Corpn. (HSBC) Ltd.
2. Andhra Bank	8. Standard Chartered Bank
3. Bank of India	9. Societe Generale
4. Bank of Maharashtra	10. Barclays Bank
5. Canara Bank	11. Royal Bank of Scotland (RBS)
6. Central Bank of India	12. Bank of Nova Scotia
7. Corporation Bank	13. Development Bank of Singapore (DBS)
8. Dena Bank	14. Credit Agricole Corporate and Investment Bank
9. Indian Bank	SCHEDULED PRIVATE BANKS
10. Indian Overseas Bank	1. Federal Bank Limited
11. Oriental Bank of Commerce	2. Kotak Mahindra Bank Limited
12. Punjab National Bank	3. Axis Bank Limited
13. Punjab & Sind Bank	4. ICICI Bank Limited
14. Syndicate Bank	5. HDFC Bank Limited
15. Union Bank of India	6. Yes Bank Limited
16. United Bank of India	7. IDFC Bank Limited
17. UCO Bank	8. IndusInd Bank
18. Vijaya Bank	9. Karur Vysya Bank
19. Bank of Baroda	10. South Indian Bank
	11. RBL

SHAREHOLDING CERTIFICATE

(To be submitted on the Letter Head of the Bidder)

Name of the Equity Holder	Type and Number of Shares Owned	% of Equity Holding	Extent of Voting Rights

Yours faithfully

(Signature and Stamp of Authorized Signatory of Bidder)

Name:

Date:

Place:

(Signature and Stamp of Company Secretary/ Director/ Chartered Accountant)

POWER OF ATTORNEY FOR BIDDING
COMPANY

*(To be stamped in accordance with Stamp Act, the Non-Judicial Stamp Paper of
Appropriate Value should be in the name of the Bidder)*

Know all men by these presents, We (name and address of the registered office of the Bidding Company as applicable) do hereby constitute, appoint and authorize Mr./Ms. (name & residential address) who is presently employed with us and holding the position of as our true and lawful attorney, to do in our name and on our behalf, all such acts, deeds and things necessary in connection with or incidental to submission of our Bid for (insert details of Tender) in response to the Tender No dated issued by Hyderabad Growth Corridor Limited (HGCL), Hyderabad including signing and submission of the Bid and all other documents related to the Bid, including but not limited to undertakings, letters, certificates, acceptances, clarifications, guarantees or any other document which the HGCL may require us to submit. The aforesaid Attorney is further authorized for making representations to the HGCL, Hyderabad and providing information/ responses to HGCL, Hyderabad representing us in all matters before HGCL, Hyderabad and generally dealing with HGCL, Hyderabad in all matters in connection with Bid till the completion of the bidding process as per the terms of the above mentioned Tender.

We hereby agree to ratify all acts, deeds and things done by our said attorney pursuant to this Power of Attorney and that all acts, deeds and things done by our aforesaid attorney shall be binding on us and shall always be deemed to have been done by us.

All the terms used herein but not defined shall have the meaning ascribed to such terms under the Tender.

Signed by the within named

..... (Insert the name of the

executant company) through the hand of

Mr.

duly authorized by the Board to issue such Power of Attorney

Dated this day of

Accepted

.....
Signature of Attorney

(Name, designation and address of the Attorney)

Attested

.....

.....

**(Signature of the
executant)**

(Name, designation and address of the executant)

.....

Signature and stamp of Notary of the place of execution

**Common seal of has been affixed in my/ our presence
pursuant to Board of Director's Resolution dated.....**

WITNESS

1.

.....

.....

(Signature)

Name.....

Designation

2.

.....

.....

(Signature)

Name.....

Designation

Notes:

The mode of execution of the power of attorney should be in accordance with the procedure, if any, laid down by the applicable law and the charter documents of the executant(s) and the same should be under common seal of the executant affixed in accordance with the applicable procedure. Further, the person whose signatures are to be provided on the power of attorney shall be duly authorized by the executant(s) in this regard.

The person authorized under this Power of Attorney, in the case of the Bidding Company/ Lead Member being a public company, or a private company which is a subsidiary of a public company, in terms of the Companies Act, 1956, with a paid-up

share capital of more than Indian Rupees Five Crores, should be the Managing Director/ Whole Time Director/ Manager appointed under section 269 of the Companies Act, 1956. In all other cases the person authorized should be a director duly authorized by a board resolution duly passed by the Company.

Also, wherever required, the executant(s) should submit for verification the extract of the chartered documents and documents such as a Board resolution/ power of attorney, in favour of the person executing this power of attorney for delegation of power hereunder on behalf of the executant(s).

Form F-25

To be executed by the successful bidder and shall form part of the Contract Agreement.

This **Non-Disclosure Agreement** ("Agreement") is made effective on this.....("Effective Date")

By and Between,

HGCL, which expression includes its successors and assigns)having its Registered Office at Nanakramguda, Hyderabad - 500 004. Telangana State, India, hereinafter referred as Disclosing Party / HGCL.

And

.....having its registered office atwhich expression includes successors and assigns, hereinafter referred as Receiving Party / Contractor.

Both collectively referred to as "Parties" and individually as "Party".

WHEREAS

- A. HGCL, Hyderabad Growth Corridor Limited(HGCL) is a CPSU under the administrative control of the Ministry of New and Renewable Energy (MNRE), set up on 20th Sept, 2011 to facilitate the implementation of JNNSM and achievement of targets set therein. It is the only CPSU dedicated to the solar energy sector. It was originally incorporated as a section-3 (not for profit) company under the Companies Act, 2013. In the present outlook of the RE sector, especially solar energy, HGCL has a major role to play in the sector's development. The company is responsible for implementation of a number of schemes of MNRE for large-scale grid-connected projects under JNNSM, solar park scheme and grid-connected solar rooftop scheme along with a host of other specialised schemes. In addition, HGCL is also developing its own Solar, Floating & Hybrid innovative RE Projects & is providing consultancy services to various major CPSUs for developing turnkey basis RE Projects. The company also has powertrading license and is active in this domain through trading of solar power from projects set up under the schemes being implemented by it.
- B. The Contractor is engaged in
- C. HGCL and Contractor are desirous of pursuing a mutually beneficial relationship through the execution of Contract awarded by HGCL to the Contractor vide No.... Dated The Contractor agrees that in the course of their association for executing the said

Contract Agreement, there may be sharing of confidential information between them. Through this Agreement, both parties define the obligations with respect to the confidential information.

- D. Contractor may receive from the other Party i.e. HGCL certain technical, non-technical, financial, business and other proprietary and confidential information in relation to their respective businesses and contract specific tasks.
- E. Due to various Information Security related risks associated with the execution of the contract, HGCL desires to mitigate the perceived risks and seeks to protect its physical and intellectual assets through defined agreements with the Contractor.

NOW THEREFORE, in consideration of the above premises the sufficiency of which is hereby acknowledged, the Contractor agrees as follows:

1. Confidential Information

“Confidential Information” shall mean any and all information disclosed to, or otherwise acquired or identified or observed by the Receiver including its subsidiaries and affiliates, and each of their respective directors, employees, representatives and agents from the Disclosing Party and its affiliated companies, relating to the business of the Disclosing Party, or received from others that the Disclosing Party is obligated to treat as confidential, and other materials and information of a confidential nature whether communicated in writing, orally, electronically, photographically, or recorded in any other form of media, including, but not limited to, all sales and operating information, contractor’s information, employee and other human resource information, existing and potential business and marketing plans and strategies, financial information, cost and pricing information, data media, know-how, designs, specifications, technical configurations, concepts, reports, methods, processes, techniques, operations, devices, , product schematics or drawings, descriptive material, patent and patent applications, trade secrets, trademarks, trade names, specifications, software (source code or object code) and the like, whether or not the foregoing information is patented, tested, reduced to practice, or subject to copyright or any other intellectual property right.

“Confidential Materials” shall mean all tangible materials containing Confidential Information, including without limitation drawings, schematics, written or printed documents, computer disks, tapes, and compact disks (CD), whether machine or user readable.

Notwithstanding the above, all Confidential Information shall be specifically marked as “CONFIDENTIAL” while disclosing the same to the Receiving Party. If the same is orally disclosed then the same to be reduced in writing and marked as “CONFIDENTIAL”. Supplier, sub-contractor and other parties engaged by the Disclosing party shall have the same rights and obligations for the Confidential Information.

2. Obligations of Receiving Party relating to Information Security

Contractor agrees to conform to the following requirements:

- a) All intelligent electronic devices (IEDs), including devices with embedded software, Automation servers Controllers, HMIs and associated network components wherein the data is routable (equipped with Ethernet/optical Ethernet, Serial/Optical Serial) must have capabilities to exceed or meet applicable technical requirements under IEEE-1686:2013 for satisfying IEC/ISO:62443-2-3, IEC/ISO:62443-2-4 and IEC/ISO:62443-3-3 requirements.
- b) Contractor agrees to submit required evidences for conformance to IEC/ISO:15408 for identified network-based systems such as routers, firewalls, SIEMs etc.
- c) Contractor agrees to provide IT architecture details such as Firmware details, Operating System, databases, middle-ware, application frameworks and related third-party drivers, software component libraries, including usage of virtualization/container technologies, of all devices qualifying under clause (a) above to facilitate vulnerability analysis of the device. HGCL reserves the right to undertake appropriate black-box testing of any system, sub-system to independently ascertain vulnerability of the product/solution.
- d) Contractor agrees to enable use of Indian Regional Navigational Satellite Constellation (IRNSS) based Time Synchronization signals through appropriate use of GPS technologies that support PTP (IEEE 1588) , if available commercially.
In case the same are not available commercially, Contractor may supply the GPS Clock as per their solution requirement. However, in case HGCL supply the Indian Regional Navigational Satellite Constellation (IRNSS) based Time Synchronization signals through appropriate use of GPS technologies that support PTP (IEEE 1588) during this contract period (up to start of the Factory System Test), Contractor shall replace the existing Clock with the HGCL supplied timing solution in the Control and Protection system, without any cost implication to HGCL.
- e) Contractor commits to ensure, its adherence to secure software development life-cycle processes as per IEC/ISO:24748-1 or a similar standard and commits itself for voluntary disclosure of vulnerabilities in the system. Contractor agrees to develop and provide patches, including those of the third-party software components, for the Contractor disclosed vulnerabilities and also for the vulnerabilities discovered/ reported by any third party organization. The Contractor agrees to ensure supply and installation of patches up to the defect liability period of the system.

- f) For all software, operating system, software patches, version upgrades, firmware images etc authorized by the Contractor to be installed during the Life-Cycle of the project, the Contractor agrees to inform HGCL through a digitally signed email, the **SHA-256** checksum of all software components.
- g) The Contractor agrees to provide a list of all equipment and processes where data encryption is used. All required details for Key Management shall be provided to HGCL. HGCL at its own cost, shall supply requisite digital certificates/keys for installation and configuration of such systems as may be required for securing its interest.
- h) HGCL shall provision Notebook PCs as per recommendations of the Contractor, which shall be only authorized device from which access to the network in use by HGCL, shall be permitted for any preventive maintenance, update and configuration.
- i) The Contractor agrees to sign an undertaking as per Annexure-A, for its commitment to ensure bug and malware-free software/ software patches/ embedded software/ firmware in systems such as PLC Cards/ Logic Cards/ other microprocessor based intelligent systems. The Contractor agrees to declare with each shipment, whether during initial supply stage or subsequent repairs, diagnostics or upgrades, that it shall be solely responsible for any Criminal and/ or Civil Liabilities arising from failures due to such malware/bug. The Contractor further agrees to send a digitally signed statement by email, detailing SHA-256 checksum of all firmware/software components installed during any field/factory activity.
- j) The Contractor agrees not to access through use of WiFi/ Bluetooth based networking to any device anywhere in the controlled network. All Bluetooth/ WiFi devices shall be disabled from associated firmware and Operating System in applicable devices of the controlled network.
- k) The Contractor agrees to submit details of all devices equipped with Serial Ports (RS232C/RS485/USB etc including with Optical interface), Virtual Serial Ports and Serial over Ethernet. Only HGCL permitted devices shall be attached to serial ports. The Contractor agrees to provide systems to log details of any serial devices connected during the operation of the equipment.

3. Protection of Confidential Information

a. Use

The Receiving Party understands and acknowledges that the Confidential Information has been developed or obtained by the Disclosing Party by the investment of significant time, effort and expense, and that Confidential Information is a valuable, special and unique asset of the Disclosing Party. Therefore, the Receiving Party agrees to hold in confidence and not to disclose the Confidential Information, to any person or entity without similar obligations agreed between the Receiving Party and such person or entity. The Receiving Party will use the same standard of care it would use to secure and

safeguard its own confidential information of similar importance, but in no event less than reasonable care.

b. No copying.

The Receiving Party will not copy or modify any Confidential Information without the prior written consent of the Disclosing Party, except where such copy or modification is required for the purpose of the execution of the contract. Any permitted reproduction of confidential information must contain all confidential or proprietary legends which appear on the original. The Receiving Party shall immediately notify the Disclosing Party in the event of any loss or unauthorized disclosure or use of the confidential information.

c. Permitted disclosures.

The Receiving Party shall permit access to the Disclosing Party's confidential information solely to the Receiving Party's Representatives and contractors who (i) have a need to know such information; and (ii) have signed the specified confidentiality agreement / similar contract conditions in favour of Receiving Party

All staff of Receiving Party (on-roll or outsourced) shall be bound by the terms of this Agreement. The Contractor agrees to individually authorize each of the member of staff assigned with the project, binding them individually with the terms of similar to this Agreement during and also post-employment.

d. Additional obligations.

The Receiving Party shall

- (i) notify the Disclosing Party promptly of any material unauthorized possession, use or knowledge, or attempt thereof, of the Disclosing Party's confidential information by any person or entity which may become known to the Receiving Party;
- (ii) promptly furnish to the Disclosing Party full details of the unauthorized possession, use or knowledge, or attempt thereof;
- (iii) use reasonable efforts to assist the Disclosing Party in investigating or preventing the recurrence of any unauthorized possession, use or knowledge, or attempt thereof, of confidential information;
- (iv) use reasonable efforts to cooperate with the Disclosing Party in any litigation and / or investigation against third parties deemed necessary by the Disclosing Party to protect its proprietary rights;
- (v) promptly use all reasonable efforts to prevent a recurrence of any unauthorized possession, use or knowledge of confidential information;
- (vi) comply with the directives of authorized agencies of Government of

India, through appropriate technical configurations and custom modifications to achieve compliance as sought by them from time to time; and

- (vii) extend its services as may be required, at least once annually, during the Information Security audits.

e. Unauthorized Disclosure of Information.

If it appears that the Receiving Party has disclosed (or has threatened to disclose) Confidential Information in violation of this Agreement, the Disclosing Party shall be entitled to an injunction to restrain the Receiving Party from disclosing, in whole or in part, the Confidential Information. The Disclosing Party shall not be prohibited by this provision from pursuing other remedies, subject to suitable notice of the same to Receiving Party and Receiving Party wilfully neglecting such notice or duties under the Agreement after such notice including a claim for losses and damages.

f. Exceptions

The following shall not be considered as Confidential Information:

- (a) Any information that the Receiving Party can show by documentary evidence was in its possession prior to the disclosure to it hereunder; or
- (b) Any information that comes into the possession of the Receiving Party's Representatives, from another party who is under no obligation to the other to maintain confidentiality of such information; or
- (c) Any information that becomes generally known other than through the fault of the Receiving Party,
- (d) Any particular portion of the Confidential Information which was developed by Receiving Party's Representatives independently of and without reference to any Confidential Information or other information that the Disclosing Party has disclosed in confidence to any third party.
- (e) Information available in the public domain whether in tangible or intangible form.
- (f) Information that is not proprietary or confidential to the Disclosing Party but an information received from third party not connected to the Project.
- (g) Information that has not been marked by the Disclosing Party as "Confidential".

The burden of proving these exceptions to the provisions of this Agreement resides with the Receiving Party.

- 4. Remote Support.** Remote Support shall be permitted only as per HGCL ISO27001 Policy and Procedures. Further, remote support will only be permitted from within geographical boundaries of India. HGCL reserves the right to only permit the remote support with the presence of HGCL's authorized representative at the remote end.

- 5. Compelled Disclosure.** In the event that Receiving Party or any of

Receiving Party's

Representatives is requested or required (by oral questions, interrogatories, requests for information or documents, subpoena, civil investigative demand or similar incidents) to disclose any of the Confidential Information to the authorities as per mandatory law, it is agreed that Receiving Party or Receiving Party's Representatives, as the case may be, will provide Disclosing Party with prompt notice of such request(s) so that Disclosing Party may seek an appropriate protective order or other appropriate remedy and/or waive compliance with the confidentiality provisions of this Agreement. In the event that such protective order or other remedy is not obtained, or Disclosing Party grants a waiver hereunder, Receiving Party or Receiving Party's Representatives may furnish that portion (and only that portion) of the Confidential Information which Receiving Party is legally compelled to disclose and will exercise reasonable efforts to obtain assurance that confidential treatment will be accorded any Confidential Information so furnished.

- 6. Information Security Audit.** HGCL reserves the right to undertake a second party / third party Information Security Audit at any point as may be required, to ascertain the risk/ vulnerability/ threats and the Contractor agrees to take necessary corrective measures in-situ or within a defined time frame, as the case may be.

- 7. Term and Termination**

This Agreement shall be valid during the contractual period w.e.f. the date of signing of the main contract agreement.

- 8. Return of Confidential Information.**

Upon the written request of the Disclosing Party, the Receiving Party shall return to the Disclosing Party all written materials / digital media containing the Confidential Information to the extent possible by the Receiving Party. The Receiving Party shall also deliver to the Disclosing Party written statements signed by the Receiving Party certifying that all materials have been returned within 05 days of receipt of the request. Any unreturned Confidential Information shall be required to be maintained with similar confidentiality obligation for 10 years or as per applicable law, whichever is longer.

- 9. Remedies.**

Receiving Party acknowledges that money damages may be incalculable and an insufficient remedy for any breach of this agreement by Receiving Party and that any such breach may cause Disclosing Party irreparable harm. Accordingly, Receiving Party also agrees that, in the event of any breach or threatened breach of this Agreement, Disclosing Party, in addition to any other remedies at law or in equity it may have, shall be entitled, without the requirement of posting a bond or other security, to equitable relief, including injunctive

relief and specific performance.

10. Relationship of Parties

Neither party has an obligation under this Agreement to purchase any service or item from the other party, or commercially offer any products using or incorporating the Confidential Information. This Agreement does not create any agency, partnership or joint venture.

11. No Grant of Proprietary Rights

The Receiving Party recognizes and agrees that, except as expressly and specifically set forth in this agreement, nothing herein shall be construed as granting any proprietary right, by license,

implication, estoppel or otherwise, to any of the Disclosing Party's, confidential information, trademark, trade name or to any invention or any patent right that has issued or that may issue based on such confidential information. All information disclosed is provided "as is" without any warranties of any kind.

12. Governing Law

This Agreement shall be governed by and interpreted in accordance with the Indian laws without regard to its conflict of law principles. In particular, the provisions of Information Technology Act 2000, and rules framed thereunder shall be applicable. Further the outline of system level requirements shall be in conformance to IS:16335-2015 standard. The applicable Information Security Policy shall be the ISO-27001:2013 policy and procedures of HGCL as modified from time to time.

13. Jurisdiction and Venue. In connection with any litigation arising hereunder, Parties hereby

- (i) irrevocably and unconditionally submit to the exclusive jurisdiction of courts in Hyderabad and
- (ii) Further that disputes if any, shall be dealt with as per the provisions of the dispute settlement clause mentioned in the contract / General Conditions of Contract (GCC).

14. General Provisions.

- (a) This Agreement sets forth the entire understanding of the Parties regarding confidentiality. Any amendments must be in writing and signed by both parties.
- (b) This Agreement is intended to facilitate only the exchange of Confidential Information and is not intended to be, and shall not be construed to create a teaming agreement, joint venture association, partnership, or other business organization or agency arrangement and no Party shall have the authority to bind the other without the separate prior written agreement thereof.
- (c) This Agreement contains the entire agreement and understanding between the Parties hereto relating to the subject matter hereof and supersedes all

other prior agreements and understandings both written and oral, between the Parties with respect to the subject matter hereof. This Agreement may be executed in several counterparts, each of which will be deemed an original, and all of which taken together will constitute one single Agreement between the Parties with the same effect as if all the signatures were upon the same instrument.

IN WITNESS WHEREOF, the parties hereto have executed this Agreement at by their duly authorized representatives as of the date first set forth above.

Hyderabad Growth Corridor

Limited Signature: _____

Signature: _

Name:

Name :

Title

Title :

FORM F-26

ANNEXURE- 4

- **PQ year is financial year at which bidder meeting qualification criteria
FY-**

SI.No.	Name of Customer	Name of the work	Gross Value of agreement, Date & Number	Gross Value of Work Done (in Rs.)	Work Start Date	Work Completion Date	Completion Years	FY for PQ	Years for PQ	Project Value at PQ FY	Project Value at current level i.e FY 2022-23
1	2	3	4	5	6	7	8=7-6	9	10	11=5/8*10	12=11*2.35
Model calculation											
1	PWD - Delhi			444117916	26.03.2013	23.09.2015	2.50	2013-14	1.00	177,647,166	417470840.10

Year before FY 2022-23	Multiplying Factor
One : FY 2021-22	1.10
Two : FY 2020-21	1.21
Three : FY 2019-20	1.33
Four : FY 2018-19	1.46
Five : FY 2017-18	1.61
Six : FY 2016-17	1.77
Seven : FY 2015-16	1.95
Eight : FY 2014-15	2.14
Nine: FY 2013-14	2.35

ANNEXURE- 5

Name of the bidder										Total Qty			For 1 FY 2013-2014		
Sl. No.	Name of Customer	Name of the work	Gross Value of agreement, Date & Number	Gross Value of Work Done	Start Date	Completion Date	Completion Years	FY for PQ	Years for PQ	Poles	LED luminaries	SITC of UG cable (km)	Poles	LED luminaries	SITC of UG cable
1	2	3	4	5	6	7	8	9	10	11	12	13	14= 11/8*10	15= 12*8/10	16= 13/8*10
Model Calculation															
1	PWD - Delhi			444117916	26.03.2013	23.09.2015	2.50	2013-14	1.00	3800	7289	185	1520	2916	74

ANNEXURE- 6

Sl.No.	Name of Customer	Name of the work	Gross Value of agreement, Date & Number	Gross Value of Work Done (in Rs.)	Work Start Date	Work Completion Date	Completion Years	FY for PQ	Years for PQ	Project Value at PQ FY	Project Value at current level i.e FY 2022-23
1	2	3	4	5	6	7	8=7-6	9	10	11=5/8*10	12=11*2.35
Model Calculation											
1	PWD - Delhi			444117916	26.03.2013	23.09.2015	2.50	2013-14	1.00	177,647,166	417470840.10

ANNEXURE- 7
KEY PERSONNEL

I / We _____ declare that the following essential key personnel required for the project during Design, Supply, Installation, Testing, Commissioning, of illumination available /hired with me as notes against each:

S. No	Key Personnel	Qualifying	Experience	No. of Persons required
1	Project Manager	B.E. (Elect.) or Equivalent	Min. 5 years as Project Manager	1
2	Site Engineer-1	B.E/ Dipoma (Elect.)/ or Equivalent	Min 3 years	1
3	Site Engineer-2	B.E./ Diploma (Civil.) or Equivalent	Min 3 years	1
4	Accident prevention Officer	Any Graduate	5 years relevant experience	1

Skilled workers and operators Minimum 5% shall have obtained certificate of proficiency from any institute.

Signature of the Tenderer

ANNEXURE-8

DECLARATION ON KEY EQUIPMENT

I / We _____ declare that the following equipment is owned or leased/hired by me as notes against each:

S.No.	Equipment Type and Characteristics	Capacity	Minimum No. Required.	Owned	Leased/Hired
1	Bucket elevator	Not less than 13.0 Mtr Height	1Nos.		
2	Escort	5 tons	1Nos.		
3	Tractors with trailer	-	1Nos.		

Signature of the Tenderer

ANNEXURE-9

INFORMATION ON LITIGATION HISTORY

S. No	Case No. / Year	Court where filed.	Subject Matter / Prayer in the case.	Respondent's i.e., SE / CE	Present Stage.
1	2	3	4	5	6

Note: A **NIL** report should be furnished if there is no litigation history

Signature of the Tenderer

ANNEXURE-10
DECLARATION ON GENUINENESS OF THE CERTIFICATES

I / We have gone through carefully all the Tender conditions and solemnly declare that I / we will abide by any penal action such as disqualification or black listing or determination of contract or any other action deemed fit, taken by, the Department against us, if it is found that the statements, documents, certificates produced by us are false / fabricated. I / We hereby declare that, I / We have not been blacklisted / debarred / Suspended / demoted in any department in Telangana state or in any State due to any reasons.

Signature of the Tenderer

SECTION - VII

TECHNICAL **SPECIFICATIONS**

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DISCLAMIER:

1. Though adequate care has been taken while preparing the Bidding documents, the Bidders/Applicants shall satisfy themselves that the document is complete in all respects. Intimation of any discrepancy shall be given to this office immediately. If no intimation is received from any Bidder within three (3) days from the date of notification of NIT/ Issue of the NIT documents, it shall be considered that the NIT documents are complete in all respects has been received by the Bidder.
2. Hyderabad Growth Corridor Limited (HGCL), the Employer, reserves the right to modify, amend or supplement this NIT documents including all formats.
3. While this bidding documents have been prepared in good faith, neither Employer or its authorized representatives nor their employees or advisors make any representation or warranty, express or implied, or accept any responsibility or liability, whatsoever, in respect of any statements or omissions herein, or the accuracy, completeness or reliability of information, and shall incur no liability under any law, statute, rules or regulations as to the accuracy, reliability or completeness of this bidding documents, even if any loss or damage is caused by any act or omission on their part.
4. The specifications mentioned for all the equipment which include Solar modules, PCU, combiner boxes, DC cables, transformer, CT, PT, LT/ HT cables, interfacing panels, switch gears & other associated equipment etc., to complete the power generation and evacuation to the designated substation, in the present bidding documents are for the **reference** only. It is subject to revise/ alters as per the design/ planning/ good engineering practices etc., to be carried out by the selected bidder, to the satisfaction of

the Employer or its authorized representatives. It is advised that the bidders must satisfy himself with the prevailing site conditions before design/ plan. The design must be optimized as per the site conditions and directed to achieve the maximum output from the installed capacity at all times. Moreover, the components not separately mentioned, but are required to complete the plant for operation is also included in the scope of bidder and shall be vetted by the Employer or its authorized representatives.

Place:

Date:

(Signature)

Name and Designation of bidder

A Design Philosophy

- 1 The main objective of the design philosophy is to construct the plant with in-built Quality and appropriate redundancy to achieve high availability and reliability with minimum maintenance efforts. In order to achieve this, the following principles shall be adopted while designing the system.
 - 1.1 Adequate capacity of SPV modules, PCUs, Junction boxes etc. to ensure generation of power as per design estimates. This will be done by applying liberal de-rating factors for the array and recognizing the efficiency parameters of PCUs, transformers, conductor losses, system losses, site conditions etc.
 - 1.2 Use of equipment and systems with proven design and performance that have high availability track records under similar service conditions.
 - 1.3 Selection of the equipment and adoption of a plant layout to ensure ease of maintenance.
 - 1.4 Strict compliance with approved and proven quality assurance (QA) systems and procedures during different stages of the project, starting from sizing, selection of make, shipment, storage (at site), during

- erection, testing and commissioning.
- 1.5 Proper monitoring of synchronization and recording, to ensure availability of power to the grid.
 - 1.6 The plant instrumentation and control system should be designed to ensure high availability and reliability of the plant to assist the operators in the safe and efficient operation of the plant with minimum effort.
 - 1.7 It should also provide the analysis of the historical data and help in the plant maintenance people to take up the plant and equipment on predictive maintenance.
 - 1.8 System design shall have intelligent protection mechanism which may include very fast responsive microprocessor-based relays etc., so that any disturbance from the grid will not cause any damage to the equipment of the Solar Power Plant.
- 2 The basic and detailed engineering of the plant shall aim at achieving high standards of operational performance especially considering following:
- 2.1 SPV power plant should be designed to operate satisfactorily in synchronization with the grid within permissible limits of high voltage and frequency fluctuation conditions. It is also extremely important to safeguard the system during major disturbances, internal and external surge conditions while ensuring safe operation of the plant.
 - 2.2 The Module Mounting Structures are already constructed by the HGCL.
 - 2.3 Shadow free plant layout to ensure minimum losses in generation during the day time.
 - 2.4 Higher system voltage and lower current options to be followed to minimize ohmic losses.
 - 2.5 Selection of PCUs with proven reliability and minimum downtime. Ready availability of requisite spares.
 - 2.6 Careful logging of operational data / historical information from the Data Monitoring Systems, and periodical analysis of the same to identify any abnormal or slowly deteriorating conditions.

- 2.7 The designed array capacity at STC shall be suitably determined to meet the proposed guaranteed generation output at the point of interconnection by the contractor in his bid. The contractor shall take care of first year degradation also by installing additional DC capacity as the CUF calculations will not factor the first-year degradation of the modules.
- 2.8 Each component offered by the bidder shall be of established reliability. The minimum target reliability of each equipment shall be established by the bidder considering its mean time between failures and mean time to restore, such that the availability of complete system is assured. Bidder's recommendation of the spares shall be on the basis of established reliability.
- 2.9 Bidder shall design the plant and equipment in order to have sustained life of 25 years with minimum maintenance efforts.
- 2.10 The work execution planning for supply, erection, commissioning and all other allied works for SPV Power Plant shall be such that it is completed within stipulated time from the date of order/ LOI/ NTP, whichever is later.
- 3 The specifications provided with this bid document are functional ones; any design provided in this document is only meant as an example. The Contractor must submit a detailed design philosophy document for the project to meet the functional requirements based upon their own design in-line with the above. The bidders are advised to visit the site and satisfy themselves before bidding.

B Electrical System

1 Photovoltaic Modules

1.1 Standards and Codes

Photovoltaic Modules shall comply with the specified edition of the following standards and codes.

Standard	Description
IEC 61215-1:2016 Ed.1	Terrestrial photovoltaic (PV) modules - Design qualification and type approval - Part 1: Test requirements
IEC 61215-1-1:2016 Ed.1	Terrestrial photovoltaic (PV) modules - Design qualification and type approval - Part 1-1: Special requirements for testing of crystalline silicon photovoltaic (PV) modules
IEC 61215-1-2:2016 Ed.1	Terrestrial photovoltaic (PV) modules - Design qualification and type approval - Part 1-2: Special requirements for testing of thin-film Cadmium Telluride (CdTe) based photovoltaic (PV) modules
IEC 61730-1:2016 Ed.2	Photovoltaic (PV) module safety qualification - Part 1: Requirements for construction
IEC 61730-2:2016 Ed.2	Photovoltaic (PV) module safety qualification - Part 2: Requirements for testing
IEC 61701:2011 Ed.2	Salt mist corrosion testing of photovoltaic (PV) modules (Applicable for coastal and marine environment)
IEC 62716:2013 Ed.1	Photovoltaic (PV) modules - Ammonia corrosion testing (if applicable)
IEC TS 62804-1:2015 Ed.1	Photovoltaic (PV) modules - Test methods for the detection of potential-induced degradation - Part 1: Crystalline silicon (under conditions of 85°C/85% RH for minimum 192 hours)
As per the Solar Photovoltaics, Systems, Devices and Components Goods (Requirements for Compulsory Registration) Order, 2017, PV Modules used in the grid connected solar power projects shall be registered with BIS and bear the Standard Mark as notified by the Bureau of Indian Standards. Further, PV Modules should be listed in the ALMM, as per MNRE Approved Models and Manufacturers of Solar Photovoltaic Modules (Requirements for Compulsory Registration) Order, 2019 including subsequent amendments/orders, if any.	

1.2 Technical Requirements

Parameter	Specification
Cell/ Module Technology	Multi-crystalline/Mono-crystalline/Thin Film
Origin of PV Modules	Cells and modules shall be domestically manufactured.
Module Efficiency	≥ 17% for Multi-crystalline ≥ 19% for Mono-crystalline ≥ 17.4% for Thin film
Rated power at STC	No negative tolerance is allowed
Temperature co-efficient of power	Not less than -0.40%/°C
Application Class as per IEC 61730	Class A

1.3 Component Specifications

1.3.1 The PV Modules glass panel shall be:

- For PV Modules with back sheet, toughened low iron glass with minimum thickness of 3.2 mm for multi or mono-crystalline modules.
- In case of glass-glass PV Modules, glass thickness shall be minimum of 2 mm on each side. It shall be laminated using a laminator with symmetrical structure, i.e., heating plates on both sides.
- The glass used shall have transmittance of above 90%.

1.3.2 The back sheet used in the PV modules shall be three-layered structure with outer (air side) layer having fluoropolymer or a material with superior UV stability properties. The back sheet shall have the following properties.

Parameter	Value
Material thickness	≥ 300 micron
Water vapour transmission rate	< 2 g/m ² /day
Partial discharge test voltage	≥ 1500 V
Elongation at break	> 100%
Adhesion strength with encapsulant	> 40 N/cm
Interlayer adhesion strength	> 4 N/cm

The Employer reserves the right to conduct Pressure Cooker (PC) test/ Highly Accelerated Stress Test (HAST) to confirm the durability of the back sheet in accelerated conditions.

- 1.3.3 The encapsulant used for the PV modules should be UV resistant and PID resistant in nature. No yellowing of the encapsulant with prolonged exposure shall occur. In case of glass-glass PV modules, encapsulant shall be polyolefin type. The

encapsulant shall have the following properties.

Parameter	Value (for PV modules with back sheet)	Value (for glass-glass PV modules)
Gel content	> 75%	> 70%
Volume resistivity	> $1 \times 10^{14} \Omega \cdot \text{cm}$	> $1 \times 10^{15} \Omega \cdot \text{cm}$
Peeling strength with glass	> 40 N/cm	> 40 N/cm

- 1.3.4 The sealant used for edge sealing of PV modules shall have excellent moisture ingress protection with good electrical insulation (Break down voltage >15 kV/mm) and with good adhesion strength. Edge tapes for sealing are not allowed.
- 1.3.5 The module frame shall be made of anodized Aluminium, which shall be electrically & chemically compatible with the structural material used for mounting the modules. It is required to have provision for earthing to connect it to the earthing grid. The anodization thickness shall not be less than 15 micron.
- 1.3.6 The material used for junction box shall be UV resistant to avoid degradation during module life. The degree of protection of the junction box shall be at least IP67. Minimum three number of bypass diodes and two number of IEC 62852/EN 50521 certified MC4 compatible connectors with appropriate length of IEC 62930/EN 50618 certified 4 sq.mm copper cable shall be provided. The cable length shall be in accordance with the PV Module wiring strategy and adequate to ensure that the cable bending radius standard is not exceeded.
- 1.3.7 Each PV Module shall be provided a RFID tag which is embedded inside the module lamination and must be able to withstand harsh environmental conditions. RFID scanner and database of all the modules containing the following information shall also be provided.
- Name of the manufacturer of PV Module
 - Name of the Manufacturer of Solar cells

- iii. Type of cell: Mono / Multi
- iv. Month and year of the manufacture (separately for solar cells and module)
- v. Country of origin (separately for solar cells and module)
- vi. I-V curve for the module
- vii. Peak Wattage, I_m , V_m and FF for the module
- viii. Unique Serial No. and Model No. of the module.
- ix. Date and year of obtaining IEC PV module qualification certificate
- x. Name of the test lab issuing IEC certificate
- xi. Other relevant information on traceability of solar cells and modules as per ISO 9000 series.

1.4 Warranty

- 1.4.1 PV modules must be warranted with linear degradation rate of power output except for first year (maximum 3% including LID) and shall guarantee 80% of the initial rated power output at the end of 25 years.
- 1.4.2 The modules shall be warranted for minimum of 10 years against all material/ manufacturing defects and workmanship.
- 1.4.3 The above warranties shall be backed by third party insurance.

1.5 Approval

- 1.5.1 The Contractor shall provide Guaranteed Technical Particular (GTP) datasheet and Bill of Materials (BOM) of the module that is submitted for approval along with the datasheets of each component. The component datasheet shall contain all the information to substantiate the compliance for component specifications mentioned above.
- 1.5.2 The Contractor shall also provide test certificates corresponding to the standards mentioned above along with complete test reports for the proposed module. The tests should have been conducted at a test laboratory compliant with ISO 17025 for testing and calibration and accredited by an ILAC/IECEE member signatory. Laboratory accreditation certificate or weblink along with scope of accreditation shall also be submitted.

- 1.5.3 The BOM proposed shall be the subset of Constructional Data Form (CDF)'s of all the test reports.
- 1.5.4 The Contractor shall submit a detailed Manufacturing Quality Plan (MQP) for the PVModule with list of checks/tests performed during incoming material inspection, production, pre-dispatch and package.
- 1.5.5 The Contractor shall obtain the approval of the proposed module make & model prior to manufacturing/ inspection call.
- 1.6 Manufacturing and Inspection
 - 1.6.1 The Contractor shall inform the module manufacturing schedule to the Employer at least 7 (seven) working days before the start of proposed schedule.
 - 1.6.2 The Employer shall perform material inspection at the Manufacturer's factory before the start of proposed dispatching schedule. Proof of procurement of components

as per the approved BOM mentioning manufacturer name, manufacturing date and relevant test certificate shall be submitted during material inspection for verification.
 - 1.6.3 The dispatching shall start only after the clearance by the Employer after the material inspection.
 - 1.6.4 The cells used for module making shall be free from all defects like edge chipping, breakages, printing defects, discoloration of top surface etc. Only Class A solar cell shall be used.
 - 1.6.5 The modules shall be uniformly laminated without any lamination defects.
 - 1.6.6 Current binning of modules shall be employed to limit current mismatch of modules. Different colour codes shall be provided on the modules as well as pallet for identification of different bins. Maximum three nos. of bins will be allowed for each module rating.
- 1.7 Pre-dispatch inspection of modules shall be performed as per the inspection protocol.
- 1.8 Transportation, Handling, Storage and Installation
 - 1.8.1 Transportation, handling, storage and installation of modules shall be in accordance with the manufacturer manual so as not to breach

warranty conditions. The Standard Operating Procedure (SOP) for the same shall be shared by the Contractor prior to dispatch.

- 1.8.2 It is required to construct a temporary platform (graded) while keeping the modules at least above the highest flood level. If the contractor scheduled/ planned to mount the modules immediately after the receipt at site, then the module shall be kept in common storage area with proper arrangement.
- 1.8.3 Modules shall be dispatched in line with the Construction schedule. If Modules are dispatched ahead of schedule, following measures shall be undertaken:
- i. Modules shall be covered with tarpaulin sheet. Alternatively, the Modules, properly stacked as per OEM recommendations, shall be stores under a temporary shed.
 - ii. Further, the temporary platform for keeping the modules shall be treated with anti-termite treatment.

2 String Combiner Box (SCB)

2.1 Standards and Codes

Standard/Code	Description
IEC 60529	Enclosure Ingress Protection
IEC 62262	Enclosure Impact Protection
IEC 60269	Fuse
IEC 61643-11	Surge Protection Device
IEC 62852 or EN 50521	Solar cable connector
IEC 60695-2-11	Fire hazard testing

2.2 Construction

- 2.2.1 SCB enclosure shall be made of UV resistant, fire retardant, thermoplastic material. Enclosure degree of protection shall be at least IP65 and mechanical impact resistance shall be at least IK08.
- 2.2.2 Not more than two strings can be connected in parallel to a single input of SCB. Onespare input terminal along with connector shall be

provided for each SCB.

- 2.2.3 Every SCB input shall be provided with fuses on both positive and negative side. In case of negative grounded system, fuse at positive side only is acceptable. The rating of the fuses shall be selected such that it protects the modules from reverse current overload. The fuses shall be 'gPV' type conforming to IEC 60269-6.
- 2.2.4 DC switch disconnecter of suitable rating shall be provided at SCB output to disconnect both positive and negative side simultaneously.
- 2.2.5 Type-II surge protective device (SPD) conforming to IEC 61643-11/IEC 61643-31/ EN 50539-11 shall be connected between positive/negative bus and earth.
- 2.2.6 Resistance Temperature Detector (RTD) type or semiconductor type temperature sensor shall be provided to monitor the cabinet temperature.
- 2.2.7 MC4 connector conforming to IEC 62852 or EN 50521 shall be provided at each SCB input. Cable gland (double compression metallic) of suitable size for DC cables shall be provided at the SCB output.
- 2.2.8 UV resistant printed cable ferrules for solar cables & communication cables and punched/ embossed aluminium tags for DC cables shall be provided at cable termination points for identification.

2.3 Warranty

The SCB unit shall be warranted against all material/ manufacturing defects and workmanship for minimum of 5 (five) years from the date of supply.

2.4 Tests

Routine tests and acceptance tests for the assembled unit shall be as per the Quality

Assurance Plan (QAP) approved by the Employer.

3 **Solar and DC Cables**

3.1 Standards and Codes

Cable	From	To	Conductor/ Insulation	Voltage Rating	Applicable Standard
Solar Cable*	Module	SCB	Copper/ XLPO	1.5 kV DC	IEC 62930 / EN 50618
DC Cable	SCB	PCU	Copper or Aluminium/ XLPE	1.5 kV DC	IS 7098 Part II
* Cable used for module interconnection shall also be referred as solar cable.					

- 3.2 Solar cable outer sheath shall be flame retardant, UV resistant and black in colour. Solar cable with positive polarity should have marking of red line on black outer sheath.
- 3.3 DC cables shall be single core, armoured, Flame Retardant Low smoke (FRLS), PVC outer sheath conforming to IS 7098-II. DC cable with positive polarity should have marking of red line on black outer sheath.
- 3.4 In addition to manufacturer's identification on cables as per relevant standard, following marking shall also be provided over outer sheath.
- Cable size and voltage grade
- Word 'FRNC/ FRLS' (as applicable) at every metre
- Sequential marking of length of the cable in metres at every metre
- 3.5 Cables shall be sized based on the following considerations:
- Rated current of module
 - In case of central inverters, average voltage drop in the cables (from PV Modules to PCU) shall be limited to 1.5 % of the rated voltage. In case of string Inverters, average voltage drop (from PV module to string inverter) shall be limited to 0.5% of the rated voltage drop. The Contractor shall provide voltage drop calculations in excel sheet.
 - Short circuit withstand capability
 - De-rating factors according to laying pattern
- 3.6 Warranty
- The cables (Solar and DC) shall be warranted against all material/ manufacturing defects and workmanship for minimum of 1 (one) year from the date of supply.
- 3.7 Tests

Type test, routine test and acceptance tests requirements shall be as per IEC 62930/EN 50618 for solar cables and IS 7098-II for DC cables.

3.8 Installation

- 3.8.1 Cable installation shall be as per IS 1255.
- 3.8.2 Only terminal cable joints shall be accepted. No cable joint to join two cable ends shall be accepted.
- 3.8.3 Solar cables shall be provided with UV resistant printed ferrules and DC cables shall be provided with punched/ embossed aluminium tags. The marking shall be done with good quality letter and numbers of proper size so that the cables can be identified easily.
- 3.8.4 Cable terminations shall be made with properly crimped lugs and passed through cable glands at the entry & exit point of the cubicles. Bimetallic lugs shall be used for connecting Cu bus bar and Al cables or vice-versa.
- 3.8.5 Solar cables shall be laid over the ground on the GI cable trays with proper support & accessories through Double Wall Corrugated (DWC) HDPE conduits over the trays. The size of the conduit or pipe shall be selected on the basis of 40% fill criteria.
- 3.8.6 Solar cables shall be aesthetically tied to Module Mounting Structure using UV resistant cable-ties suitable for outdoor application.
- 3.8.7 A.C and D.C cables shall be kept in cable trays supported by free standing columns or structures enabling easy installation or replacement of cables. The horizontal and vertical clearances between power and communication cable shall not be less than 300mm. However HT cables shall be buried underground as per IS 1255.

4 Power Conditioning Unit

4.1 Standards and Codes

Power Conditioning Unit (PCU) shall comply with the specified edition of the following standards and codes.

Standard	Description
IEC 61683 Ed. 1	Photovoltaic systems - Power conditioners - Procedure for measuring efficiency
IEC 62109-1 Ed. 1	Safety of power converters for use in photovoltaic power systems - Part 1: General requirements
IEC 62109-2 Ed. 1	Safety of power converters for use in photovoltaic power systems - Part 2: Particular requirements for inverters
IEC 61000-6-2 Ed. 2	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
IEC 61000-6-4 Ed. 2.1	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
IEC 62116 Ed. 2	Utility-interconnected photovoltaic inverters - Test procedure of islanding prevention measures
IEC 60068-2-1:2007	Environmental testing - Part 2-1: Tests - Test A: Cold
IEC 60068-2-2:2007	Environmental testing - Part 2-2: Tests - Test B: Dry heat
IEC 60068-2-14:2009	Environmental testing - Part 2-14: Tests - Test N: Change of temperature
IEC 60068-2-30:2005	Environmental testing - Part 2-30: Tests - Test Db: Damp heat, cyclic (12 h + 12 h cycle)
CEA Technical Standards for Connectivity to the Grid Regulations 2007 with 2013 and 2019 Amendment	
As per the Solar Photovoltaics, Systems, Devices and Components Goods (Requirements for Compulsory Registration) Order, 2017, Inverters used in the grid connected solar power projects shall be registered with BIS and bear the Standard Mark as notified by the Bureau of Indian Standards.	

4.2 Supplier Qualification Criteria

The Inverter Supplier should have supplied minimum 5 GW capacity globally or 500MW in India in the past 5 years (as on last date of the bid submission).

4.3 Technical Requirements

Parameter	Specification
Rated AC power	As per design
Maximum input voltage	1500 V

Rated AC output voltage	As per design
Tolerance on rated AC output voltage	+/-10%
Rated frequency	50 Hz
Operating frequency range	47.5 Hz to 52 Hz
Power factor control range	0.9 lag to 0.9 lead
European efficiency	Minimum 98%
Maximum loss in Sleep Mode	0.05% of rated AC power
Total Harmonic Distortion	Less than 3% at 100% load
Degree of protection	IP 54(Outdoor), String Inverter – IP 65

- 4.3.1 The rated/ name plate AC capacity of the PCU shall be AC power output of the PCU at 50°C.
- 4.3.2 Maximum power point tracker (MPPT) shall be integrated in the PCU to maximize energy drawn from the Solar PV array. The MPPT voltage window shall be sufficient enough to accommodate the output voltage of the PV array at extreme temperatures prevailing at site.
- 4.3.3 The PCU output shall always follow the grid in terms of voltage and frequency. The operating voltage and frequency range of the PCU shall be sufficient enough to accommodate the allowable grid voltage and frequency variations.
- 4.4 Construction
 - 4.4.1 Power Conditioning Unit (PCU) shall consist of an electronic three phase inverter along with associated control, protection, filtering, measurement and data logging devices.
 - 4.4.2 Every DC input terminal of PCU shall be provided with Fuse/MCB/MCCB of appropriate rating with lockout feature for isolation of DC input. The combined DC feeder shall have suitably rated isolators for safe start up and shut down of the system. One spare DC input terminal shall be provided for each PCU. String inverters without DC fuse may be acceptable in case not more than two strings are connected to the same MPPT.
 - 4.4.3 Type-II surge protective device (SPD) conforming to IEC 61643-11 / IEC 61643-31 / EN 50539-11 shall be connected between positive/negative bus and earth.
 - 4.4.4 In case external auxiliary power supply is required, UPS shall be used to meet auxiliary power requirement of PCU. It shall have a backup storage capacity of 2 hours.
 - 4.4.5 Circuit Breaker or Relay of appropriate voltage and current rating shall be provided at the output to isolate the PCU from grid in case of faults.
 - 4.4.6 The PCU shall be tropicalized and the design shall be compatible with conditions prevailing at site. Suitable number of exhaust fan with proper ducting shall be

provided for cooling keeping in mind the extreme climatic condition of the site as per the recommendations of OEM to achieve desired performance and life expectancy.

- 4.4.7 In case of central inverter, PCU cabinet shall have manual switches (on front side) controlling the inverter & its contactors. START/STOP, DC Disconnect ON/OFF, AC Disconnect ON/OFF, Emergency STOP shall be provided to stop the inverter operation & disconnect PCU from DC input & the grid during emergency conditions.
- 4.4.8 All the conducting parts of the PCU that are not intended to carry current shall be bonded together and connected to dedicated earth pits through protective conductor of appropriate size. DC negative terminal shall be grounded. In case DC negative grounding is not possible, appropriate anti-PID device shall be provided.
- 4.4.9 Dedicated communication interface shall be provided to monitor the PCU from SCADA. It should also include remote operation control (STOP) from the SCADA itself.
- 4.4.10 PCU front panel shall be provided with LCD/ LED to display all the relevant parameters related to PCU operation and fault conditions. It shall include, but not limited to, the following parameters.
 - i. DC input power
 - ii. DC input voltage
 - iii. DC input current (for each terminal)
 - iv. AC output power
 - v. AC output voltage (all the 3 phases and line)
 - vi. AC output current (all the 3 phases and line)
 - vii. Frequency
 - viii. Power Factor

Note: In case of outdoor PCU, PCU without LCD display with provision for Data access over Bluetooth / Wi-Fi shall be acceptable.

- 4.4.11 String inverter, if installed in open, shall be placed inside a canopy shed with at least 15 cm in all directions. Alternatively, the Contractor may install the inverter on the column post of the Module Mounting

Structure, below the modules. In such case, the canopy is not required, and the column and foundation shall be designed accordingly.

- 4.4.12 AC combiner box for string inverter configuration shall comply with Clause 9 of the Technical Specifications with exception of rated system voltage.

4.5 Operating Modes

Operating modes of PCU shall include, but not limited to, the following modes. These operating modes and conditions for transition are indicative only. The Contractor shall provide the detailed flow chart indicating the various operating modes and conditions for transition during detailed engineering.

4.5.1 Standby Mode

The PCU shall continuously monitor the input DC voltage and remain on Standby Mode until it reaches the pre-set value.

4.5.2 MPPT Mode

When the input DC voltage is above the pre-set value and AC grid connection conditions are fulfilled, the PCU shall enter into MPPT mode.

4.5.3 Sleep Mode

When the AC output power/DC input voltage decreases below the pre-set value for pre-set time delay, the PCU shall switch into Sleep Mode.

4.6 Protection Features

The PCU shall include appropriate self-protective and self-diagnostic feature to protect itself and the PV array from damage in the event of PCU component failure or from parameters beyond the PCU's safe operating range due to internal or external causes. The self-protective features shall not allow signals from the PCU front panel to cause the PCU to be operated in a manner which may be unsafe or damaging. Faults due to malfunctioning within the PCU, including commutation failure, shall be cleared by the PCU protective devices.

The PCU shall provide protection against the following type of faults, among others.

- i. DC/AC over current
- ii. DC/AC over voltage

- iii. DC reverse polarity
- iv. DC earth fault
- v. AC under voltage
- vi. AC under frequency/over frequency
- vii. Islanding
- viii. Over temperature
- ix. Lightning surges

4.7 Grid Support Functions

4.7.1 Active power regulation

The PCU shall be able to limit the active power exported to the grid based on the set point provided through PCU front control panel. The PCU shall also be able to automatically limit the active power after an increase in grid frequency above a pre-set value. The ramp rate shall be adjustable during operation and start-up after fault. The applicability of the requirement shall be as per CEA regulation and compliance.

4.7.2 Reactive power control

The PCU shall be able to inject /absorb reactive power to/ from the grid based on the set point provided through PCU front control panel. The same shall be performed automatically with adjustable ramp rate based on dynamic changes in grid voltage or reactive power reference.

Note: As per TSTRANSCO guidelines, after synchronization, the PCU shall not draw reactive power from the grid during generation.

4.7.3 Voltage Ride Through

The PCU shall remain connected to the grid during temporary dip or rise in grid voltage as per the LVRT and HVRT requirements of CEA Technical Standards for Connectivity to the Grid Regulations. The PCU shall also be able to inject reactive power during the period of voltage dip.

4.8 Warranty

The complete Power Conditioning Unit shall be warranted against all material/ manufacturing defects and workmanship for minimum of 5 (five) years.

4.9 Tests

4.9.1 Type Tests

The type test certificates as per the standards mentioned above should be from any of the ILAC/IECEE member signatory accredited test centers. Laboratory accreditation certificate or weblink along with scope of accreditation shall also be submitted. It is the responsibility of the Contractor to substantiate the compliance for CEA Regulations using test reports.

4.9.2 Routine Tests

Routine tests and acceptance tests shall be as per the Quality Assurance Plan (QAP) approved by the Employer.

5 **Inverter Transformer and Auxiliary Transformer**

5.1 Standards and Codes

Inverter transformer and auxiliary transformer, wherever applicable, shall comply with the latest edition of the following standards and codes including amendments.

Standard	Description
IS 2026, IEC 60076	Specification of Power Transformers
IS 11171, IEC 60076	Dry-Type Power Transformers
IS 2099, IEC 60137	Bushings for alternate voltage above 1000 V
IS 335, IEC 60296	Insulating oil
IS 3639	Fittings and Accessories for Power Transformers
IS 12063	Degree of protection provided by enclosures
CBIP publication no. 295	
Indian Electricity rules and other statutory regulations	

5.2 Technical Requirements

Parameters	Inverter Transformer	Auxiliary Transformer
VA Rating	As per system design requirement	
Voltage Ratio	33 kV / Inverter output voltage	As per system design

Duty, Service & Application	Continuous Solar Inverter application and converter Duty (Outdoor)	Continuous application (Outdoor/Indoor)
Winding	As per system design requirement	2
Frequency	50 Hz	50Hz
Nos. of Phase	3	3
Vector Group & Neutral earthing	As per system/inverter manufacturer requirement	Dyn11
Cooling	ONAN	ONAN / AN
Tap Changer	OCTC, No. of steps shall be as per system requirement	
Impedance at 75°C	As per Inverter Manufacturer requirement	As per system requirement
Permissible Temperature rise over an ambient of 50°C (irrespective of tap)		
Top Oil	50°C	As per IS/IEC
Winding	55°C	As per IS/IEC
SC withstand time (thermal)	2 second	2 second
Short Circuit Apparent power	As per system requirement	
Termination	As per system requirement	
Bushing rating, Insulation class (Winding & bushing)	36 kV – porcelain bushings 1.1 kV – epoxy bushings	As per the system requirement
Noise level	As per NEMA TR-1	
Loading Capability	Continuous operation at rated MVA on any tap with voltage variation of +/-3%, also transformer shall be capable of being loaded in accordance with IEC 60076-7	
Flux density	Not to exceed 1.9 Wb/sq.m. at any tap position with combined frequency and voltage variation from rated V/f ratio by 10% corresponding to the tap. Transformer shall also withstand following over fluxing conditions due to combined voltage and frequency fluctuations: a) 110% for continuous rating b) 125% for at least one minute c) 140% for at least five seconds. Bidder shall furnish over fluxing characteristic up to 150%	

Air Clearance	As per CBIP
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For 3000KVA Power Transformer

1. Rated Output kVA 3000kVA
2. Application Step-up Transformer suitable for Inverter operation and bi directional flow of Power application- Continuous operation in Solar Power Plant LV Input Source Inverter
3. Number of phases 3
4. Number of windings: LV-2,HV-1
5. Degree of Protection :IP55
6. Applicable Standards IS: 2026/ IEC 60076
7. Type of Cooling ONAN
- Rated Voltage
 1. LV1, LV2 800V
 2. HV 33 kV
- 8 Rated Frequency 50Hz
9. Connection
 - i. LV Delta
 - ii. HV Delta
- 3000kVA, 33kV/800-800, Oil filled three winding transformer
10. Percentage impedance- 6.25%

For 2000KVA Power Transformer

1. Rated Output kVA 2000kVA
2. Application Step-up Transformer suitable for Inverter operation and bi directional flow of Power application- Continuous operation in Solar Power Plant LV Input Source Inverter
3. Number of phases 3
4. Number of windings: LV-2,HV-1
5. Degree of Protection :IP55
6. Applicable Standards IS: 2026/ IEC 60076
7. Type of Cooling ONAN
Voltage
- Rated
1. LV1, LV2 800V
2. HV 33 kV
- 8 Rated Frequency 50Hz

9. Connection

i. LV Delta

ii. HV Delta

iii. LV Neutral Unearthed.

2000kVA, 33kV/800-800, Oil filled three winding transformer

10. Percentage impedance- 6.25%

5.3 Construction

5.3.1 The transformer shall be provided with conventional single compartment conservator with prismatic toughened glass oil gauge. The top of the conservator shall be connected to the atmosphere through indicating type cobalt free silica gel breather with transparent enclosure. Silica gel shall be isolated from atmosphere by an oil seal. Inverter transformers shall be provided with Magnetic Oil Gauge (MOG) with low oil level alarm contact.

5.3.2 It is the responsibility of the Contractor to ensure that the inverter transformer comply with all the requirements of inverter provided by the inverter manufacturer.

5.3.3 Inverter Transformer shall be designed for at least 5% total harmonic distortion (THD) to withstand distortion generated by the inverter as well as possible outside harmonics from the network.

5.3.4 The transformer shall be suitable for continuous operation with a frequency variation of $\pm 2.5\%$ from nominal frequency of 50 Hz without exceeding the specified temperature rise.

5.3.5 Inverter Transformer shall have shield winding between LV & HV windings. Each LV winding must be capable of handling non-sinusoidal voltage with voltage gradient as specified by the inverter manufacturer. Also, shield winding shall be taken out from tank through shield bushing and the same shall be brought down to the bottom of the tank using copper flat and support insulator for independent grounding.

5.3.6 Neutral bushing of Inverter duty transformer shall be brought outside the tank for the testing purpose. It shall be covered with MS sheet and a sticker "For testing purpose only. Do not earth". Neutral bushing of

- auxiliary transformer shall be brought outside the tank for earthing.
- 5.3.7 Transformer shall have 150 mm dial type Oil Temperature Indicator (OTI) and Winding Temperature Indicator (WTI) with alarm and trip contacts. All indicators shall have accuracy of 1.5%. For inverter transformers, WTI shall be provided for all the windings.
- 5.3.8 The radiators shall be detachable type, mounted on the tank with shut off valve at each point of connection to the tank, lifts, along with drain plug/ valve at the bottom and air release plug at the top.
- 5.3.9 Marshalling Box shall be of sheet steel, dust and vermin proof provided with proper lighting and thermostatically controlled space heaters. The degree of protection shall be IP 55. Marshalling Box of all transformers shall be preferably Tank Mounted. One dummy terminal block in between each trip wire terminal shall be provided. At least 10% spare terminals shall be provided on each panel. The gasket used shall be of neoprene rubber. Wiring scheme (TB details) shall be engraved in a stainless-steel plate with viewable font size and the same shall be fixed inside the Marshalling Box door.
- 5.3.10 Buchholz relay, double float type with alarm and trip contacts, along with suitable gas collecting arrangement shall be provided.
- 5.3.11 Inverter transformer shall be provided with spring operated Pressure Relief Device (with trip contacts) with suitable discharge arrangement for oil. For Auxiliary transformers, diaphragm type explosion vent shall be provided.
- 5.3.12 Filter valve at top the tank and drain cum sampling valve at bottom of the tank shall be provided.
- 5.3.13 All external surface of the transformer shall be painted with two coats of epoxy-based paint of colour shade RAL 7032. Internal surface of cable boxes and marshalling box shall be painted with epoxy enamel white paint. The minimum dry film thickness (DFT) shall be 100 microns.
- 5.3.14 LV and HV cable box shall be provided with disconnecting chamber to facilitate the movement of transformer without disturbing cable

box and termination.

- 5.3.15 Air release plug, bi-directional wheel/skids, cover lifting eyes, transformer lifting lugs, jacking pads, towing holes, core and winding lifting lugs, inspection cover, rating plate, valve schedule plate, accessories and terminal marking plates, two nos. of earthing terminals shall be provided.
- 5.3.16 Rain hoods to be provided on Buchholz, MOG & PRD. Entry points of wires shall be suitably sealed.
- 5.3.17 The accessories listed above are indicative only. Accessories which are not mentioned above but required for satisfactory operation of the transformers are deemed to be included in the contract without extra charges.
- 5.3.18 Fire-protection for inverter transformer shall be provided in accordance with relevant CEA regulations as amended time to time.

5.4 Dry Type Auxiliary Transformer

- 5.4.1 Transformer shall be cast resin encapsulated dry type transformer, made of cold rolled grain-oriented silicon steel laminations of M4 grade or better. Winding conductor shall be electrolytic grade Copper/Aluminium and insulation shall be Class F or better.
- 5.4.2 The transformers shall be housed in a metal protective housing, having a degree of protection of IP 23 suitable for indoor installation. The enclosure shall be provided with suitable hardware and accessories required for satisfactory operation of the transformer per the relevant standard.

5.5 Warranty

The transformer shall be warranted against all material/ manufacturing defects and workmanship for minimum of 5 (five) years from the date of supply.

5.6 Testing and Inspection

5.6.1 Type Tests and Special Tests

The following type test and special test reports shall be submitted during detailed engineering. The tests should have been conducted on the similar transformer by NABL accredited laboratory.

5.6.1.1 Type Tests

- i. Lightning impulse (Full & Chopped Wave) test on windings as per IEC60076-3
- ii. Temperature Rise test at a tap corresponding to maximum losses as per IEC 60076-2

5.6.1.2 Special Tests

- i. Measurement of zero-sequence impedance as per IEC 60076-1
- ii. Measurement of harmonics of no-load current as per IEC 60076-1
- iii. Measurement of acoustic noise level as per NEMA TR-1
- iv. Short-circuit withstand test as per IEC 60076-5

Note: In case the contractor is not able to submit the test reports during detailed engineering, the contractor shall submit the reports of type/special tests either conducted by NABL accredited laboratory or witnessed by Employer.

5.6.1.3 Type and Special tests are not required for auxiliary transformers of rating including

100 kVA and below. However, auxiliary transformer energy efficiency shall correspond to 3-star BEE rating under BIS guidelines or better. The Contractor shall demonstrate the same during the routine tests.

5.6.2 Routine Tests

Each completed transformer shall be subjected to following routine tests as per the latest edition of IEC 60076 unless specified otherwise.

- i. Measurement of winding resistance at each tap
- ii. Measurement of voltage ratio between HV and LV windings at each tap
- iii. Check of vector group
- iv. Measurement of no-load loss and no-load current
- v. Measurement of short-circuit impedance and load loss
- vi. Magnetic balance test as per CBIP manual publication no. 295
- vii. Separate source voltage withstand test
- viii. Induced over voltage withstand test
- ix. Measurement of insulation resistance
- x. Marshalling box functional test

- xi. IR Measurement on wiring of marshalling box
- xii. Breakdown voltage test on transformer oil as per IS 335
- xiii. Oil leakage test on completely assembled transformer along with radiators

5.6.3 Tests at Site

After erection at site all transformer(s) shall be subjected to the following tests.

- i. Measurement of voltage ratio
- ii. Check of vector group
- iii. Magnetic balance test
- iv. Measurement of insulation resistance
- v. Breakdown voltage test on transformer oil

Note: In case the equipment is not found as per the requirements of the Technical Specifications of NIT, all expenses incurred during site testing will be to the Contractor's account and the equipment shall be replaced by him at free of cost.

6 HT Switchgear

6.1 Standards and Codes

All equipment provided under HT switchgear shall comply with latest editions and amendments of the relevant IEC standards and IS codes. In particular, the switchgear shall comply with the following standards and codes.

Standard/Code	Description
IS/IEC 62271-1	High Voltage Switchgear and Control gear - Part 1: Common Specifications
IS/IEC 62271-100	High Voltage Switchgear and Control gear - Part 100: AC Circuit Breakers
IS/IEC 62271-102	High Voltage Switchgear and Control gear - Part 102: AC Disconnectors and Earthing Switches
IS/IEC 62271-200	High Voltage Switchgear and Control gear - Part 200: AC Metal Enclosed Switchgear and Control gear for Rated Voltages Above 1 kV and Up to and Including 52 kV
IEC 61869	Instrument Transformers
IS 3231	Electrical relays for power systems protection
IEC 60255	Measuring relays and protection equipment

IEC 61850	Communication networks and systems for power utility automation
IEC 61131-3	Programmable controllers - Part 3: Programming languages
IS 9385	High voltage fuses
IS 9431	Indoor post insulators of organic material for systems with nominal voltages greater than 1000 V up to and including 300 kV
IEC 60099-4	Surge arresters - Part 4: Metal-oxide surge arresters without gaps for A.C. systems
IS 3070-3	Lightning Arresters for Alternating Current Systems - Part 3: Metal Oxide Lightning Arresters Without Gaps
IEC 62052-11	Electricity metering equipment (A.C.) - General requirements, tests and test conditions - Part 11: Metering equipment
IEC 62053	Electricity metering equipment (A.C.) - Particular requirements
IS 14697	AC Static Transformer Operated Watthour and Var-hour Meters, Class 0.2S and 0.5S

6.2 Technical Parameters

Parameter	Specification
System Parameters	
Highest system voltage	36 kV
Rated system voltage	33 kV
Rated frequency	50 Hz
Number of phases	3
Power frequency withstand voltage	70 kV (r.m.s.)
Lightning impulse withstand voltage	170 kV (peak)
System fault current	As per system requirement
Internal Arc Classification Rating	System fault current for 1 s
Circuit Breaker	
Type	Vacuum type
Operating duty cycle	O – 0.3sec – CO – 3min – CO
Short circuit breaking current	As per system requirement
Short circuit making current	2.5 times S.C. breaking current
Re-strike performance class	C2
Mechanical endurance class	M1

Current Transformer	
Accuracy class	0.2 for metering (0.2s for metering at outgoing feeder), 5P20 for protection
Rated VA burden	As per requirement
Insulation class	Class E or better
Voltage Transformer	
Accuracy class	0.2 for metering, 3P for protection
Rated VA burden	As per requirement
Insulation class	Class E or better

6.3 Switchgear Panel

- 6.3.1 The switchgear panel shall be free standing, floor mounted, single front, single tier fully compartmentalized, metal enclosed construction. Each panel shall have separate compartments for circuit breaker, bus bars, cable termination and auxiliary circuit.
- 6.3.2 The circuit breakers shall be mounted on horizontally withdrawable trucks with locking facility in SERVICE and TEST positions.
- 6.3.3 The panel enclosure shall be constructed with CRCA steel/Aluzinc sheet. The thickness of load bearing members shall be minimum 3 mm and that of non-load bearing members shall be minimum 2 mm.
- 6.3.4 All surfaces shall be painted with two coats of epoxy-based paint of colour shade RAL 7032. The minimum dry film thickness (DFT) shall be 100 microns.
- 6.3.5 The circuit breaker and auxiliary circuit compartments provided on the front side shall have separate concealed hinged doors. Cable and bus bar compartments provided on the rear side shall have separate bolted covers. All doors and covers shall be provided with neoprene/synthetic rubber gaskets to prevent entry of vermin and dust.
- 6.3.6 Pressure relief device shall be provided in each high voltage compartment of a panel to safely vent the gases in the event of internal arc. Seal-off bushing arrangement shall be provided between the breaker compartment and bus bar/cable compartments to prevent transfer of arc from one compartment to other.

- 6.3.7 Automatic safety shutters shall be provided to cover up the fixed high voltage contacts on bus bar and cable sides when the truck is moved to TEST position.
- 6.3.8 Degree of protection shall not be less than IP 5X for auxiliary circuit compartment. However, for remaining compartments it shall not be less than IP 4X. For outdoor panels, degree of protection shall not be less than IP 55.
- 6.3.9 Mechanical /Electrical interlocks shall be provided to prevent mal-operation and in particular to ensure the following:
- i. The breaker shall be operated only if it is in SERVICE or TEST position.
 - ii. Movement of the breaker truck between SERVICE and TEST positions shall be possible only if the breaker is OFF.
 - iii. It shall be possible to open the door only when the breaker is in TEST position.
 - iv. Rear cover OPENING SAFETY INTERLOCK shall be provided to prevent opening of rear covers of 33 kV HT switchgear panels when the panel is in charged condition.
- 6.3.10 Panel shall be provided with local bus-bar protection
- 6.3.11 Each switchgear panel shall be provided with thermostatically controlled space heaters, separately for breaker, cable and bus bar compartments, to prevent condensation within the compartment. The space heater shall be connected to 240 V, 50 Hz, single phase AC supply through suitable switch and fuse.
- 6.3.12 240 V, 5 A, SPN industrial socket-outlet with ON/OFF switch shall be provided in each panel.
- 6.3.13 Each panel shall be provided with LED lamp rated for 240 V, 50 Hz, single phase AC supply for interior illumination controlled by door switch.
- 6.3.14 Gapless, metal-oxide surge arrestors shall be provided between line and earth in cable compartment of the switchgear panel.
- 6.3.15 Suitable lifting hooks shall be provided for each panel.

6.4 Circuit Breakers

- 6.4.1 Circuit breakers shall be of vacuum type. It shall comprise of three separate identical single pole units operated through the common shaft and shall be fully interchangeable both electrically and mechanically.
- 6.4.2 The circuit breaker operating mechanism shall be based on motor operated spring charging and it shall be re-strike free, trip free both electrically and mechanically, with anti-pumping feature.
- 6.4.3 The rated control voltage of the spring charging motor shall be 110 VDC/230 VAC. Closing coil shall operate at all values of voltages between 85% and 110% of rated voltage. Opening coil shall operate correctly under all operating conditions of the circuit breaker up to the rated breaking capacity and at all values of supply voltage between 70% and 110% of rated voltage.
- 6.4.4 The spring charging motor shall have adequate thermal rating such that continuous sequence of the closing and opening operations is possible as long as power supply is available to the motor. It shall also be possible to charge the spring manually and close the breaker in the event of failure of motor / control supply to motor. Operating handle shall be provided for charging the operating mechanism. After failure of control supply to the motor, one open-close-open operation shall be possible with the energy contained in the operating mechanism.
- 6.4.5 The motor rating shall be such that it requires not more than 30 seconds for full charging of the closing spring. Closing action of the circuit breaker shall compress the opening spring ready for tripping. When closing springs are discharged after closing the breaker, they shall be automatically charged for the next operation.
- 6.4.6 Mechanical indicators shall be provided to indicate OPEN/CLOSED positions of the circuit breaker and CHARGED/ DISCHARGED positions of the closing spring. An operation counter shall also be provided. These indicators and counter shall be visible from the panel front door without opening it.

6.5 Relays

- 6.5.1 All relays shall be microprocessor based numerical type. However, auxiliary relays can be static or electromechanical type. The relays shall be flush mounted on panel front with connections from the inside.
- 6.5.2 The relays shall be capable of operating continuously between 80 – 120% of auxiliary voltage.
- 6.5.3 All numerical relays shall have adequate number of freely configurable, optically isolated, Binary Inputs (BI) and potential free Binary Outputs (BO).
- 6.5.4 All numerical relays shall have minimum four no. of current inputs, three for phase current and one for earth current, suitable for CT secondary current of 1A. The current inputs shall be compatible with both residual connected CT and Core Balance CT (CBCT). In addition, numerical relay in main outgoing feeder shall have three no. of voltage inputs for Under Voltage/Over Voltage protection.
- 6.5.5 All I/O's shall have galvanic isolation. Analog inputs shall be protected against switching surges and harmonics.
- 6.5.6 Making, breaking and continuous capacity of the relay contacts shall be adequate enough for the circuits in which they are used.
- 6.5.7 The numerical relay shall have the following protection functions with at least two independent protection setting groups. The protection functions shall be selectable from any of the IEC characteristic curves.
- i. Definite time (DT) phase over current protection
 - ii. Inverse Definite Minimum Time (IDMT) phase over current protection
 - iii. Definite time (DT) earth fault current protection
 - iv. Inverse Definite Minimum Time (IDMT) earth fault current protection
 - v. Under Voltage protection
 - vi. Over Voltage protection
- 6.5.8 As per TSTRANSO guidelines, following protection functions shall be provided:
- i. For 33 kV feeders:
 - a. (Three) directional IDMT over current protection relays
 - b. 1 (One) directional IDMT E/L protection relay

- ii. For 132 kV feeders with line length > 10 Kms:
 - a. Main: Numerical line distance relay
 - b. Back-up: Numerical directional IDMT O/C & E/F relays
 - iii. For 132 kV feeders with line length < 10 Kms:
 - a. Main: Numerical line differential relay with inbuilt distance protection features with fibre optic communication.
 - b. Back-up: Numerical directional IDMT O/C & E/F relays
- 6.5.9 Transformer feeder protection relay shall have provision for the following protection functions.
 - i. Buchholz alarm & trip
 - ii. Oil Temperature Indicator (OTI) alarm & trip
 - iii. Winding Temperature Indicator (WTI) alarm & trip
 - iv. Pressure Relief Valve (PRV) trip
 - v. Magnetic Oil Gauge (MOG) alarm
- 6.5.10 All numerical relays shall have provision for measurement and storage of electrical parameters such as voltage, current, frequency, active power, reactive power etc.
- 6.5.11 The numerical relay shall be able to record faults and events in non-volatile memory.
 - i. Fault record – At least 5 recent faults including the protection function operated, operating phase(s), voltages and currents along with date and time stamp.
 - ii. Event record – At least 200 events with date and time stamp.
- 6.5.12 The numerical relay shall have trip circuit supervision facility to monitor the circuit breaker trip circuit both in pre-trip and post-trip conditions. The relay shall also be able to provide circuit breaker monitoring, CT and VT supervision.
- 6.5.13 The numerical relay shall have self-diagnostic feature with separate output contact for indication of any internal relay failure.
- 6.5.14 The numerical relay shall have RS-232/RS-485/RJ-45/USB ports on front side for local communication with PC and on rear side for remote communication to SCADA system.

- 6.5.15 The numerical relay shall have feature for time synchronization through the SCADA System / networking.
- 6.5.16 The numerical relay shall be provided with backlit alphanumeric LCD to access protection settings, measurement parameters, fault and event records. Read and write access to protection settings shall be password protected.

6.6 Instrument Transformers

- 6.6.1 Instrument transformers shall be completely encapsulated cast resin type, suitable for continuous operation at the ambient temperature prevailing inside the switchgear enclosure, when the switchgear is operating at its rated load and the outside ambient temperature is 50°C.
- 6.6.2 Polarity marks shall indelibly be marked on each instrument transformer and at the lead terminals at the associated terminal block.
- 6.6.3 Voltage transformers shall be single phase units. Bus voltage transformers shall be housed in a separate panel on withdrawable truck.
- 6.6.4 HRC fuses of suitable rating shall be provided on primary side of voltage transformers. For secondary side, four pole Miniature Circuit Breakers (MCB) shall be provided with its supervision facility.

6.7 Earthing

- 6.7.1 An earth bus made of copper shall be provided throughout the length of the panel. It shall be bolted to the framework of each panel and brazed to each breaker earthing contact bar.
- 6.7.2 The earth bus shall have sufficient cross section to carry maximum fault current without exceeding the allowable temperature rise.
- 6.7.3 All non-current carrying conductors of the panel shall be connected to the earth bus. All joints to the earth bus shall be made through at least two bolts. Hinged doors shall be earthed through flexible earthing braid of adequate cross section. Suitable provision shall be provided at each end of the earth bus for connection with Owner's Earth conductor.

- 6.7.4 Positive earthing of the breaker truck and frame shall be maintained when it is in the connected position and in all other positions whilst the auxiliary circuits are not totally disconnected.
- 6.7.5 All metallic cases of relays, instruments and other panel mounted equipment shall be connected to earth bus by independent copper wires of size not less than 2.5 sq. mm with green colour insulation.
- 6.7.6 Instrument transformer secondary neutral point shall be earthed at one place only on the terminal block. Such earthing shall be made through links so that earthing of one circuit may be removed without disturbing the earthing of other circuits.
- 6.7.7 Separate earthing trucks shall be provided for earthing of busbars and incoming/outgoing feeders. The trucks shall have voltage transformer to indicate presence of voltage prior to earthing. An audible alarm shall also be provided in case of voltage on the earthing terminal. Integral earth switches may also be considered instead of earthing trucks. The earthing truck/switch shall have short circuit withstand capability equal to that of the associated switchgear panel.
- 6.7.8 The interlocks shall be provided to ensure the following.
- i. It is not possible to rack-in the earthing truck/close the earthing switch when the breaker truck is in SERVICE position.
 - ii. It is not possible to rack-in the breaker truck into SERVICE position when earthing truck is connected/earthing switch is in closed position.

6.8 Bus bar

- 6.8.1 Bus bar shall be made of copper or aluminium with uniform cross section throughout their length. They shall be adequately supported on insulators to withstand electrical and mechanical stresses due to specified short circuit current.
- 6.8.2 All bus bars joints shall be thoroughly cleaned and anti-oxide grease shall be applied. Plain and spring washers shall be provided to ensure good contacts at the joints and taps. Wherever aluminium to copper connections are required, suitable bimetallic connectors or clamps

shall be used.

6.8.3 Bus bars shall be provided with heat shrinkable sleeves of suitable insulation class throughout their length with proper colour coding. All bus bar joints and taps shall be shrouded.

6.8.4 Bus bar support insulators shall be made of non-hygroscopic, arc and track resistant, high strength material suitable to withstand stresses due to over voltage and short circuit current.

6.8.5 The Contractor shall submit busbar sizing calculation for specified continuous and short time current ratings during detailed engineering.

6.9 Measuring Instruments

6.9.1 All the measuring instruments shall be digital, flush mounting type with communication facility.

6.9.2 All feeders except main outgoing feeder shall be provided with digital Multi-Function Meter (MFM). Tri Vector Meter (TVM) shall be provided for the main outgoing feeder (in the HT Panel). Accuracy class of MFM shall be 0.2 and that of TVM shall be 0.2S.

6.9.3 Measuring instruments shall have provision to display the following parameters.

- i. Line and phase voltages
- ii. Line and phase currents
- iii. Active power, Reactive power, Apparent power
- iv. Frequency
- v. Power factor
- vi. Total Harmonic Distortion (THD)

6.10 Wiring and Terminal blocks

6.10.1 All internal wiring shall be done with 650 V grade, 1.5 sq.mm. PVC insulated stranded flexible copper wire. For CT secondary circuits, 2.5 sq.mm copper wire shall be used.

6.10.2 Wire terminations shall be made with solderless crimping type tinned copper lugs, which shall firmly grip the conductor. Insulation sleeves shall be provided at all the wire terminations.

6.10.3 Printed identification ferrules, marked to correspond with panel

wiring diagram shall be provided at both ends of each wire. The ferrules shall be firmly located on each wire so that they cannot move or turn freely on the wire. Wire identification shall be done in accordance with IS 11353.

- 6.10.4 The Contractor shall be solely responsible for the completeness and correctness of the internal wiring and for the proper functioning of the connected equipment.
- 6.10.5 All internal wiring to be connected to the external equipment shall terminate on terminal blocks. Terminal blocks shall be rated for 650 V, 10 A and made of non- inflammable material.
- 6.10.6 CT and VT secondary circuits shall be terminated on stud type, non-disconnecting terminal blocks.
- 6.10.7 At least 10% spare terminals shall be provided on each panel and these spare terminals shall be distributed on all terminal blocks.

6.11 Warranty

The HT panel unit shall be warranted against all material/ manufacturing defects and workmanship for minimum of 2 (Two) years from the date of supply.

6.12 Testing and Inspection

6.12.1 Type Tests

The switchgear panel shall be of type tested design. The following type test report shall be submitted during detailed engineering. The tests should have been conducted on the similar equipment by NABL accredited laboratory.

Test	Standard	Relevant IEC Clause
Switchgear Panel		
Dielectric tests		
Power frequency voltage test	IEC 62271-200	6.2.6.1
Lightning impulse voltage test	IEC 62271-200	6.2.6.2
Dielectric tests on auxiliary and control circuits	IEC 62271-200	6.2.10

Measurement of the resistance of the main circuit	IEC 62271-200	6.4.1
Temperature-rise tests	IEC 62271-200	6.5
Short-time withstand current and peak withstand current tests	IEC 62271-200	6.6
Verification of the IP coding	IEC 62271-200	6.7.1
Verification of making and breaking capacities	IEC 62271-200	6.101
Mechanical operation test	IEC 62271-200	6.102
Internal arc test	IEC 62271-200	6.106
Circuit Breaker		
Mechanical operation test at ambient air temperature (M2 Class)	IEC 62271-100	6.101.2
Basic short-circuit test-duties	IEC 62271-100	6.106
Relays		
Vibration tests	IEC 60255-21-1	
Shock and bump tests	IEC 60255-21-2	

Seismic tests	IEC 60255-21-3	
Electromagnetic compatibility requirements	IEC 60255-26	
Product safety requirements	IEC 60255-27	
Common requirements	IEC 60255-1	
Functional requirements	Relevant parts of IEC 60255-100 series	
Communication requirements	IEC 61850	
Current Transformers		
Temperature-rise test	IEC 61869-2	7.2.2
Impulse voltage withstand test on primary terminals	IEC 61869-2	7.2.3
Tests for accuracy	IEC 61869-2	7.2.6
Short-time current tests	IEC 61869-2	7.2.201
Voltage Transformer		

Temperature-rise test	IEC 61869-3	7.2.2
Impulse voltage withstand test on primary terminals	IEC 61869-3	7.2.3
Test for accuracy	IEC 61869-3	7.2.6
Short-circuit withstand capability test	IEC 61869-3	7.2.301

Note: In case the contractor is not able to submit the test reports during detailed engineering, the contractor shall submit the reports of type/special tests either conducted by NABL accredited laboratory or witnessed by Employer.

6.12.2 Routine Tests

Routine tests and acceptance tests shall be as per the Quality Assurance Plan (QAP) approved by the Employer.

7 AC Cables

7.1 Standards and Codes

All AC Cables shall conform to the following standards and codes.

IS 7098	Crosslinked polyethylene insulated PVC sheathed cables, Part 1: For working voltage up to and including 1100 V
IS 7098	Crosslinked Polyethylene Insulated Thermoplastics Sheathed Cables Part 2: for Working Voltages from 3.3 kV up to and Including 33 kV

7.2 All AC cables shall be flame retardant, low smoke (FRLS) type designed to withstand all mechanical, electrical and thermal stresses develop under steady state and transient operating conditions.

7.3 Only terminal cable joints shall be accepted. No cable joint to join two cable ends shall be accepted. However, cable joints may be allowed if the route length is more than maximum available drum length subject to Employer's approval.

7.4 In addition to manufacturer's identification on cables as per relevant standard, following marking shall also be provided over outer sheath.

- i. Cable size and voltage grade
- ii. Word 'FRLS' at every metre
- iii. Sequential marking of length of the cable in metres at every metre

7.5 Cables shall be sized based on the following considerations:

- i. Rated current the equipment
- ii. In case of central inverters, maximum voltage drop in LT cable (from PCU to inverter transformer) shall be limited to 0.5% of the rated voltage. In case of string inverters, maximum voltage drop (from string inverter to LT combiner panel and from LT combiner panel to Inverter duty transformer) shall be limited to 1.5%. For HT cables (from inverter transformer to plant take off point), maximum voltage drop shall be limited to 0.5 % of the rated voltage. The Contactor shall provide voltage drop calculations in excel sheet.
- iii. Short circuit withstand capability as per design for 1s.
- iv. De-rating factors according to laying pattern

7.6 Warranty

All cables shall be warranted for minimum of 1 (one) year against all material/ manufacturing defects and workmanship.

7.7 Testing

Type tests, routine tests and acceptance tests requirements shall be as per relevant standards for all cable sizes.

7.8 Installation

7.8.1 Cable installation shall be as per IS 1255.

7.8.2 Cables within transformer yard and switchyard shall be laid through RCC cable trench with supports.

7.8.3 Cable terminations shall be made with properly crimped lugs and passed through cable glands at the entry & exit point of the cubicles. Bimetallic lugs shall be used for connecting Cu bus bar and Al cables or vice-versa.

7.8.4 All AC cables shall be provided with punched/embossed aluminium tags. The marking shall be done with good quality letter and numbers of proper size so that the cables can be identified easily.

8 Auxiliary Supply System

- 8.1 Scheme for Auxiliary supply system shall be submitted by contractor during detailed engineering for the approval by Employer.
- 8.2 It shall mainly comprise of auxiliary transformer, AC distribution board(s) (ACDB), Battery & battery charger system, emergency lighting network, Uninterrupted power supply (UPS), distribution cables and metering & protective devices.
- 8.3 Auxiliary system shall be provided with two independent sources for reliable auxiliary power supply.
- 8.4 Following consideration shall be taken into account while sizing the auxiliary transformer:
 - i. 20% future load margin
 - ii. 20% design margin
 - iii. Total connected load at 0.8 power factor

9 LT Switchgear

The LT switchgear specifications mentioned in this section are applicable for auxiliary supply distribution panel, AC combiner box and LT switchgear panels in case of string inverter configuration.

9.1 Standards and Codes

All equipment provided under LT switchgear shall comply with latest revisions and amendments of the relevant IEC standards and IS codes. In particular, the switchgear shall comply with the following standards and codes.

Standard/Code	Description
IS/IEC 61439-1	Low-voltage switchgear and control gear assemblies - Part 1: General rules
IS/IEC 61439-2	Low-voltage switchgear and control gear assemblies - Part 2: Power switchgear and control gear assemblies
IEC 60947-1	Low-voltage switchgear and control gear - Part 1: General rules
IEC 60947-2	Low-Voltage Switchgear and Control gear: Circuit Breakers
IEC 60947-3	Low voltage switchgear and control gear: Part 3 Switches, disconnectors, switch-disconnectors and fuse combination units

IEC 60947-4-1	Low-voltage switchgear and control gear - Part 4-1: Contactors and motor-starters - Electromechanical contactors and motor-starters
IEC 60947-5-1	Low-voltage switchgear and control gear - Part 5-1: Control circuit devices and switching elements - Electromechanical control circuit devices
IEC 62052-11	Electricity metering equipment (a.c.) - General requirements, tests and test conditions - Part 11: Metering equipment
IS 694	Polyvinyl chloride insulated unsheathed-and sheathed cables/ cords with rigid and flexible conductor for rated voltages - up to and including 450/750V
IEC 61869	Instrument Transformers
IS 3043	Code of practice for earthing
IEC 60255	Measuring relays and protection equipment - Part 1: Common requirements

9.2 Technical Parameters

System Details	
Rated system voltage	800 V $\pm$ 10%, 3 Phase, 4 wire, Neutral Solidly Earthed
Rated frequency	50 Hz $\pm$ 5%
System fault current	As per system requirement
Air Circuit Breaker (ACB)	
Type	Air-break
Rated Current	As per system requirement
Operating Duty	O – 3min – CO – 3min – CO

Rated Ultimate Short-Circuit Breaking Capacity & Rated Service Short-Circuit Breaking Capacity	As per system fault current
Rated Short-Circuit Making Capacity	As per rated ultimate short-circuit breaking capacity
Moulded case circuit breaker (MCCB)	
Rated Voltage	415 V
Release	Thermal-Magnetic/Microprocessor

Rated current	As per system requirement
Poles	4 poles
Rated insulation level	690 V
Rated Ultimate Short-Circuit Breaking Capacity & Rated Service Short-Circuit Breaking Capacity	As per system requirement
Rated Short-Circuit Making Capacity	2.1 X Short circuit breaking Capacity
Utilization category	A
Current transformer (CT)	
Type	Cast Resin Bar Primary
Voltage class and frequency	650 V, 50 Hz
CT Secondary Current	1 A
Class of insulation	Class E or better
Accuracy class & burden	
a) For Protection	5P20, 5 VA PS Class for REF and core balance CT (CBCT)
b) For Metering	Class 0.5, 5 VA (min)
Minimum primary earth fault current to be detected by CBCT	1 A
Instrument Security Factor for metering CT	5
Voltage Transformer (VT)	
Type	Cast Resin
Accuracy Class	0.5
Rated Voltage Factor	1.1 Continuous, 1.5 for 30 seconds
Class of Insulation	E or better
Digital Multifunctional Meter (MFM)	
Accuracy class	0.5 class
Communication with SCADA	RS485 communication with Modbus RTU

9.3 Constructional Details

9.3.1 The panel shall be metal enclosed, free standing, floor mounted,

modular type with compartmentalized construction having degree of protection of IP 2X (Indoor) and IP54 (Outdoor) as per IS/IEC 60529. All doors and covers shall be provided with neoprene gaskets to prevent entry of vermin and dust.

- 9.3.2 All switches, push buttons etc. shall be operated front and shall be flush/semi-flush mounted.
- 9.3.3 The panel shall be fabricated from 2 mm CRCA sheet steel for frame & load bearing surfaces. Partitions may be fabricated from 1.6 mm CRCA if no components are mounted on them.
- 9.3.4 Cable entries shall be from bottom. The opening of cable entry shall be covered by 3mm thick gland plates with proper sealing to avoid water and rodent entry.
- 9.3.5 Earthing bus bar of suitable cross section shall be provided throughout the length of panel.
- 9.3.6 The panel shall be duly wired with suitable size of 1.1kV, PVC insulated cable and terminals shall be brought out for cable connections. 10% spare terminals subjected to minimum one of each rating shall be provided on each distribution switchgear. All wire shall have ferrules as per wiring diagram.
- 9.3.7 The panel shall be painted with 2 coats of primer after pre-treatment and 2 coats of Polyurethane / epoxy paint with shade as decided by the Owner.
- 9.3.8 The panel shall be of dead front construction suitable for front operated and back maintained functioning.
- 9.3.9 240 V, 5 A, 3 pin industrial socket-outlet with ON/OFF switch shall be provided in each panel.
- 9.3.10 Each panel shall be provided with LED lamp rated for 240 V, 50 Hz, single phase AC supply for interior illumination controlled by door switch.
- 9.3.11 Suitable lifting hooks shall be provided for each panel.
- 9.3.12 Each switchgear panel shall be provided with thermostatically controlled space heaters to prevent condensation within the enclosure. The space heater shall be connected to 240 V, 50 Hz, single phase AC supply through suitable switch and

fuse.

- 9.3.13 Earth leakage relay with Core balance CTs (CBCT) shall be provided on main incoming feeders having phase CT ratio more than 50/1A. CBCT's shall be circular window type with window size based on the overall diameter of the cables, to be finalized during detailed engineering.

9.4 Air Circuit Breaker

- 9.4.1 The circuit breaker shall be three pole, air break, horizontal draw-out type.
- 9.4.2 The circuit breaker shall have three positions, i.e. SERVICE, TEST and ISOLATED.
- 9.4.3 The circuit breaker operating mechanism shall be based on motor operated spring charging and it shall be re-strike free, trip free both electrically and mechanically, with anti-pumping feature.
- 9.4.4 The rated control voltage of the spring charging motor shall be 110 VDC. Closing coil shall operate at all values of voltages between 85% and 110% of rated voltage. Opening coil shall operate correctly under all operating conditions of the circuit breaker up to the rated breaking capacity and at all values of supply voltage between 70% and 110% of rated voltage.
- 9.4.5 The spring charging motor shall have adequate thermal rating such that continuous sequence of the closing and opening operations is possible as long as power supply is available to the motor. It shall also be possible to charge the spring manually and close the breaker in the event of failure of motor / control supply to motor. Operating handle shall be provided for charging the operating mechanism. After failure of control supply to the motor, one open-close-open operation shall be possible with the energy contained in the operating mechanism.
- 9.4.6 The motor rating shall be such that it requires not more than 30 seconds for full charging of the closing spring. Closing action of the circuit breaker shall compress the opening spring ready for tripping. When closing springs are discharged after closing the breaker, they shall be automatically charged for the next operation.
- 9.4.7 Mechanical indicators shall be provided to indicate OPEN/CLOSE,

SERVICE/TEST positions of the circuit breaker and CHARGED/DISCHARGED positions of the closing spring. An operation counter shall also be provided.

9.4.8 The circuit breaker shall be provided with microprocessor based front adjustable protection release for overload, short circuit and earth fault.

9.4.9 Mechanical/Electrical interlocks shall be provided to prevent mal-operation and in particular to ensure the following.

- (i) It shall be possible to close the circuit breaker only if it is in SERVICE or TEST position.
- (ii) It shall be possible to open the door only when the breaker is in TEST position.
- (iii) Movement of the circuit breaker between SERVICE and TEST positions shall be possible only if the breaker is OFF.
- (iv) Racking in the circuit breaker from TEST to SERVICE position shall be possible only if door is closed.

9.4.10 Telescopic trolley or suitable arrangement shall be provided for maintenance of circuit breaker. The trolley shall be such that the top most breaker module can be withdrawn on the trolley and can be lowered for maintenance purpose. The telescopic trolley shall be such that all type, size and rating of breaker can be withdrawn/inserted.

9.4.11 The circuit breaker shall have suitable provision for integration with SCADA.

9.5 Instrument Transformers

9.5.1 Instrument transformers shall be completely encapsulated cast resin type, suitable for continuous operation at the ambient temperature prevailing inside the switchgear enclosure, when the switchgear is operating at its rated load and the outside ambient temperature is 50°C.

- 9.5.2 Polarity marks shall indelibly be marked on each instrument transformer and at the lead terminals at the associated terminal block.
- 9.5.3 HRC fuses of suitable rating shall be provided on primary side of voltage transformers. For secondary side, four pole Miniature Circuit Breakers (MCB) shall be provided.
- 9.5.4 For auxiliary supply switchgear, earth leakage relay with Core balance CTs (CBCT) shall be provided on main incoming feeders having phase CT ratio more than 50/1A. CBCT's shall be circular window type with window size based on the overall diameter of the cables, to be finalized during detailed engineering.

9.6 Bus bar

- 9.6.1 Bus bar shall be made of copper or aluminium with uniform cross section throughout their length. They shall be adequately supported on insulators to withstand electrical and mechanical stresses due to specified short circuit current.
- 9.6.2 All bus bars joints shall be thoroughly cleaned and anti-oxide grease shall be applied. Plain and spring washers shall be provided to ensure good contacts at the joints and taps. Wherever aluminium to copper connections are required, suitable bimetallic connectors or clamps shall be used.
- 9.6.3 Bus bars shall be provided with heat shrinkable sleeves of suitable insulation class throughout their length with proper colour coding. All bus bar joints and taps shall be shrouded.
- 9.6.4 Bus bar support insulators shall be made of non-hygroscopic, arc and track resistant, high strength material suitable to withstand stresses due to over voltage and short circuit current.
- 9.6.5 The Contractor shall submit busbar sizing calculation for specified continuous and short time current ratings during detailed engineering.

9.7 Earthing

- 9.7.1 An earth bus made of copper or aluminium shall be provided throughout the length of the panel. It shall be bolted to the framework of each panel and brazed to each breaker earthing contact bar.
- 9.7.2 The earth bus shall have sufficient cross section to carry maximum fault current without exceeding the allowable temperature rise.
- 9.7.3 All non-current carrying conductors of the panel shall be connected to the earth bus. All joints to the earth bus shall be made through at least two bolts. Hinged doors shall be earthed through flexible earthing braid of adequate cross section. Suitable provision shall be provided at each end of the earth bus for connection with Owner's Earth conductor.
- 9.7.4 Positive earthing of the carriage and breaker frame shall be maintained when it is in the connected position and in all other positions whilst the auxiliary circuits are not totally disconnected.
- 9.7.5 All metallic cases of relays, instruments and other panel mounted equipment shall be connected to earth bus by independent copper wires of size not less than 2.5 sq. mm with green colour insulation.
- 9.7.6 Instrument transformer secondary neutral point shall be earthed at one place only on the terminal block. Such earthing shall be made through links so that earthing of one circuit may be removed without disturbing the earthing of other circuits.

9.8 Multi-Function Meter

- 9.8.1 Digital, flush mounting type Multi-Function Meter (MFM) of 0.5 accuracy class shall be provided. It shall have provision for integration with SCADA.
- 9.8.2 MFM shall have provision to display the following parameters.
 - (i) Line and phase voltages
 - (ii) Line and phase currents
 - (iii) Active power, Reactive power, Apparent power

- (iv) Frequency
- (v) Power factor
- (vi) Total Harmonic Distortion (THD)

9.9 Wiring and Terminal blocks

- 9.9.1 All internal wiring shall be done with 650 V grade, 1.5 sq.mm. PVC insulated stranded flexible copper wire. For CT secondary circuits, 2.5 sq.mm copper wire shall be used.
- 9.9.2 Wire terminations shall be made with solderless crimping type tinned copper lugs, which shall firmly grip the conductor. Insulation sleeves shall be provided at all the wire terminations.
- 9.9.3 Printed identification ferrules, marked to correspond with panel wiring diagram shall be provided at both ends of each wire. The ferrules shall be firmly located on each wire so that they cannot move or turn freely on the wire. Wire identification shall be done in accordance with IS 11353.
- 9.9.4 The Contractor shall be solely responsible for the completeness and correctness of the internal wiring and for the proper functioning of the connected equipment.
- 9.9.5 All internal wiring to be connected to the external equipment shall terminate on terminal blocks. Terminal blocks shall be rated for 650 V, 10 A and made of non- inflammable material.
- 9.9.6 CT and VT secondary circuits shall be terminated on stud type, disconnecting terminal blocks.
- 9.9.7 At least 10% spare terminals shall be provided on each panel and these spare terminals shall be distributed on all terminal blocks.

9.10 Warranty

LT Switchgear shall be warranted against all material/ manufacturing defects and workmanship for minimum of 1 (one) year from the date of supply.

9.11 Testing and Inspection

9.11.1 Type Tests

The switchgear panel shall be of type tested design. Type test reports as per the following standards shall be submitted during detailed

engineering. The tests should have been conducted on the similar equipment by NABL accredited laboratory.

Equipment	Standard
Switchgear Panel	Relevant parts of IEC 61439
Air Circuit Breaker	IEC 60947-2
Moulded Case Circuit Breaker	IEC 60947-2
Current Transformer	Relevant parts of IEC 61869
Voltage Transformer	Relevant parts of IEC 61869

In case the contractor is not able to submit the test reports during detailed engineering, the contractor shall submit the reports of type/special tests either conducted by NABL accredited laboratory or witnessed by Employer.

9.11.2 Routine Tests

Routine tests and acceptance tests shall be as per the Quality Assurance Plan (QAP) approved by the Employer.

10 **Uninterrupted Power Supply**

10.1 Standards and Codes

Standard/Code	Description
IEC 62040-1	Uninterruptible power systems (UPS) - Part 1: General and safety requirements for UPS
IEC 62040-2	Uninterruptible power systems (UPS) - Part 2: Electromagnetic compatibility (EMC) requirements
IEC 62040-3	Uninterruptible power systems (UPS) - Part 3: Method of specifying the performance and test requirements

10.2 General Requirements

10.2.1 The Uninterrupted Power Supply (UPS) system shall be designed to supply power to following loads (but not limited to).

- i. Data logger / SCADA
- ii. Fire Detection/ Alarm Panel
- iii. HMI of SCADA
- iv. Emergency Lighting

- v. Inverter's Auxiliary supply (if applicable)
- vi. HT panel auxiliary
- vii. CCTV

10.2.2 Sizing of UPS shall be done considering the above-mentioned load at power factor of 0.8 lagging inclusive of 10% design margin at 50 °C.

10.3 System Description

10.3.1 The UPS shall automatically provide continuous, regulated AC power to critical loads under normal and abnormal conditions, including loss of input AC power. The UPS system shall consist of the following major equipment.

- i. UPS Module
 - a. Insulated Gate Bipolar Transistor (IGBT) Converter
 - b. Insulated Gate Bipolar Transistor (IGBT) Inverter
 - c. Digital Signal Processor (DSP) using Pulse Width Modulation (PWM) for Direct Digital Control (DDC) of all UPS control and monitoring functions
 - d. Static bypass switch
- ii. Battery system for 2 hours
- iii. Battery protective and disconnect device
- iv. Maintenance bypass switch
- v. LCD display panel and LED indications
- vi. Integrated UPS Communications Protocols capable of communicating with SCADA system

10.3.2 The UPS shall meet the following minimum specifications.

Parameter	Specification
Topology	Online double conversion UPS
Input	
Voltage	230 V $\pm$ 10% AC
Frequency	50 $\pm$ 5 Hz
Power factor	0.95

Output	
Voltage	230V $\pm$ 1% AC
Frequency	50 Hz
Power factor	0.8
Battery	
Type	Sealed, Maintenance-Free (AGM) battery

Capacity	100% UPS load for 4 hours
Monitoring and communication	
LED Indicators	Load on Inverter, Battery operation, Load on Bypass, Overload, LCD Fault, UPS Fault
Electrical contacts	Closing contacts for each of the following conditions: 1. Unit on Battery 2. Low Battery 3. Summary Alarm 4. UPS On 5. Input Fail
Local Display	LCD/ LED
SCADA communications	RS-485 Interface Port
Overall efficiency	>90%
Electrical Protection	Input/ output under voltage, over temperature, overload, Short circuit, battery low trip

10.3.3 The UPS shall be forced air cooled by internally mounted fans. The fans shall be redundant in nature to ensure maximum reliability. The fans shall be easily replaceable without the use of special tools.

10.3.4 Contractor shall provide the Operation & Maintenance Manual and mandatory spareparts list along with the equipment

10.4 Warranty

UPS and batteries shall be warranted for minimum of 5 (five) years against all material/ manufacturing defects and workmanship from the date of supply.

10.5 Tests

- 10.5.1 Routine tests and acceptance tests on final product shall be done as per QAP approved by the Employer.
- 10.5.2 On completion of installation and commissioning of the equipment on site tests shall be carried out with the max. available load, which does not exceed the rated continuous load. An on-site test procedure shall be submitted by contractor include a check of controls and indicators after installation of the equipment.

11 **Battery and Battery Charger**

11.1 Standards and Codes

Standard/Code	Description
IEC 60896-22:2004	Stationary lead-acid batteries - Part 22: Valve regulated types - Requirements
IEC 60896-21:2004	Stationary lead-acid batteries - Part 21: Valve regulated types - Methods of test
IS 1652	Specification for stationary cells and batteries, lead acid type (with plante positive plates)
IS 8320	General requirements and methods of tests for lead acid storage batteries.
IS 15549	Stationary Regulated Lead Acid Batteries

11.2 General

110 V DC system (Battery, Battery Charger & DCDB) in accordance with this specification and standards stated herein, shall comprise of the following.

- i. Sealed Maintenance Free (VRLA) Battery complete with racks & accessories.
- ii. One No. Float charger.
- iii. One No. Float cum Boost charger.
- iv. DC Distribution Board (DCDB)

11.3 Battery

11.3.1 Battery shall be used to supply the following loads with back up of four hours in case of complete power failure:

- i. Trip and closing coil of HT circuit breaker
- ii. Spring charging motors for HT circuit breaker
- iii. Annunciator and Indication circuit of HT panel
- iv. Auxiliary supply to protection relays

11.3.2 The battery sizing shall account for suitable temperature correction factors, ageing factors of 1.25, design margin of 1.25 & depth of discharge of 80%.

11.3.3 The design of the battery bank and sizing calculation along with the data sheet for the battery and battery charger shall be submitted for approval.

11.3.4 Battery voltage – 220V dc or 110V dc

11.4 Battery Charger

11.4.1 The Float Charger shall be used to supply normal DC loads and float charging current of charged battery. The Float cum Boost charger shall be designed to supply boost charging current requirement of the associated battery as well as to supply normal DC load. After full discharge of battery bank, the Float Cum boost charger shall be capable of charging the battery to its full capacity in 8 hours duration while supplying normal DC load.

11.4.2 The float charger shall have both auto and manual voltage regulation arrangements with provision of selector switch.

11.4.3 Suitable filter circuits shall be provided in all the chargers to limit the ripple content (peak to peak) in the output voltage and current to 2% and 5% respectively.

11.4.4 Digital Outputs shall be configured for connection to the SCADA to monitor the outputs like charger output current, output voltage, float/boost mode, etc.

11.4.5 The charging equipment shall be housed in a free standing, floor mounted compartmentalized panels. Panel shall have provision for

bottom cable entry with removable undrilled cable gland plate of 3.0 mm thickness.

11.4.6 The panel shall be of CRCA sheet steel construction having thickness of at least 2.0mm. Degree of protection provided by the enclosure to the internals of charger shall be IP-42.

11.4.7 The instruments, switches and indicating lamps shall be flush mounted on the front panel.

11.5 DC distribution board (DCDB)

11.5.1 DCDB shall be an integral part of battery charger panel board.

11.5.2 Doors and covers shall be provided with neoprene gaskets to prevent entry of vermin and dust. Also, door shall be provided with lock and key arrangement to prevent unauthorized access to the board.

11.5.3 DCDB shall have adequate number of outgoing feeders with double pole, DC MCBs. At least 20% feeders shall be provided as spare.

11.6 Warranty

Batteries and battery charger shall be warranted for minimum of 5 (Five) years against material/ manufacturing defects and workmanship.

11.7 Tests

Routine tests and acceptance tests shall be as per the Quality Assurance Plan (QAP) approved by the Employer.

12 **Earthing**

12.1 Standards and Codes

Earthing system shall comply with latest revisions and amendments of the relevant IEC standards and IS codes. In particular, earthing system shall comply with the following standards and codes.

Standard/Code	Description
IS 3043	Code of Practice for Earthing
IEC 62561-2	Requirements for conductors and earth electrodes
IEC 62561-7	Requirements for earthing enhancing compounds
IEEE 80	IEEE Guide for Safety in AC Substation Grounding

IEEE 142	IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems
Indian Electricity Rules	

12.2 General Requirements

- 12.2.1 Earthing system shall be designed based on system fault current and soil resistivity value obtained from geo-technical investigation report. Earth grid shall be formed consisting of number of earth electrodes sufficient enough to dissipate the system fault current interconnected by earthing conductors.
- 12.2.2 The earth electrode shall be made of high tensile low carbon steel rod, molecularly bonded by high conductivity copper on outer surface with coating thickness not less than 250 micron as per relevant standards. Suitable earth enhancing material shall be filled around the electrode to lower the resistance to earth. Inspection chamber and lid shall be provided as per IS 3043.
- 12.2.3 Earth conductors shall be made of copper bonded steel or galvanized steel of sufficient cross section to carry the fault current and withstand corrosion.
- 12.2.4 Earth conductors buried in ground shall be laid minimum 600 mm below ground level unless otherwise indicated in the drawing. Back filling material to be placed over buried conductors shall be free from stones and harmful mixtures.
- 12.2.5 Earth electrodes shall not be situated within 1.5m from any building whose installation system is being earthed. Minimum distance between earth electrodes shall be two times the driven depth of the electrode.
- 12.2.6 Transformer yard and switchyard fence shall be connected to the earth grid by one GS flat and gates by flexible lead to the earthed post.
- 12.2.7 All welded connections shall be made by electric arc welding. For rust protection, the welds should be treated with red lead compound and afterwards thickly coated with bitumen compound.

12.3 Earthing of PV array field

- 12.3.1 All PV Modules, Module Mounting Structures (MMS) and String Combiner Box (SCB) structures in the PV array field shall be bonded to the earthing system by two distinct connections.
- 12.3.2 Each PV Module frame shall be earthed using copper wire of sufficient cross section. The copper wire shall be connected to the earth hole provided in the module frame using suitable arrangement in line with the manufacturer recommendation. The earthing arrangement shall use stainless washers to prevent galvanic corrosion between aluminium frame and copper wire. In order to achieve effective earthing, serrated washers shall be employed to penetrate the anodization layer of the module frame.
- 12.3.3 Continuous copper earthing wire shall be run to connect a group of modules and both ends of the loop shall be bolted to the DC earth grid using bimetallic lugs and stainless-steel fasteners. The copper earthing wire shall be routed in such a way to avoid physical contact with the module aluminium frame.
- 12.3.4 The connection between MMS and DC earth grid shall be bolted or welded. Portion of the MMS which undergoes welding at site shall be coated with two coats of cold galvanising and anti-corrosion paint afterwards.
- 12.3.5 Earth electrodes of the DC earth grid shall be uniformly distributed throughout the PV array field so that optimum earth resistance is offered to leakage current flowing from any module frame or MMS.
- 12.3.6 For SCB equipment earthing, SCB shall be provided with two exclusive earth pits as per CEIG instructions.

12.4 PCU Earthing

DC negative bus bar of the PCU shall be earthed to avoid Potential Induced Degradation (PID). DC negative bus bar and PCU equipment earth shall be bonded to the PCU earth bus and connected to earth electrodes through flexible copper cable of sufficient cross section as mentioned by the manufacturer. The interconnection of PCU earth electrodes with DC earth grid shall be as per PCU manufacturer recommendation. In case DC negative grounding is not possible, appropriate anti-PID device shall be provided.

12.5 Transformer Earthing

- 12.5.1 Inverter transformer neutral shall be floating, not to be earthed. However, recommendation of inverter manufacturer shall also be taken into account.
- 12.5.2 Transformer tank, cable box, marshalling box and all other body earth points shall be earthed.
- 12.5.3 Inverter transformer shield shall be earthed separately using minimum two no. of earth electrodes. Earthing conductor between shield bushing and earth electrodes shall be copper flat of suitable size not less than 25 x 6 mm.
- 12.5.4 Neutral and body of the auxiliary transformer shall be earthed.

12.6 Inverter Room and Main Control Room Earthing

- 12.6.1 Metallic enclosure of all electrical equipment inside the inverter room and main control room shall be connected to the earth grid by two separate and distinct connections.
- 12.6.2 Cable racks and trays shall be connected to the earth grid at minimum two places using galvanized steel flat.
- 12.6.3 SCADA and other related electronic devices shall be earthed separately using minimum two no. of earth electrodes.

12.7 Switchyard Earthing

The metallic frame work of all switchyard equipment and support structures shall be connected to the earth grid by means of two separate and distinct connections.

Note: Switchyard shall be shielded against direct lightning strike by provision of overhead shield wire or earth wire or spikes(masts) or a combination thereof as per CEA regulations 2010 (Technical standards)- 42(2)(C).

12.8 Tests

Type test reports for earthing electrode, earth enhancing compound and its associated accessories shall be submitted during detailed engineering for approval.

On completion of installation, continuity of earth conductors and efficiency of all bonds and joints shall be checked. Earth resistance at earth terminations shall be measured and recorded.

The earth plate shall be provided to facilitate its identification and for carrying out periodical inspection.

13 Lightning Protection System

13.1 Lightning Protection System (LPS) for entire plant against direct lightning strokes shall be provided as per IEC 62305:2010 or NFC 17-102:2011.

13.2 Protection level for the entire plant shall be Level-III.

13.3 LPS as per IEC 62305

Location of air terminals shall be designed as per rolling sphere method.

13.4 LPS as per NFC 17-102

Lightning Protection System shall consist of following accessories.

- i. Early Streamer Emission (ESE) air terminal
 - ii. Highly insulated poly-plastic adaptor to fix the ESE air terminal with the FRPmast
 - iii. Fiberglass Reinforced Plastic (FRP) mast
 - iv. Coupler to connect FRP mast with GI mast
 - v. Galvanized Iron mast with base plate and guy wire kit
 - vi. Down-conductor: PVC insulated flexible copper cable of suitable size complying with EN 50164-2 or equivalent standard. It shall be routed along the mast with suitable fixings and connectors
 - vii. Test joint with each down conductor
 - viii. Lightning event counter complying with EN 50164-6 or equivalent standard. It shall be fixed at suitable height in series with the down conductor.
 - ix. Earth termination system in accordance with NFC 17-102. Earth electrodes shall comply with the EN 50164-2 or equivalent standard. Earth enhancing compounds complying with EN 50164-7 or equivalent standard, may be used where soil resistivity is higher and making it impossible to achieve system resistance within specified limit.
- 13.5 Accessories listed above are indicative only and any other fittings or accessories, which are usual or necessary for satisfactory operation of the lightning protection shall be provided by the Contractor without

extra charges.

- 13.6 Necessary foundation/anchoring for holding the lightning mast in position to be made after giving due consideration to shadow on PV array, maximum wind speed and maintenance requirement at site in future.
- 13.7 As per CEIG instructions, 2 (two) earth pits shall be constructed & 25x6 mm copper flats shall be connected from counter of lightening conductors to earth pits. Earth pit chambers shall be constructed for earth rod of lightening conductors.
- 13.8 The product shall be warranted for minimum of 5 (Five) years against all material/manufacturing defects and workmanship.
- 13.9 Type test reports as per IEC 62305:2010 or NFC 17-102:2011 shall be submitted during detailed engineering for approval.

14 Communication Cables

14.1 Optical Fibre Cables

- 14.1.1 Optic Fibre cable shall be 4/8/12 core, galvanized corrugated steel taped armoured, fully water blocked with dielectric central member for outdoor/ indoor application so as to prevent any physical damage.
- 14.1.2 The cable shall have multiple single-mode or multimode fibres on as required basis so as to avoid the usage of any repeaters.
- 14.1.3 The outer sheath shall have Flame Retardant, UV resistant properties and are to be identified with the manufacturer's name, year of manufacturing, progressive automatic sequential on-line marking of length in meters at every meter on outer sheath.
- 14.1.4 The cable core shall have suitable characteristics and strengthening for prevention of damage during pulling.
- 14.1.5 All testing of the optic fibre cable being supplied shall be as per the relevant IEC, EIA and other international standards.
- 14.1.6 The Contractor shall ensure that minimum 100% cores are kept as spare in all types of optical fibre cables.
- 14.1.7 Cables shall be suitable for laying in conduits, ducts, trenches, racks and underground buried installation.

14.1.8 Spliced/ Repaired cables are not acceptable. Penetration of water resistance and impact resistance shall be as per IEC standard.

14.2 Communication Cable (Modbus)

14.2.1 Data (Modbus) Cable to be used shall be shielded type with stranded copper conductor. Cable shall have minimum 2 pair each with conductor size of 0.5 Sq.mm. Cable shall be flame retardant according to IEC 60332-1-2.

14.2.2 Cable shall be tested for Peak working voltage of not less than 300 V and shall be suitable for serial interfaces (RS 422 and RS 485).

14.2.3 Communication cable shall be laid through underground with suitable HDPE ducts.

15 SCADA

15.1 General Requirements

15.1.1 The Contractor shall provide complete SCADA system with all accessories, auxiliaries and associated equipment and cables for the safe, efficient and reliable operation and monitoring of entire solar plant and its auxiliary systems.

15.1.2 The Contractor shall provide all the components including, but not limited to, Hardware, Software, Panels, Power Supply, HMI, Laser Printer, Gateway, Networking equipment and associated Cables, firewall etc. needed for the completeness.

15.1.3 SCADA System shall have the provision to perform the following features and/or functions:

- i. Web enabled Operator Dashboards: Showing key information on Generation, Performance and Current Status of various equipment in Single Line Diagram(SLD) format with capability to monitor PV array Zone level (i.e. SCB level) parameters.
- ii. Real time Data Logging with Integrated Analytics & Reporting: Logging of all parameters - AC, DC, Weather, System Run Hours, Equipment Status and Alarms as well as derived/ calculated/ integrated values. The SCADA User interface shall be customizable and enable Report Generation and Graphical Analysis.

- iii. Fault and System Diagnostics with time stamped event logging.
- iv. Support for O&M Activities: The interface shall allow integration with Surveillance System(s), Module Cleaning System and various other O&M support systems to provide a Data Analysis and Decision Support System for smooth and efficient Plant Operations.
- v. AI based Distributed Analytics for Predictive Maintenance, trend analysis and Alerts.
- vi. Generate, store and retrieve user configurable Sequence of Event (SOE) Reports.
- vii. Interface with different field equipment in the plant and work seamlessly with field equipment supplied by different companies.
- viii. Transfer of plant data reliably, to a Cloud server on any kind of remote network including low bandwidth and wireless links such as 2G/3G/VSAT

Note: Telecom Lease line connection, if required for transferring data from Plant over internet shall be taken by Contractor in the name of Employer for O&M period

- 15.1.4 The Control system shall be designed to operate in non-air-conditioned area. However, the Contractor shall provide a Package/ Split AC of suitable capacity decided by heat load requirement in SCADA room at Main Control Room.

15.2 Architecture

- 15.2.1 The SCADA System shall be built over Industrial IoT architecture with integrated Analytics, secure web access, enterprise software and Database.
- 15.2.2 Data acquisition shall be distributed across MCR and LCRs while plant level data aggregation shall be done in both local and remote server (as specified by Owner).
- 15.2.3 Analog and Digital IO modules shall have integrated processor for distributed IO processing and control.

- 15.2.4 Data communication system shall be built over fibre optic cables/ wireless network with high bandwidth TCP/IP communication (Fast Ethernet or 802.11a/b/g/n) across all Inverter and Control Rooms with Internet/Intranet access at Main Control Room. Firewall shall be provided for network security.
- 15.2.5 Plant SCADA Server shall have Industrial Grade server hardware running SCADA & Monitoring Software with data storage (complete plant data) space for 2 years.
- 15.2.6 Plant data for monitoring and control operations should be accessible without dependence on external network.
- 15.2.7 A virtual/cloud server running SCADA & Monitoring Software shall be configured in parallel with Plant Server to enable easy access to plant data from outside the plant without having to login to plant server. Effectively, the plant data shall be replicated in both places i.e. between systems at the Plant Server and Remote Server to provide data redundancy for complete plant data.
- Note:** Configuration of Cloud server and procurement of associated subscription services shall be in the scope of the EPC Contractor.
- 15.2.8 Operator Workstation/PC shall be of Industrial Grade for browser-based access to plant data from Plant or remote server. Plant control & SLDC/Utility related operations shall only be initiated through browser-based interface requiring no client software or database to be installed on the Workstation. All critical software and Plant Data shall be installed/stored on local and remote servers only with user access control for protecting the software and data assets from accidental deletion or corruption.
- 15.2.9 Internet/Intranet at Plant: Public or private network access shall be provided at the plant through any broadband/VSAT connectivity of 2Mbps or higher bandwidth. In case no broadband/VSAT connectivity can be provided at the plant, a 3G/4G data

card from any Internet Service Provider (ISP) may be provided. SCADA system shall be capable of sending all plant data in real time to the Remote Server.

- 15.2.10 GPS based Time Synchronization System: The SCADA system shall have a Master/Slave Clock system along with antenna, receiver, cabinet and internal interconnection cables. All SCADA controllers, servers, OWS and communicating equipment shall be synchronized to the GPS clock.

15.3 Industrial IoT Controllers & Data Acquisition

The Plant SCADA and Monitoring System may use one or more IIoT Controllers at each Inverter Control Room and MCR for the purpose of data acquisition and data forwarding to the Local and Remote SCADA Servers. The IIoT Controllers shall meet the following minimum requirements:

- 15.3.1 The IIoT Controllers shall be distributed in nature and work independently of other IIoT Controllers or any central controller in the system.
- 15.3.2 Shall be capable of supporting wide range of field protocols to communicate with different field equipment (Modbus over RS485/Ethernet, etc.)
- 15.3.3 Shall have local storage for a minimum of 2 weeks (in case of network failure).
- 15.3.4 Provide web-based interface to configure the controller for various equipment in the field.
- 15.3.5 IO Functionality: Shall support status monitoring of VCBs & Trip relays on RMU/HT & Transformer panels through distributed DI/AI modules.
- 15.3.6 Controls: Shall be capable of Controlling breakers (ON/OFF). Both ON/OFF and Parameter control of inverters shall be supported.
- 15.3.7 Data Communication with Servers: Shall send the data collected, from all the equipment at Inverter Control Room and/or Main Control Room, to the Monitoring & Control Server.
- 15.3.8 Controllers shall be capable of sending data over Internet connections USB data cards.

15.3.9 Shall not require a static public IP address, at the plant for the purpose of remote access.

15.4 Functionalities

15.4.1 In case of central inverter, SCADA system shall enable PV array Zone monitoring i.e. the total current from each String Combiner Box shall be monitored on the DC side.

15.4.2 The SCADA system shall monitor instantaneous and cumulative electrical parameters from all DC & AC Equipment including inverters, weather station, MFM, Transformer and Switchgear (LT & HT Panels) at regular intervals not greater than one minute.

15.4.3 The SCADA system shall monitor Instantaneous and cumulative environment parameters from weather sensors or data loggers at same interval as electrical parameters and provide PR, CUF on the fly.

15.4.4 The SCADA system shall provide Alarms and Alerts on equipment faults and failure in less than 5 seconds. Alarms on status change of hardwired DI shall also be provided.

15.4.5 The SCADA system shall provide configurable alerts on any parameter crossing settable thresholds. The list of such parameters shall be finalised in consultation with the Owner.

15.4.6 The SCADA system shall enable integration with other sub-systems at the plant for supporting O&M activities. The list shall include but not limited to:

- i. Surveillance Cameras,
- ii. Module Cleaning System – For monitoring of water usage and efficacy of cleaning process (in case of Wet Type Cleaning System).

15.4.7 The SCADA system shall have user-friendly browser-based User Interface for secure access from anywhere, for minimum ten concurrent connections from the Operator PC or other securely connected laptop/mobile, for plant monitoring, O&M, daily reporting, and analysis. A dashboard providing summary details of total plant generation, day's export, irradiance, Inverter Control Room level

generation and performance indicators like PR and CUF.

- 15.4.8 Reporting: The SCADA system shall provide downloadable reports in Excel/PDF, configurable for equipment parameters across the plant.
- 15.4.9 The system shall have Configurable Analysis page for self-configured as well as on demand Analytics charts.
- 15.4.10 The SCADA system shall be extensible to include maintenance of O&M schedules and related activities for plant equipment as per the O&M Manual.
- 15.4.11 Connectivity shall be provided to Owner's Data Monitoring Centre. Data collected by Plant SCADA shall be replicated in real-time, using industry standard interfaces such as Web Services, OPC-UA, data files, as required – with Owner's Central Monitoring System in Hyderabad. The data recording intervals for different parameters from different devices in the solar plant shall be considered when creating schedules to “push” the data from Plant SCADA to data receivers stationed at Hyderabad.
- 15.4.12 Mobile User Interface: summary of plant performance and issues should be accessible in a mobile Native UI or browser UI.
- 15.4.13 Data Communication to SLDC: SCADA system shall provide required interface to integrate with TRANSCO-SLDC, in compliance with grid code, to send any parameters specified by SLDC.

Note: The methodology and specification of SLDC interface will be provided separately by SLDC/TRANSCO and it shall be the responsibility of the Contractorto determine the same.
- 15.4.14 Power Plant Control: SCADA system shall provide required interface to the local SCADA operator to set various power control modes (active/reactive power/frequency/PF) through the inverters over industry standard communication protocols like Modbus over TCP/IP.
- 15.4.15 Forecasting and Scheduling: SCADA shall provide day ahead and week ahead forecasting and scheduling for power generation at the plant as per SLDC/Utility stipulations.

- 15.4.16 Predictive Maintenance: SCADA system shall have in-built or pluggable frameworks to support AI based Predictive Maintenance for all key equipment including inverters, transformers and switchgear at the plant.
- 15.4.17 All programming functionalities shall be password protected to avoid unauthorized modification.
- 15.4.18 The Contractor shall provide software locks and passwords to Employer for all operating & application software. Also, the Contractor shall provide sufficient documentation and program listing so that it is possible for the Employer to carry out modification at a later date.

15.5 Earthing

- 15.5.1 Two isolated electronic earth pits near to SCADA panel at every Inverter and Control Room with < 1 Ohm resistance shall be provided. One earth pit shall be used for protective/body earth and the other to be used for Signal Earth.
- 15.5.2 Apart from providing separate earth pits, manufacturer specified earthing recommendations shall be followed for all communicating equipment connected to SCADA. This includes but is not limited to SMBs, Inverters, WMS and Switchgear panels.

15.6 Communication Cable Laying

- 15.6.1 All RS485, IO and CAT6 cables shall be laid in separate conduits with a minimum separation of 1.5ft from AC/DC power cables all along.
- 15.6.2 Power cables shall be laid deep in the trenches first. Data cables shall be laid in separate conduits after partially filling the trenches to ensure minimum 1.5 ft separation between power and communication cables all along the trench.
- 15.6.3 IO Cables between switch gear panels and SCADA panel shall be laid on separate cable trays, with a minimum of 1.5ft separation from trays carrying AC Power cables.

- 15.6.4 RS485 & CAT6 cables between switch gear panels or Inverters and SCADA panel shall be laid on separate cable trays, with a minimum of 1.5ft separation from trays carrying AC Power cables.

15.7 Control Cabinets / Panels / Desks at Main Control Room

- 15.7.1 The cabinets shall be IP-22 protection class. The Contractor shall ensure that the temperature rise is well within the safe limits for system components even under the worst condition and specification requirements for remote I/O cabinets.
- 15.7.2 The cabinets shall be totally enclosed, free standing type and shall be constructed with minimum 2 mm thick steel plate frame and 1.6 mm thick CRCA steel sheet or as per supplier's standard practice for similar applications.

15.8 Software Licences

The Contractor shall provide software license for all software being used in Contractor's System. The software licenses shall be provided for the project and shall not be hardware/ machine-specific.

15.9 Hardware at Main Control Room

- 15.9.1 The Hardware as specified shall be based on latest state of the art Workstations and Servers and technology suitable for industrial application & power plant environment.
- 15.9.2 The Local Monitoring & Control Server and the Operating Work station, to be deployed in the Plant Control Room, shall have the following server hardware and operating system along with accessories:

Plant Server	
Server Hardware	Hex/Octal Core Xeon, 32GB RAM (expandable to 64 GB RAM), 4 X 2TB SATA hard discs in RAID 5 configuration, 2TB external USB hard disc (for backup), dual power supplies, 2 LAN ports, LCD console, keyboard & mouse.

	The Server hardware shall be housed in a rugged fan-cooled, and rodent-proof Server Rack.
Operating System	Operating System and Database shall be of enterprise scale (preferably RedHat Linux or equivalent Linux OS, Oracle/MySQL or equivalent DB), with required AMC for 5 years.
Accessories	i. Monitor: Min 22" LED Flat Monitor with non-interfaced refresh rate min. 75 Hz. ii. Keyboard: ASCII type iii. Pointing Device: Mouse iv. Intelligent UPS (on line): Minimum 2 hour battery backup.
Operator Workstation	
Hardware	i7 CPU running at 3.0 GHz or faster with 8GB RAM, 500GB hard disk, 25" LED monitor, keyboard and mouse, 4 USB ports, LAN port
Operating System	Windows operating system with necessary tools, anti-virus software.
Accessories	i. Screen Display Unit: Min 50" LED Flat Monitor with wall mounted arrangement for the display of SCADA screen ii. A4 size monochrome laser printer. iii. UPS of required capacity with 2 hour battery backup.

15.9.3 All network components of LAN and Workstations shall be compatible to the LAN, without degrading its performance.

15.10 Factory Acceptance Test (FAT)

FAT procedure shall be submitted by bidder for approval. SCADA shall communicate with all third devices which are part of solar plant and same shall be demonstrated during the FAT.

16 Energy Management System

16.1 Energy Management System (EMS) system shall be a computerized system for real time monitoring, operation, control, reliable & efficient operation of the Plant facilities. EMS shall be able to acquire real time data of various equipment of Plant facilities, have in built logic/programming to monitor, control, and optimize the performance

of Plant facilities as per specification. Contractor shall provide complete EMS system with all accessories, auxiliaries and associated equipment and cables for the safe, efficient and reliable operation of entire Plant facilities and its auxiliary systems. Contractor shall include in his proposal all the Industrial Grade Hardware, Software, Panels, Power Supply, HMI, Laser Printer, Gateway, Networking equipment and associated Cable etc. needed for the completeness even if the same are not specifically appearing in this specification.

16.2 Standards and Codes

- 16.2.1 The EMS shall comply with IEC 61970 for interoperability.
- 16.2.2 The EMS shall have the functionality to ensure compliance to the CEA Technical standards for Connectivity (2019) regulations.
- 16.2.3 The EMS shall comply with cyber security guidelines issued by the Central Government, from time to time, and the technical standards for communication system in Power Sector laid down by the Authority.

16.3 EMS functionality for the Plant Control

- 16.3.1 The EMS monitors grid and Plant facility variables and should be programmable for selecting the optimum-operating mode of the whole plant w.r.t. active and reactive power, grid voltage, grid frequency, etc. Additionally, it can receive external set points and automatically adapt the Plant Facility behaviour to the new settings.
- 16.3.2 The EMS shall perform following functionality to Control the Plant facilities:
 - i. Communication with grid or SCADA
 - ii. Communications with PV Inverters and other power units
 - iii. Measuring and processing of the electrical magnitudes at EMS (voltage, current, PF)
 - iv. Control capability of PV Inverters and other power units
 - v. The EMS shall allow following operation modes for the Plant facilities:
 - vi. Reactive Control (Q Control, setting point of reactive power Q at EMS)
 - vii. Power Factor Control (PF Control, setting point of $\cos(L)$ at EMS)
 - viii. Voltage Control (V closed loop control, setting point of V at EMS)
 - ix. Voltage Droop (Reactive power vs Voltage programmable curve or droop)

- x. Apparent Power Control (S Lim, setting point of S Lim at EMS)
- xi. Active Power Limitation (P Lim, setting point of P Lim at EMS)

16.3.3 The EMS shall have the functionality to receive the target values specified by operators using a standard protocol (i.e. Modbus TCP/IP).

16.4 Control & Power Supply Scheme

Contractor shall provide the UPS/ DC Power supply of suitable rating to cater all the load requirements of EMS system and its auxiliaries

17 **Power Transformer**

17.1 Standards and Codes

Power Transformer shall comply with the latest edition of the following standards and codes including amendments.

Standard	Description
IS 2026, IEC 60076	Specification of Power Transformers
IS 2099, IEC 60137	Bushings for alternate voltage above 1000 V
IS 8468	On-load tap changers
IS 335, IEC 60296	Insulating oil
IS 3639	Fittings and Accessories for Power Transformers

17.2 Technical Requirements

Parameter	Specification
Rated Capacity	As per system design
Rated Voltage	33 kV / 0.8 kV or 33 kV / 0.415 kV
Duty & Service	Continuous duty & Outdoor
Number of phases	3
Frequency	50 Hz
Vector group	As per system requirement
Cooling	ONAN / ONAF
Impedance at principal tap and 75°C	10%
Tap changer	On Load Tap Changer (OLTC) on HV side, +/-10% with steps of 2.5%

Winding Insulation Level	HV	HVN (If applicable)	LV	LVN (If applicable)
One min power frequency withstand voltage (kV)	275	38	70	38
Full wave lightning impulse withstand voltage (kVp)	650	95	170	95
Chopped wave lightning impulse withstand voltage (kVp)	715	-	187	-
Switching impulse withstand voltage (kVp)	540	-	-	-
Permissible temperature rise over an ambient of 50°C (irrespective of tap)				
Top oil	50°C			
Winding	55°C			
Fault level & duration	As per system requirement			
Short-circuit withstand time (Thermal)	2 second			
Bushing Insulation Level	HV	HVN (If applicable)	LV	LVN (If applicable)
Rated voltage (kV)	145	36	36	36
One min power frequency withstand voltage (kV)	305	77	77	77
Full wave lightning impulse withstand voltage (kVp)	650	170	170	170
Switching impulse withstand voltage (kVp)	-	-	-	-
Termination	As per system requirement			
Noise level	As per NEMA TR-1			
Loading capability	Continuous operation at rated MVA on any tap with voltage variation of +/- 10%, also transformer shall be capable of being loaded in accordance with IEC 60076-7			

Flux density	Not to exceed 1.9 Wb/sq.m. at any tap position with +/- 10% voltage variation of voltage corresponding to the tap. Transformer shall also withstand following over fluxing conditions due to combined voltage and frequency fluctuations: i. 110% for continuous rating ii. 125% for at least one minute iii. 140% for at least five seconds The Contractor shall furnish over fluxing characteristic up to 170%
Air clearance	As per applicable standards

17.3 Tank

- 17.3.1 The Transformer tank and cover shall be fabricated from high grade low carbon plate steel of adequate thickness. The tank and the tank cover shall be of welded construction. All seams and joints shall be welded and where practicable, they shall be double welded. The tank so welded shall be reinforced by stiffener of structural steel for general rigidity.
- 17.3.2 The transformer top shall be provided with a detachable tank cover with bolted flanged gasket joint. Lifting lugs shall be provided for removing the cover. The surface of the cover shall be suitably sloped so that it does not retain rain water.
- 17.3.3 The main tank body of the transformer, excluding tap changing compartments and radiators, shall be capable of withstanding pressure of 760mm of Hg.
- 17.3.4 Inspection hole(s) with welded flange(s) and bolted cover(s) shall be provided on the tank cover. The inspection hole(s) shall be of sufficient size to afford easy access to the lower ends of the bushings, terminals etc.
- 17.3.5 Suitable guides shall be provided for positioning the various parts during assembly or dismantling. Adequate space shall be provided between the cores and windings and the bottom of the tank for collection of any sediment.
- 17.3.6 All bolted connections to tank shall be fitted with suitable oil-tight gasket, which shall give satisfactory service under the operating conditions. All gaskets shall be closed design (without open ends)

and shall be of one piece only. Gasket of nitrile rubber or equivalent shall be used. Gaskets of neoprene and / or any kind of impregnated / bonded core or cork only which can easily be damaged by over-pressing are not acceptable. Use of hemp as gasket material is also not acceptable.

17.3.7 Lifting lugs shall be provided on all parts of the transformer requiring independent handling during assembly or dismantling. In addition, the transformer tank shall be provided with lifting lugs and bosses properly secured to the sides of the tank for lifting the complete transformer assembly with oil either by crane or by jacks.

17.3.8 The transformer tank shall be supported on a structural steel base equipped with forged steel single flanged wheels suitable for moving the transformer completely with oil. The wheels shall be bi-directional and mounted on swivels which may be turned through 90° when the tank is jacked up and capable of being locked in position parallel to and at right angles to the longitudinal axis.

17.4 Core

17.4.1 The transformer core shall be built up with high-grade non-ageing cold rolled grain oriented (CRGO) silicon steel laminations having high permeability and low hysteresis loss. The thickness of lamination shall be 0.27 mm or less.

17.4.2 The transformer shall be so designed that the flux density in the core shall not exceed 1.7 tesla at rated voltage and rated frequency. The maximum flux density in any part of core or yoke at 10% continuous over voltage condition shall not exceed 1.9 tesla.

17.4.3 The laminations shall be free of all burrs and sharp projections. Each sheet shall have an insulating coating resistant to the action of hot oil.

17.4.4 The core shall be rigidly clamped to ensure adequate mechanical strength and to prevent vibration during operation and transportation. The clamping structure shall be designed to minimize eddy current loss.

- 17.4.5 The design of magnetic circuit shall be such as to avoid static discharges, development of short circuit paths within itself or to the earthed clamping structure and production of flux components at right angles to the plane of the laminations which may cause local heating.
- 17.4.6 The core shall be provided with lugs suitable for lifting the complete CCA of the transformer. The CCA shall be fixed with the tank so that it does not shift when transformer is moved or during short circuit.
- 17.4.7 The insulation of core to bolts and core to clamp plates shall be able to withstand a voltage of 2 kV RMS for one minute.
- 17.4.8 The core shall not be earthed at multiple locations. Terminal shall be brought on top of tank and earthed through link. Core and Frame terminals should be brought out on transformer top so as to enable megger.

17.5 Winding

- 17.5.1 The conductor for winding shall be made of electrolytic grade copper. The winding shall be so designed that all coil assemblies of identical voltage ratings shall be interchangeable and field repairs can be readily done without special equipment.
- 17.5.2 The coils shall be supported between adjacent sections by insulating spacers and barriers. Bracings and other insulation used in the assembly of the windings shall be arranged to ensure a free circulation of the oil and to reduce hot spots in the windings.
- 17.5.3 The insulation paper shall be of high quality and the value of degree of polymerization shall not be less than 1200 Pv.
- 17.5.4 Materials used for insulation and assembly of the windings shall be insoluble, non- catalytic and chemically inactive in the hot transformer oil and shall not soften or otherwise get affected under the operating conditions.
- 17.5.5 All threaded connections shall be provided with locking facilities. All leads from the winding to the terminal board and bushings shall be rigidly supported to prevent injury from vibration. Guide tubes shall be used where practicable.

- 17.5.6 The conductor shall be transposed at sufficient intervals in order to minimize eddy currents and equalize the distribution of currents and temperature along the windings.
- 17.5.7 Windings shall be subjected to a shrinkage treatment before final assembly, so that no further shrinkage occurs during service. Adjustable device shall be provided for taking up any possible shrinkage of coils in service if required.
- 17.5.8 The windings shall be clamped securely in place so that they will not be displaced or deformed during short circuits. The assembled core and windings shall be vacuum dried and suitably impregnated before removal from the treating tank. The copper conductors used in the coil structure shall be best suited to the requirements and all permanent current carrying joints in the windings and the locks shall be welded or brazed.

17.6 Insulating Oil

The oil for first filling together with 10% extra shall be supplied with the transformer. The oil shall comply in all respects with the provisions of the latest edition of IS 335 (as amended up to date). Particular attention shall be paid to deliver the oil free from moisture having uniform quality throughout in non-returnable steel drums.

17.7 On-Load Tap Changer

- 17.7.1 On-Load Tap Changer (OLTC) shall be designed for remote control operation from Remote Tap Change Control (RTCC) Panel in the control room in addition to being capable of local manual as well as local electrical operation. The OLTC shall include the following.
- i. An oil immersed tap selector and arcing switch or arc suppressing tap selector, provided with reactor or resistor for reduction of make and break arcing voltages and short circuits.
 - ii. Motor driven mechanism
 - iii. Control and protection devices
 - iv. Local /Remote tap changer position indicator

- v. Manual/Electrical operating device
- vi. Pressure relief device

17.7.2 The OLTC shall be so designed that the contacts do not interrupt arc within the maintank of the transformer. The tap selector and arcing switch or arc suppressing selector switch shall be located in oil filled compartment. The compartment shall be provided with Oil Surge Relay. It shall be designed so as to prevent oil in the tap selector compartment from mixing with the oil in the transformer tank.

17.7.3 The contactors and associated gear for the driving motor shall be housed in a local kiosk mounted adjacent to or on the transformer. The degree of protection of the complete arrangement shall be IP 55 or better. The motor shall be suitable for operation with three phase, 415 V, 50 Hz external power supply.

17.7.4 RTCC Panel

Remote Tap Change Control (RTCC) Panel shall include, but not limited to, the following.

- i. Automatic Voltage Regulator with SCADA compatibility
- ii. Under voltage relay to monitor the taper changer control voltage
- iii. Raise and lower push button
- iv. Tap position indicator
- v. Indication lamp showing tap changing in progress
- vi. Alarms and Annunciation
- vii. Any other accessory required for satisfactory operation or required during detail engineering

17.8 Bushing

17.8.1 The bushings shall have high factor of safety against leakage to ground and shall be so located as to provide adequate electrical clearances between bushings and grounded parts. Bushings of identical voltage rating shall be interchangeable.

17.8.2 All bushings shall be equipped with terminals suitable for bimetallic connection. Each bushing shall be so coordinated with the transformer insulation that all flash over will occur outside the tank.

- 17.8.3 Bushings of rated voltage below 52 kV shall be porcelain insulator of oil- communicating type or OIP (non-oil communicating) or RIP type.
- 17.8.4 Bushings of rated voltage 52 kV and above shall be OIP condenser type (non-oil communicating) with porcelain insulator with following fittings.
- i. Oil level gauge
 - ii. Oil filling plug and drain valve if not hermetically sealed
 - iii. Tap for capacitance and tan delta test

17.9 Radiators

- 17.9.1 Radiators provided shall have sufficient cooling surface to limit the temperature rise to the values as specified in the 'Technical Requirements'. The radiators shall be seamless and made of mild steel/CRCA with minimum thickness not less than 1.2 mm. It shall be suitably braced to protect them from mechanical shocks.
- 17.9.2 The radiators shall be connected to the tank by machined steel flanges with adequate gaskets to avoid oil leakage. Each radiator unit shall be provided with butterfly type or positive operated gate type oil leak proof shut-off valve which can be fastened in either closed or open position and separate oil tight flange for each tank connection for use when the radiator unit is detached. Each radiator unit shall have a lifting arrangement and oil drain at the bottom and a vent at the top.
- 17.9.3 It shall be possible to take out any of the radiator unit without disturbing the transformer. The radiators shall be so designed as to prevent any accumulation of water on the outer surface or formation of gas pockets when the tank is being filled.

17.10 Accessories

17.10.1 Conservator

The conservator shall have air cell type constant oil preservation system to prevent oxidation and contamination of oil due to contact with moisture. The conservator shall be provided with separate compartment for OLTC. No separate conservator tank shall be provided for OLTC.

The conservator shall be fitted with oil filling hole, cap and drain valve. Prismatic toughened glass oil level gauge and 150 mm Magnetic Oil Gauge (MOG) with low oil level alarm contact shall also be provided.

17.10.2 Silica gel breather

The top of the conservator shall be connected to the atmosphere through indicating type cobalt free silica gel dehydrating breather with transparent enclosure. Silica gel shall be isolated from atmosphere by an oil seal. The capacity of breather should be such that it can contain minimum 5 kg silica gel for main conservator compartment and minimum 1 kg silica gel for OLTC conservator compartment. The GI pipe connecting breather with conservator should be seamless and no joint is permitted.

17.10.3 Buchholz relay

Buchholz relay, double float type with alarm and trip contacts, along with suitable gas collecting arrangement shall be provided. The relay shall be provided with a test cock suitable for a flexible pipe connection for checking its operation and taking gas sample. A copper or stainless-steel tube shall be connected from the gas collector to a valve located about 1200 mm above ground level to facilitate sampling when the transformer is in service. The relay shall be provided with shut off valve on the conservator side as well as on the tank side.

17.10.4 Pressure Relief Device

Pressure Relief Device shall be provided on main tank and OLTC for rapid release of any pressure in transformer which may endanger the equipment. The device shall operate at a static pressure of less than hydraulic test pressure of transformer tank/OLTC chamber. The terminal box of the PRD shall be water tight with protection class IP 56 or better as per IEC 60529. Electrically insulated contact shall be provided for trip signal.

17.10.5 Temperature Indicators

17.10.5.1 Oil Temperature Indicator (OTI)

150 mm dial type temperature indicator with 'Maximum' reading pointer and resetting device shall be provided. The indicator shall have adjustable, electrically independent, potential free alarm and trip

contacts. A temperature sensing elements suitably located in a pocket on top oil shall be provided. Accuracy class of OTI shall be 1.5% or better.

17.10.5.2 Winding Temperature Indicator (WTI)

A device for measuring the hot spot temperature of each of the winding shall be provided. It shall comprise the following.

- i. Temperature sensing elements, one each on HV and LV winding.
- ii. Image coil.
- iii. Auxiliary CTs, if required to match the image coil.
- iv. 150 mm dial type temperature indicator with 'Maximum' reading pointer and resetting device with adjustable, electrically independent, potential free alarm and trip contacts.
- v. Calibration device.

The winding temperature indicator shall be responsive to the combination of top oil temperature and winding current, calibrated to follow the hottest spot temperature of the transformer winding.

Accuracy class of WTI shall be 1.5% or better.

17.10.6 Marshalling Box

Marshalling Box shall be of sheet steel, dust and vermin proof provided with proper lighting and thermostatically controlled space heaters. The degree of protection shall be IP 55. One dummy terminal block in between each trip wire terminal shall be provided. At least 10% spare terminals shall be provided on each panel. The gasket used shall be of neoprene or synthetic rubber. Wiring scheme (TB details) shall be engraved in a stainless-steel plate with viewable font size and the same shall be fixed inside the marshalling box door.

17.10.7 Valves

The transformer shall be provided with the following (but not limited to) valves.

- i. Two nos. of filter valves, one at top and another at bottom on diagonally opposite corners
 - ii. Two nos. of sampling valves at top and bottom of the tank
 - iii. Drain valve on main tank
 - iv. Drain valves on main and OLTC compartment of conservator
 - v. Valves (for nitrogen injection and oil drain) as required by firefighting system
- All valves shall be constructed of stainless steel, brass or gun metal except of shutoff valve for radiator and cooler. For radiator and cooler, valve shall be made up of gunmetal or cast iron.

17.11 Painting

- 17.11.1 Before painting or filling with oil, un-galvanized parts shall be completely cleaned and free from rust, scale and grease. All external rough surfaces on casting shall be filled by metal deposition. The interior of transformer tank and other filled chambers and internal structural steel work shall be cleaned of all scale and rust by sand blasting or other approved method. These surfaces shall be painted with an oil resisting varnish or paint.
- 17.11.2 Except for nuts, bolts and washers, all external surfaces shall receive a minimum of three coats of paint. The primary coat shall be applied immediately after cleaning. The second coat shall be of oil paint of weather resisting nature. The final coat shall be of a glossy, oil and weather resisting non-fading paint. The paint shade shall be as provided by the Employer during detailed engineering.
- 17.11.3 All internal surfaces of mechanism chambers and kiosk except those which have received anticorrosion treatment, shall receive three coats of paint applied to the thoroughly cleaned metal surface. The final coat shall be of light coloured anti-condensation mixture.
- 17.11.4 Any damage to paint work incurred during transport and erection shall be made good by thoroughly cleaning the damaged portion and by applying full number of coats of paints.

17.12 Transportation

- 17.12.1 Transformer tank is filled with oil or pure dry nitrogen/ air depending upon the transport weight limitations. Necessary arrangement shall be ensured to take care of pressure drop of nitrogen or dry air during transit and storage till completion of oil filling during erection. A gas pressure testing valve with necessary pressure gauge and adaptor valve shall be provided.
- 17.12.2 Bushings shall be crated, packed and transported as per standard guide lines of the Bushing Manufacturer. All care should be taken to avoid any damage of the porcelain due to vibration during transport.
- 17.12.3 Special attention shall be paid in packing the accessories & spares to avoid moisture ingress. All parts shall be adequately marked to facilitate field erection.

17.13 Warranty

The power transformer shall be warranted for minimum of 5 (five) years against all material/ manufacturing defects and workmanship.

17.14 Testing and Inspection

17.14.1 Type Tests and Special Tests

The following type test and special test reports shall be submitted during detailed engineering. The tests should have been conducted on the similar transformer by NABL accredited laboratory.

17.14.1.1 Type Tests

- i. Lightning impulse (Full & Chopped Wave) test on windings as per IS 2026-3/IEC 60076-3
- ii. Temperature Rise test at a tap corresponding to maximum losses as per IS 2026-2/IEC 60076-2. Dissolved Gas Analysis (DGA) shall be conducted on oil sample taken before and immediately after temperature rise test. Gas analysis shall be as per IS 9434/IEC 60567 and results will be interpreted as per IS 10593/IEC 60599.

17.14.1.2 Special Tests

- i. Short circuit withstand test as per IS 2026-5/IEC 60076-5

- ii. Measurement of zero-sequence impedance as per IS 2026-1/IEC 60076-1
- iii. Measurement of harmonics of no-load current as per IS IEC 60076-1
- iv. Measurement of acoustic noise level as per NEMA TR-1

Note: In case the contractor is not able to submit the test reports during detailed engineering, the contractor shall submit the reports of type/special tests either conducted by NABL accredited laboratory or witnessed by Employer.

17.14.2 Routine Tests

Each completed transformer shall be subjected to following routine tests as per the latest edition of IEC 60076 unless specified otherwise.

- i. Measurement of winding resistance at each tap
- ii. Measurement of voltage ratio between HV and LV windings at each tap
- iii. Check of vector group
- iv. Measurement of no-load loss and no-load current at 90%, 100% & 110% of rated voltage
- v. Measurement of short-circuit impedance and load loss at principal and extreme taps
- vi. Magnetic balance test & magnetizing current test as per CBIP manual publication no. 295
- vii. Separate source voltage withstand test
- viii. Induced over voltage withstand test
- ix. Measurement of insulation resistance and polarization index
- x. Measurement of tan delta and capacitance of winding
- xi. Core isolation test
- xii. Marshalling box functional test
- xiii. IR Measurement on wiring of marshalling box
- xiv. Test on on-load tap changer
- xv. Breakdown voltage test on transformer oil as per IS 335
- xvi. Jacking test followed by D.P. test
- xvii. Oil leakage test on completely assembled transformer along with radiators

18 Control and Relay Panel

18.1 Standards and Codes

All equipment provided under Control and Relay Panel shall comply with latest editions and amendments of the relevant IEC standards and IS codes. In particular, the C&R Panel shall comply with the following standards and codes.

Standard/Code	Description
IS 3231	Electrical relays for power systems protection
IEC 60255	Measuring relays and protection equipment
IEC 61850	Communication networks and systems for power utility automation
IEC 61131-3	Programmable controllers - Part 3: Programming languages
IS 9385	High voltage fuses
IS 9431	Indoor post insulators of organic material for systems with nominal voltages greater than 1000 V up to and including 300 kV
IEC 60099-4	Surge arresters - Part 4: Metal-oxide surge arresters without gaps for A.C. systems
IS 3070-3	Lightning Arresters for Alternating Current Systems - Part 3: Metal Oxide Lightning Arresters Without Gaps
IEC 62052-11	Electricity metering equipment (A.C.) - General requirements, tests and test conditions - Part 11: Metering equipment
IEC 62053	Electricity metering equipment (A.C.) - Particular requirements
IS 14697	AC Static Transformer Operated Watthour and Var-hour Meters, Class 0.2S and 0.5S

18.2 Construction

- 18.2.1 The control and relay panel shall be free standing, floor mounted, simplex type, metal enclosed construction. The panel enclosure shall be made of CRCA steel sheet. The thickness of load bearing members shall be minimum 3 mm and that of non-load bearing members shall be minimum 2 mm.
- 18.2.2 All external surface shall be painted with two coats of epoxy-based paint of colour shade RAL 7032. Internal surface shall be painted

with epoxy enamel white paint. The minimum dry film thickness (DFT) shall be 100 micron.

- 18.2.3 Controls, indications, relays, meters and other instruments shall be flush mounted on the front of the panel. Door shall be provided at the rear of the panel. All doors and removable covers shall be provided with neoprene or synthetic rubber gasket.
- 18.2.4 The panel shall be dust, moisture and vermin proof with degree of protection not less than IP 4X as per IEC 60529.
- 18.2.5 Cable entry shall be through the bottom of the panel. Gland plate of thickness not less than 3 mm shall be provided.

18.3 Relays

- 18.3.1 All relays shall be microprocessor based numerical type. However, auxiliary relays can be static or electromechanical type. The relays shall be flush mounted on panel front with connections from the inside.
- 18.3.2 Auxiliary voltage of the relays shall be 110 VDC and the relays shall be capable of operating continuously between 80 – 120% of auxiliary voltage.
- 18.3.3 All numerical relays shall have adequate number of freely configurable, optically isolated, Binary Inputs (BI) and potential free Binary Outputs (BO). All I/O's shall have galvanic isolation. Analog inputs shall be protected against switching surges and harmonics.
- 18.3.4 All numerical relays shall have sufficient number of current and voltage inputs required for all the required protection functions.

- 18.3.5 The numerical relay shall provide choice of ANSI/IEC/IEEE relay characteristic curves with wide protection setting ranges through a minimum of two protection setting groups.
- 18.3.6 Making, breaking and continuous capacity of the relay contacts shall be adequate enough for the circuits in which they are used.
- 18.3.7 All numerical relays shall have provision for measurement and storage of electrical parameters such as voltage, current, frequency, active power, reactive power etc.
- 18.3.8 The numerical relay shall be able to record faults and events in non-volatile memory.
- i. Fault record – At least 5 recent faults including the protection function operated, operating phase(s), voltages and currents along with date and time stamp.
 - ii. Event record – At least 200 events with date and time stamp.
- 18.3.9 The numerical relay shall have trip circuit supervision facility to monitor the circuit breaker trip circuit both in pre-trip and post-trip conditions. The relay shall also be able to provide circuit breaker monitoring, CT and VT supervision.
- 18.3.10 The numerical relay shall have self-diagnostic feature with separate output contact for indication of any internal relay failure.
- 18.3.11 The numerical relay shall have two serial communication ports, one on front side for local communication with PC and another on rear side for remote communication with SCADA system as per IEC 61850.
- 18.3.12 The numerical relay shall have feature for time synchronization through the SCADA System / networking.
- 18.3.13 The numerical relay shall be provided with backlit alphanumeric LCD or LED to access protection settings, measurement parameters, fault and event records. Read and write access to protection settings shall be password protected.
- 18.3.14 Necessary software and hardware to up/down load the data to/from the relay from/to the PC shall also be provided.

18.4 Protection Scheme

The following protection schemes shall be implemented for the protection of power transformer and its feeder.

- i. Biased Differential Protection with Second Harmonic Restraint
- ii. Non-directional Over Current and Earth Fault Protection
- iii. Restricted Earth Fault Protection
- iv. Under Voltage and Over Voltage Protection
- v. Buchholz Alarm and Trip
- vi. OTI Alarm and Trip
- vii. WTI Alarm and Trip
- viii. PRV Trip
- ix. MOG Alarm
- x. OSR Trip

Note: The above-mentioned protection schemes are indicative only. All the protection schemes required for safe and reliable operation of power transformer and the feeders shall be provided.

18.5 Measuring Instruments

- 18.5.1 All measuring instruments shall be enclosed in dust proof, moisture resistant cases and flush mounted on the panel.
- 18.5.2 Analog Ammeter and Voltmeter with selector switch shall be provided. Accuracy class shall be 0.5 or better. Instrument dial shall be with white scale, black pointer and black numerals.
- 18.5.3 Digital Multi-Function Meter (MFM) of accuracy class 0.2 or better shall be provided. It shall have communication capability for integration with SCADA. MFM shall be able to measure line & phase voltages, line & phase currents, active power, reactive power, apparent power, power factor and frequency.

18.6 Control Switches

All control switches shall be rotary operated type with adequate making, carrying and breaking current ratings. The control switches shall be pistol grip type, lockable with spring return to normal position. They shall be flush mounted on the panel with shrouded terminals.

18.7 Indications

All indicating lamps shall be flush mounted LED type with supply voltage

of 110 VDC. Lamp covers shall preferably be screwed type and moulded from heat resisting material. Indicating lamps shall be provided for R, Y, B PT supply, Breaker ON & OFF, Auto trip, Spring charged, Trip circuit healthy, etc.

18.8 Annunciation

Flush mounted static type annunciator with sufficient number of windows to accommodate all trip and alarm signals shall be provided. Separate audible annunciation for alarm and trip shall be provided by means of buzzer and hooter. Visual

annunciation shall be by flickering of facia. Push buttons for test, accept and reset shall also be provided.

18.9 Earthing

18.9.1 An earth bus made of copper or aluminium shall be provided throughout the length of the panel and bolted to the framework of the panel. The earth bus shall have sufficient cross section to carry maximum fault current without exceeding the allowable temperature rise.

18.9.2 All non-current carrying conductors of the panel shall be connected to the earth bus. All joints to the earth bus shall be made through at least two bolts. Hinged doors shall be earthed through flexible earthing braid of adequate cross section. Suitable provision shall be provided at each end of the earth bus for connection with earth grid.

18.9.3 All metallic cases of relays, instruments and other panel mounted equipment shall be connected to earth bus by independent copper wires of size not less than 2.5 sq. mm with green colour insulation.

18.9.4 Instrument transformer secondary neutral point shall be earthed at one place only on the terminal block. Such earthing shall be made through links so that earthing of one circuit may be removed without disturbing the earthing of other circuits.

18.10 Mimic Diagram

Colored mimic diagram made of metal or plastic with symbols to facilitate exact representation of the system shall be fixed on the front of control panel. Semaphore indicators shall be incorporated in the mimic diagram

for indicating position of circuit breakers, isolators and earthing switches.
The rated control voltage of semaphore indicator shall be 110 / 220 VDC.

18.11 Wiring and Terminal Blocks

- 18.11.1 All internal wiring shall be done with 1100 V grade, 2.5 sq.mm. PVC insulated stranded flexible copper wire. For CT secondary circuits, 4 sq.mm copper wire shall be used.
- 18.11.2 Wire terminations shall be made with solderless crimping type tinned copper lugs, which shall firmly grip the conductor. Insulation sleeves shall be provided at all the wire terminations.
- 18.11.3 Printed identification ferrules, marked to correspond with panel wiring diagram shall be provided at both ends of each wire. The ferrules shall be firmly located on each wire so that they cannot move or turn freely on the wire. Wire identification shall be done in accordance with IS 11353.
- 18.11.4 The Contractor shall be solely responsible for the completeness and correctness of the internal wiring and for the proper functioning of the connected equipment.
- 18.11.5 All internal wiring to be connected to the external equipment shall terminate on terminal blocks. Terminal blocks shall be rated for 1100 V, 10 A and made of non- inflammable material.
- 18.11.6 CT and VT secondary circuits shall be terminated on stud type, non-disconnecting terminal blocks.
- 18.11.7 At least 10% spare terminals shall be provided on each panel and these spare terminals shall be distributed on all terminal blocks.
- 18.11.8 Screw driver operated stud type test terminal block shall be provided.

18.12 Accessories

- i. Thermostatically controlled space heater with switch for isolation
- ii. 240 V, 15 A industrial socket with ON/OFF switch
- iii. LED lamp controlled by door switch

18.13 Warranty

The control and relay panel unit shall be warranted for minimum of 2 (Two) years against all material/ manufacturing defects and workmanship.

18.14 Testing and Inspection

18.14.1 Type Tests

The Contractor shall submit type test report of the panel for degree of protection as required by the Technical Specifications as per IEC 60529. The test should have been conducted by NABL accredited laboratory.

18.14.2 Routine Tests

Routine tests and acceptance tests shall be as per the Quality Assurance Plan (QAP) approved by the Employer.

19 **33 kV Switchyard Equipment**

19.1 Standards and Codes

All equipment provided shall comply with latest editions and amendments of the relevant IEC standards and IS codes. In particular, the switchyard equipment shall comply with the following standards and codes.

Standard/Code	Description
IS/IEC 62271-100	High Voltage Switchgear and Control gear - Part 100: AC Circuit Breakers
IEC 60376, IS 13072	Specification of technical grade sulfur hexafluoride (SF6) for use in electrical equipment
IS/IEC 62271-102	High Voltage Switchgear and Control gear - Part 102: AC Disconnectors and Earthing Switches
IEC 61869	Instrument Transformers
IS 2099	Bushings for alternating voltages above 1000 Volts
IS 2544	Porcelain post insulators for systems with nominal voltage greater than 1000 Volts
IS 335, IEC 60296	Insulating oil
IS/IEC 60034	Rotating electrical machines
IS 996	Single-phase AC industrial motors for general purpose
IS 3070, IEC 60099-4	Surge arresters - Part 4: Metal-oxide surge arresters without gaps for A.C. systems
Indian Electricity Act, CBIP manual, CEA rules and guidelines	

19.2 General Technical Parameters

System Parameters	Specification
Highest system voltage	145 kV
Rated system voltage	132 kV
Rated frequency	50 Hz
Number of phases	3
One minute power frequency withstand voltage	275 kV (rms)
Full wave impulse withstand voltage (1.2 / 50 μ s)	650 kV (peak)
Maximum Radio Interference Voltage between 0.5 MHz and 2.0 MHz	500 μ V at 92 kV rms
Rated short-time withstand current	31.5 kA for 1 s
Rated peak withstand current	80 kAp
System neutral earthing	Effectively earthed
Minimum creepage distance	As per site pollution level
Minimum clearance	
(i) Phase to phase clearance	1300 mm
(ii) Phase to earth clearance	1300 mm
(iii) Sectional clearance	4000 mm
(iv) Ground clearance	4800 mm

19.3 Supplier Qualification Criteria

Only TSTRANSCO/TSSPDCL approved components shall be used for construction of 33 kV switchyard.

19.4 Circuit breaker

19.4.1 Technical Parameters

Parameters	Specification
Type	Outdoor SF6/ VCB, single pressure
Operating duty cycle	O – 0.3sec – CO – 3min – CO
Rated break time	60 ms
Total break time	65 ms
Total closing time	Not more than 150 ms
Re-strike performance class	C2
Mechanical endurance class	M2

First pole to clear factor	1.3
Reclosing	Three phase high speed auto reclosing
Rated terminal load	Adequate to withstand 100 kg static load as well as wind, seismic and short circuit forces without impairing reliability or current carrying capacity
Noise level	Maximum 140 dB at 50 m distance from base of circuit breaker
Seismic level	0.5 g horizontal for the site location under Zone-V as per IS 1893 0.3 g horizontal for the site location under other than Zone-V as per IS 1893
Auxiliary contacts	
No. of contacts	As required plus 10 NO and 10 NC contacts per pole as spare
Thermal rating	10 A at 220 V DC
Breaking capacity	2 A DC with circuit time constant not less than 20 ms

19.4.2 Duty Requirements

- 20.4.2.1 The circuit breaker shall be capable of performing their duties without opening resistors. The circuit breaker shall meet the duty requirements for any type of fault or fault location and shall be suitable for line charging and dropping when used on effectively grounded or ungrounded systems and perform make and break operations as per the stipulated duty cycles satisfactorily.
- 20.4.2.2 The circuit breaker shall be capable of breaking the steady and transient magnetizing current corresponding to power transformers of applicable rating. It shall be capable of breaking line charging currents as per IEC 62271-100 with a voltage factor of 1.4. The rated transient recovery voltage for terminal fault and shortline faults shall be as per IEC 62271-100.
- 20.4.2.3 The total break time of the breaker shall not be exceeded under any duty conditions specified such as with the combined variation of the trip coil voltage, pneumatic/hydraulic pressure and arc extinguishing medium pressure, etc. While furnishing the proof of the total break time of complete circuit breaker, the effect of non-simultaneity between contacts within a pole or between poles shall be brought out to establish the guaranteed total break time. While

furnishing particulars regarding the D.C. component of the circuit breaker, the Contractor shall note that IEC 62271-100 requires that this value should correspond to the guaranteed minimum opening time under any condition of operation.

19.4.3 Construction

20.4.3.1 Circuit breakers shall be SF6 insulated, single pressure type. The design and construction of the circuit breaker shall be such that there is a minimum possibility of gas leakage and entry of moisture. There should not be any condensation of SF6 gas on the internal insulating surfaces of the circuit breaker.

20.4.3.2 Each pole shall form an enclosure filled with SF6 gas independent of two other poles and the SF6 density of each pole shall be monitored individually.

20.4.3.3 The SF6 gas density monitor shall be adequately temperature compensated to model the density changes due to variations in ambient temperature within the body of circuit breaker as a whole. It shall be possible to dismantle the monitor without removal of gas. Temperature compensated SF6 pressure gauge shall be provided which will be visible from ground level.

20.4.3.4 Sufficient SF6 gas shall be supplied to fill all the circuit breakers installed plus an additional 20% of the quantity as spare.

20.4.3.5 All making and breaking contacts shall be sealed and free from atmospheric effect. In the event of leakage of extinguishing medium to a value, which cannot withstand the dielectric stresses specified in the open position, the contacts shall preferably self-close. Main contacts shall be easily accessible for inspection and replacement.

If there are no separately mounted arcing contacts, then the main contacts shall be easily accessible for inspection and replacement. Main contacts shall have ample area and contact pressure for carrying the rated current under all conditions.

20.4.3.6 All the three poles of the breaker shall be linked together either electrically/pneumatically or electro hydraulically.

20.4.3.7 Circuit breakers shall be provided with two (2) independent trip coils, suitable for trip circuit supervision. The trip circuit supervision

relay would also be provided. Necessary terminals shall be provided in the central control cabinet of the circuit breaker.

19.4.4 Operating Mechanism and Control

20.4.4.1 Circuit breaker shall be operated by pneumatic mechanism or electrically spring charged mechanism or electro-hydraulic mechanism or a combination of these. It shall be gang operated for 3-phase reclosing operation.

20.4.4.2 The pneumatically operated mechanism shall offer unit compressor with each circuitbreaker with the breaker local air receivers having a capacity for two 'CO' operations of the breaker at the lowest pressure for reclose duty without refilling.

20.4.4.3 The spring-operated mechanism shall be complete with motor, opening spring & closing spring with limit switch for automatic charging and other necessary accessories to make the mechanism a complete operating unit. As long as power is available to the motor, a continuous sequence of closing and opening operations shall be possible. The motor shall have adequate thermal rating for this duty. After failure of power supply to the motor, one close-open operation shall be possible with the energy contained in the operating mechanism. Motor ratings shall be such that it requires not more than 30 seconds for fully charging the closing spring.

20.4.4.4 The hydraulic mechanism shall be suitable for at least two close open operations after failure of ac supply to the motor starting at pressure equal to lowest pressure of auto-reclose duty. All hydraulic joints shall have no oil leakage under the site conditions and joints shall be tested at factory against oil leakage at a minimum of 1.5 times maximum working pressure.

19.5 Disconnecter

19.5.1 Technical Parameters

System Parameters	Specification
Service	Outdoor
Type	Gang operated, Double break type
Rated short-time withstand current for isolator & earth switch	31.5 kA for 1 s
Rated peak withstand current for isolator & earth switch	80 kAp
Operating Mechanism	AC / DC / Universal motor operated
Maximum operating time	12 s
Control Voltage	110 / 220 V DC
Auxiliary contacts	
No. of contacts for isolator	As required plus 8 NO and 8 NC contacts per pole as spare
No. of contacts for earth switch	Total 6 NO and 6 NC
Thermal rating	10 A at 220 V DC
Breaking capacity	2 A DC with circuit time constant not less than 20 ms
Mechanical endurance class a) Isolator b) Earth switch	M2 M0

19.5.2 Duty Requirements

20.5.2.1 Isolators and earth switches shall be capable of withstanding the dynamic and thermal effects of the maximum possible short circuit current of the system in their closed position. They shall be constructed such that they do not open under influence of short circuit current and wind pressure together.

20.5.2.2 The earth switches, wherever provided, shall be interlocked so that the earth switches can be operated only when the isolator is open and vice versa. In addition to the constructional interlock, isolator and earth switches shall have provision to prevent their electrical and manual operation unless the associated and other interlocking conditions are met. All these interlocks shall be of failsafe type.

Suitable individual interlocking coil arrangements shall be provided. The interlocking coil shall be suitable for continuous operation from DC supply and within stipulated variation range. The interlock coil shall be provided with adequate contacts for facilitating permissive logic for DC control scheme of the isolator as well as for AC circuit of the motor to prevent opening or closing of isolators when the interlocking coil is not energized.

- 20.5.2.3 The earthing switches shall be capable of discharging trapped charges of the associated lines. Isolators and earth switches shall be able to bear on the terminal the total forces including wind loading and electrodynamic forces on the attached conductor without impairing reliability or current carrying capacity.
- 20.5.2.4 The isolator shall be capable for making/breaking normal currents when no significant change in voltage occurs across the terminals of each pole of the isolator on account of making/breaking operation.

19.5.3 Construction

20.5.3.1 Contacts

- i. The contacts shall be self-aligning and self-cleaning type and shall be so designed that binding cannot occur after remaining in closed position for prolonged period in a heavily polluted atmosphere.
- ii. No undue wear or scuffing shall be evident during the mechanical endurance tests. Contacts and spring shall be designed so that readjustments in contact pressure shall not be necessary throughout the life of the isolator or earthing switch. Each contact or pair of contacts shall be independently sprung so that full pressure is maintained on all contacts at all time.
- iii. Contact springs shall not carry any current and shall not lose their characteristics due to heating effects.
- iv. The moving contact of double break isolator shall preferably be turn-and-twist type or other suitable type of locking arrangement to ensure adequate contact pressure.
- v. Flexible braided copper, where used, shall have corrosion

resistant coatings such as tinning or silvering.

20.5.3.2 Base

Each single pole of the isolator shall be provided with a complete galvanised steel base provided with holes and designed for mounting on a standard supporting structure.

20.5.3.3 Blades

- i. All metal parts shall be of non-rusting and non-corroding material. All current carrying parts shall be made from high conductivity electrolytic copper/aluminium. Bolts, screws and pins shall be provided with lock washers. Keys or equivalent locking facilities if provided on current carrying parts shall be made of copper silicon alloy or stainless steel or equivalent. The bolts or pins used in current carrying parts shall be made of non-corroding material. Ferrous parts, other than stainless steel shall not be used in close proximity of main current path. All ferrous castings, if used elsewhere shall be made of malleable cast iron or cast-steel. No grey iron shall be used in the manufacture of any part of the isolator.
- ii. The live parts shall be designed to eliminate sharp joints, edges and other corona producing surfaces. Where this is impracticable, adequate corona rings shall be provided. Corona shields are not acceptable. Corona rings shall be made up of aluminum/aluminum alloy.
- iii. Isolators and earthing switches including their operating parts shall be such that they cannot be dislodged from their open or closed positions by short circuit forces, gravity, wind pressure, vibrations, shocks, or accidental touching of the connecting rods of the operating mechanism.
- iv. The isolator and earth switch shall be designed such that no lubrication of any part is required except at very infrequent intervals. i.e., after every 1000 operations or after 5 years whichever is earlier.

20.5.3.4 Insulator

- i. The insulator shall conform to IS / IEC 60168 and IS 16683 / IEC TS 60815.
- ii. In addition to all type, routine and acceptance tests, as per IS / IEC 60168, the following additional routine/ acceptance tests shall also be carried out.
 - a. Bending load test in four directions at 50% of minimum bending load guaranteed on all insulators, as routine test
 - b. Bending load test in four directions at 100% of minimum bending load guaranteed as a sample test on each lot
 - c. Torsional test on sample insulator of a lot
 - d. Ultrasonic test as a routine test
- iii. The porcelain of the insulator shall have minimum cantilever strength of 600kg.
- iv. Pressure due to the contact shall not be transferred to the insulators after the main blades are fully closed.

20.5.3.5 Earthing Switches

- i. Where earthing switches are specified, these shall include the complete operating mechanism and auxiliary contacts. The earthing switches shall form an integral part of the isolator and shall be mounted on the base frame of the isolator.
- ii. Earthing switches shall only be locally operated.
- iii. Each earth switch shall be provided with flexible copper/aluminum braids for connection to earth terminal. These braids shall have the same short time current carrying capacity as the earth blade. The transfer of fault current through swivel connection will not be accepted.

19.5.4 Operating Mechanism and Control

20.5.4.1 The Contractor shall offer motor operated switches having padlock arrangement for both ON and OFF positions.

20.5.4.2 Limit switches for control shall be fitted on the isolator / earth switch shaft within the cabinet to sense the open and close positions of the

isolators and earth switches.

20.5.4.3 It shall not be possible, after final adjustment has been made, for any part of the mechanism to be displaced at any point in the travel sufficient enough to allow improper functioning of the isolator when the isolator is opened or closed at any speed.

20.5.4.4 Control cabinet / operating mechanism box shall conform to requirements stipulated elsewhere in the document and IS/IEC 61439 as applicable.

19.5.5 Operation

20.5.5.1 Isolator shall be electrically/mechanically gang operated for main blades and earthswitches. The operation of all the three poles shall be well synchronized and interlocked.

20.5.5.2 The design shall be such as to provide maximum reliability under all service conditions. All operating linkages carrying mechanical loads shall be designed for negligible deflection. The length of inter insulator and interpole operating rods shall be capable of adjustments.

20.5.5.3 The isolator and earth switches shall be provided with 'dead centre mechanism' to prevent accidental opening by wind, vibration, short circuit forces or movement of the support structures.

20.5.5.4 The design of linkages and gears be such so as to allow one man to operate the handle with ease for isolator and earth.

19.6 Surge Arrester

19.6.1 Technical Parameters

Parameter	Specification
Arrester Classification	Station Medium (SM)
Nominal discharge current (8/20 μ s)	10 kA
Repetitive charge transfer rating	1.6 coulomb
Rated thermal energy rating	7 kJ/kV
Rated arrester voltage	120 kV
Continuous operating voltage at 50°C	102 kV

Maximum Residual Voltage (i) At 30/60 μ s, 1 kA current (ii) At 8/20 μ s, 5 kA current (iii) At 8/20 μ s, 10 kA current	280 kVp 310 kVp 330 kVp
High-current short duration test value (4/10 μ s)	100 kAp
Current for pressure relief test	40 kA
Partial discharge at 1.05 times the continuous operating voltage	≤ 10 pC

19.6.2 Duty Requirements

- 20.6.2.1 The Surge Arresters shall be capable of discharging over-voltages occurring due to switching of unloaded transformers, reactors and long lines.
- 20.6.2.2 The reference current of the arresters shall be high enough to eliminate the influence of grading and stray capacitance on the measured reference voltage.
- 20.6.2.3 The Surge Arresters shall be capable of withstanding meteorological and shortcircuit forces under site conditions.
- 20.6.2.4 The SAs shall protect power transformers, circuit breakers, disconnecting switches, instrument transformers, etc. with insulation levels specified in this specification.

19.6.3 Construction

- 20.6.3.1 Each surge arrester shall be hermetically sealed single-phase unit. The non-linear blocks shall be made of sintered metal oxide material. The surge arrester construction shall be robust with excellent mechanical and electrical properties.
- 20.6.3.2 Surge Arresters shall be fitted with pressure relief devices and arc diverting ports suitable for preventing shattering of polymer housing and to provide path for flow of rated fault current in the event of SA failure.
- 20.6.3.3 Outer insulator of surge arrester shall be made of porcelain/polymer. The outer insulator housing shall be so coordinated that external flashover will not occur due to application of any impulse or switching surge voltage up to the maximum design

value for arrester. Arresters shall not fail due to insulator contamination.

- 20.6.3.4 Seals shall be provided in such a way that they are always effectively maintained even when discharging rated lightning current.
- 20.6.3.5 The cantilever strength of the insulator shall be minimum 150 kg.
- 20.6.3.6 The following details shall be furnished for quality checks.
 - i. The heat treatment cycle details along with necessary quality checks used for individual blocks and insulation layer formed across each block.
 - ii. Metalizing coating thickness for reduced resistance between adjacent discs.

19.6.4 Fittings and Accessories

- 20.6.4.1 Surge arrester shall be complete with insulating base having provision for mounting to structure.
- 20.6.4.2 Grading/corona rings shall be provided on each surge arrester unit, as required.
- 20.6.4.3 The end fittings shall be made of corrosion proof material and preferably be nonmagnetic.
- 20.6.4.4 Self-contained discharge counters, suitably enclosed for outdoor use and requiring no auxiliary or battery supply for operation shall be provided for each single pole unit along with necessary connection arrangement. Suitable leakage current meters shall also be provided in the same enclosure. The reading of ammeter and counter shall be visible through an inspection glass panel to maintenance personnel standing on ground. The terminals shall be robust and of adequate size and shall be so located that incoming and outgoing connections are made with minimum possible bends. The surge counter shall be provided with a potential free contact rated for 220 V DC which shall close whenever a surge is recorded by the surge monitor. Necessary arrangement shall be provided for extending the contact information to Substation Automation System/RTU.

19.7 Instrument Transformer

19.7.1 Technical Parameters

Parameter	Specification
Current Transformer	
Accuracy class	Metering – 0.2S Protection – PS / 5P20
Rated VA burden	As per requirement
Insulation class	Class E
One minute power frequency withstand voltage between secondary terminals & earth	5 kV
Rated short time thermal withstand current	31.5 kA for 1 s
Rated dynamic current	80 kAp
Partial discharge level	10 pico Coulomb (max)
No. of terminals	All terminals of control circuits wired up to marshalling box plus 20% spare
Capacitive Voltage Transformer	
Accuracy class	Metering – 0.2 Protection – PS / 3P
Rated VA burden	As per requirement
Insulation class	Class E
Standard reference range of frequencies for which the accuracies are valid	96% to 102% for protection and 99% to 101% for measurement
High frequency capacitance for entire carrier frequency range	Within 80% to 150% of rated capacitance
Equivalent series resistance over entire carrier frequency range	< 40 ohm
One minute power frequency withstand voltage between secondary terminals & earth	
(i) Between LV (HF) terminal and earth terminal	10 kV for exposed terminals 4 kV for terminals enclosed in a weather proof box
(ii) For secondary winding	3 kV
Partial discharge level	10 pico Coulomb (max)
Rated voltage factor	1.2 continuous and 1.5 for 30 sec.
No. of terminals	All terminals of control circuits wired up to marshalling box plus 20% spare

19.7.2 General Requirements

20.7.2.1 Instrument transformers shall be hermetically sealed single-phase units, oil immersed, self-cooled suitable for outdoor installations and shall be supplied with common marshalling box for a set of three single phase units.

20.7.2.2 The external surface of instrument transformer, if made of steel, shall be hot dip galvanized or painted with color shade as decided by the Employer during detailed engineering.

20.7.2.3 Insulating oil to be used for instrument transformers shall be of EHV grade and shall conform to IS 335 / IEC-60296. Non-PCB based synthetic insulating oil conforming to IEC 60867 shall be used in the capacitor units of CVT.

20.7.2.4 Polarity marks shall indelibly be marked on each instrument transformer and at the lead terminals at the associated terminal block.

20.7.2.5 The insulators shall have cantilever strength of more than 350 kg.

20.7.2.6 Marshaling box shall conform to all requirements given elsewhere in the document. The wiring diagram for the interconnection of three phase instrument transformer shall be pasted inside the box. Terminal blocks in the marshaling box shall have facility for star/delta formation, short circuiting and grounding of secondary terminals. The box shall have enough terminals to wire all control circuits plus 20 spare terminals.

19.7.3 Current Transformer

20.7.3.1 Current transformer shall have single primary of either ring type or hair pin type or bar type. Wound type primary is acceptable only for metering CTs of ratio less than 400/1. In case of inverted type/live tank CT, the following requirements shall be met.

(i) The secondaries shall be totally encased in metallic shielding providing a uniform equipotential surface for even electric field distribution.

(ii) The lowest part of the insulation assembly shall be properly secured to avoid any risk of damage due to transportation stresses.

- (iii) The upper part of insulation assembly resting on primary bar shall be properly secured to avoid any damage during transportation due to relative movement between insulation assembly & top dome.
 - (iv) The insulator shall be one piece without any metallic flange joint.
- 20.7.3.2 Core lamination shall be of cold rolled grain-oriented silicon steel or other equivalent alloys. The cores shall produce undistorted secondary current under transient conditions at all ratios with specified parameters.
- 20.7.3.3 The CT shall be provided with oil filling plug, drain plug, and oil sight glass which should be clearly visible to maintenance personnel standing on ground.
- 20.7.3.4 The secondary terminals of CT shall be terminated to suitable number of stud type non-disconnecting and disconnecting terminal blocks as required inside the terminal box of degree of protection IP 55 at the bottom of CT.
- 20.7.3.5 Different ratios shall be achieved by secondary taps only; primary reconnection shall not be accepted.
- 20.7.3.6 The Instrument Security Factor (ISF) at all ratios shall be less than five (5) for metering core. If any auxiliary CTs/reactors are used, then all parameters specified shall be met treating auxiliary CTs as an integral part of the CT. The auxiliary CTs/reactors shall preferably be in-built construction of the CT. In case these are to be mounted separately, these shall be mounted in the central marshalling box suitably wired up to the terminal blocks.
- 20.7.3.7 Current transformers shall be suitable for high-speed auto reclosing.
- 19.7.4 Capacitor Voltage Transformer
- 20.7.4.1 Capacitor Voltage Transformer shall consist of a capacitor divider and an electromagnetic unit housed in independent, non-oil communicating hermitically sealed compartments.
- 20.7.4.2 The capacitor divider shall consist of primary and secondary capacitance housed in high quality porcelain insulators filled with oil. The electromagnetic unit shall comprise of compensating reactor,

intermediate transformer, protective and damping devices.

- 20.7.4.3 Suitable damping device shall be permanently connected to one of the secondary windings and shall be capable of suppressing ferro-resonance oscillations.
- 20.7.4.4 All the secondary windings of the CVT shall be protected by HRC cartridge type fuses or MCBs. In addition, fuses/MCBs shall also be provided for protection and metering windings for connection to fuse monitoring scheme.
- 20.7.4.5 The secondary terminals of the CVT shall be terminated to stud type non- disconnecting terminal blocks via fuses/MCBs inside the terminal box of degree of protection IP 55. It should be ensured that access to secondary terminals is without any danger of access to high voltage circuit.
- 20.7.4.6 CVTs shall be suitable for High Frequency (HF) coupling required for Power Line Carrier Communication (PLCC). Carrier signals must be prevented from flowing into EMU circuit by means of RF choke/reactor over the entire frequency range of 40 to 500 kHz. HF terminal shall be brought out through a suitable bushing and shall be easily accessible for connection to the coupling filters of the carrier communication equipment. Further, earthing link with fastener to be provided for HF terminal.
- 20.7.4.7 A protective surge arrester/spark gap shall preferably be provided to prevent breakdown of insulation by incoming surges and to limit abnormal rise of terminal voltage of shunt capacitor, tuning reactor, RF choke, etc. due to short circuit in transformer secondary. The details of this arrangement (or alternative arrangement) shall be furnished by Contractor for Employer's review.
- 20.7.4.8 The accuracy of metering core shall be maintained through the entire burden range up to rated value without any adjustments during operations.
- 20.7.4.9 The protection cores shall not saturate at about 1.5 times the rated voltage for a minimum duration of 30s.

19.8 Warranty

All switchyard equipment shall be warranted for minimum of 5 (Five) years against all material/ manufacturing defects and workmanship.

19.9 Testing and Inspection

19.9.1 Type Tests

All switchyard equipment shall be of type tested design. Type test reports as per the relevant IEC/IS standards shall be submitted during detailed engineering. The tests should have been conducted on the similar equipment by NABL accredited laboratory. In case the contractor is not able to submit the test reports during detailed engineering, the contractor shall submit the reports of type/special tests either conducted by NABL accredited laboratory or witnessed by Employer.

19.9.2 Routine Tests

Routine tests and acceptance tests shall be as per the Quality Assurance Plan (QAP) approved by the Employer.

20 **Weather Monitoring System**

As a part of weather monitoring system, the Contractor shall provide the following measuring instruments with all necessary software and hardware required to integrate with SCADA.

20.1 Pyranometer

20.1.1 The Contractor shall provide Class-A pyranometers (ISO 9060:2018 classification) along with necessary accessories for measuring incidental solar radiation at horizontal and inclined plane of array.

20.1.2 Specification of the pyranometer shall be as follows.

Parameter	Specification
Spectral Response (50% points)	0.31 to 2.8 micron
Operating temperature range	0°C to +80°C
Ingress Protection	IP 67
Resolution	Minimum +/- 1W/m ²
Output	Analog output: 4 – 20 mA Serial output: RS485

20.1.3 Each instrument shall be supplied with necessary cables. Calibration certificate with calibration traceability to World Radiation Reference (WRR) or World Radiation Centre (WRC) shall be furnished along with the equipment. The signal cable length shall not exceed 20m. The Contractor shall provide instrument manual in hard and soft form.

20.2 Temperature Sensor

The Contractor shall provide minimum 3 (three) temperature sensors (1 (one) for ambient temperature measurement with shielding case and 2 (two) for module temperature measurement) at each site. The temperature sensor shall be Resistance Temperature Detector (RTD)/ Semiconductor type with measurement range of 0°C to 80°C. The instrument shall have valid calibration certificate.

20.3 Anemometer

Contractor shall provide minimum one no. ultrasonic wind sensor (no moving parts) for wind speed and direction monitoring.

Parameter	Specification
Velocity range with accuracy limit	0-60m/s with +/-2% accuracy @12 m/s; Resolution: 0.01m/s
Wind direction range with accuracy limit	0 to 360° (No dead band) with +/-2° accuracy @12 m/s; Resolution: 1°
Mounting Bracket	Anodized Aluminium bracket to reduce corrosion, all mounting bolts of SS
Protection Class	IP 66
Output	RS 485

20.4 Data logger and Data Acquisition System

Data logger for the weather monitoring station should have the following features:

20.4.1 Provision for analog, digital and counter type inputs for interfacing with various type of sensors

- i. Analog Input
 - Adequate nos. for all analog sensors with redundancy

- Provision for operation in different current and voltage ranges as per connected sensors
- Accuracy of +/-0.1% of FS
- ii. Digital Inputs
 - Adequate no. of Digital inputs and outputs for the application
- iii. Provision for RS232 and RS485 serial outputs
- iv. Built-in battery backup
- v. Connectivity and Data transmission:
 - RS485 MODBUS interface for data collection and storage on SCADA
 - Communication protocol should support fast data transmission rates, enable operation in different Frequency bands and have an encryption- based data security layer for secure data transmission
- vi. Display Settings: Graphic LCD screen which should be easily accessible and should display relevant details like all sensor values, battery strength, network strength etc.
- vii. Provision of Time synchronization from telecom time or server time
- viii. Data Storage: Provision for at least 2 MB internal Flash Memory and at least 8 GB Micro SD card (expandable)
- ix. Protection level: IP65

21 Fire Alarm System

21.1 Standards and Codes

Standard/Code	Description
IS 2189	Selection, Installation and Maintenance of Automatic Fire Detection and Alarm System Code of Practice
IS 2171	Portable Fire Extinguishers, Dry Powder (Cartridge Type)
IS 8149	Functional requirements for twin CO ₂ fire extinguishers (trolley mounted)
IS 2546	Galvanized mild steel fire bucket
National Building code 2016	

- 21.2 Contractor shall ensure the compliance of fire detection and alarm system as per relevant standards and regulations. The installation shall meet all applicable statutory requirements and safety regulations of state/central fire department/body or any other competent authority in terms of fire protection.
- 21.3 Firefighting system for the proposed power plant for fire protection shall be consisting of but not limited to:
- i. Sand buckets
 - ii. Portable fire extinguishers (CO₂ and dry powder type)
 - iii. Microprocessor based fire alarm panel
 - iv. Multi sensor smoke detectors
 - v. Hooter cum strobe
 - vi. Manual call points
 - vii. Cables from sensor to fire Panel.
- 21.4 Minimum two numbers of fire extinguishers (CO₂ and Foam type each, of capacity 9 kg having BIS certification marking as per IS: 2171) shall be provided at every building/ enclosure, transformer yard and switchyard. However, contractor must comply with existing building code for fire protection and relevant IS codes.
- 21.5 Four numbers of stand with four sand buckets on each stand shall be provided in the Transformer Yard. Sand buckets inside the building shall be provided at strategic locations as decided during detailed engineering.
- 21.6 Digital output from the fire detection system shall be integrated with SCADA
- 21.7 Contractor shall submit the plan for fire and smoke detection system for the Employer's approval.

22 Testing Instruments

The Contractor shall provide the following set of instruments for on-site testing.

22.1 Earth resistance tester

Parameter	Specification
Display	Backlit LCD or LED display
Range	Earth Resistance: up to 2000 Ω Earth Voltage: 200 V

Accuracy	± (2% + 5)
Safety Ratings	IP 56
Programmable Limits setting	Enabled
Accessories	
Earth Ground Stakes – 4 Nos.	
Cable Reels – 3 Nos.	
Battery - 2 set	
Carry Case with sufficient space for accommodating accessories	

22.2 Array tester

Parameter	Specification
Display	Backlit LCD or LED display
Functionality	All electrical tests required by IEC 62446-1:2016
Memory	Up to 200 records & USB downloadable to Computer
Accessories	
A set of two, 4mm fused leads for extra protection during installation tests.	
Leads which enable the array tester to connect directly to PV arrays	
Battery - 2 set	
Carry Case with sufficient space for accommodating accessories	

22.3 Insulation tester

22.4 Digital Multimeter

Parameter	Specification
Voltage Range	1500 V DC / 1000 V AC (True RMS)
Display	4 ½ digits, Backlit LCD or LED
Measuring Category	1000 V CAT III as per IEC Standard 61010-1
Additional Functions	Resistance, Temperature, Continuity, Diode, Capacitance, Frequency, Duty cycle measurement
Accessories	
Temperature Probe – 1	
Test Leads with Alligator Clips - 1 set	
Battery – 2 set	
Carry Case with sufficient space for accommodating accessories	

22.5 Clamp meter

Parameter	Specification
Current Range	400 A DC / 1000 A AC (True RMS)
Display	Backlit LCD or LED display
Measuring Category	1000V CAT III as per IEC 61010-1
Additional Functions	Active, Reactive and Apparent Power, THD, PF
Accessories	
Test Leads - 1 set	
Battery – 2 set	
Carry Case with sufficient space for accommodating accessories.	

22.6 Infra-red thermal imaging camera

Parameter	Specification
Spectral response	8 μm to 14 μm (LW)
Temperature-sensitivity and calibration range	–20 °C to +120 °C
Atmospheric air temperature	-10 °C to +40 °C
Thermal sensitivity	NETD $\leq$ 0.1 K at 30 °C
Geometric resolution	640 x 480 pixels
Absolute error of measurement	$< \pm 2$ K
Adjustable parameters	Emissivity, Reflected temperature
Adjustable functions	Focus, temperature level and span
Measurement functions	Measuring spot, measuring area with average and maximum temperature
Calibration	The measuring system (Camera, lens, aperture and filter): The thermographic camera has to be traceably calibrated at least every two years. The calibration has to be documented. If the camera is not compliant (absolute temperature and/or temperature differences), it has to be readjusted by the manufacturer.

Documentation	Storing of the infrared picture with the radiometric data to be able to determine absolute temperatures
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22.7 Digital lux meter

Parameter	Specification
Range	0 – 1000 lux
Accuracy	± (2% + 5)
Resolution	1 lux
Display	3½ digits, Backlit LCD/LED
Accessories	
Battery – 2 set	
Carry Case with sufficient space for accommodating accessories.	

22.8 All testing equipment shall possess valid calibration certificate issued from approved NABL labs.

22.9 Instruments of superior rating is allowed after seeking consent of the Employer.

22.10 Maintenance, calibration, up keeping, repair & replacement of these tools will be in the scope of the Contractor during O&M.

22.11 It is Contractor's responsibility to arrange for tools, tackles, logistics, test kits, manpower, experts etc. required for trouble free operation of Plant.

23 Power Evacuation System

23.1 The Contractor has to do the power evacuation and integration to and with the designated substation via underground cables at specified grid voltage with all necessary infrastructure such as protection switchgears and metering systems as per the requirement of the STU/Employer.

23.2 The Contractor shall get the route approval from the Employer/TSTRANSCO prior to start of the construction. Any changes in the route or scheme due to ROW issues at any point of the time prior to commissioning shall be complied without any additional cost to the Employer.

- 23.3 The ROW for the TL/UG cable shall be obtained prior to the construction of the line from the concerned authorities.
- 23.4 Only TSTRANSCO/TSSPDCL approved components shall be used for construction of underground cables.
- 23.5 Underground cable
Power evacuation is planned with underground cable for plant internal evacuation, the cable shall be approved by the Employer. However, in case of external power evacuation, the evacuation plan shall be as per STU/DISCOM requirement and the same shall be submitted to Employer for approval/ accord.

C Civil, Mechanical and Plumbing Works

1 General Requirement

- 1.1 This section of Technical Specifications describes detailed technical and functional requirements of all civil, structural, mechanical & plumbing works included in the scope.
- 1.2 This excludes design, supply and installation of 11 kV, 22 kV & 33 kV transmission poles & accessories which shall be designed following latest guidelines of TSSPDCL (State electricity Discom) and got approved from TSSPDCL before execution. In absence of TS Discom guidelines REC (Rural electrification corporation) standards shall be followed. Poles at corner with angle > 100 shall be provided with 4-pole structure or lattice tower. Use of Pre-stressed cement concrete spun poles is not acceptable. Approved copies of these designs & drawings shall be submitted to the employer for reference and record.
- 1.3 Standards & Codes
- 1.3.1 All design and construction of civil works shall conform to relevant Indian standards such as BIS, IRC, MORTH, NBC etc.
- 1.3.2 Design of steel structures shall conform to IS: 800, 801 or 802 as applicable. Design of concrete structures shall conform to IS: 456. For design of liquid retaining structure IS: 3370 shall be followed. Only in case of non-availability of Indian standard, equivalent American or

British standard may be used for design with prior approval of the Engineer and the contractor shall submit proper justification for the same alongwith his request to the Engineer for review and approval, and the decision of the Engineer shall be final and binding.

- 1.3.3 All the design/ drawings shall be prepared/ approved either by in-house Engineering Team of the contractor (or by his Engineering Consultant) with qualified engineering staff with relevant experience in successful design of solar SPV plants.
- 1.3.4 The design calculations for RCC structure, Steel structure, Foundation system etc. shall be submitted for prior approval of Engineer before commencement of construction.
- 1.3.5 As per project requirements, the Employer may ask for approval of all civil designs and drawings by a Chartered Civil/ Structural Engineer.

2 Transformer Yard Civil Works

- 2.1 Transformer and equipment foundations shall be founded on piles/isolated spread footings or block foundation depending on the final geotechnical investigation report and functional requirements.
- 2.2 In case of transformer oil tank capacity ≥ 2000 litres, the transformer foundation shall have its own soak pit which would cover the area of the transformer and cooler banks, so as to collect any spillage of oil in case of emergency. The retention capacity of the soak pit shall be equal to volume of the transformer oil (excluding free space above gravel) and it shall be filled with granite stone gravel of size 40mm, uniformly graded, with 200 mm free space above gravel fill.
- 2.3 In case of transformer oil tank capacity more ≥ 20000 litres, the soak pit shall be connected to a separate burnt oil pit through discharge pipe (300 mm dia) and shall be suitably sized to accommodate full oil volume (excluding free board above inlet pipe) of the transformer connected to it, without backflow. In this case the capacity of the soak pit may be reduced to min. $1/3^{\text{rd}}$ of the total transformer oil volume. The burnt oil pit shall be further connected to oily water drainage system. The water shall be discharged into the nearest drain by gravity flow or

pumping after suitable treatment as per statutory and code provisions.

- 2.4 Both, the transformer soak including side walls and the burnt oil pit shall be of RCC and shall be provided with sump (min. 500 mm x 500 mm x 400mm deep) and slope of 1:50 in concrete screed of 1:1 – ½:3 to the floor slab towards the sump pit. The oil collection pit shall be provided with 20mm dia. MS rung ladder with 2 coats of epoxy paint over 2 coats of primer, a manhole & removable RCC cover. The inside of oil collection pit shall be plastered with 6 mm thick CM 1:6 and painted with 2 coats of epoxy paint over 2 coats of primer.
- 2.5 The area around the transformer and equipment shall be covered with uniformly graded granite stone gravel of size 40mm.
- 2.6 The area shall be provided with galvanized chain link fence of height min 1.8m with 3.5m wide gate. The specifications for fencing shall be similar to those specified under Cl. No. 31.3 except fence post which shall be of MS angle (ISA 65x65x6) spaced at 2.5m c/c.
- 2.7 The Gate of size 3.5m shall be of MS pipe (medium class conforming to IS: 1161) frame with hard drawn steel wire fabric mesh (50x50mmx3mm thick conforming to IS: 1566) including all accessories and fittings. MS angle posts shall conform to IS 2062.
- 2.8 In addition to main gate a wicket gate of MS pipe (medium class conforming to IS: 1161) frame with 1.0 m width with hard drawn steel wire fabric (50x50x3mm thick conforming to IS: 1566) shall be provided for man entry for maintenance purpose.
- 2.9 The transformer yard fencing work shall conform to CEIG requirements.
- 2.10 The requirement of fire barrier wall between transformers shall be as per Electricity Rules and IS: 1646 recommendations. Minimum wall thickness shall be 230mm for RCC wall and 300mm for masonry wall.

3 PV Module Cleaning System – Wet or Dry Type Wet type Cleaning System

- 3.1 The contractor shall design and install the effective module cleaning system.

- 3.2 A regular supply of suitable quantity of water shall be ensured by the contractor to cater day-to-day requirement of drinking water and for cleaning of PV modules during entire O&M period.
- 3.3 The Contractor shall estimate the water requirements for cleaning the photovoltaic modules at least thrice in month or at closer frequency as per the soiling conditions prevailing at site, in order to operate the plant at its guaranteed plant performance. Also, the contractor is required to plan the water storage accordingly with provision of a tank of suitable capacity for this purpose. However, min. consumption of 2 Ltr / Sqm of surface area of SPV module shall be considered in estimation of required quantity of water storage.
- 3.4 Water used for drinking & PV module cleaning purpose shall generally be of potable quality and fit for cleaning the modules with TDS generally not more than 75 PPM. In case of higher salt contents, the water shall be thoroughly squeezed off to prevent salt deposition over module surface. However, water with TDS more than 200 PPM shall not be used directly for module cleaning without suitable treatment to control the TDS within acceptable limits. The water must be free from any grit and any physical contaminants that could damage the panel surface.
- 3.5 If required, for settlement of any grit/ unacceptable suspended particles in the water a settling tank shall be installed before the inlet of the storage tank. Suitable arrangement for discharge/ disposal of sediment/ slush shall be provided in silting chamber by gravity disposal in surface drain or with provision of sludge sump and pump of adequate capacity.
- 3.6 The module cleaning system shall include construction of RCC tank or supply and installation of Ground mounted PVC tank (s) of required storage capacity, pumps (including 1 No. standby pump), water supply mains and flexible hose pipes, taps, valves (NRV, Butterfly valve, Ball valve, Gate valve, PRV, scour valve etc.), Water hammer arrester(s), pressure gauge, flow meter etc. as per the planning & design.
- 3.7 In case of over ground water storage tank, the contractor shall check its effect on plant performance through shadow analysis. The PVC storage tank shall conform to IS: 12701. The valves shall conform to

IS: 778. A suitable metal sheet canopy for protection from direct sunlight shall be provided over the tank area.

- 3.8 The water supply mains could be either of GI, uPVC or HDPE, however, the vertical pipe connecting supply main to the discharge point shall be of GI.

3.9 Masonry chamber shall be provided for Main gate valve at pump end. Whereas, as per requirements, at other locations either a masonry or GI/ HDPE pipe chamber may be provided.

- 3.10 Module cleaning procedure and pressure requirement at discharge point shall be as per the recommendation of PV module manufacturer. However, discharge pressure at outlet shall not be less than 50kg/cm² (5 MPa)

- 3.11 All the pipes thus laid shall be buried in ground at least 150mm below FGL or laid above ground clamping on suitable concrete support blocks. In case of above ground piping only GI pipes shall be used.

Dry Cleaning System

- 3.12 Alternate to Wet type Module Cleaning System as stipulated in the above Clauses, the Contractor may propose Robotic type dry Cleaning system with micro-fibre based brushes to avoid scratches on the PV Module. The Contractor shall also ensure following considerations during design:

- 3.12.1 The necessary design considerations for the mounting the robotic system shall be incorporated in the Module Mounting Structure design, foundation design as well as PV array layout, in addition to the specifications provided elsewhere in this Section.
- 3.12.2 The system shall be designed for operation under the climatic conditions at site.
- 3.12.3 The Robotic Cleaning system shall be self-powered, with battery backup (no external supply). The battery shall be compliant with IEC 62133: Secondary cells and batteries containing alkaline or other non-acid electrolytes - Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications.

3.12.4 The Robotic Cleaning system shall be integrated with the Plant SCADA.

4 Miscellaneous structures

4.1 Support structure for weather monitoring device

- 4.1.1 Weather monitoring device shall be mounted on tubular steel pole of required height. The pole shall conform to IS: 2713.
- 4.1.2 The pole shall be secured to an independent RCC foundation structure through Baseplate and Anchor bolt assembly.
- 4.1.3 200 long 20 dia. rods shall be welded to the pole at 300 mm C/c for access to the device for maintenance purpose.
- 4.1.4 The support structure shall be hot dip galvanized.

4.2 Support structures for SCB

- 4.3 When supported independently, the SCB shall be mounted on a structural steel supporting frame of galvanized ISMC 75.
- 4.3.1 Column post and bracings shall be supported with 300 mm (min.) diameter and 850 mm (min.) deep below GL piles in cement concrete (nominal mix 1:1:2). The column post and bracings shall be extended into the piles upto 800 mm with 50mm cover at the bottom.
- 4.3.2 The pile shall project 200 mm above GL.

4.3.3 The support structure shall hot-dip galvanized and of adequate height to ensure min. ground clearance of .8 m to SCB unit.

4.4 LA Mast and Foundation

4.4.1 The LA mast shall be a self-supporting structure with GI tubular pole of required height. The pole shall conform to IS: 2713.

4.4.2 The pole shall be supported on RCC pedestal and foundation structure through Baseplate & Anchor bolt assembly.

4.4.3 200 mm long, 20 dia rods shall be welded to the pole at 300 mm c/c for access to the device for maintenance purposes.

4.4.4 The support structure shall be hot-dip galvanized. The minimum depth of foundation shall be 1000 mm below GL.

TECHNICAL SPECIFICATION 33KV VACUUM CIRCUIT BREAKER WITH CONTROL & RELAY PANELS AND CURRENT TRANSFORMERS

I. SCOPE :

This specification covers the manufacture, assembly, supply and delivery at Destination stores of 33 KV (Vacuum) Out Door porcelain clad circuit breakers including current transformers, meters, relays and control panels for outdoor installations. The power system is with neutral solidly earthed. **The equipment shall be guaranteed for 5 years.**

II. APPLICABLE STANDARDS:

Unless otherwise modified in this specification, the Circuit Breakers, Current Transformers etc. shall comply with the following Standards with latest amendments.

IS	13118/1991 – High Voltage A.C. Circuit Breakers.
IEC	62271-100 or latest version – Circuit Breakers.
IS	2705 (Part I to IV/1992) – Current Transformers.
IS	2099/1986 – Bushings for Alternating Voltages.
IS	13010/2003, IS 13779/1999 – Energy Meters.
IS	3231/1986 & 87 – Relays.
IS	1248/2003 – Ammeters & Voltmeters.
IS	13947/(Pt-1993 – Degree of protection provided by Enclosures.

III. CLIMATIC CONDITIONS:

The climatic conditions under which the equipment shall operate satisfactorily are as indicated at clause No. 22 of General and Financial terms and conditions for supply of materials.

IV. PRINCIPAL PARAMETERS :

1. CIRCUIT BREAKERS	:
1.1. Nominal System Voltage	: 33KV
1.2 Type	: Vacuum porcelain clad
1.3 Service	: --- Outdoor ---
1.4 No. of Poles/Phases	: --- THREE ---
1.5 Highest System Voltage	: 36KV

- 1.6 Rated Frequency : ----- 50 Hz -----
 1.7 System of earthing : --- Neutral solidly grounded ---
 1.8 Insulation level :
 1.8.1. Lightning Impulse withstand Voltage : 170 KV peak
1.8.2 One minute power frequency : 70 KV (rms)

withstand voltage.

- 1.8.3 Power frequency withstand : --- 2 KV (rms) ---/1 minute
 voltage on Auxiliary circuit

Note : Please note that the above insulation levels supply at the reference conditions of temperature, pressure and humidity specified below :

Temperature : 20 degree C.
 Pressure : 1013 millibars
 Humidity : 11 g/m³

- 1.9 Rated Thermal Current : **1250 A**
 1.10. Rated short circuit :
 breaking current
1.10.1. Symmetrical : 25 KA
 1.10.2. Asymmetrical : As per IS : 13118/1991 or IEC – 62271-100
 or latest version
 1.11. Rating making capacity : 2.50 times of rated short circuit breaking
 Current (Symmetrical)
 1.12. Rated short time withstand
 Current for : **25 KA/3Sec**
 1.13. Total break time : **80 m.sec (Max)**
 1.14. Bushing Insulator
 Creepage distance : Not less than 900 mm
 1.15. Mounting :--- Steel Structure --- (GI)
 1.16. Rated Operating sequence : --- O-0.3 Sec. – CO– 3 Min – CO.
 1.17. Operating Mechanism : Motor operated/ Manual spring Charged. The standard DC
 Voltage for the operating devices shall be 24V DC.
 Operating Voltage for motor spring charging mechanism
 shall be 250V AC single phase. Normally the breaker shall
 be operated by Power and there shall be provision for
 manual operation.
 1.18. Terminal Connector :
 1.18.1. Material : **--- Aluminium ---**
 1.18.2. Suitable for : Panther ACSR
 1.19. Limits of temperature : The limits of temperature shall be in accordance with.
 IS : 13118/IEC- 62271-100 or latest version.
 1.20. Requirement of Simultaneity
 of Poles : The maximum difference between instants of contact
 touching during closing and the maximum difference
 between the instant of contacts separation during opening
 between 3 poles shall not exceed one half cycle of the
 rated frequency. The Breaker shall be open and close

simultaneously on all three phases for fault on any phase and or all the phases.

- 1.21 Protection : The Breaker shall be provided with
 (1) 3 Nos. Over Current (O/C) and
 (2) One Earth Fault (1 E/L) IDMT relays.
 (3) Anti pumping relay
 (4) DC Supervision relay
 (5) Trip Circuit supervision relay
 (6) Master trip relay (Hand Reset Type)

Note : Over current Relay settings ranging from 50% to 200% in steps of 25% and earth fault relay settings ranging from 20% to 80% in steps of 10%.

“The over current relays shall be Numerical draw out type and all other mechanical relays are of non-draw out type”.

The over current relays should be of reputed make with IDMT characteristics and also with high instantaneous elements for high fault currents and shall comply with IS 3231.

The relays shall have the facility of minimum 4 Nos. fault tripping data such as :

(a) Fault magnitude (b) Date (c) Time (d) Number of trippings

- 1.22 : Auxiliary Power Supply
 1.22.1 A.C. Supply : 250 V + or – 10% (Phase to neutral),
 3 phase, 50 HZ + or – 5%,
 Effectively earthed system.
 1.22.2 D.C. Supply : 24V + or – 10%, 2 wire
 ungrounded system.
 1.22.3. Supply point:
 1.22.3.1 : Auxiliary power supplies listed above will be made
 available to each circuit breaker as below :
 A.C. Supply : Single Feeder
 D.C. Supply : Single feeder
 1.22.3.2 M.C.B. shall be provided at the circuit breaker for each
 Incoming A.C. Supply. For D.C. supply double pole
 M.C.B. shall be provided. (with different colours for each
 identification).

2. CURRENT TRANSFORMERS : 33 KV

- 2.1 Rated voltage : 33 KV
 2.2. Type : --- Single phase outdoor live tank oil
 Cooled Vacuum impregnated type –
 2.3. Earthing : --- Solidly earthed ---
 2.4 Insulation level :
 2.4.1. Nominal system voltage : 33 KV
 2.4.2. Highest system voltage : 36 KV

- 2.4.3. Lighting Impulse withstand voltage : 170 KV
- 2.4.4. One minute power frequency
Withstand voltage
(a) Primary (HV) : 70 KV
(b) Secondary (LV) : 3 KV
- 2.5. Frequency : --- 50 Hz ---
- 2.6. Transformation ratio : 400-200-100/1-1A
- 2.7. Rated secondary current (Amp) :
i) Core I (Protection) : 1
ii) Core II (Metering) : 1
- 2.8. Rated output (VA):
i) Protection : 15 VA
ii) Metering : **5 VA**
- 2.9. Class of accuracy :
- 2.9.1. i) Protection (Core I) : 5P10
ii) Metering (Core II) : **0.2S**
- 2.10. Short time thermal current : 25KA/3 Sec
And its duration : The short time thermal current should
Suit the breaker rupturing capacity and
Duration to suit the maximum tripping
Time.
- 2.11. Accuracy Limit factor : --- ≤ 10 for protective core ---
- 2.12. Class of Insulation : Immersed in new insulating oil.
- 2.12.1 Limit of Temp. Rise (Max) : 55 Degree C
- 2.13. Rated Continuous Thermal Current: 1250 A

NOTE : The CTs shall be live tank outdoor type and shall be offered with first filling of Oil. The Oil shall conform to the latest IS 335.

TECHNICAL REQUIREMENTS :

I. CIRCUIT BREAKERS

1.1 DESIGN CRITERIA

The equipment will be used in High Voltage system having characteristics as listed in the specification.

The equipment will be installed outdoor in hot, humid and tropical atmosphere.

All equipment, accessories and wiring shall have tropical protection, involving special treatment of metal and insulation against fungus, insects and corrosion.

The maximum temperature in any part of the equipment at specified rating shall not exceed the permissible limits as stipulated in the relevant standards.

The equipment shall be capable of withstanding the dynamic and thermal stresses of listed short circuit current without any damage or deterioration.

The safety clearances of all live parts of the equipment shall be as per relevant standards.

1.2 TYPE AND DUTY:

The circuit breaker shall be outdoor, 3-pole, vacuum type **M2 class (minimum 10,000 operations)** in case of 11KV having internal isolation without any sequential interlock.

In the event of order, the tenderer have to offer at least one breaker tested in respect of Temperature rise test and Mechanical endurance test of 10,000 Mechanical operations (M2 Class) in presence of purchaser representative as an acceptance test.

The duty of the circuit breaker shall involve satisfactory interruption of short circuit currents as listed in the clause-IV (Principal Parameters).

The breaker shall be capable of interruption of reactive current (lagging/leading) without under/over voltage.

1.3 CONSTRUCTION FEATURES :

1.3.1 GENERAL ARRANGEMENTS :

The circuit breaker shall have fixed type construction consisting three single identical poles, complete with a gang operated mechanism for specified duty. The interrupters shall be enclosed in a sealed porcelain housing conforming to protection to IS:13947 (PT-1) equivalent to IP-65 protection (IS:2147). All three poles of circuit breaker shall be linked together electrically / mechanically to ensure simultaneous closing/tripping of all poles.

The trip free operating mechanism, 3 phase inter connection links shall be completely accommodated in the base. There shall be sufficient clearance between live parts of the circuit breakers and the ground. The circuit breaker shall be complete with operating mechanism, other accessories/materials to ensure complete assembly and proper functioning. The current

transformers shall be externally mounted on the supporting structure integrated with circuit Breaker structure. **Aluminium Terminal connectors** suitable for Panther ACSR conductor for 33 KV should be supplied for Circuit Breakers and CTs. The circuit breaker shall be provided with proper standard earthing and with terminal earth bar for earthing connection. Suitable interconnection cable from breaker terminal, CTs terminal to control relay panel is 2.5 Sq.mm. PVC copper control cable (un-armored) are to be provided.

Neither the circuit breaker nor any part of the switchgear or its supporting structures shall be permanently strained due to vibration etc. when making or breaking the rated short circuit currents.

The details of any device incorporated in the circuit breakers to limit or control the rate of restriking voltage across the circuit breaking contacts shall be stated.

The vacuum interrupter assembly used in the circuit breakers shall be interchangeable with indigenously available vacuum interrupters (make and type shall be mentioned).

Flexible laminations used in VCB shall be electrolytic copper. The ends of the laminations shall be solidified with hot pressed/electro fusion/Electro solidification method to ensure good electrical contacts and achieve minimum contact resistance.

1.3.2. INTERLOCK :

All electrical and mechanical interlocks which are necessary for safe and satisfactory operation of the circuit breaker shall be furnished.

1.3.3. AUXILIARY CONTACTS :

Each breaker shall be provided with 4 normally open and 4 normally closed electrically separate spare Auxiliary contacts in addition to those required for its own operation and indication exclusively for purchaser's use.

The auxiliary contacts shall be convertible type so that normally open contacts can be converted into normally close contacts and vice verse at site.

The auxiliary contacts shall be rated for 10A for AC and 5A for DC.

- Note : 1. 1no Spare tripping coil and 1no closing coil shall be clamped in the Breaker.
2. Spring charging multiplier with **2 NO+2NC** shall be available and shall be wired to the spare terminal blocks.
3. Auxiliary switch contact multiplier shall also be incorporated in the control panel and shall be wired to the spare terminal blocks.

1.3.4. INSULATORS :

Bushing insulators for circuit Breakers shall comply with IS2099-1986 specification for High

Voltage porcelain bushings.

Insulators shall be wet process porcelain, brown glazed and free from all blemishes. Ferrous metal parts and hardware shall be hot-dip galvanised.

Insulators shall have adequate mechanical strength and rigidity to withstand the duty involved.

When operated at maximum system voltage, there shall be no electrical discharge. Shielding rings, if necessary, shall be provided. Insulation shall be coordinated with basic impulse level of the system. The creepage distance shall correspond to heavily polluted atmosphere.

1.4. **OPERATING MECHANISM:**

The operating mechanism shall be motor operated and manual operated for spring closing mechanism with trip free features complete with shut trip coils. All three poles of the breaker shall operate simultaneously. It shall operate in principle in such a way that the closing springs after each closing operation, are automatically charged by the motor and locked in the charged position by a latch. Means shall be provided to charge the springs manually also. Provision shall be made for the slow closing of the VCB without spring charging.

The contact loading spring shall be designed in such a way that the closing stroke be completed and the opening stroke is commenced only from fully closed position. **All the breakers shall be suitable for manual operation as well as slow closing irrespective of spring charge position.**

Operation counter and mechanically operated indicator to show whether the circuit breaker is open or closed shall be provided on the circuit breaker operating mechanism.

All manually operating gear shall be so designed that the circuit breaker can be operated by one movement. The mechanism shall be such that the tripping spring can be charged while the circuit breaker is closed and the closing mechanism when charged shall not be operated by vibration caused by the circuit breaker opening under fault conditions. The spring shall be suitable for 4 short reclosing operation.

The mechanism shall be designed for electrical control from remote. Local manual close/trip (lever/button) shall be provided in the mechanism box only.

Mechanical components other than spring charging mechanism link linkages shaft etc. shall have minimum plating thickness of 10-15 microns. **The surface finishing ensures zinc plating trivalent passivation (ROHS compliance) and should withstand for salt spray test in artificial atmosphere up to 192 hours without white rust.**

All fasteners exposed to air should be of hot dip galvanized of MS 8.8 Grade for M10 and above and below M10 grade shall be of stainless steel.

The maximum power required for closing coil and shunt trip coil should not be more than 200 watts.

1.5 **CONTROL CUBICLE (MECHANISM BOX)**

A common control cubicle shall be furnished to house electricals, controls, monitoring devices and all other accessories except those which must be located on individual poles.

The cubicle shall have protection as per IS-13947 PT-1 equivalent to IP-55 protection (IS-2147) of gasketed weather proof construction, fabricated from sheet steel minimum 2.5mm thick.

The cubicle shall have front access door with lock and keys, and removable gland plate at the bottom for cable entry.

Additional locking arrangement (pad locking facility) is to be provided

1.6 WIRING & TERMINAL BLOCKS :

1.6.1 WIRING

Wiring shall be complete in all respects to ensure proper functioning of the control, protection, monitoring and interlocking schemes.

Wiring shall be done with flexible **1100 V** grade, PVC insulated switch board wires with 2.5 sq.mm. stranded copper conductor.

CT wiring. TB must be connected/disconnected type of ledger type 12 way and control wiring terminal ledger must be 16A open type

All CT wiring must be done with 2.5mm/ 4 mm dia ring type lugs only.

Each wire shall be identified at both ends with permanent markers bearing wire numbers as per wiring diagram. The wiring schematic may conform to relevant standards.

Wire termination shall be done with crimping type connectors with insulating sleeves. Wire shall not be spliced between terminals.

All spare contacts of relays, push buttons, auxiliary switchers etc. shall be wired upto terminal blocks in the control cubicle.

1.6.2. TERMINAL BLOCKS :

Terminal blocks shall be **1100 V grade**, box clamp type (Nut & Bolt Type).

Not more than two wires shall be connected to any terminal. Spare terminals equal in number to 20% active terminals shall be furnished.

Terminal blocks shall be located to allow easy access. Wiring shall be so arranged that individual wires of an external cable can be connected to constructive terminals.

For CTPT – disconnected type terminal blocks are required.

1.7 NAME/RATING PLATE:

BREAKERS : Each circuit-breaker shall be provided with a name plate or plates legibly and indelibly marked with atleast the following information:

- a) Name of manufacturer.
- b) Type of designation and serial number.
- c) Rated Voltage and current.
- d) Rated frequency.
- e) Rated symmetrical breaking capacity.
- f) Rated making capacity and
- g) Rated short-time current and its duration.
- h) P.O.No. with Date
- i) Period of guarantee

(which shall be either one or three seconds).

- Note : 1. The word "Rated" need not appear on the name plate, recognised abbreviations may be used to express the above quantities.
- 2. When the circuit breaker is fitted with closing and/or tripping devices necessitating any auxiliary supply the nature of the auxiliary supply shall be stated either on the circuit breaker name-plate or in any other acceptable position.
 - 3. The purchase order No. and date and year of supply and the words TSSPDCL must be etched on the name plate.

CONTROL & RELAY PANELS:

2.1 CONSTRUCTION :

The Control panels to house meters, relays and other items as specified in clause VI(3) shall

be weather proof and vermin proof and of rust-free pressed steel cubicle type with hinged door and locking device. Ventilation louvers shall be provided with GI mech. The frame shall be made of angle iron or structural steel of sufficient weight and strength to ensure alignment. The control panel shall be provided with inter connections, small wiring leads, terminal bolts, fuses, earth bar, multi core cable glands, earth connections etc, The panel should be provided with locking handle with built-in door lock and lock shall be provided with duplicate keys.

The outdoor panels shall be preferable of the following dimensions :-

Height	1750 mm
Width	600 mm
Depth	600 mm

The exterior of the panels shall be painted with dark admiral grey colour and interior should be painted with half white colour. The panel shall be provided with fuse units/MCBs for AC Circuit (both phase and neutral) for all phases of potential Circuit (Potential coil connections to energy meters etc.) DC Control Circuit etc. The internal wiring of the panels shall be with 2.5 sq.mm 1100 V grade standard copper PVC Insulated wiring of reputed make.

The Gauge of the sheet steel for the front of the panels (where the meters etc. are fixed) and supporting members of the panels shall be of 3 mm thick and for the other members of the panel sheet steel thickness shall be 2 mm.

Wherever control panels are fixed to the breaker structure necessary clearance from the ground to read the meters and to operate the switches shall be maintained. The suitable interconnection cable from breaker terminal, CTs terminal to control relay panel is 2.5 Sq.mm. PVC copper control cable (un-armored) shall be provided.

2.2 PAINTING:

Before painting all non-galvanized parts shall be completely cleaned and made free of rust, scale and grease and all external rough surface cavities on casting shall be filled by metal deposition. It shall be ensured that Phosphate coating is done by a suitable process on both the external and internal surfaces. The painting to be done on surface shall be by powder coating process

The interior parts and internal structural steel work shall be cleaned of all scale and rust by sand blasting or other approved method.

All metal enclosures shall be treated in 7 tank Pre-treatment process & should be painted with UV Resistant Pure Polyester Powder coating. The powder coated sheet steel fabrication shall fulfill 700Hrs of Salt spray test. The thickness of painting/powder coating shall be of 70-90 microns to withstand tropical heat and extremes of weather. The paint shall be guaranteed for 5 years from the date of receipt of material.

CURRENT TRANSFORMERS :

3.1 CONSTRUCTION

The core shall be high grade non-ageing electrical silicon-laminated steel of low hysteresis loss and high permeability to ensure high accuracy, at both normal and over current.

The secondary terminals shall be brought into a compartment on one side of current transformer for easy access. The Secondary terminal shall be provided with short circuiting arrangements. The secondary taps shall be adequately reinforced to withstand normal handling, without damage.

The current transformers shall be suitable for mounting on steel structures or concrete pedestals. The necessary flanges, bolts, etc, for the base of the Current Transformer shall be supplied and these shall be galvanised. The current transformer tank and other metal parts shall be galvanised.

The primary of the current transformers shall have double cotton covered roped conductors with insulation by Kraft paper. The secondary shall be super enameled copper wire. Details of winding and core including sectional drawings shall be furnished.

The Current Transformers shall be complete in all respects with filling of oil conforming to IS:335 and with oil level indicator with minimum/maximum oil levels. **The top cover and terminal box cover should be such that rain water does not enter even through the gaskets.**

The cover of the CT should be designed to avoid the stagnation of water. The creepage distance should be 300mm (min).

The minimum clearance required between poles in air required should be 300mm.

Note : Facility for selecting the CT ratio in the control panel by closing the appropriate links in the Control and Relay Panel shall be provided. In no case changing of CT ratio with primary links shall be provided.

3.2. PRIMARY & SECONDARY TERMINALS: Primary terminals of Current Transformers to which the line connections are to be made shall have dimensions as per IS: 10601:1983 and **material shall be of tinned cooper.**

The secondary terminals shall be brought out into suitable compartment, which shall have a removable cover. The terminal box with the cover closed and tightened and the cable/conduit in position when supplied shall have a degree of protection conforming to IP 54 of IS: 2147.**The secondary terminals will be M6 Tinned Brass studs.**

3.3. TERMINAL AND EARTH CONNECTORS: Terminal connectors suitable for Panther ACSR Conductor shall be supplied. Suitable earth connectors for earthing connections shall also be supplied.

Thickness of the clamp must be minimum of 12mm and the stud clamp will be bimetallic.

3.4. EARTHING: The assembly comprising of the chassis, frame work and the fixed parts of the metal casing of the CT, shall be provided with two separate earthing terminals. The earthing terminals shall be adequate size protected against corrosion and metallicity clean and identified by means of the sign marked in a legible and indelible manner on or adjacent to the terminals.

3.5. SEALING ARRANGEMENT: Provision for sealing secondary terminal compartment, primary ratio change strips (if any) and tank effectively shall be made such that no fraud etc. such as tampering of the ratio or circuit (current) is possible. The holes provided for the above sealing provision shall be of adequate size and pass the sealing wire of about 14 SWG.

3.6. Each instrument Transformer shall be provided with prismatic type oil sight window at suitable location so that the oil level is clearly visible with naked eye to an observer standing at ground level.

3.7. For compensation of variation in volume of the oil due to temperature variation nitrogen cushion or stainless steel bellows shall be used. Rubber diaphragms shall not be permitted for this purpose.

3.8. The units shall be vacuum filled with oil, after processing and thereafter hermetically sealed to eliminate air and moisture from entering the tank.

3.9 NAME/RATING PLATE :

Each Current Transformer shall have the following particulars indelibly marked on it or on a label permanently secure to it or its casing.

- a) Manufacturer's Name :
- b) Manufacturer's Sl.No. and /or type of designation:
- c) Rated transformation ratio :
- d) Rated Frequency
- e) Highest system voltage
- f) Insulation level and
- g) Rated short time thermal current with the associated rate time and rated dynamic current.

The Purchase Order No. and date and year of supply and the words "TSSPDCL" must be etched on the name plate.

VI. SCHEDULE OF EQUIPMENT

1. CIRCUIT BREAKERS :

Vacuum Circuit Breaker complete with suitable painted steel support structure (with anchor bolts & nuts) for mounting 1 No. circuit breaker – 3 poles. Mechanism box, control and relay panel and Current Transformers.

- Note :
- 1. Earth strips as per IS shall be provided for proper earthing of equipments.
 - 2. Earth bar of copper (suitable for termination of 2 nos. 40 * 6 mm flats) shall be provided on circuit breaker support structure.
 - 3. Connecting Cable from Breaker to Control panel and Breaker to CTs are to be provided.
 - 3 (i) Connecting Cable from Breaker to CTs 6 Core cable of 5 mtrs length from each CT i.e., total 3 x 5 = 15 mtrs / Breaker and from Breaker to Control panel 10 core cable of 15 mtrs length.
 - 3 (ii) Size of interconnection cable from breaker terminal, CTs terminal to control relay panel is 2.5 Sq.mm. PVC copper control cable (un-armored).

2. MECHANISM BOX CONTAINING :

- a) Operating mechanism
- b) Mechanical indicator for "ON" and "OFF" coupled to the Circuit breaker operating mechanism.

- c) Mechanical close and trip (with protective flap) lever/push button.
- d) Terminal blocks for control wiring and a spare terminal block (with 20% of the active terminals) (preferably nut & bolts type)..
- e) Operation Counter.
- f) Operating handle for manual charging of springs and for slow closing.**
- g) 2 Nos. cable glands over and above those provided for control cables with suitable dummies.
- h) Not less than **6 Nos. normally open and 6 Nos. normally lose spare auxiliary contacts** over and above those required for normal operation.
- i) 230 V single phase AC Motor/Manual operated spring charging mechanism complete with electrical spring release coil, 2 Nos. shunt trip coil and 1 No. closing coil.
- j) Local – Remote selector switch.
- k) Earth bar (suitable for termination of 2 Nos. 50 x 6 mm class)
- l) **6 Nos. Terminal connectors suitable for panther conductor for incoming and outgoing connector for outdoor VCB.**
- m) Set of 2 pole MCBs for AC and DC supply with different colours.
- n) **1 No. of reputed make anti-pumping relay. (If offered breaker is with anti-pumping mechanism plug-in relay is acceptable).**
- o) TNC Switch heavy duty piston grip type.

3. CONTROL AND RELAY PANEL :

Weather proof vermin proof control and relay panel shall be either structure mounted / floor mounted having equipped as follows:

- a) 3 ½ Digit LED Display CT operated digital A.C. Ammeter **with Class of Accuracy 0.5** to suit the CT ranges indicated in Section-IV Schedule of materials. (110V AC Auxiliary supply).
- b) One ammeter selector switch with "R", "Y", "B", "Neutral" and "OFF" position of size (96x96) mm.
- c) One Heavy duty Pistol grip type Control Switch for VCB trip/Neutral/Close positions with spring return to neutral position.
- d) One Red and one Green pilot LED lamps for indicating close and open positions respectively.
- e) One Yellow LED lamp for healthy trip indication with push button control.
- f) One white LED lamp for spring charged indication.
- g) One amber LED Lamp for auto-trip indication.
- h) One 3 phase 4 wire static HT Trivector CMRI Compatible as per TSSPDCL standards **class 0.2S** accuracy suitable scaled to work on 110 V AC 50 Hz 1 Amp to suit the CT ranges indicated in Section-IV Schedule of materials.
- i) Numeric Relays shall be of reputed make (drawn out type) with the following features.
 - a) IDMT relay with three poles arranged for O/L and one pole for E/L with high set instantaneous elements for high fault currents for 33 KV Circuit Breakers.

- j) Two Nos. cable glands over and above those provided for control CT, PT, Control Circuit cables and auxiliary supplies (AC&DC) with suitable dummies.
- k) 3 ½ Digit LED Display digital A.C. Voltmeter with selector switch (Seven position) for 33KV Circuit Breakers (110V AC Auxiliary supply) of size (96x96) mm.
- l) One 3 Phase Mw meter and one 3 phase MVAR Meter to work on 110V AC 50 Hz 1 Amp and scaled to suit the ranges of CTs. **(Size (144x144 mm))**
- m) One set alarm bell and push button for acceptance of alarm. **Audio alarms shall be static Hooters. Voltage rating shall be 24V DC.**
- n) Necessary LT Fuses/MCBs for Control Circuit. Set of fine wiring with ferrules with standard code number of respective terminals and with suitable terminal connectors.
- o) **230 V** three pin socket with plug & switch.
- p) 3Ph 4wire Test terminal block with CT and PT terminals with screw type for testing ammeter and energy meter.**
- q) Master trip relay of HR (fast acting)
- r) Trip circuit supervision & DC fail supervision relays are required.
- s) Stud terminal blocks (Bolt and nut type) for CT and PT control cables suitably wired with 6 Nos. spare terminals.
- t) One suitable copper earth strip of 12 x 3 mm size with adequate number of holes with suitable nuts and bolts.

- NOTE :**
1. All the instruments and relays to be provided on control panel shall be flush mounted unless otherwise specified. The relays are to be worked on 1 amp secondary current of CTs and with DC Voltage of 24 V
 2. All the indicating meters (Ammeter, Voltmeter, MW meter, MVAR meter) shall be of 96 x 96 mm and 144 x 144 mm or standard size and preferable of reputed make.
- u) 1 No. LED for TC1, through TS Relay, 1 No. LED for TC2 through TS relay.
 - v) 1 No. plug point with switch (10A, 250 VAC)
 - w) 1 No. Illuminator lamp with switch (40W, 250 VAC)
 - x) 1 No. panel space heater with heater switch thermostat.

4. CURRENT TRANSFORMERS.

Three numbers outdoor CTs as specified.

5. Terminals connectors suitable for Panther ACSR conductor for 33 KV and panther ACSR conductor for 11KV shall be supplied for Circuit Breaker (6 Nos.) and CT terminals (6 Nos.) Suitable inter connection between circuit breaker terminals and CT terminals are to be provided. Cables from circuit breaker and CTs to control panel shall be provided. As specified in the Section VI Schedule of equipment.

Note : Other standard accessories which are not specifically mentioned but supplied with breakers of similar type and rating for efficient and trouble free operation shall be provided.

SECTION - VIII

LIST OF APPROVED MAKES

List of approved makes of Electrical materials to be used for the work

S.No.	Materials	Make
1	Polycrystalline PV Modules	as per ALMM (approved list of manufacturers by the MNRE).
2	AC & DC cables	Polycab/ Havels/ Universal / KEI
3	MCBS/Isolators	L&T / Schreder / Siemens.
4	MCB Distribution boards	L&T / Schreder / Siemens.
5	Power Transformers	As approved by HGCL
6	Connectors	Elmex/ Connect well/ Raychem/ Andrew Yule
7	PVC FRLS / HFFR insulated flexible Copper cable	Polycab/ Finolex / RR kabel/ Gloster Cable
8	A.B. Switch/H.G. Fuse Set/Lightning arrestor/Insulators	TSSPDCL approved make
9	GI pipes/ GI wire / GI flat	TATA/JINDAL with ISI marked
10	HT termination kits	Reychem/M seal/Denson/Multy
11	HDPE pipes	Sudhakar,Dura Line (Dura guard) or its equivalent make
12	Powe contactors	L&T make ML type contactors only
13	UPVC	Ashirwad/finolex/Sudhakar
14	Inverters	As per BIS approved make
15	HT Switch Gear	Siemens/ S&S/ABB as approved by HGCL
16	ICGU	As approved by HGCL

Note: For all the Electrical items the successful bidder has to submit three makes for each item for Source approval by HGCL

SECTION – IX

(Special conditions for O&M)

- 1 Operation & maintenance shall be for period of 5 years from the date of successful commissioning of the project is deemed to be included in the prices quoted by the bidder. No extra amount shall be payable to the contractor on account of O&M.
- 2 The contractor shall be paid for O&M on quarterly basis shown under special conditions.
- 3 The O&M shall necessarily mean managing all kind of day to day operations and maintenance of entire Solar Power plant including supply of all kind of materials, and cleaning etc. Contractor shall arrange qualified man power, tools & tackles, equipment's, spares, statutory clearances, transportation to staff etc. without any extra cost implications to the client. O&M staff shall work as per the instructions of the Engineer.
- 4 The contractor shall depute qualified Electrical engineer cum supervisor having minimum 3 years of relevant experience Skilled electricians, helpers in adequate numbers to meet the requirement of O&M. The contractor shall engage only license holder electricians and helper. Necessary safety equipment shall be provided by the contractor to his employees/Staff.
- 5 The Contractor shall have to give the list of names of the personnel to be engaged for the work with their pass port size photographs, certificates & qualifications proof etc. The list will have to be updated once in every six months.
- 6 The Contractor shall fulfill all provision under various acts like contract labour act, minimum wages act, workman compensation act, Indian Electricity act etc. as applicable to such work. Any financial implication arising due to breach of any act shall be fully borne by the contractor. The contractor shall also be liable to pay service tax or any other taxes to Govt. for carrying out the O&M work.
- 7 The Contractor shall deploy adequate numbers of mobile ladders/cranes of suitable heights for O&M purpose.
- 8 The Solar panels and switch gear which are out of order shall have to be repaired within 48 hours shall attend the same immediately and inform to the Engineer. The Contractor shall also however, for critical locations as defined by the Engineer, corrective actions need to be taken up within 24 hours. In case of any electrical shock/ emergency/theft, the Contractor lodge police complaint, if required. The Contractor shall be fully responsible for any damage and /or loss of life, accident occurring during the work.

- 9 All required maintenance spares such as controllers, contactors, wires, cables, junction box, cable jointing kits, hard ware, etc. all kinds of T&P etc. shall be in the scope of supply of the contractor. HGCL shall not provide any materials at all. The Contractor shall be required to maintain good aesthetics of the equipment's like Solar panels, switch gear etc.
- 10 The Contractor shall provide office accommodation with communication to his engineers/Supervisors. The Maintenance staff of the contractor shall be available round the clock on all days including holidays/Sundays.
- 11 After successful completion of 5 years of O&M period, The Contractor needs to hand over the complete installation to the Engineer in good working condition. At the time of handing over, The Contractor needs to submit complete item wise inventory as given in the priced BOM format at the time of Bid submission.
- 12 To watch the day to day functioning of Solar PV modules and to maintain day to day power Generation record and the same has to be Submitted to HGCL. The daily reports, weekly and monthly reports has to be prepared and submitted accordingly.
- 13 To attend the day to day complaints as per the routine maintenance schedule and to see that there should not be any dark spot at any place over the ORR.
- 14 To attend day to watch over the Solar Power plant and to see that there should not be any loose connections and any problem noticed with MCBs/MCCBs/relays should be attended immediately.
- 15 To keep patrol vehicle along with O&M staff as per the specifications of the agreement on shift duties so as keep Solar power plant in to work in condition at all times.
- 16 To pour the water in to the earth pits at the transformer yards, including D.P. Structure and to see that earth resistance is maintained as per the IE rules and there should not be any earth fault at any place.
- 17 To coordinate with civil O&M agency and Toll agency and to see that there should not be any power supply interruptions and to attend the complaints noticed by the both civil O&M agency and Toll agency immediately.
- 18 To coordinate with TSCEIG authorities during their inspections and to see that there all observations have to be attended as per IE rules immediately and necessary compliance report have to be furnished through Licensed Electrical Contractor under intimation to the both The Engineer and HGCL.
- 19 To conduct periodical check over the underground cable along the Solar Power plant and to test IR values and to see that there should not be any cable failures and cable end terminations.

- 20 To maintain record over the time to time Solar Power Generation and to intimate the same to the Engineer and HGCL.
- 21 To ensure that only genuine and good quality with ISI marked of materials are used during the course of replacement of worn out parts and break downs.
- 22 To keep the crane with skilled operator and to see that there should not be any traffic hindrance during the break downs and necessary road safety precautions have to be followed in consultation with civil O&M agency and Toll agency.
- 23 To keep sufficient stock of required electrical consumables at the site stores so to avoid all possible delays in attending the complaints.
- 24 To arrange periodical inspection of transformers and switch gear.
- 25 To furnish the day to day record, periodical checks record to the Engineer and HGCL during their inspections and to obtain their counter signatures in token of verification. No payment can be considered in the absence of above record.
- 26 It is the complete responsibility of the agency to interact with local authorities and pay necessary charges towards ESI and PF contributions regularly and No payment can be considered in the absence of above record.
- 27 The separate window should provide on the dash board to show in Scada day to day Generation and their load amps and voltage should be recorded every hour. The report has to be generated on hourly interval, remarks has to be recorded if any load current increasing/ decreasing. The same should be submitted to HGCL on daily basis.
- 28 In the event of mal function of automation system during construction period or O&M period the double the excess charges will be recovered from the contractor.

SECTION – X

(Special conditions for Payment of O&M)

- 1 The quoted rate is for finished item of work shall be inclusive of transportation, vehicles, labour, etc. except GST. The same will be allowed with bill as per the prevailing govt rules

Payment in respect of O&M Works (Part-B):

- 2 The payment during O&M will be considered as under after successful completion of each quarter subject to fulfillment of service level requirements. Daily reports, weekly reports and monthly reports are to be generated and to be submitted regularly. In case of deficiencies in service levels, the penalties will be levied as stipulated and the balance amount will be considered for payment.

Period	Payment
During 1 st year	Quarterly Basis as per O&M BOQ
During 2 nd Year	Quarterly Basis as per O&M BOQ
During 3 rd Year	Quarterly Basis as per O&M BOQ
During 4 th year	Quarterly Basis as per O&M BOQ
During 5 th year	Quarterly Basis as per O&M BOQ

SECTION – XI

Service levels

Service Level Requirements:

Net Minimum Guarantee Generation:

ESTIMATED SOLAR POWER GENERATION

Client : HGCL

Project : 13MW AC (16MW DC) Grid Connected Solar Power Plant Location : 02 Stretches of ORR Cycling Track, Hyderabad, Telangana.

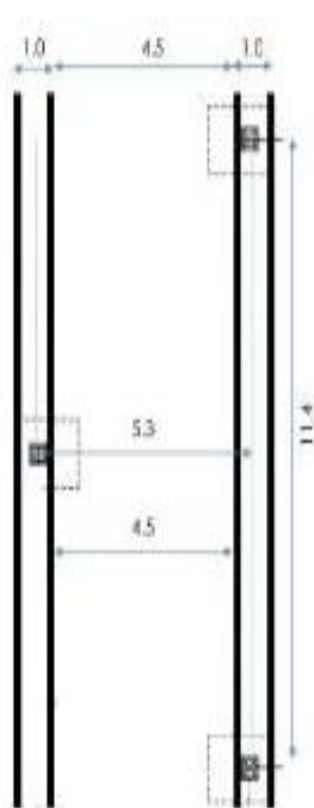
(Net Minimum Guarantee Generation)

End of year	Yearly Degradation MWH (Modules and system)- Degradation Consider in PV System Generation data	Generation units per KW per Annum	Generation units per MW per Annum
1	1%	1401.6	1401600
2	1%	1387.58	1387580
3	1%	1373.71	1373708
4	1%	1359.97	1359971
5	1%	1346.37	1346371

In case of not meet in the above generation, for the short fall units the contractor has to pay as per the prevailing tariff of TSSPDCL to the HGCL or the same may be deducted from the O&M bills.

SECTION – XII

DRAWINGS

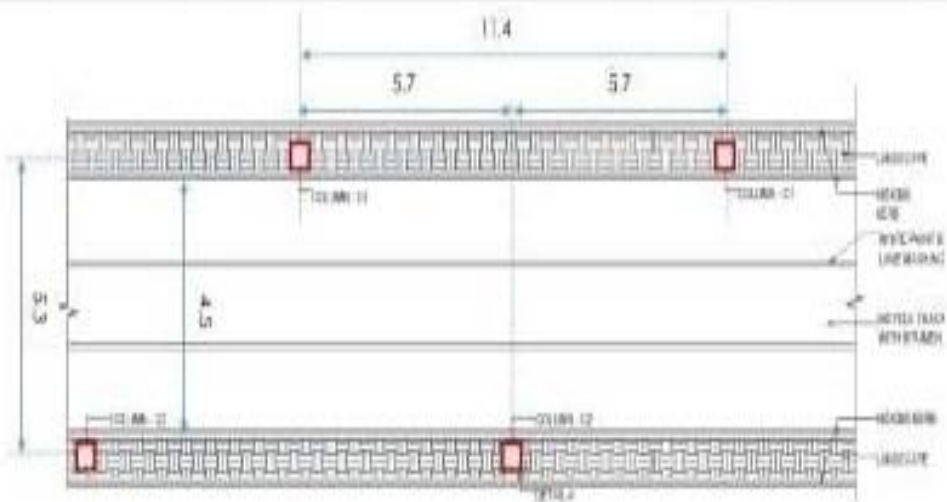


Track Details

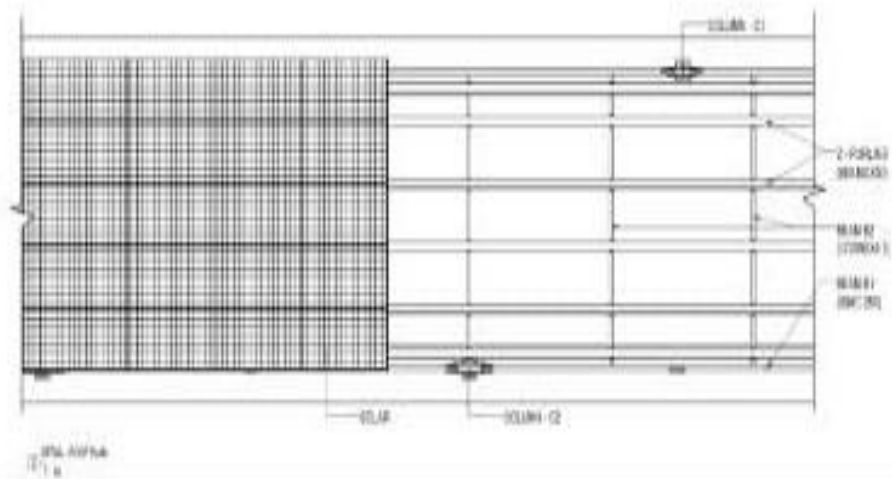


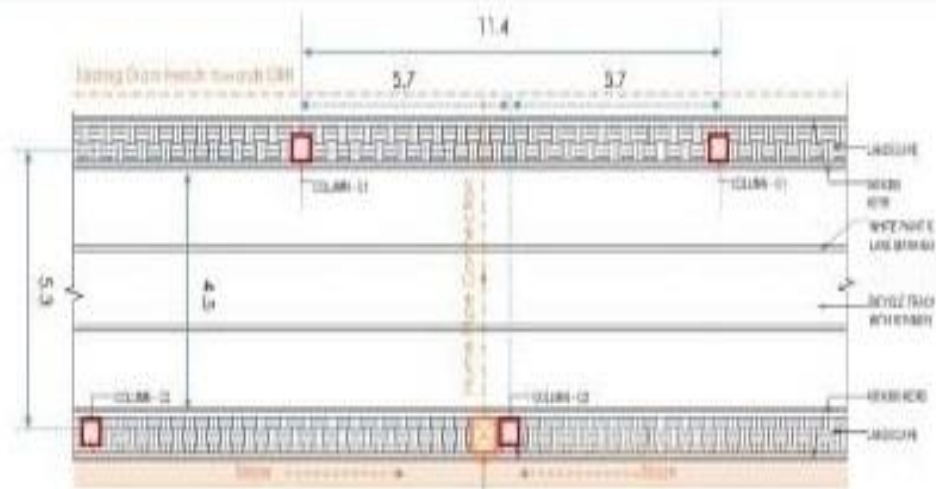
Pedestal

- Track Clear Width : 4.5 Mt
- Column c/c : 5.3 Mt
- Distance Between 2 Columns : 11.4 Mt

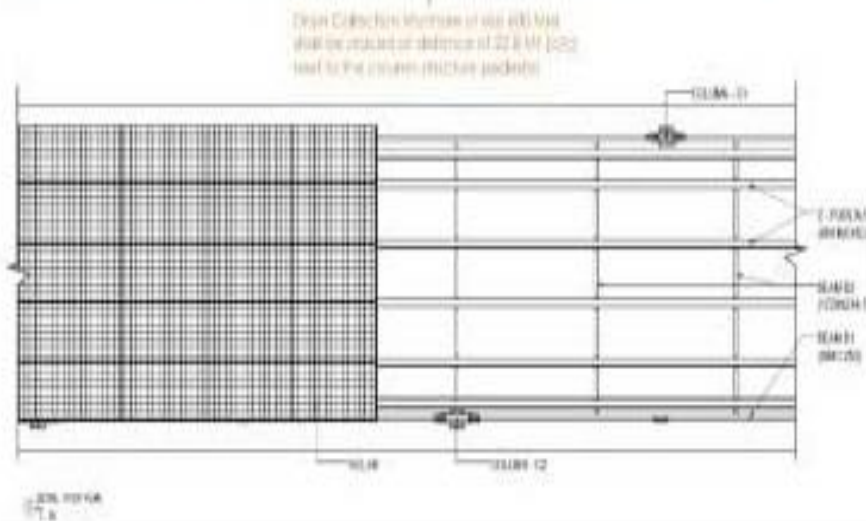


- Track Clear Width : 4.5 m
- Column C1 : 5.3 m
- Distance Between 2 Columns : 11.4 m
- Column Section : 115.5 x 115.5 x 8 mm
- Column C1 H (Final) : 3350 mm
- Column C2 H (Large) : 4000 mm
- Beam B1 : SMC 250
- Beam B2 : 172 X 92 X 4.5 mm
- 2 Purlins : 65 x 100 x 50 mm

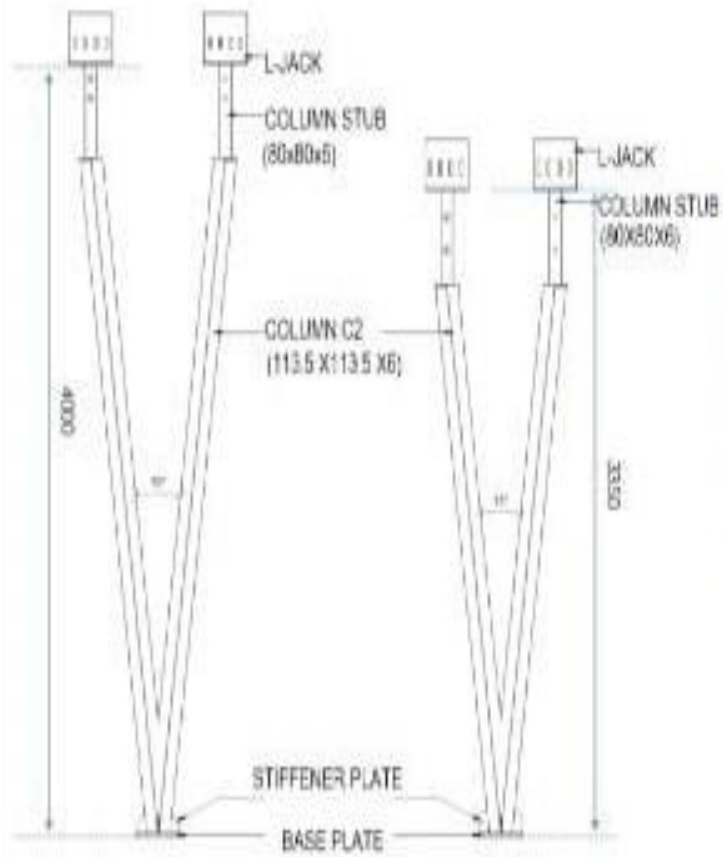




- Track Clear Width : 4.5 M
- Column c/c : 5.3 M
- Distance Between 2 Columns : 11.4 M
- Column Section : 1'3.5 x 1'3.5 x 4 MM
- Column C1 (H (small)) : 3350 MM
- Column C2 (H (large)) : 4000 MM
- Beam B1 : (SAC 250)
- Beam B2 : 172 X 92 X 4.5 MM
- 2 Furlis : 65 x 100 x 50 MM



Column



SECTION – XIII
BILL OF QUANTITIES
PREAMBLE

1. The Bill of Quantities shall be read in conjunction with the instructions to Tenderers, General and Special conditions of Contract Technical Specifications and Drawings.
2. The quantities given in the Bill of Quantities are estimated and provisional and are given to provide common basis for tendering. *The quantities here given are those upon which the lumpsum tender cost of the work is based but they are subject to alterations, omissions, deductions or additions as provided for in the conditions of this contract and do not necessarily show the actual quantities of work to be done.* The basis of payment will be actual quantities of work ordered and carried out as measured by the Contractor and verified by the Engineer and valued at the estimate rate plus or minus tender percentage quoted in the Bill of Quantities where applicable, and otherwise at such rates and prices as the Engineer-in-Charge may fix within the terms of Contract.
3. The estimate rates in the Bill of Quantities shall, except in so far as it is otherwise provided under the Contract include cost of all constructional material, labour, machinery, transportation, erection, maintenance, profit, taxes and duties together with all general risks, liabilities and obligations set out or implied in the Contract.
4. The plans enclosed with the tender are liable to be altered during execution of work as per necessity of site conditions. The Tender percentage quoted by the Tenderer shall hold good for execution of work even with altered plans.
5. The whole cost of complying with the provisions of the Contract shall be included in the estimated rates for items provided in the Bill of Quantities and where no items are provided in the Bill of Quantities, their cost shall be deemed to be distributed among the estimate rates entered for the related items of work.
6. General directions and descriptions of work and materials are not necessarily repeated nor summarised in the Bill of Quantities. References to the relevant sections of the Contract documentation shall be made before entering estimate rate against each item in the Bill of Quantities.
7. The method of measurements of completed work for payment shall be in accordance with the relevant B.I.S. Codes & A. P. S. Specifications.
8. All items of work are to be executed as per the drawings / specifications supplied with the contract documents. If there is any contradiction between the drawings and the text of the specifications, the later shall prevail.
9. The Tenderer should inspect and select the quarries of his choice before he quotes the tender percentage in the Schedule of Bill of Quantities and satisfy himself about the availability of required quantum of materials.
10. Diversion drains should be excavated before completion of the embankments and the useful soils should be used in the nearby embankments.

11. The actual mix proportion by weight to be adopted during execution will be got designed in the laboratories to suit the grade of concrete and mortar to be used. It will be the responsibility of the contractor to manufacture concrete and mortar of required strength.
12. The quantum of measurement for all items of earthwork involving conveyance manually or by machinery shall be as assessed by level measurement. The measurements for the embankment will be for the consolidated banks only.
13. Wherever bailing out of water is involved either for excavate ion or for foundations or for constructions, the percentage quoted shall take into account the de-watering charges necessary. No separate payment will be made for de-watering.
14. Wherever embankment work is involved, useful soils approved by the Engineer-in-Charge from the cutting reaches and diversion drains shall be taken and used for forming nearby embankments soils used for constructions will be at free of cost.
15. The quoted tender percentage shall also include the work of any kind necessary for the due and satisfactory construction, completion and maintenance of the works according to the drawings and these specifications and further drawings and orders that may be issued by the Engineer-in-Charge from time to time. The quoted tender percentage shall include compliance by the Contractor with all the general conditions of contract, whether specifically mentioned or not in the various clauses of these specifications, all materials, machinery, plant, equipment, tools,
fuel, water, strutting, timbering, transport, offices, stores, workshop staff, labour and the provision of proper and sufficient protective works, diversions, temporary fencing and lighting. It shall also include safety of workers, first aid equipment suitable accommodation for the staff and workmen, with adequate sanitary arrangements, the effecting and maintenance of all insurance, the payment of all wages, salaries, fees, royalties / Taxes, duties or other charges arising out of the execution of works and the regular clearance of rubbish, reinstatement and clearing-up of the site as may be required on completion of works safety of the public and protection of the works and adjoining land. The work of Building in quality control / assurance shall be deemed to be covered in the quoted percentage.
16. The Contractor shall ensure that, the quoted tender percentage shall cover all stages of work such as setting out, selection of materials, selection of construction methods, selection of equipment and plant, deployment of personnel and supervisory staff, quality control testing etc. The work quality assurance shall be deemed to be covered in the tender percentage.
17. a) The special attention of the Tenderer is drawn to the conditions in the tender notices wherein reference has been made to the Telangana state Standard Specifications [APSS] and the Standard preliminary specifications containing therein. These preliminary specifications shall apply to the agreement to be entered into between the contractor and the Government of Telangana and shall form an in-separable condition of the contract along with the estimate. All these documents taken together shall be deemed to form one contract and shall be complimentary to another.
b) The Tenderer shall examine, closely the T.S.S.S. / MORT&H and also the standard preliminary specifications contained therein and sign the Chief General Manager's office copy of the TSSS / MORT&H and its addenda volume in token of such study before submitting his overall tender percentage which shall be for finished work in-situ. He shall also carefully study the drawings and additional specifications and all the documents, which form part of the agreement to be entered into by the successful Tenderer. The TSSS / MORT&H and other documents connected with contract such as estimate plans,

specifications, can be seen on all working days in the office of the Chief General Manager, HGCL.

18. The Tenderers attention is directed to requirements for materials under the clause 'materials and workmanship' in the preliminary specifications of APSS. Materials conforming to the Bureau of Indian Standards specifications, TSSS etc., shall be used on the work and the Tenderers shall quote his overall tender percentage accordingly.
19. The Tenderer has to do his own testing of materials and satisfy himself that they conform to the specifications of respective I.S.I. Codes before tendering.
20. The contractor shall himself procure the required construction materials of approved quality including the earth for formation of embankment and water from quarries / sources of his choice. All such quarries / sources of materials required for the work shall be got approved by the Engineer-in-Charge in writing well before their use of the work.
21. The contractor shall himself procure the steel, cement, Bitumen, Blasting materials, sand, metal, soils, etc., and such other materials required for the work well in advance. The contractor has to bear the cost of materials for conveyance. The department will not take any responsibility for fluctuations in market in cost of the materials, transportation and for loss of materials etc.
22. Inspection of site and quarries by the Tenderer: Every Tenderer is expected before quoting his overall tender percentage, to inspect the site of proposed work. He should also inspect the quarries and satisfy himself about the quality, and availability of materials. The best class of materials to be obtained from quarries, or other sources shall be used on the work. In every case the materials must comply with the relevant standard specifications. Samples of materials as called for in the standard specifications or in this tender notice, or as required by the Executive Engineer, in any case, shall be submitted for the Executive Engineer's approval before the supply to site of work is begun.
23. The Tenderer's particular attention is drawn to the sections and clauses in the T.S. standard specification dealing with
 - a) Test, inspection and rejection of defective materials and work.
 - b) Carriage
 - c) Construction plant
 - d) Water and lighting
 - e) Cleaning up during the progress and for delivery.
 - f) Accidents
 - g) Delays
 - h) Particulars of payments.

The contractor should closely peruse all the specification clauses, which govern the overall tender percentage he is tendering.

24. The defect liability period of contract in terms of GO Ms.No.8, T,R&B Dept., dt:8.1.2003 is twenty four months.
25. The estimate rates for items shown in the Schedule "A" include all construction materials. No escalation in rates will be paid unless specified in the tender document. The Tenderer has to quote an overall tender percentage considering all the aspects of the tender to complete the finished item of work as per the TSSS / MORT&H / B.I.S. specifications, the special specifications appended, Drawings etc.

26. If there is any contradiction between TSSS / MORT&H and B.I.S. specifications, listed and detailed technical specifications, the latter shall prevail.
27. In case of a job for which specifications are not available with the Schedule or in APSS / MORT&H or B.I.S. code and are required to be prescribed, such work shall be carried out in accordance with the written instructions of the Engineer-in-charge.
28. The contractor should use the excavated useful soils and stone for construction purpose. Soils used for construction either for homogeneous section in hearting or in casing zone based on the suitability will be at free of cost and the cost of stone used for construction purpose will be recovered from the contractor's bill.
The contractor should quote his tender percentage keeping in view of the above aspects.
29. Additions and alternations by the Tenderer in the Schedule of quantities will disqualify the tender.
30. In the case of discrepancies between the written description of the item in the Schedule "A" and the detailed description in the specification of the same item, the latter shall be adopted.
31. The Unit rates noted below are those governing payment of extras or deductions for omissions according to the conditions or the contract as set-forth in the preliminary Specifications of the T.S. standard specifications and other conditions of specification of this contract.
32. It is to be expressly understood that the measured work is to be taken according to the actual quantities when in place and finished according to the drawings or as may be ordered from time to time by the Deputy General Manager and the cost calculated by measurement or weight at their respective rates without any additional charge for any necessary or contingent works connected herewith. The Percentage Excess or less on ECV quoted are for works in situ and complete in every respect.
33. For all items of work in excess of the quantities indicated the rates payable for such excess quantities will be tendered rates i.e., estimate rates plus or minus tender percentage.
34. For all items of work, intermediate payment will be made provisionally as per relevant clause. Full-accepted agreement rates will be paid only after all the items of works are completed.
35. The contractor is bound to execute all supplemental works that are found essential incidental and inevitable during execution of main work.
36. The contractor has to arrange to &pro flight tickets and accommodation with lodging, boarding & local travelling for inspection of material for all materials at manufacture /procurement place for three persons from HGCL including consultant from his residence/requested place to inspection places for each inspection .
37. Bill of Quantities called Schedule "Part-A & Part -B " and the bid offer accompanies the tender document as Volume-II. The Bill of Quantities contains not only the quantities but also the rates worked out by the Department and the amount for each item and total value of the estimated contract.
38. The tenderer should workout his own rates keeping in view the work, site conditions and quote his overall tender percentage with which he intends to execute the work.
39. The payment of rates for supplement items of work will be regulated as under.

A) Extra Items:

- a. Extra items of work shall not vitiate the contract. The contractor shall be bound to execute extra items of work as directed by the Engineer - in – Charge. The rates for extra items shall be worked out by the Deputy General Managers per the conditions of the Contract and the same are binding on the Contractor.
- b. The contractor shall before the 15th day of each month, submit in writing to the Deputy General Manager a statement of extra items if any that they have executed during the preceding month failing which the contractor shall not be entitled to claim any.

B) Entrustment of additional items:

- i. Where ever additional items not contingent on the main work and outside the scope of original agreement are to be entrusted to the original contractor dispensing with bids and if the value of such items exceeds the limits up to which the officer is empowered to entrust works initially to contractor without calling for tenders, approval of **next higher authority** shall be obtained. Entrustment of such items on nomination shall be at rates not exceeding the estimated rates.
- ii. Entrustment of the additional items contingent on the main work will be authorised by the officers up to the monetary limits up to which they themselves are competent to accept items in the original agreement so long as the total amounts up to which they are competent to accept in an original agreement rates for such items shall be worked out in accordance with the procedure (I) For all items of work in excess of the quantities shown in the Bill of Quantities of the Tenders, the rate payable for such items shall be estimate rates for the items (+) or (-) over all tender percentage accepted by the competent authority.
- iii. Entrustment of either the additional or supplemental items shall be subject to the provisions of the agreement entered into by a Competent Authority after the tender is accepted. The Chief General Manager being the higher authority who entered into the agreement approves the rate for the items / variation in quantity in the current agreement. The items shall not be ordered by an officer on his own responsibility if the revised estimate or deviation statement providing for the same requires the sanction of higher authority.

Note: It may be noted that the term Estimate Rate used above means the rate in the sanctioned estimate with which the tenders are accepted, or if no such rates is available in the estimate, the rate derived will be with reference to the Standard Schedule of Rates adopted in the sanctioned estimate with which tenders are accepted.

STATEMENT - I

Work Programme of Achieving the Mile stones:

Mile stone No.	Period from the date of signing the agreement	Minimum percentage of work to be completed (Cumulative)
For the Milestone-1	End of 2 months	Completion of SITC of Stretch-1 ie., from Nanakramguda to TSPA Junction
For the Milestone-2	End of 3 months	SITC of two stretches ie., Completion of 100% of work

BILL OF QUANTITIES**[Part-I]**

Name of work Given in the Estimate:

S.No	Approximate Quantity In figures/ words	Description of work	Specification No. / APSS / BIS / MORT&H	Unit In figures / words	Estimate Rate In figures / words	Amount in Rs.
		Schedule - A Enclosed				

BILL OF QUANTITIES

Part-II

- i) **The rates mentioned in Schedule “A” are inclusive of the following including seigniorage charges etc and also Contractors over heads, profit and other provisions i.e., overhead and contract profit at 13.615%. The overhead charges includes**
- Site accommodation, setting up plant, access road, water supply, electricity and general site arrangements.
 - Office furniture, equipment and communications
 - Expenditure on Corporate office of contractor
 - site supervision
 - Documentation and “as built” drawings
 - Mobilisation/ de-mobilisation of resources
 - Labour camps with minimum amenities and transportation to work sites.
 - Light vehicles for site supervision including administrative and managerial requirements.
 - Laboratory equipment and quality control including field and laboratory testing.
 - T & P and survey instruments and setting outworks, including verification of line, dimensions, trial pits and bore holes, where required.
 - Watch and ward
 - Traffic management and providing hydra with bucket during core cutting on bridge side wall at the height of 15mts on ground level.
 - Expenditure on safeguarding environment
 - Sundries
 - Financing Expenditure
 - Sales/ Turn over tax
 - Work Insurance/ compensation
- ii) The Reimbursable amounts are as follows
GST will be paid as per the rules in vogue

PRICE BID
[to be placed in the COVER -‘B’]

Name of work Given in the Estimate: Design, Engineering, Supply, Construction, Erection, Testing & Commissioning of cumulative 13 MW (AC) Solar PV Power Plant on rooftop of Cycle track at 2 stretches along the ORR with 5 years Plant O&M.

Certificate ESTIMATED CONTRACT VALUE (IN FIGURES&WORDS) : **Rs.99.71/- crores**
(Rupees Ninety Nine Crores and Seventy One Lakhs only)

I Sri / Smt./M/s. do
hereby express my willingness to execute the aforesaid work as per the conditions,
standards, specifications, rules, regulations, etc., stipulated in the tender documents.

a) at an overall tender percentage of (in
figures).....
.....(in words) **excess / less** over estimated value.

OR

b) at estimate value.

SIGNATURE, NAME OF THE TENDERER / AUTHORISED SIGNATORY.

SECTION - XIV

**Any Other Documents Listed as
forming part of the contract**